

DILATED HANKEL DETERMINANTS

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ABSTRACT. For a sequence $\mathbf{a} = (a_0, a_1, \dots)$ we define its *dilated Hankel determinant* $\ddot{H}_n(\mathbf{a}) = \det(a_{2i+j})_{0 \leq i, j \leq n-1}$, the minor of the infinite Hankel matrix (a_{i+j}) formed from the even-indexed rows and the first n columns. We prove that, for a broad class of sequences, $\ddot{H}_n$ admits a remarkably simple product evaluation. This mirrors the behaviour of the classical Hankel determinant H_n , but with two key distinctions: the class of sequences for which such formulas are known is far larger in the classical case; and, whereas H_n enjoys a single universal evaluation — the Heilmann formula via the Jacobi continued fraction — no analogous general method exists for the dilated determinant, which is therefore considerably more challenging. Our evaluations instead rest on six reduction principles developed here, four of general scope and two of a more specialised nature. The cases treated include the factorial numbers, the Catalan and central binomial coefficients; the Euler numbers and a one-parameter secant family; the involution numbers; the Springer numbers along with elliptic and derivative deformations; the reciprocal-sine function, whose evaluation rests on a new Catalan determinant proved by condensation; a Bessel analogue of the Euler numbers; and a multiplicative Bessel family. As an application, we settle a conjecture of Chapoton and the author on the roots of the Poupard and Kreweras polynomials.

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1. INTRODUCTION

Let $\mathbf{a} = (a_0, a_1, a_2, \dots)$ be a sequence over a field, and let $f(x) = \sum_{n \geq 0} a_n x^n / n!$ be its exponential generating function. The (*classical*) *Hankel determinant* of order n is

$$H_n = H_n(f) = H_n(\mathbf{a}) = \det(a_{i+j})_{0 \leq i, j \leq n-1} = \begin{vmatrix} a_0 & a_1 & a_2 & \cdots & a_{n-1} \\ a_1 & a_2 & a_3 & \cdots & a_n \\ a_2 & a_3 & a_4 & \cdots & a_{n+1} \\ \vdots & & & & \vdots \\ a_{n-1} & a_n & a_{n+1} & \cdots & a_{2n-2} \end{vmatrix}. \quad (1.1)$$

Hankel determinants are a meeting point of orthogonal polynomials, the moment problem, continued fractions, total positivity and enumerative combinatorics. When the a_k are the moments of a linear functional μ , the determinants H_n are the Gram determinants of the monomial basis; their positivity governs the classical moment problem and the total positivity of the Hankel matrix [43, 39, 36], and the theorem of Heilermann and Stieltjes evaluates them through the Jacobi continued fraction of the generating function [20, 47, 46, 12, 26, 27, 15]. Combinatorially, H_n counts families of non-intersecting lattice paths [30, 13, 46, 34]; Flajolet’s theory [12] reads the continued fraction off weighted Motzkin paths and gives, in one stroke, the continued-fraction expansions — hence the Hankel determinants — of the Catalan, Bell, Stirling, Euler, tangent and secant numbers, and of the elliptic functions. The Jacobi continued fraction, however, exists only when all $H_n \neq 0$; as soon as some H_n vanish it breaks down, and one turns instead to the *Hankel* continued fraction [17, 16], which is defined for every sequence.

A recurring theme of the subject is the contrast between sequences whose Hankel determinants have a *closed product form* — a formula of the NICE(n) type $\xi^n \text{Rat}(n) \prod_i (a_i n + b_i)^{\pm 1}$ in Krattenthaler’s sense [27] — and those whose determinants carry sporadic large prime factors and admit no such formula. The first class is recognised by its arithmetic, the prime factors of H_n growing only linearly in n ; it is this “niceness” that combinatorialists prize and that the determinant calculus [26, 27] is designed to certify. Which sequences fall on which side is a subtle matter, and it is this question that the present paper takes up — for a Hankel-type determinant of a new kind.

Definition 1.1. The *dilated Hankel determinant* of order n of the sequence $\mathbf{a} = (a_0, a_1, \dots)$ is

$$\ddot{H}_n = \ddot{H}_n(f) = \ddot{H}_n(\mathbf{a}) = \det(a_{2i+j})_{0 \leq i, j \leq n-1} = \begin{vmatrix} a_0 & a_1 & a_2 & \cdots & a_{n-1} \\ a_2 & a_3 & a_4 & \cdots & a_{n+1} \\ a_4 & a_5 & a_6 & \cdots & a_{n+3} \\ \vdots & & & & \vdots \\ a_{2n-2} & a_{2n-1} & a_{2n} & \cdots & a_{3n-3} \end{vmatrix}, \quad (1.2)$$

with the convention $\ddot{H}_0 = 1$.

More generally, replacing the row index i by ri gives the r -step *Hankel determinant*

$$\det(a_{ri+j})_{0 \leq i, j \leq n-1};$$

the classical Hankel determinant (1.1) is the case $r = 1$, and the *dilated* one the case $r = 2$. This paper is devoted to $r = 2$. We call it the *dilated Hankel determinant*, or simply the *dilated determinant*, throughout, and reserve the name *2-step* for the rare occasions when it must be set against the general r -step determinant (Section 3.1).

Two readings of (1.2) place it within known theory. First, $\ddot{H}_n$ is the *minor of the infinite Hankel matrix* $(a_{p+q})_{p, q \geq 0}$ that selects the even-indexed rows $\{0, 2, 4, \dots, 2n-2\}$ and the first n columns; general minors of Hankel matrices fall under the determinant calculus of Krattenthaler [26, 27]. Second, if $a_k = \mu(z^k)$ are moments then $a_{2i+j} = \mu(z^{2i} \cdot z^j)$, so

$$\ddot{H}_n = \det(\langle z^{2i}, z^j \rangle_\mu)_{0 \leq i, j \leq n-1}, \quad \langle p, q \rangle_\mu := \mu(pq), \quad (1.3)$$

the Gram determinant of the bilinear form $\langle \cdot, \cdot \rangle_\mu$ on the two bases $\{1, z^2, z^4, \dots\}$ and $\{1, z, z^2, \dots\}$. The factor 2 in a_{2i+j} is thus the quadratic map $z \mapsto z^2$ together with the even/odd splitting of μ — the realm of the quadratic decomposition and symmetrisation of orthogonal polynomials [5] and of biorthogonal and multiple orthogonal polynomials [32, 21]. The closely related but distinct *shifted* Hankel determinants $\det(a_{i+j+k})$ of [17] translate the index by a constant rather than scaling it; the interaction of the two operations — shifts of a dilated determinant — is itself a source of clean evaluations (Sections 8, 14, 22 and 24).

Results. Our central finding is that, for a wide range of classical sequences, $\ddot{H}_n$ has a surprisingly simple product formula even when the ordinary H_n does not. We organise the evaluations into a few families.

- The *Beta family* $a_{m+1}/a_m = \rho(m+\alpha)/(m+\beta)$ (Section 4) — the sequences whose term ratio has degree less than or equal to 1 — has a single closed form (Theorem 4.1) specialising to the factorials, the Catalan numbers, the central binomial coefficients and the double factorials. The resulting dilated Hankel determinants, such as the Catalan values 1, 3, 32, 1232, 172032, $\dots$ and the central binomial values 1, 8, 224, 22528, 8200192, $\dots$, appear not to have been recorded before.
- The involution numbers and the Gaussian family $\exp(cx+bx^2)$ give $\ddot{H}_n = (4bc)^{\binom{n}{2}} \prod_{k < n} k!$ (Theorem 5.1, Section 5); the Gaussian is the unique exponential weight for which this simplification occurs.
- The *Euler number family* $1/\cos(x)^{s+1} + \int_0^x dy/\cos(y)^{t+1}$ (Proposition 7.1, Section 7), whose $(s, t) = (0, 1)$ member has the Euler (secant–tangent) numbers [2] as its moments, has a closed product evaluation for every admissible (s, t) , even though the relevant odd moment functional is not classical.
- The *secant-number family* $(1+x)/\cos(x)^{s+1}$, a separate family, has the product evaluation $\ddot{H}_n = c_n \prod_{i=1}^{n-1} (s+1)_i$ valid for *every* s (Theorem 12.1, Section 12), even though its odd moment functional is not classical once $s \geq 1$.
- The *Springer number family* $1/(\cos x - t \sin x)^r$ [40, 39], whose $t = 1, r = 1$ member has the Springer numbers as its moments, a derivative $(\cos x + \sin x)/(\cos x - \sin x)^s$ of it, and an elliptic Euler family all have product formulas (Theorems 16.1, 17.1, 19.1).
- An *algebraic family* $(1+x)/(1-x^2)^{s/2}$ (Theorem 20.1, Section 20) also has a product formula for every s , even though *neither* of its two moment functionals is classical; the proof accordingly abandons orthogonal polynomials altogether in favour of an elementary contiguous relation in s .
- A *rank-one perturbation* $(\sin x + 1)/\cos^2 x + s \sin x$ of the Euler number family (Proposition 15.1, Section 15) factors for every s , with $\ddot{H}_n$ affine in s ; at the member $s = -1$, where it becomes $(1 + \sin^3 x)/\cos^2 x$, the seemingly sporadic prime factors 29, 37, 23 at $n = 8, 9, 10$ turn out to be $\binom{n}{2} + 1$.
- The *reciprocal-sine function* $(1+x)x/\sin x$ (Theorem 18.1, Section 18) is evaluated as a product of factorials — the deepest evaluation of the paper. Its two moment functionals turn out to be Wilson functionals differing in a *single* parameter; the connection

coefficients of the biorthogonal reduction then collapse to a single hypergeometric term, and the evaluation reduces to a determinant of products of two Catalan numbers, established by Desnanot–Jacobi condensation (Theorem 18.8) — a determinant evaluation of independent interest.

- A *Bessel (s, t) family* $\text{cosb}_s(x) + \int_0^x \text{cosb}_t(y) dy$ (Theorem 23.1, Section 23), built from the normalised Bessel function $\text{cosb}_s = \Gamma(s+1)(2/x)^s J_s(x)$, has a closed product evaluation for *every* (s, t) : the signed factor of the Euler number family reappears at doubled argument, so the determinants are nonzero exactly on the half-integer offsets $t - s \in \frac{1}{2} + \mathbb{Z}$ and vanish eventually on the integer ones — the Bessel ladder walks in half steps.
- A *multiplicative Bessel family* $(1+x)\text{cosb}_\nu^2$ (Theorem 25.4, Section 25), the Bessel analogue of the secant-number family with the exponent ladder replaced by the order ladder, has a closed product evaluation at every *even* order $n = 2N$, complete with quarter-integer zeros — the blocks $(2\nu + \frac{1}{2})_k$ — that have no trigonometric counterpart. Although its even moment functional is not classical, Watson’s product formula makes the even moments a single hypergeometric term, and the evaluation falls, unexpectedly, to the elementary Vandermonde reduction, whose only other application is the Beta family; at odd orders the degree count fails by one and an irreducible carrier appears.

Origin and application. As is so often the way with determinant evaluations [26], the object of this paper first announced itself in the course of another problem. While studying a conjecture of Chapoton and the author on the roots of the Poupard and Kreweras polynomials [4], the author was confronted with a determinant that did not yield to the ordinary continued-fraction machinery and whose first row was the secant-type sequence $1, 1, 1, 3, 5, 25, 61, 427, \dots$ of $(1+x)/\cos x$; this clue is what led to the dilated determinant studied here. We close the circle in Section 26: Conjecture 5.4 of [4] is, after a change of basis and up to an explicit power of two, the dilated Hankel determinant of $(1+x)/\cos x$ (Theorem 26.3), whose evaluation (Proposition 11.1) settles it.

Methods. The classical Hankel determinant enjoys a *universal* method of evaluation: the Jacobi continued fraction of the generating function produces the whole sequence $H_1, H_2, \dots$ at once, from a single list of coefficients [20, 43, 47]. For the dilated determinant there is no such tool. $\ddot{H}_n$ is in general a substantially harder object: no one method evaluates every family. We therefore develop *six* reductions of $\ddot{H}_n$, collected and abstracted in Section 3 and tagged **M1**–**M6** throughout the paper. Four of them run through the paper: the elementary Vandermonde (row-factorisation) reduction **M1**; the biorthogonal reduction **M2**, which splits **a** into its even and odd moment functionals — the determinantal face of the even/odd contraction of continued fractions [47, 12] — and reduces $\ddot{H}_n$ to a determinant of connection coefficients; the one-functional reduction **M3**; and the divisor method **M4**, which evaluates a one-parameter family by its degenerations through a dilated Cauchy–Binet expansion. Each of these four comes with a sharp rigidity boundary marking where the product form must fail. The remaining two reductions are of a different, more specialised character, each used only once rather than running through the paper: the algebraic family (Section 20) is settled by a contiguous relation (**M5**), and the rank-one perturbation (Section 15) by the matrix-determinant lemma (**M6**). Finally, the reciprocal-sine function of Section 18 is reached by **M2** followed by a new finishing step: the connection determinant turns out to be a determinant of products of Catalan numbers, which we evaluate by Desnanot–Jacobi condensation; the same finishing step closes the Bessel (s, t) family of Section 23, whose connection determinant carries the parameters in its entries. In the opposite direction, the multiplicative Bessel family of Section 25 is closed by **M1** alone — its even moments are a single hypergeometric term, and at even orders the row factorisation lands exactly on the Vandermonde degree bound: the one evaluation of the paper where the most elementary method reaches beyond the Beta family. The classical background needed — orthogonal polynomials, the moment problem, J - and S -fractions and their contractions — is gathered, in self-contained algebraic form, in Section 2.

Organisation. Section 2 fixes notation and recalls the classical toolbox; Section 3 presents the six reductions **M1**–**M6** and closes with a table matching methods to families. The evaluations

then proceed family by family: the Beta family (Section 4); the Gaussian family — containing the involution numbers — and its derivative rule (Sections 5 and 6); the Euler number family with its double and single shifts (Sections 7–10); the secant-number family $(1+x)/\cos(x)^{s+1}$ with its shifts (Sections 11–14); the rank-one perturbation of the Euler number family (Section 15); the Springer number family and a derivative of it (Sections 16–17); the reciprocal-sine case (Section 18); the elliptic Euler family (Section 19); the algebraic and squared algebraic families with their shifted determinants (Sections 20–22); the Bessel (s, t) family with its double shift (Sections 23 and 24); and the multiplicative Bessel family at even orders (Section 25). Section 26 settles the Chapoton–Han conjecture, and Section 27 collects, family by family, the explicit corollaries and their shifts. Section 28 then sketches the Lindström–Gessel–Viennot point of view — a natural method which, for the dilated determinant, yields no evaluation and which no proof of this paper uses; it is recorded for orientation only. Section 29 records the classical Hankel determinants H_n of several of the moment sequences studied here. Finally Section 30 gathers the concluding remarks and open problems. All closed forms were discovered and checked with SageMath.

2. ALGEBRAIC PRELIMINARIES: ORTHOGONAL POLYNOMIALS AND CONTINUED FRACTIONS

This section collects the classical material we use, in the self-contained algebraic form suited to formal power series; no analysis or positivity is needed. Standard references are [44, 5, 21] for orthogonal polynomials, [47, 12, 46, 26] for continued fractions and Hankel determinants. A reader fluent in this material may skip to the reductions of Section 3 and refer back as needed.

Conventions. Throughout the paper, $m!! = m(m-2)(m-4)\cdots$, down to 1 or 2, is the double factorial, with $0!! = (-1)!! = 1$; empty products equal 1; $(\alpha)_m = \alpha(\alpha+1)\cdots(\alpha+m-1)$ is the Pochhammer symbol; and $\binom{\alpha}{d} = 0$ for $d < 0$. For sequences given by a generating function we use the *exponential* convention $f(x) = \sum_{n \geq 0} a_n x^n / n!$, i.e. $a_n = n! [x^n] f$.

2.1. Moments, the Hankel determinant, and orthogonal polynomials. A *linear functional* $\mathcal{L}$ on the polynomial ring $\mathbb{Q}[y]$ (more generally over any field) is fixed by its values on the monomial basis,

$$\mu_k := \mathcal{L}[y^k] \quad (k \geq 0),$$

the *moments* of $\mathcal{L}$. The *Hankel determinant* of the moment sequence is

$$H_n := \det(\mu_{i+j})_{0 \leq i, j \leq n-1} = \det(\mathcal{L}[y^i \cdot y^j])_{0 \leq i, j \leq n-1}, \quad H_0 = 1,$$

i.e. the Gram determinant of $1, y, \dots, y^{n-1}$ under the symmetric bilinear form $\langle p, q \rangle := \mathcal{L}[pq]$. We assume $\mathcal{L}$ is *quasi-definite*, $H_n \neq 0$ for all n ; this is exactly the condition under which the Gram–Schmidt process never breaks down.

Under this hypothesis there is a unique sequence of *monic orthogonal polynomials* $P_0 = 1, P_1, P_2, \dots$ with $\deg P_n = n$ and

$$\mathcal{L}[P_n P_m] = h_n \delta_{nm}, \quad h_n := \mathcal{L}[P_n^2] \neq 0,$$

where h_n is the *squared norm* of P_n (a formal one — no positivity is assumed, so h_n need not be a square; we abbreviate it to *norm* throughout). They are the Gram–Schmidt orthogonalisation of $1, y, y^2, \dots$, and have the explicit determinantal form

$$P_n(v) = \frac{1}{H_n} \begin{vmatrix} \mu_0 & \mu_1 & \cdots & \mu_n \\ \mu_1 & \mu_2 & \cdots & \mu_{n+1} \\ \vdots & & & \vdots \\ \mu_{n-1} & \mu_n & \cdots & \mu_{2n-1} \\ 1 & y & \cdots & y^n \end{vmatrix}.$$

Expanding along the last row shows P_n is monic of degree n , and $\mathcal{L}[y^j P_n] = 0$ for $j < n$ because the determinant then has a repeated row; this is orthogonality. Taking $\mathcal{L}$ of $y^n P_n$ and comparing minors gives the basic bridge between the norms and the Hankel determinants,

$$h_n = \frac{H_{n+1}}{H_n}, \quad \text{equivalently} \quad H_n = \prod_{k=0}^{n-1} h_k \quad (h_0 = \mu_0). \quad (2.1)$$

Thus *evaluating a Hankel determinant is the same as computing the norms h_k of the orthogonal polynomials*, and the latter are read off a continued fraction (Section 2.4).

Orthogonal expansion. Since $\deg P_n = n$, the set $\{P_0, \dots, P_n\}$ is a basis of $\mathbb{Q}[y]_{\leq n}$, the polynomials of degree $\leq n$. Hence every R with $\deg R \leq n$ has a unique expansion $R = \sum_m r_m P_m$, and pairing against P_m recovers the coefficient,

$$\mathcal{L}[R P_m] = r_m h_m, \quad \text{i.e.} \quad r_m = \frac{\mathcal{L}[R P_m]}{h_m}. \quad (2.2)$$

2.2. Change of family. Every reduction in this paper replaces the row and column families of a moment matrix by monic families of the same degrees. That this leaves the determinant unchanged is the following elementary but pivotal fact, which we isolate once and cite throughout.

Lemma 2.1 (change of family). *Let $\mathcal{L}$ be a linear functional on a space of polynomials, and let $(R_i)_{0 \leq i < n}$ and $(C_j)_{0 \leq j < n}$ be two families of polynomials. If $(\tilde{R}_i)$ and $(\tilde{C}_j)$ are unitriangular recombinations,*

$$\tilde{R}_i = R_i + \sum_{k < i} p_{ik} R_k, \quad \tilde{C}_j = C_j + \sum_{l < j} q_{jl} C_l,$$

then

$$\det(\mathcal{L}[\tilde{R}_i \tilde{C}_j])_{0 \leq i, j < n} = \det(\mathcal{L}[R_i C_j])_{0 \leq i, j < n}.$$

In particular, if the matrix $(\mathcal{L}[\tilde{R}_i \tilde{C}_j])$ is triangular, then $\det(\mathcal{L}[R_i C_j]) = \prod_{i=0}^{n-1} \mathcal{L}[\tilde{R}_i \tilde{C}_i]$.

In every application the new families are monic of strictly increasing degree—typically the orthogonal polynomials of the relevant functional—hence unitriangular recombinations of the monomial (or split-monomial) families they replace.

2.3. The three-term recurrence and Favard's theorem. Because yP_n has degree $n+1$, expansion (2.2) writes it in the basis $P_0, \dots, P_{n+1}$,

$$yP_n = \sum_{m=0}^{n+1} r_m P_m, \quad r_m = \frac{\mathcal{L}[yP_n P_m]}{h_m}.$$

For $m < n-1$ we have $\deg(yP_m) = m+1 < n$, so $\mathcal{L}[yP_n P_m] = \mathcal{L}[P_n yP_m] = 0$ by orthogonality; hence $r_m = 0$ for all $m < n-1$. This leaves a *three-term recurrence*

$$P_{n+1} = (y - c_n) P_n - \lambda_n P_{n-1}, \quad P_0 = 1, \quad P_{-1} = 0, \quad (2.3)$$

with

$$c_n = \frac{\mathcal{L}[y P_n^2]}{h_n} \quad (n \geq 0), \quad \lambda_n = \frac{h_n}{h_{n-1}} \quad (n \geq 1).$$

In particular $\lambda_n \neq 0$, and telescoping $\lambda_n = h_n/h_{n-1}$ with (2.1) gives the norms and the Hankel determinant directly from the recurrence coefficients,

$$h_n = h_0 \prod_{k=1}^n \lambda_k, \quad H_n = \prod_{k=0}^{n-1} h_k = \mu_0^n \prod_{k=1}^{n-1} \lambda_k^{n-k}. \quad (2.4)$$

Conversely, *Favard's theorem* states that any monic sequence defined by a recurrence of the shape (2.3) with all $\lambda_n \neq 0$ is orthogonal for a (unique up to scale) quasi-definite functional. So the data (c_n, λ_n) and the moment data (μ_k) determine each other.

2.4. Continued fractions: the J-fraction and S-fraction. The translation between moments and recurrence coefficients is most compact as a continued fraction. The *Jacobi continued fraction* (J-fraction) of $\mathcal{L}$ is

$$\sum_{k \geq 0} \mu_k x^k = \frac{\mu_0}{1 - c_0 x - \frac{\lambda_1 x^2}{1 - c_1 x - \frac{\lambda_2 x^2}{1 - \ddots}}}, \quad (2.5)$$

where c_n, λ_n are *exactly* the recurrence coefficients of (2.3); this is the theorem of Stieltjes [47, 12], and with (2.4) it is the standard route to closed-form Hankel determinants [20]. Combinatorially

μ_n is the generating polynomial of weighted Motzkin paths of length n , a level step at height k weighted c_k and a fall from height k (paired with the matching rise) weighted λ_k [12, 46].

When $\mu_0 = 1$, the series often admits the narrower *Stieltjes continued fraction* (S-fraction):

$$\sum_{k \geq 0} \mu_k x^k = \frac{1}{1 - \frac{b_1 x}{1 - \frac{b_2 x}{1 - \frac{b_3 x}{1 - \ddots}}}}, \quad (2.6)$$

with coefficients $b_1, b_2, \dots$; here μ_n counts weighted Dyck-type paths.

2.5. The even contraction $S \rightarrow J$. The S-fraction (2.6) and the J-fraction (2.5) are *different* continued fractions — one has only first-order terms $b_j x$, the other both $c_n x$ and $\lambda_n x^2$ — yet they can expand the *same* power series. Indeed a series given by an S-fraction also has a J-fraction: the *even contraction* of (2.6) is, by definition, the continued fraction whose sequence of convergents is the even-indexed subsequence $\mathcal{C}_0, \mathcal{C}_2, \mathcal{C}_4, \dots$ of the convergents $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2, \dots$ of (2.6); it turns out to be a J-fraction. The b_j are therefore not themselves recurrence coefficients, but determine the c_n, λ_n through this contraction, which yields

$$\sum_{k \geq 0} \mu_k x^k = \frac{1}{1 - b_1 x - \frac{(b_1 b_2) x^2}{1 - (b_2 + b_3) x - \frac{(b_3 b_4) x^2}{1 - \ddots}}}}, \quad (2.7)$$

or uniformly $c_n = b_{2n} + b_{2n+1}$, $\lambda_n = b_{2n-1} b_{2n}$ ($n \geq 1$), $\lambda_0 = 0$ [47, 12]. Equation (2.4) then gives the norms in the clean telescoped form

$$h_n = \prod_{k=1}^n \lambda_k = \prod_{k=1}^n b_{2k-1} b_{2k} = \prod_{j=1}^{2n} b_j, \quad H_n = \prod_{i=0}^{n-1} \prod_{j=1}^{2i} b_j. \quad (2.8)$$

2.6. The odd contraction. The *odd contraction* of (2.6) is, by definition, the continued fraction whose sequence of convergents is the odd-indexed subsequence $\mathcal{C}_1, \mathcal{C}_3, \mathcal{C}_5, \dots$ of the convergents $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2, \dots$ of (2.6); it yields

$$\sum_{k \geq 0} \mu_k x^k = 1 + \frac{b_1 x}{1 - (b_1 + b_2) x - \frac{(b_2 b_3) x^2}{1 - (b_3 + b_4) x - \frac{(b_4 b_5) x^2}{1 - \ddots}}}},$$

which is not a J-fraction. We observe, however, that the shifted moment sequence does have a J-fraction [47, 12],

$$\sum_{k \geq 0} \mu_{k+1} x^k = \frac{b_1}{1 - (b_1 + b_2) x - \frac{(b_2 b_3) x^2}{1 - (b_3 + b_4) x - \frac{(b_4 b_5) x^2}{1 - \ddots}}}},$$

with

$$\mu'_0 = b_1, \quad c'_n = b_{2n+1} + b_{2n+2} \quad (n \geq 0), \quad \lambda'_n = b_{2n} b_{2n+1} \quad (n \geq 1). \quad (2.9)$$

The shifted moment sequence $(\mu_{k+1})_{k \geq 0}$ is the *Christoffel transform* of $\mathcal{L}$ by the variable y : if $\mathcal{L}'[P] := \mathcal{L}[y P]$ denotes the functional obtained by multiplying $\mathcal{L}$ by y , then $\mathcal{L}'[y^k] = \mathcal{L}[y^{k+1}] =$

μ_{k+1} . Thus the odd contraction is exactly the passage from the S -fraction of $\mathcal{L}$ to the J -fraction of its Christoffel transform $\mathcal{L}'$, and the squared norms of the latter telescope to

$$h'_n = h'_0 \prod_{k=1}^n \lambda'_k = b_1 \prod_{k=1}^n b_{2k} b_{2k+1} = \prod_{j=1}^{2n+1} b_j, \quad H'_n = \prod_{i=0}^{n-1} \prod_{j=1}^{2i+1} b_j, \quad (2.10)$$

where the factor $h'_0 = \mu'_0 = b_1$ must be kept since the Christoffel sequence is not normalised ($\mu'_0 = b_1 \neq 1$), unlike the even case (2.8). The Christoffel transform recurs in the shifted determinants of Sections 9, 13, 14 and 24. There the sequence is advanced by one index, $a_k \mapsto a_{k+1}$; on the functional side this is exactly the passage from $\mathcal{L}$ to its y -multiple $\mathcal{L}' = \mathcal{L}[y \cdot]$, since $\mathcal{L}'[y^k] = \mathcal{L}[y^{k+1}] = a_{k+1}$.

3. SIX REDUCTIONS OF THE DILATED HANKEL DETERMINANT

Six reductions of $\ddot{H}_n$ are developed in this paper. Four of them run through it, listed here from the most elementary to the most structured; each evaluates the dilated determinant under a different structural hypothesis on $\mathbf{a}$, and each has a boundary beyond which the product form fails; Subsection 3.5 collects, in tabular form, the hypotheses, the collapse conditions and the applications of all the methods.

- M1.** The *Vandermonde reduction* (Section 3.1) factors each row and uses no functional at all; it applies to the sequences whose term ratio has degree ≤ 1 (the Beta family) and, through the dilation, to the even orders of the multiplicative Bessel family (Section 25), whose *even* moments have a hypergeometric term ratio of degree $(2, 2)$.
- M2.** The *biorthogonal reduction* (Section 3.2) splits $\mathbf{a}$ into its even and odd functionals $\mathcal{S}, \mathcal{T}$ and reduces $\ddot{H}_n$ to a determinant of connection coefficients; it is exact whenever both are quasi-definite, and collapses to a product when the connection coefficients are themselves explicit — notably when $\mathcal{S}$ and $\mathcal{T}$ belong to a single classical family and differ in one parameter, so that the connection array is a single hypergeometric term (see *The collapse mechanism* at the end of Section 3.2).
- M3.** The *one-functional reduction* (Section 3.3), the mechanism of Cigler and Krattenthaler [6, 29], uses the orthogonal polynomials of the single functional $\mathcal{E}$ on z ; the Gaussian is the only Appell family for which it collapses.
- M4.** The *divisor method*, with the dilated Cauchy–Binet lemma (Section 3.4), treats $\ddot{H}_n(f_\theta)$ as a polynomial in a parameter and locates its zeros at the degenerations of f_θ .

Methods **M1** and **M4** are elementary; **M2** and **M3** rest on the orthogonal-polynomial preliminaries of Section 2. We tag results by the method used throughout the paper.

The remaining two reductions are of a more specialised character, each used only once rather than running through the paper, and are tagged accordingly.

- M5.** The *contiguous-relation method* (Section 20) applies when neither moment functional is classical: it relates $\ddot{H}_n$ at a parameter s to $\ddot{H}_n$ at $s-2$ by an elementary column operation, solves the resulting relation by a periodicity argument, and fixes the constant at a single special value. No orthogonal polynomials are used.
- M6.** The *rank-one/matrix-determinant-lemma method* (Section 15, Remark 3.12) applies to a rank-one perturbation of a family already evaluated by another method: the perturbed determinant is the unperturbed one times an explicit scalar, computed by the matrix-determinant lemma rather than by the divisor method **M4**.

3.1. Row factorisation and the Vandermonde reduction (M1). The most elementary reduction uses no functional, no positivity and no continued fraction. It evaluates the *generalised* Hankel determinant $\det(a_{x_i+j})$ with *arbitrary* row indices $x_0, x_1, \dots, x_{n-1}$ — the dilated determinant $\ddot{H}_n$ being the case $x_i = 2i$, and the r -step Hankel determinant $\det(a_{ri+j})$ the case $x_i = ri$ — whenever the rows of the matrix separate into a row-dependent prefactor times a polynomial in the row index.

Lemma 3.1 (Vandermonde reduction). *Let (a_m) be a sequence over a field K , let $n \geq 1$, and suppose there are a scalar function $x \mapsto w(x)$ and polynomials $Q_0, \dots, Q_{n-1} \in K[x]$ with $\deg Q_j \leq n-1$ such that*

$$a_{x+j} = w(x) Q_j(x) \quad (0 \leq j \leq n-1) \quad (3.1)$$

for every row index x used below. Then, for any $x_0, \dots, x_{n-1}$,

$$\det(a_{x_i+j})_{0 \leq i,j \leq n-1} = \det M \prod_{0 \leq i < j \leq n-1} (x_j - x_i) \prod_{i=0}^{n-1} w(x_i), \quad (3.2)$$

where $M = (m_{kj})_{0 \leq k,j \leq n-1}$ is the coefficient matrix of the Q_j , $Q_j(x) = \sum_{k=0}^{n-1} m_{kj} x^k$. The constant $\det M$ is independent of the x_i ; since (3.2) is a polynomial identity in $x_0, \dots, x_{n-1}$, it may be evaluated at any convenient values to determine $\det M$. In particular, if $\deg Q_j = j$ with leading coefficient ℓ_j for each j , then M is triangular and $\det M = \prod_{j=0}^{n-1} \ell_j$.

Proof. By (3.1) the (i, j) entry is $a_{x_i+j} = w(x_i) \sum_{k=0}^{n-1} m_{kj} x_i^k$, so the matrix factors as $(a_{x_i+j}) = D V M$ with $D = \text{diag}(w(x_i))$, $V = (x_i^k)_{0 \leq i,k \leq n-1}$ and $M = (m_{kj})$. Taking determinants and using the Vandermonde evaluation $\det V = \prod_{i < j} (x_j - x_i)$ gives (3.2). $\square$

The reduction (3.2) is classical determinant calculus: it is the mechanism behind Krattenthaler's determinant lemma [24, Lemma 2.2], [26]. What the present section adds is the delimitation of its exact range — the term-ratio boundary of degree at most one (3.3) below — and the even-index escape that reaches the multiplicative Bessel family of Section 25.

The factorisation (3.1) is available precisely for the sequences whose term ratio is rational of degree at most one. Indeed, if $a_{m+1}/a_m = R(m)$ is rational with numerator and denominator of degrees p and q , then $a_{x+j}/a_x = \prod_{l=0}^{j-1} R(x+l)$; clearing the common denominator over the shifts $l = 0, \dots, n-2$ leaves $a_{x+j} = w(x) Q_j(x)$ with $\deg Q_j = pj + q(n-1-j)$, which is $\leq n-1$ for all $j < n$ iff $p \leq 1$ and $q \leq 1$, i.e.

$$\frac{a_{m+1}}{a_m} = \rho \frac{m + \alpha}{m + \beta}. \quad (3.3)$$

This is the three-parameter *Beta family*: here $w(x) = a_x/(x + \beta)_{n-1}$ and $Q_j(x) = \rho^j (x + \alpha)_j (x + \beta + j)_{n-1-j}$, and Lemma 3.1 yields the closed form, computed in Section 4, of which the factorial, Catalan and central binomial determinants are specialisations; the triangular case $\deg Q_j = j$ is the degeneration $q = 0$ (denominator absent). Conversely, a term ratio of degree ≥ 2 inflates some $\deg Q_j$ beyond $n-1$, the collapse onto the Vandermonde determinant fails, and the determinants acquire large sporadic prime factors (Section 4). One escape from this boundary exists and is used once: in the dilated determinant only *even* row indices occur, so (3.1) is needed only in the contracted row variable i (with $x_i = i$), where a hypergeometric term ratio of degree $(2, 2)$ on the even-indexed half of $\mathbf{a}$ still produces polynomials of admissible degree — fitting the bound $n-1$ exactly when n is even; this closes the multiplicative Bessel family of Section 25.

3.2. The biorthogonal reduction (M2). Let $\mathbf{a} = (a_n)_{n \geq 0}$ be a sequence over a field, split into its even and odd parts. We reduce $\ddot{H}_n = \det(a_{2i+j})_{0 \leq i,j \leq n-1}$ to a determinant of connection coefficients (Lemma 3.4) under the sole hypothesis that its even and odd parts are quasi-definite moment sequences; no continued-fraction assumption is needed at this stage, and the closed-form evaluation follows once the data are specialised (Proposition 7.1). Throughout write $\bar{n} = \lceil n/2 \rceil$ and $\underline{n} = \lfloor n/2 \rfloor$, so $\bar{n} + \underline{n} = n$ and $\bar{n} \in \{\underline{n}, \underline{n} + 1\}$.

Define two linear functionals on $\mathbb{Q}[y]$ by

$$\mathcal{S}[y^k] = a_{2k}, \quad \mathcal{T}[y^k] = a_{2k+1}.$$

Every polynomial $\phi(z)$ decomposes as $\phi(z) = G(z^2) + zK(z^2)$, and the functional $\mathcal{E}[z^m] = a_m$ satisfies $\mathcal{E}[\phi] = \mathcal{S}[G] + \mathcal{T}[K]$. The entries of our dilated matrix are $a_{2i+j} = \mathcal{E}[z^{2i} \cdot z^j]$; in other words, writing $y = z^2$,

$$\ddot{H}_n = \det(\mathcal{E}[y^i \cdot \chi_j])_{0 \leq i,j \leq n-1}, \quad \chi_{2l} = y^l, \quad \chi_{2m+1} = zy^m. \quad (3.4)$$

We assume throughout this section that $\mathcal{S}$ and $\mathcal{T}$ are *quasi-definite*, so each carries its monic orthogonal polynomials— P_i for $\mathcal{S}$ and Q_m for $\mathcal{T}$ —with three-term recurrences

$$P_{i+1} = (y - c_i^S) P_i - \lambda_i^S P_{i-1}, \quad Q_{m+1} = (y - c_m^T) Q_m - \lambda_m^T Q_{m-1} \quad (P_0 = Q_0 = 1), \quad (3.5)$$

and nonzero squared norms $h_i^S = \mathcal{S}[P_i^2]$ and $h_m^T = \mathcal{T}[Q_m^2]$. Equivalently, the generating functions $\sum_k a_{2k} x^k$ and $\sum_k a_{2k+1} x^k$ have J -fractions with coefficients c_i^S, λ_i^S and c_m^T, λ_m^T respectively.

Lemma 3.2 (one-sided reduction). *For this reduction only the even functional need be quasi-definite. Let $P_0, P_1, \dots$ be the monic $\mathcal{S}$ -orthogonal polynomials, with squared norms $h_l^S = \mathcal{S}[P_l^2]$, and let $\mathcal{T}$ be an arbitrary linear functional. Then*

$$\ddot{H}_n = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^S \right) \det(\mathcal{T}[P_{\bar{n}+r} y^m])_{0 \leq r, m \leq \bar{n}-1}. \quad (3.6)$$

Proof. Start from (3.4). Replace the row family $(y^i)_{i < n}$ by $(P_i)_{i < n}$ and the even-column family $(y^l)_{l < \bar{n}}$ by $(P_l)_{l < \bar{n}}$ — each monic of the same degree, hence a unitriangular recombination, which leaves the determinant unchanged by Lemma 2.1; the odd columns $(z y^m)_{m < \underline{n}}$ are left untouched. The $(i, 2l)$ entry becomes $\mathcal{S}[P_i P_l] = \delta_{il} h_l^S$ and the $(i, 2m+1)$ entry becomes $\mathcal{T}[P_i y^m]$. Reorder the columns so that the $\bar{n}$ even indices $0, 2, \dots, 2\bar{n}-2$ precede the $\underline{n}$ odd indices $1, 3, \dots, 2\underline{n}-1$; this permutation has $\sum_{l=0}^{\bar{n}-1} \min(l, \underline{n}) = \binom{\bar{n}}{2}$ inversions (as $l \leq \bar{n}-1 \leq \underline{n}$), hence sign $(-1)^{\binom{\bar{n}}{2}}$. In the reordered matrix the first $\bar{n}$ columns are diagonal on the rows $i < \bar{n}$ and vanish on the rows $i \geq \bar{n}$, so a Laplace expansion along them contributes $\prod_{l=0}^{\bar{n}-1} h_l^S$ and leaves the complementary rows $i = \bar{n}, \dots, n-1 = \bar{n}+r$ against the odd columns, giving (3.6). No property of $\mathcal{T}$ was used. $\square$

When $\mathcal{T}$ is also quasi-definite it carries its own monic orthogonal polynomials Q_m , and since $\{Q_0, \dots, Q_i\}$ is a basis of $\mathbb{Q}[y]_{\leq i}$ we may expand the $\mathcal{S}$ -orthogonal P_i in the $\mathcal{T}$ -orthogonal basis,

$$P_i = \sum_{m=0}^i \kappa_{i,m} Q_m; \quad (3.7)$$

the $\kappa_{i,m}$ are the *connection coefficients*, lower-triangular in (i, m) with $\kappa_{i,i} = 1$ (both families monic). Applying the projection (2.2) for $\mathcal{T}$ isolates a single one,

$$\mathcal{T}[P_i Q_m] = \sum_{m'} \kappa_{i,m'} \mathcal{T}[Q_{m'} Q_m] = \kappa_{i,m} \mathcal{T}[Q_m^2] = \kappa_{i,m} h_m^T, \quad \kappa_{i,m} = \frac{\mathcal{T}[P_i Q_m]}{h_m^T}. \quad (3.8)$$

One further column operation turns (3.6) into a determinant of these $\kappa_{i,m}$.

Lemma 3.3 (connection coefficients). *Write $P_i = \sum_{m=0}^i \kappa_{i,m} Q_m$ for the expansion of the $\mathcal{S}$ -orthogonal polynomials in the $\mathcal{T}$ -orthogonal basis. Then $\kappa_{i,i} = 1$, while $\kappa_{i,m} = 0$ for $m > i$ (and we set $\kappa_{i,m} = 0$ for $m < 0$), and*

$$\kappa_{i+1,m} = \kappa_{i,m-1} + (c_m^T - c_i^S) \kappa_{i,m} + \lambda_{m+1}^T \kappa_{i,m+1} - \lambda_i^S \kappa_{i-1,m}. \quad (3.9)$$

Proof. Since $\deg P_i = i$ and $\{Q_0, \dots, Q_i\}$ is a basis of the polynomials of degree $\leq i$, the expansion $P_i = \sum_{m \leq i} \kappa_{i,m} Q_m$ exists; comparing leading coefficients (both monic) gives $\kappa_{i,i} = 1$, while $\kappa_{i,m} = 0$ for $m > i$ (and we set $\kappa_{i,m} = 0$ for $m < 0$).

For the recurrence, substitute $P_i = \sum_m \kappa_{i,m} Q_m$ into $P_{i+1} = (y - c_i^S) P_i - \lambda_i^S P_{i-1}$ from (3.5), and expand the $y Q_m$ that appears using the Q -recurrence $y Q_m = Q_{m+1} + c_m^T Q_m + \lambda_m^T Q_{m-1}$:

$$\sum_m \kappa_{i+1,m} Q_m = \sum_m \kappa_{i,m} (Q_{m+1} + c_m^T Q_m + \lambda_m^T Q_{m-1}) - c_i^S \sum_m \kappa_{i,m} Q_m - \lambda_i^S \sum_m \kappa_{i-1,m} Q_m.$$

Now read off the coefficient of Q_m on both sides: the term Q_{m+1} contributes through $\kappa_{i,m-1}$, and the term $\lambda_m^T Q_{m-1}$ contributes (at index $m+1$) through $\lambda_{m+1}^T \kappa_{i,m+1}$. This gives (3.9),

$$\kappa_{i+1,m} = \kappa_{i,m-1} + (c_m^T - c_i^S) \kappa_{i,m} + \lambda_{m+1}^T \kappa_{i,m+1} - \lambda_i^S \kappa_{i-1,m},$$

which together with $\kappa_{0,0} = 1$ determines all $\kappa_{i,m}$. $\square$

Lemma 3.4 (determinant reduction). *With the squared norms $h_l^S = \mathcal{S}[P_l^2]$ and $h_m^T = \mathcal{T}[Q_m^2]$,*

$$\ddot{H}_n = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^S \right) \left(\prod_{m=0}^{n-1} h_m^T \right) \det(\kappa_{\bar{n}+r, m})_{0 \leq r, m \leq \underline{n}-1}.$$

Proof. Since $\mathcal{T}$ is now quasi-definite, carry out one further column operation in the one-sided reduction (3.6): expand each odd-column monomial in the $\mathcal{T}$ -orthogonal basis, $y^m = Q_m +$ (lower-degree Q 's), a unitriangular recombination of the odd columns that leaves the determinant unchanged (Lemma 2.1) and turns the entry $\mathcal{T}[P_{\bar{n}+r} y^m]$ into $\mathcal{T}[P_{\bar{n}+r} Q_m]$. By the projection (3.8) this equals $\kappa_{\bar{n}+r, m} h_m^T$, so

$$\ddot{H}_n = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^S \right) \cdot \det(\kappa_{\bar{n}+r, m} h_m^T)_{0 \leq r, m \leq \underline{n}-1}.$$

Pulling the column factor h_m^T out of column m gives the claim. $\square$

Specialisation: even and odd S -fractions. The general reduction **M2** (Lemma 3.4) needs only the recurrence and norm data $(c_i^S, \lambda_i^S, h_i^S)$ and $(c_m^T, \lambda_m^T, h_m^T)$. These become fully explicit, and the norms telescope into products, as soon as the even and odd parts of $\mathbf{a}$ admit Stieltjes S -fractions:

$$\sum_{k \geq 0} a_{2k} x^k = \frac{1}{1 - \frac{u_1 x}{1 - \frac{u_2 x}{1 - \ddots}}}, \quad \sum_{k \geq 0} a_{2k+1} x^k = \frac{1}{1 - \frac{v_1 x}{1 - \frac{v_2 x}{1 - \ddots}}}. \quad (3.10)$$

The passage from these S -fractions to the recurrence coefficients and norms of Section 3.2 is the even contraction (Section 2.5).

Lemma 3.5 (continued fractions, recurrences, norms). *The monic orthogonal polynomials P_i of $\mathcal{S}$ and Q_m of $\mathcal{T}$ satisfy the three-term recurrences*

$$P_{i+1} = (y - c_i^S) P_i - \lambda_i^S P_{i-1}, \quad Q_{m+1} = (y - c_m^T) Q_m - \lambda_m^T Q_{m-1} \quad (P_0 = Q_0 = 1),$$

where

$$c_i^S = u_{2i} + u_{2i+1}, \quad \lambda_i^S = u_{2i-1} u_{2i}, \quad c_m^T = v_{2m} + v_{2m+1}, \quad \lambda_m^T = v_{2m-1} v_{2m},$$

and orthogonality and norms

$$\mathcal{S}[P_i P_l] = \delta_{il} \prod_{j=1}^{2i} u_j, \quad \mathcal{T}[Q_m Q_{m'}] = \delta_{mm'} \prod_{j=1}^{2m} v_j.$$

Proof. The two series in (3.10) are Stieltjes S -fractions with coefficient sequences $(u_1, u_2, \dots)$ and $(v_1, v_2, \dots)$. The passage from an S -fraction with coefficients $(b_1, b_2, \dots)$ to the monic three-term recurrence of the associated orthogonal polynomials is the *even contraction* of Section 2.5, which gives $c_i = b_{2i} + b_{2i+1}$, $\lambda_i = b_{2i-1} b_{2i}$ (with $b_0 = 0$) and squared norms $\prod_{k=1}^{2i} b_k$. Taking $b = u$ for $\mathcal{S}$ and $b = v$ for $\mathcal{T}$ yields the stated coefficients and norms. $\square$

The collapse mechanism. Lemma 3.4 is exact, but it leaves the determinant of the connection array to be evaluated, and for a generic pair of classical functionals this array is a balanced ${}_3F_2$ -type *double sum*, out of reach of the determinant calculus. In every case where **M2** closes in this paper, one and the same two-stage mechanism is at work. First, $\mathcal{S}$ and $\mathcal{T}$ lie in a *single* classical family and differ in a *single* parameter; for such a pair the connection coefficients collapse from a double sum to one hypergeometric term, which splits into a row factor, a column factor, and a kernel depending only on $i - m$ and (possibly) $i + m$. Second, the resulting kernel determinant — which carries the shift $\bar{n}$ of Lemma 3.4, hence free integer parameters — is evaluated in closed form. The paper contains three instances. For the Euler (s, t) family the two functionals differ in the exponent parameter, the kernel is the binomial coefficient $\binom{(t-s)/2}{i-m}$, and the shifted binomial determinant is evaluated by the dual Jacobi–Trudi identity (Proposition 7.1). For

the reciprocal-sine function the two functionals are Wilson functionals differing in one Wilson parameter, the kernel is the product of Catalan numbers $C_{i-m-1} C_{i+m}$, and the shifted Catalan determinant is evaluated by Desnanot–Jacobi condensation (Theorem 18.8). For the Bessel (s, t) family the two functionals are Jacobi–Gegenbauer functionals differing in one parameter, the kernel is $\binom{t-s}{i-m} / ((t+2m+1)_{i-m} (s+i+m)_{i-m})$ — binomial in $i-m$ but carrying the parameters through $i+m$ — and the shifted connection determinant is again evaluated by condensation (Theorem 23.7). When the two classical functionals do *not* form such a one-parameter pair, the array remains a genuine double sum and no product formula is to be expected.

3.3. A one-functional reduction (M3). The biorthogonal reduction of Section 3.2 splits $\mathcal{E}$ into its even and odd parts $\mathcal{S}, \mathcal{T}$ and passes to the contracted variable $y = z^2$. A parallel reduction keeps the single functional $\mathcal{E}[z^k] = a_k$ on z and uses *its own* orthogonal polynomials. It needs $\mathcal{E}$ itself to be quasi-definite—a hypothesis independent of the quasi-definiteness of $\mathcal{S}$ and $\mathcal{T}$ —and it underlies the Hermite evaluation of the Gaussian family in Section 5.

Assume $\mathcal{E}$ is quasi-definite, with monic orthogonal polynomials $(\pi_m)_{m \geq 0}$. By (1.2), $\ddot{H}_n = \det(\mathcal{E}[z^{2i} \cdot z^{2j}])_{0 \leq i, j \leq n-1}$.

The reduction collapses to a product as soon as, alongside the columns, the rows too can be replaced by a one-sidedly orthogonal family. Suppose there are monic polynomials Φ_i of degree $2i$, each a unitriangular recombination of the even monomials $1, z^2, \dots, z^{2i}$, orthogonal to every polynomial of degree $< i$:

$$\mathcal{E}[\Phi_i \pi_j] = 0 \quad (j < i).$$

Such a Φ_i exists and is unique whenever the corresponding Gram minor is nonzero.

Lemma 3.6 (one-functional reduction). *If the even-orthogonal family $(\Phi_i)_{i < n}$ above exists, then*

$$\ddot{H}_n = \prod_{i=0}^{n-1} \mathcal{E}[\Phi_i \pi_i]. \quad (3.11)$$

Proof. Replace the column family $(z^j)_{j < n}$ by the orthogonal family $(\pi_j)_{j < n}$ and the row family $(z^{2i})_{i < n}$ by $(\Phi_i)_{i < n}$; both are unitriangular recombinations, so Lemma 2.1 gives

$$\ddot{H}_n = \det(\mathcal{E}[\Phi_i \pi_j])_{0 \leq i, j < n}.$$

By orthogonality $\mathcal{E}[\Phi_i \pi_j] = 0$ for $j < i$, so this matrix is upper triangular and its determinant is the product of the diagonal entries $\mathcal{E}[\Phi_i \pi_i]$. $\square$

Equation (3.11) is the one-functional counterpart of Lemma 3.4. Section 5 carries it out for the Gaussian functional, whose orthogonal polynomials are the Hermite polynomials and whose Φ_i has a closed umbral form. The passage from a Hankel determinant to a triangular product built from the orthogonal polynomials of a single functional is precisely the mechanism developed by Cigler and Krattenthaler [6, 29], who reduce the Hankel determinant of a linear combination of moments of orthogonal polynomials to a smaller determinant of those polynomials and, as we do throughout, close several of their evaluations by Desnanot–Jacobi (Dodgson) condensation.

3.4. The divisor method and the dilated Cauchy–Binet lemma (M4). The reductions M1–M3 evaluate $\ddot{H}_n$ from the moment data directly. The divisor method (M4) instead applies to a *family* f_θ depending on a parameter θ : one treats $\ddot{H}_n(f_\theta)$ as a polynomial in θ , bounds its degree, and locates its zeros at the *degenerate* values of θ where f_θ collapses to a geometric/exponential function or to an even function. At such a value the moments form an exponential sum, and the dilated determinant is evaluated, to leading order, by a single Cauchy–Binet expansion into two Vandermonde determinants. Three general facts drive the method; the case-specific inputs (the degree bound and the precise orders) are supplied in each application (Sections 16, 17 and 19).

The first is a homogeneity, used to rescale and to take limits.

Lemma 3.7 (homogeneity). *For scalars β, c , $\ddot{H}_n(f(\beta x)) = \beta^{3\binom{n}{2}} \ddot{H}_n(f)$ and $\ddot{H}_n(cf) = c^n \ddot{H}_n(f)$.*

Proof. The coefficients of $f(\beta x)$ are $\beta^n a_n$, so the (i, j) entry a_{2i+j} is scaled by β^{2i+j} ; pulling β^{2i} from row i and β^j from column j multiplies the determinant by $\beta^{\sum_i 2i + \sum_j j} = \beta^{2\binom{n}{2} + \binom{n}{2}}$. The factor c multiplies every entry, hence the determinant by c^n . $\square$

The second is the bookkeeping principle that turns “enough vanishing” into an exact product.

Observation 3.8 (divisor principle). Let $D(\theta)$ be a polynomial of degree $\leq N$, not identically zero, that vanishes to order $\geq d_k$ at distinct points $\theta_1, \dots, \theta_s$. If $\sum_k d_k \geq N$, then $\sum_k d_k = N$, every order is exact, and $D(\theta) = c \prod_{k=1}^s (\theta - \theta_k)^{d_k}$ with $c \neq 0$ fixed by the value of D at any further point.

The third, and the engine of the method, expands a dilated determinant whose moments are an exponential sum.

Lemma 3.9 (dilated Cauchy–Binet). *Let $a_m = \sum_{\nu \in I} c_\nu \zeta_\nu^m$ be a (formal or convergent) exponential sum over an index set I . Then*

$$\ddot{H}_n(a) = \sum_T \left(\prod_{\nu \in T} c_\nu \right) \det(\zeta_\nu^{2i})_{0 \leq i < n, \nu \in T} \det(\zeta_\nu^j)_{0 \leq j < n, \nu \in T},$$

the sum over the n -subsets $T \subseteq I$; each term is a product of two Vandermonde determinants in the nodes $\{\zeta_\nu : \nu \in T\}$, the first in their squares ζ_ν^2 . If moreover the index set is symmetric, $-I = I$ with $\zeta_{-\nu} = -\zeta_\nu$ (so $\zeta_{-\nu}^2 = \zeta_\nu^2$), then only the n -subsets on which $\nu \mapsto |\nu|$ is injective survive: a subset containing a pair $\{\nu, -\nu\}$ makes the first determinant have two equal columns.

Proof. Writing $a_{2i+j} = \sum_\nu c_\nu \zeta_\nu^{2i} \zeta_\nu^j$, we factor the $n \times n$ matrix as $X \operatorname{diag}(c_\nu) Y^\top$ with $X = (\zeta_\nu^{2i})_{0 \leq i < n, \nu \in I}$ and $Y = (\zeta_\nu^j)_{0 \leq j < n, \nu \in I}$. Here X and Y are rectangular $n \times |I|$ matrices — one column per node, and $|I|$ (possibly infinite) need not equal n — so this is the general (non-square) Cauchy–Binet formula, which gives the displayed sum over all n -subsets $T \subseteq I$. In the symmetric case $\zeta_{-\nu}^{2i} = \zeta_\nu^{2i}$, so on a subset containing both ν and $-\nu$ the matrix (ζ_ν^{2i}) has two equal columns and the first determinant vanishes; on the survivors $\nu \mapsto \zeta_\nu^2$ is injective and the first determinant is a genuine Vandermonde. $\square$

The first determinant is a Vandermonde in the *squared* nodes ζ_ν^2 , and this squaring is the algebraic imprint of the dilation. Indeed the even rows enter only through $\zeta_\nu^{2i} = (\zeta_\nu^2)^i$, which depends on ζ_ν only through ζ_ν^2 ; hence these rows cannot tell a node ζ_ν apart from its negative $-\zeta_\nu$ (the two share the same square). This is exactly why, in the symmetric case above, a node and its negative collapse the term.

We use Lemma 3.9 in the next sections in the following way. When the exponential sum depends on a parameter and the nodes and coefficients degenerate as the parameter tends to a special value, the same expansion lets us read off the *order* to which $\ddot{H}_n$ vanishes there, and its leading coefficient. Suppose then that the exponential sum of Lemma 3.9 depends on a parameter ϵ ,

$$a_m(\epsilon) = \sum_{\nu \in I} c_\nu(\epsilon) \zeta_\nu(\epsilon)^m, \quad (3.12)$$

each coefficient and each node being a (formal or convergent) power series in ϵ ; write $\bar{\zeta}_\nu := \zeta_\nu(0)$ for the limiting nodes. Lemma 3.9 expands $\ddot{H}_n(a(\epsilon))$ termwise,

$$\ddot{H}_n(a(\epsilon)) = \sum_T \left(\prod_{\nu \in T} c_\nu(\epsilon) \right) \det(\zeta_\nu(\epsilon)^{2i})_{0 \leq i < n, \nu \in T} \det(\zeta_\nu(\epsilon)^j)_{0 \leq j < n, \nu \in T}, \quad (3.13)$$

the sum over n -subsets $T \subseteq I$; call T *surviving* if its term is not identically zero — in the symmetric case of Lemma 3.9 exactly the subsets on which $\nu \mapsto |\nu|$ is injective. Given integers $o_\nu \geq 0$ with $\operatorname{ord}_\epsilon c_\nu(\epsilon) \geq o_\nu$ for every $\nu \in I$ (and, if I is infinite, $\{\nu : o_\nu \leq N\}$ finite for every N , so that each power of ϵ in (3.13) receives only finitely many contributions), put

$$M := \min_{T \text{ surviving}} \sum_{\nu \in T} o_\nu.$$

Corollary 3.10 (order at a degeneration). *With the data and notation above:*

- (i) $\text{ord}_\epsilon \ddot{H}_n(a(\epsilon)) \geq M$.
- (ii) If moreover a unique surviving T^* attains M , the orders on T^* are exact — $c_\nu(\epsilon) = \bar{c}_\nu \epsilon^{o_\nu} + O(\epsilon^{o_\nu+1})$ with $\bar{c}_\nu \neq 0$, for $\nu \in T^*$ — and the limiting squared nodes $\bar{\zeta}_\nu^2$ ($\nu \in T^*$) are pairwise distinct, then

$$\ddot{H}_n(a(\epsilon)) = C \epsilon^M + O(\epsilon^{M+1}), \quad C = \left(\prod_{\nu \in T^*} \bar{c}_\nu \right) \det(\bar{\zeta}_\nu^{2i})_{0 \leq i < n, \nu \in T^*} \det(\bar{\zeta}_\nu^j)_{0 \leq j < n, \nu \in T^*} \neq 0,$$

so that $\text{ord}_\epsilon \ddot{H}_n(a(\epsilon)) = M$ exactly.

Proof. (i) Every surviving term of (3.13) has order $\geq \sum_{\nu \in T} o_\nu \geq M$: the coefficient product has order $\geq \sum_{\nu \in T} o_\nu$, and the two determinants, being polynomials in the $\zeta_\nu(\epsilon)$, have order ≥ 0 . Hence so has the sum.

(ii) By (i) it remains to read off the coefficient of ϵ^M . Since T^* is the unique minimiser, every other surviving term is $O(\epsilon^{M+1})$, so only T^* contributes; taking the leading part $\bar{c}_\nu \epsilon^{o_\nu}$ of each coefficient and setting $\epsilon = 0$ in the two determinants gives C . It is nonzero: the first determinant is a Vandermonde in the pairwise-distinct $\bar{\zeta}_\nu^2$, hence the $\bar{\zeta}_\nu$ are themselves pairwise distinct and the second determinant, a Vandermonde in the $\bar{\zeta}_\nu$, is nonzero as well; and $\prod_{\nu \in T^*} \bar{c}_\nu \neq 0$ because the orders on T^* are exact. $\square$

The corollary above governs the *distinct*-node case, where the limiting nodes $\bar{\zeta}_\nu$ stay separate. The opposite extreme is *confluent*: the nodes merge to a single point, and a Vandermonde degenerates into a Wronskian — the determinant of a column together with its successive derivatives. This is what happens when an *even* generating function is translated by a parameter ϕ , that is $g_\phi(x) = g(x + \phi)$: as the next lemma shows, each column of the dilated matrix of g_ϕ is the ϕ -derivative of the one before it, so $\ddot{H}_n(g_\phi)$ is literally a Wronskian in ϕ .

Lemma 3.11 (confluent case: a translated even series). *Let $g(x) = \sum_{m \geq 0} \mu_m x^m / m!$ be even ($\mu_{2k+1} = 0$), write $\nu_k := \mu_{2k}$ and $g_\phi(x) := g(x + \phi)$, with coefficients $\beta_m(\phi) = \sum_{k \geq 0} \frac{\phi^k}{k!} \mu_{m+k}$. Then*

$$\ddot{H}_n(g_\phi) = \det(\beta_{2i+j}(\phi))_{0 \leq i, j < n} = W(\beta_0, \beta_2, \dots, \beta_{2n-2})(\phi),$$

the Wronskian of n even functions; consequently $\text{ord}_{\phi=0} \ddot{H}_n(g_\phi) \geq \binom{n}{2}$.

Proof. Since $\partial_\phi \beta_m = \beta_{m+1}$, column j is the j -th derivative of column 0, so $\beta_{2i+j} = \beta_{2i}^{(j)}$ and the determinant is the Wronskian of the rows $\beta_{2i}(\phi)$. As g is even, $\beta_{2i}(\phi) = \sum_{l \geq 0} \nu_{i+l} \phi^{2l} / (2l)!$, so by the Cauchy–Binet identity for Wronskians

$$W(\beta_0, \dots, \beta_{2n-2}) = \sum_{0 \leq l_0 < \dots < l_{n-1}} \det(\nu_{i+l_a})_{i,a} W\left(\frac{\phi^{2l_0}}{(2l_0)!}, \dots, \frac{\phi^{2l_{n-1}}}{(2l_{n-1})!}\right),$$

and the Wronskian of the $\phi^{2l_a} / (2l_a)!$ is a multiple of $\phi^{2 \sum_a l_a - \binom{n}{2}}$, of order $\geq 2 \binom{n}{2} - \binom{n}{2} = \binom{n}{2}$. $\square$

Finally, the same squared-node collapse explains, in its most degenerate form, why a single conjugate pair of frequencies perturbs a dilated determinant only by *rank one*.

Remark 3.12. A two-element symmetric set $I = \{\pm 1\}$ ($\zeta_{-1} = -\zeta_1$) gives, by the same identity $\zeta_{-1}^{2i} = \zeta_1^{2i}$, a matrix of *rank one*: $a_{2i+j} = \zeta_1^{2i} (c_1 \zeta_1^j + c_{-1} \zeta_{-1}^j)$. This is the structure behind the rank-one perturbation of Section 15 (there $\zeta_{\pm 1} = \pm i$, the perturbing term being $\sin x$), evaluated by the matrix-determinant lemma (M6) rather than by the divisor method M4.

3.5. Which method for which family. Table 1 summarises the six methods: the structural input each one requires, the condition under which the reduction collapses to a closed product, and where in the paper it is applied.

	input	collapse condition	applications
M1	term ratio a_{m+1}/a_m rational of degree ≤ 1	always: Vandermonde $\times$ triangular coefficient matrix (Lemma 3.1); fails from degree 2 on, except through the dilation (§25)	Beta family: factorials, Catalan, central binomial, double factorials (§4); shifts of the $(1+x)$ algebraic family (§22); even orders of $(1+x)\cos b_\nu^2$ (§25)
M2	even and odd functionals $\mathcal{S}, \mathcal{T}$ quasi-definite	connection array a single term $(\mathcal{S}, \mathcal{T})$ a one-parameter classical pair) and the kernel determinant evaluable	Euler number family (§7); the secant-number family $(1+x)/\cos^{s+1}$ (§11–12); reciprocal sine (§18); Bessel (s, t) family (§23); and their double shifts (§8, 14 and 24)
M3	single functional $\mathcal{E}$ quasi-definite	even-orthogonal Φ_i exist (3.11); among Appell families: the Gaussian only	involutions, $\exp(cx + bx^2)$ (§5)
M4	family $f_\theta, \ddot{H}_n(f_\theta)$ polynomial in θ of bounded degree	degenerations of f_θ account for the full degree (Observation 3.8)	Springer (§16), derivative of Springer (§17), elliptic (§19)
M5	contiguous relation $s \mapsto s - 2$ by column operations	two-periodicity in s plus one special value, or a single contiguous step closed by Cauchy–Binet	algebraic families $(1+x)/(1-x^2)^{s/2}$ and $(1+x)^2/(1-x^2)^{s/2}$ (§20 and 21), and their shifted determinants (§22)
M6	rank-one perturbation of an already evaluated family	matrix-determinant lemma (15.1)	$(\sin x + 1)/\cos^2 x + s \sin x$ (§15)

TABLE 1. The six reductions of Section 3.

4. THE BETA FAMILY

Throughout this section $\mathbf{a} = (a_m)_{m \geq 0}$ is a *plain* sequence: no exponential normalisation is applied, and the entries of the determinants are the a_m themselves. We assume that the term ratio a_{m+1}/a_m is a rational function of m with numerator and denominator of degree *at most* one — the exact hypothesis under which the Vandermonde reduction **M1** of Section 3.1 operates. The generic case, numerator and denominator both of degree one, is the Beta family proper (Theorem 4.1); the degenerate case with denominator of degree zero — β absent — is treated separately (Proposition 4.2). Such sequences form the three-parameter family

$$a_m = a_0 \rho^m \frac{(\alpha)_m}{(\beta)_m}, \quad \frac{a_{m+1}}{a_m} = \rho \frac{m + \alpha}{m + \beta}. \quad (4.1)$$

The name is explained by the case $\beta > \alpha > 0$, $a_0 = \rho = 1$: there (4.1) is the moment sequence of the *Beta distribution* on $[0, 1]$ with density $x^{\alpha-1}(1-x)^{\beta-\alpha-1}/B(\alpha, \beta-\alpha)$, since $\int_0^1 x^m d\mu = (\alpha)_m/(\beta)_m$. Beta distributions are the model solutions of the Hausdorff moment problem on a bounded interval, the polynomials orthogonal to them are the Jacobi polynomials [5, 21], and the associated Jacobi continued fraction [47, 12] governs the classical (one-step) Hankel determinants of the family. None of this positivity is needed below: the scale ρ and the total mass a_0 are inessential, and α, β, ρ may be taken as arbitrary parameters, the determinant identities being polynomial (or rational) in them. Members of (4.1) include the factorials $(m+r)! = r!(r+1)_m$ (degenerate, β absent), the Catalan numbers $C_m = \frac{1}{m+1} \binom{2m}{m} = 4^m (\frac{1}{2})_m / (2)_m$, and the central binomial coefficients $b_m = \binom{2m}{m} = 4^m (\frac{1}{2})_m / (1)_m$.

The section is organised as follows. Subsection 4.1 proves a single closed form for $\det(a_{x_i+j})$ with *arbitrary* row indices x_i (Theorem 4.1); Subsection 4.2 specialises it to the dilated, shifted

and r -step determinants of the classical members; Subsection 4.3 records where the mechanism stops.

4.1. The general evaluation.

Theorem 4.1. *Let $n \geq 1$, let $\rho \neq 0$ and α, β be parameters with $\beta \notin \{0, -1, -2, \dots\}$, let a_m be as in (4.1), and let $0 \leq x_0 < x_1 < \dots < x_{n-1}$ be integers. Then*

$$\det(a_{x_i+j})_{0 \leq i, j \leq n-1} = \rho^{\binom{n}{2}} \prod_{i=0}^{n-1} (\beta - \alpha)_i \prod_{0 \leq i < j \leq n-1} (x_j - x_i) \prod_{i=0}^{n-1} \frac{a_{x_i}}{(x_i + \beta)_{n-1}}. \quad (4.2)$$

Proof. Step 1: row factorisation. Iterating the term ratio in (4.1) gives $a_{x+j} = a_x \rho^j (x + \alpha)_j / (x + \beta)_j$, and over the common denominator $(x + \beta)_{n-1}$, for $0 \leq j \leq n-1$,

$$a_{x+j} = \frac{a_x}{(x + \beta)_{n-1}} Q_j(x), \quad Q_j(x) = \rho^j (x + \alpha)_j (x + \beta + j)_{n-1-j}, \quad (4.3)$$

each Q_j being a polynomial in x of degree exactly $j + (n-1-j) = n-1$, with coefficients independent of the x_i .

Step 2: reduction to a Vandermonde determinant. The row factorisation (4.3) is exactly the hypothesis of the Vandermonde reduction **M1** (Lemma 3.1), with $w(x) = a_x / (x + \beta)_{n-1}$ and the Q_j above, each of degree $n-1$. Writing $Q_j(x) = \sum_{k=0}^{n-1} m_{kj} x^k$ and $M = (m_{kj})_{0 \leq k, j \leq n-1}$, it gives

$$\det(a_{x_i+j}) = \prod_{i=0}^{n-1} \frac{a_{x_i}}{(x_i + \beta)_{n-1}} \cdot c_n \prod_{0 \leq i < j \leq n-1} (x_j - x_i), \quad c_n = \det M, \quad (4.4)$$

where the constant c_n does not depend on $x_0, \dots, x_{n-1}$.

Step 3: the triangulating substitution. Since $\det(Q_j(x_i)) = c_n \prod_{i < j} (x_j - x_i)$ is an identity of polynomials in the indeterminates $x_0, \dots, x_{n-1}$, we may compute c_n by any convenient substitution — the integrality and ordering of the x_i are irrelevant here. Take $x_i = 1 - \beta - i$. Then

$$(x_i + \beta + j)_{n-1-j} = (1 + j - i)(2 + j - i) \cdots (n - 1 - i),$$

which vanishes if and only if $i > j$: the matrix $(Q_j(x_i))$ is upper triangular. Its diagonal entries are

$$Q_i(x_i) = \rho^i (1 - \beta - i + \alpha)_i (1)_{n-1-i} = \rho^i (-1)^i (\beta - \alpha)_i (n - 1 - i)!,$$

while the Vandermonde product at this substitution is $\prod_{i < j} (i - j) = (-1)^{\binom{n}{2}} \prod_{i=0}^{n-1} i!$. In the quotient the signs cancel, and so does $\prod_i (n - 1 - i)! = \prod_i i!$, leaving $c_n = \rho^{\binom{n}{2}} \prod_{i=0}^{n-1} (\beta - \alpha)_i$. $\square$

Note the case $\alpha = \beta$: the sequence is geometric, every such determinant with $n \geq 2$ is zero, and the formula sees this through the factor $(\beta - \alpha)_i$, which vanishes for $i \geq 1$.

When the denominator parameter is absent the situation degenerates further: the reduction is triangular from the start, and no substitution is needed.

Proposition 4.2. *Let $a_m = a_0 \rho^m (\alpha)_m$ with $\rho \neq 0$. Then*

$$\det(a_{x_i+j})_{0 \leq i, j \leq n-1} = \rho^{\binom{n}{2}} \prod_{0 \leq i < j \leq n-1} (x_j - x_i) \prod_{i=0}^{n-1} a_{x_i}. \quad (4.5)$$

Proof. Here $a_{x+j} = a_x P_j(x)$ with $P_j(x) = \rho^j (x + \alpha)_j$ of degree exactly j and leading coefficient ρ^j . This is the triangular case of the Vandermonde reduction (Lemma 3.1), with $w(x) = a_x$ and $\det M = \prod_{j=0}^{n-1} \rho^j = \rho^{\binom{n}{2}}$, so $\det(a_{x_i+j}) = \rho^{\binom{n}{2}} \prod_{i < j} (x_j - x_i) \prod_i a_{x_i}$. $\square$

4.2. Specialisations: factorials, Catalan and central binomial numbers. The substitution $x_i = 2i$ in (4.2) gives the dilated Hankel determinant of every member of the family at one stroke. More generally, for an integer $r \geq 1$ we call $\det(a_{ri+j})_{0 \leq i, j \leq n-1}$ the r -step Hankel determinant of $\mathbf{a}$ — the classical case is $r = 1$, the dilated case $r = 2$ — and the substitution $x_i = ri + t$ evaluates every shifted r -step determinant $\det(a_{ri+j+t})$ of the family in closed form.

Corollary 4.3. *With a_m as in (4.1),*

$$\ddot{H}_n(\mathbf{a}) = \det(a_{2i+j})_{0 \leq i,j \leq n-1} = (2\rho)^{\binom{n}{2}} \prod_{i=0}^{n-1} i! (\beta - \alpha)_i \frac{a_{2i}}{(2i + \beta)_{n-1}}. \quad (4.6)$$

We now harvest the classical members. No further proofs are needed: each corollary is a substitution of parameters into Theorem 4.1 or Proposition 4.2, followed by a conversion of Pochhammer symbols into factorials, and we record these conversions as *derivations*.

Corollary 4.4 (factorials). *Let $r \geq 0$ and $a_m = (m+r)!$ — in the exponential convention of the other sections, the coefficient sequence of the derivative $f^{(r)}$ of $f(x) = 1/(1-x)$. Then*

$$\ddot{H}_n((m+r)!) = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k! \cdot \prod_{i=0}^{n-1} (2i+r)!, \quad \text{hence} \quad \frac{\ddot{H}_n(f^{(r+1)})}{\ddot{H}_n(f^{(r)})} = \prod_{i=0}^{n-1} (2i+r+1). \quad (4.7)$$

The first three ratios ($r = 0, 1, 2$) are $(2n-1)!!$, $(2n)!!$, $(2n+1)!!$. In particular

$$\begin{aligned} \ddot{H}_n(m!) &= \prod_{k=1}^{n-1} 2^k k! (2k)! = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!, \\ \ddot{H}_n((m+1)!) &= \prod_{k=1}^{n-1} 2^k k! (2k+1)!, \quad \ddot{H}_n((m+2)!/2) = \prod_{k=1}^{n-1} 2^{k-1} k! (2k+2)!. \end{aligned}$$

Derivation. $(m+r)! = r!(r+1)_m$ is the case $\rho = 1$, $\alpha = r+1$ of Proposition 4.2; with $x_i = 2i$, $\ddot{H}_n = \prod_{i < j} 2(j-i) \prod_i (2i+r)! = 2^{\binom{n}{2}} \prod_{k < n} k! \prod_i (2i+r)!$. The last evaluation follows by scaling rows by $1/2$. $\square$

Proposition 4.2 holds for arbitrary α , so the factorial evaluation (4.7) is the integer specialisation of a continuous one-parameter family.

Corollary 4.5 (the Gamma/Laguerre continuation). *For the moments $a_m = \Gamma(m + \alpha + 1) = \Gamma(\alpha + 1)(\alpha + 1)_m$ of the Laguerre weight $x^\alpha e^{-x}$ on $[0, \infty)$ (here $\rho = 1$),*

$$\det(\Gamma(2i+j+\alpha+1))_{0 \leq i,j \leq n-1} = \Gamma(\alpha+1)^n 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k! \prod_{m=1}^{n-1} [(\alpha+2m-1)(\alpha+2m)]^{n-m}, \quad (4.8)$$

which reduces to (4.7) at a non-negative integer $\alpha = r$.

Derivation. With $x_i = 2i$, $\rho = 1$ and $a_{2i} = \Gamma(\alpha+1)(\alpha+1)_{2i}$, Proposition 4.2 gives $\det(a_{2i+j}) = 2^{\binom{n}{2}} \prod_{k < n} k! \prod_i a_{2i}$. In $\prod_{i=0}^{n-1} (\alpha+1)_{2i} = \prod_{i=0}^{n-1} \prod_{l=1}^{2i} (\alpha+l)$ the factor $(\alpha+l)$ occurs once for each i with $2i \geq l$, hence with multiplicity $n - \lceil l/2 \rceil$; writing $l = 2m-1$ and $l = 2m$ gives the exponent $n-m$. $\square$

Corollary 4.6 (Catalan numbers). *Let $C_m = \frac{1}{m+1} \binom{2m}{m}$ and let $0 \leq x_0 < x_1 < \dots < x_{n-1}$ be integers. Then*

$$\det(C_{x_i+j})_{0 \leq i,j \leq n-1} = \prod_{0 \leq i < j \leq n-1} (x_j - x_i) \prod_{i=0}^{n-1} \frac{(2x_i)!(n+i)!}{(2i)! x_i! (x_i+n)!}. \quad (4.9)$$

In particular, $x_i = i$ recovers the well-known $\det(C_{i+j}) = 1$ [26]; $x_i = 2i$ gives the dilated Hankel determinant

$$\ddot{H}_n(C) = 2^{\binom{n}{2}} \prod_{i=0}^{n-1} \binom{4i}{2i} \frac{i!(n+i)!}{(n+2i)!} = 1, 3, 32, 1232, 172032, \dots; \quad (4.10)$$

and $x_i = ri$ gives the r -step analogue

$$\det(C_{ri+j})_{0 \leq i,j \leq n-1} = r^{\binom{n}{2}} \prod_{i=0}^{n-1} \frac{i!(2ri)!(n+i)!}{(2i)!(ri)!(ri+n)!}, \quad (4.11)$$

e.g. the 3-step Hankel values 1, 9, 990, 1363230, ... for $r = 3$.

Derivation. $C_m = 4^m (\frac{1}{2})_m / (2)_m$ is the member $(\alpha, \beta, \rho) = (\frac{1}{2}, 2, 4)$ of (4.1). The row factors of (4.2) are $C_x / (x+2)_{n-1} = \frac{(2x)!}{x!(x+1)!} \cdot \frac{(x+1)!}{(x+n)!} = \frac{(2x)!}{x!(x+n)!}$, and the constant is $4^{\binom{n}{2}} \prod_i (\frac{3}{2})_i = \prod_i \frac{(2i+1)!}{i!} = \prod_i \frac{(n+i)!}{(2i)!}$, using $(\frac{3}{2})_i = (2i+1)! / (4^i i!)$ and the fact that both $\prod_{i < n} (2i+1)! / i!$ and $\prod_{i < n} (n+i)! / (2i)!$ equal 1 at $n = 1$ and are multiplied by $(2n+1)! / n!$ when $n \mapsto n+1$. $\square$

The general Catalan evaluation (4.9) is not new: it is in the 1989 Gessel–Viennot preprint [14], proved in a journal by Krattenthaler [28, Theorem 3] via [24, Lemma 2.2], with earlier special cases in [35] and related evaluations in the compendia [26, 27]. The specialisations (4.10) and (4.11) appear to be unrecorded.

Corollary 4.7 (central binomial coefficients). *Let $b_m = \binom{2m}{m}$ and let $0 \leq x_0 < x_1 < \dots < x_{n-1}$ be integers. Then*

$$\det(b_{x_i+j})_{0 \leq i, j \leq n-1} = 2^{n-1} \prod_{0 \leq i < j \leq n-1} (x_j - x_i) \prod_{i=0}^{n-1} \frac{(2x_i)! (n-1+i)!}{(2i)! x_i! (x_i + n-1)!}. \quad (4.12)$$

In particular, $x_i = i$ recovers the classical $\det(b_{i+j}) = 2^{n-1}$; $x_i = 2i$ gives

$$\ddot{H}_n(b) = 2^{\binom{n}{2}+n-1} \prod_{i=0}^{n-1} \binom{4i}{2i} \frac{i! (n-1+i)!}{(n-1+2i)!} = 1, 8, 224, 22528, 8200192, \dots; \quad (4.13)$$

and $x_i = ri$ gives

$$\det(b_{ri+j})_{0 \leq i, j \leq n-1} = 2^{n-1} r^{\binom{n}{2}} \prod_{i=0}^{n-1} \frac{i! (2ri)! (n-1+i)!}{(2i)! (ri)! (ri + n-1)!}. \quad (4.14)$$

Derivation. $b_m = 4^m (\frac{1}{2})_m / (1)_m$ is the member $(\frac{1}{2}, 1, 4)$. The row factors are $b_x / (x+1)_{n-1} = (2x)! / (x! (x+n-1)!)$, and the constant is $4^{\binom{n}{2}} \prod_i (\frac{1}{2})_i = \prod_i \frac{(2i)!}{i!} = 2^{n-1} \prod_i \frac{(n-1+i)!}{(2i)!}$, using $(\frac{1}{2})_i = (2i)! / (4^i i!)$ and checking, as before, that the last two products agree at $n = 1$ and are both multiplied by $(2n)! / n!$ when $n \mapsto n+1$. $\square$

Remark 4.8. The case $x_i = i$ of (4.12), $\det(b_{i+j}) = 2^{n-1}$, is classical and is usually proved by continued fractions: from $\sum_m b_m z^m = (1-4z)^{-1/2} = 1/(1-2zc(z))$, with $c(z)$ the Catalan generating function and its Stieltjes fraction $c(z) = 1/(1-z/(1-z/(1-\dots)))$, the contracted Jacobi fraction has $\lambda_1 = 2$ and $\lambda_k = 1$ for $k \geq 2$, so the Heilermann–Stieltjes theorem [20, 47, 26] gives $\prod_k \lambda_k^{n-k} = 2^{n-1}$. Theorem 4.1 re-proves it without continued fractions.

Example 4.9. Three more integer members, with their dilated Hankel determinants (immediate from Corollary 4.3 and Proposition 4.2):

$$a_m = (2m+1)!! = 2^m (\frac{3}{2})_m : \quad \ddot{H}_n = 4^{\binom{n}{2}} \prod_{i=0}^{n-1} i! (4i+1)!! , \quad (4.15)$$

$$a_m = \frac{(2m)!}{m!} = 4^m (\frac{1}{2})_m : \quad \ddot{H}_n = 8^{\binom{n}{2}} \prod_{i=0}^{n-1} \frac{i! (4i)!}{(2i)!} , \quad (4.16)$$

$$a_m = \binom{2m+1}{m} = 4^m \frac{(\frac{3}{2})_m}{(2)_m} : \quad \ddot{H}_n = 2^{\binom{n}{2}} \prod_{i=0}^{n-1} \frac{i! (4i+1)! (n+i)!}{(2i+1)! (2i)! (2i+n)!} . \quad (4.17)$$

For the last sequence the ordinary Hankel determinant ($x_i = i$) equals 1.

4.3. The boundary of the mechanism.

Remark 4.10. Theorem 4.1 also delimits the mechanism. If the term ratio a_{m+1}/a_m is a rational function of degree $d \geq 2$ — as for the Fuss–Catalan numbers $\frac{1}{2m+1} \binom{3m}{m}$ or for $\binom{3m}{m}$ — the polynomials Q_j acquire degree $d(n-1) > n-1$ and the collapse onto the Vandermonde determinant fails; and indeed the corresponding dilated Hankel determinants exhibit large sporadic prime factors. The same holds for sequences that are not of hypergeometric type at all (Motzkin, Delannoy, Apéry, Franel, Bell, ...).

Remark 4.11. The formulas of this section explain the experimental observation that these determinants factorise into very small primes: a ratio of factorials of arguments at most $4n$ has no prime factor beyond $4n$.

5. INVOLUTIONS AND THE GAUSSIAN FAMILY

The Hankel determinants of the Gaussian (normal) moment sequence — and, more generally, of moment sequences of orthogonal polynomial families such as the Hermite polynomials — are classical, and are treated in the standard references on the moment problem and orthogonal polynomials: Shohat and Tamarkin [38], Akhiezer [1] and Szegő [44]. The evaluation below is a *dilated* (row-shifted) refinement of that classical Hankel determinant.

The Beta family of Section 4 is a plain sequence, and its evaluation was the Vandermonde reduction **M1**. In this section the sequence is given by an exponential generating function, $f(x) = \sum_n a_n x^n / n!$, and the tool is the one-functional reduction **M3** (Lemma 3.6). We carry it out on its simplest instance, the Gaussian family: a single orthogonal family (the Hermite polynomials) suffices, and the dilated Hankel determinant is triangularised directly.

The bookkeeping principle behind every evaluation here is Lemma 2.1, the change-of-family lemma: replacing the row and column families of a moment matrix by monic families of the same degrees leaves the determinant unchanged, and the triangular outcome is then read off the diagonal. Here the new families are the Hermite polynomials He_j and the even family Φ_i of Lemma 5.4, both monic of strictly increasing degree, hence unitriangular recombinations of the (split-)monomial families they replace.

Theorem 5.1. *Let c be an indeterminate (or any scalar) and let (a_n) be defined by*

$$f(x) = \sum_{n \geq 0} a_n \frac{x^n}{n!} = e^{cx+x^2/2}, \quad \text{i.e.} \quad a_n = \sum_{j \geq 0} \binom{n}{2j} (2j-1)!! c^{n-2j}.$$

Then for all $n \geq 1$,

$$\ddot{H}_n(f) = \det(a_{2i+j})_{0 \leq i,j \leq n-1} = (2c)^{\binom{n}{2}} \prod_{k=1}^{n-1} k!.$$

Corollary 5.2. *Let $f(x) = e^{x+x^2/2}$, so that a_n is the number of involutions of n letters. Then for all $n \geq 1$,*

$$\ddot{H}_n(f) = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k!.$$

Corollary 5.2 is the case $c = 1$ of Theorem 5.1: an involution of n letters is obtained by choosing the $2j$ non-fixed letters and one of the $(2j-1)!!$ perfect matchings on them, so $a_n = \sum_j \binom{n}{2j} (2j-1)!!$ is the number of involutions.

We use the *Gaussian moment functional* $\mathcal{L}$, defined on polynomials in z (with coefficients in $\mathbb{Q}[c]$) by linearity and

$$\mathcal{L}[z^m] = \begin{cases} (m-1)!! & \text{if } m \text{ is even,} \\ 0 & \text{if } m \text{ is odd,} \end{cases}$$

so that $a_n = \mathcal{L}[(c+z)^n]$. For an indeterminate α one has, as a formal power series in α (the computation is finite in each degree),

$$\mathcal{L}[e^{\alpha z}] = \sum_{j \geq 0} \frac{\alpha^{2j} (2j-1)!!}{(2j)!} = \sum_{j \geq 0} \frac{\alpha^{2j}}{2^j j!} = e^{\alpha^2/2}. \quad (5.1)$$

Let He_m denote the monic Hermite polynomials,

$$e^{tz-t^2/2} = \sum_{m \geq 0} \text{He}_m(z) \frac{t^m}{m!},$$

so that He_m is monic of degree m and $\text{He}_m(-z) = (-1)^m \text{He}_m(z)$.

Lemma 5.3 (mixed Hermite moments). *For all $p, q \geq 0$,*

$$\mathcal{L}[\text{He}_p(c+z) \text{He}_q(z)] = q! \binom{p}{q} c^{p-q}.$$

(In particular the value is 0 for $q > p$.)

Proof. By the generating function and (5.1),

$$\begin{aligned} \sum_{p,q \geq 0} \mathcal{L}[\text{He}_p(c+z) \text{He}_q(z)] \frac{t^p}{p!} \frac{s^q}{q!} &= \mathcal{L}\left[e^{t(c+z)-t^2/2} e^{sz-s^2/2}\right] = e^{tc-\frac{t^2+s^2}{2}} \mathcal{L}[e^{(t+s)z}] \\ &= e^{tc-\frac{t^2+s^2}{2}+\frac{(t+s)^2}{2}} = e^{t(c+s)} = \sum_{p \geq 0} \frac{t^p (c+s)^p}{p!}. \end{aligned}$$

Extracting the coefficient of $t^p s^q / (p! q!)$ gives the claim. $\square$

Lemma 5.4 (triangularising family). *For $i \geq 0$ put*

$$\Phi_i(z) := \sum_{k=0}^i \binom{i}{k} (-c^2)^k \text{He}_{2i-2k}(c+z).$$

Then Φ_i is a monic even polynomial of degree $2i$ in $u := c+z$, and for all $j \geq 0$

$$\mathcal{L}[\Phi_i(z) \text{He}_j(z)] = j! \binom{i}{j-i} (2c)^{2i-j}.$$

In particular this vanishes for $j < i$ and equals $i! (2c)^i$ for $j = i$.

Proof. Each $\text{He}_{2i-2k}(u)$ is an even monic polynomial in u of degree $2i-2k$, so Φ_i is even in u , of degree $2i$, with leading coefficient 1. By Lemma 5.3,

$$\mathcal{L}[\Phi_i \text{He}_j] = \sum_{k=0}^i \binom{i}{k} (-c^2)^k j! \binom{2i-2k}{j} c^{2i-2k-j} = j! c^{2i-j} \sum_{k=0}^i (-1)^k \binom{i}{k} \binom{2i-2k}{j}.$$

The last sum is the coefficient of y^j in

$$\sum_{k=0}^i (-1)^k \binom{i}{k} (1+y)^{2i-2k} = ((1+y)^2 - 1)^i = y^i (y+2)^i,$$

namely $2^{2i-j} \binom{i}{j-i}$, which gives the claim (and the vanishing for $j < i$, since $y^i (y+2)^i$ has valuation i). $\square$

Remark 5.5 (where the definition comes from). The formula for Φ_i is not guessed; it is forced. We seek a monic even polynomial of degree $2i$ in $u = c+z$ that is orthogonal to $\text{He}_0, \dots, \text{He}_{i-1}$ under the pairing $(f, g) \mapsto \mathcal{L}[f(z)g(z)]$. Writing it in the even Hermite basis, $\Phi_i = \sum_{k=0}^i \gamma_k \text{He}_{2i-2k}(c+z)$ with $\gamma_0 = 1$, and applying Lemma 5.3, the whole row assembles into one generating function,

$$\sum_{j \geq 0} \frac{\mathcal{L}[\Phi_i \text{He}_j]}{j!} y^j = \sum_{k=0}^i \gamma_k (c+y)^{2i-2k}.$$

The triangularity condition $\mathcal{L}[\Phi_i \text{He}_j] = 0$ for $j < i$ says this series has valuation $\geq i$ at $y = 0$; in the variable $w = c+y$ it says $\sum_k \gamma_k w^{2i-2k}$ is divisible by $(w-c)^i$. Being even and monic of degree $2i$, it is then divisible by $(w+c)^i$ as well, hence equal to $(w^2 - c^2)^i$. Matching coefficients gives $\gamma_k = \binom{i}{k} (-c^2)^k$. Equivalently, Φ_i is the Hermite-umbral image of $(u^2 - c^2)^i = (z(z+2c))^i$, obtained by replacing each power u^{2m} by $\text{He}_{2m}(u)$.

Proof of Theorem 5.1. Write $u = c+z$, so that

$$\ddot{H}_n = \det(a_{2i+j})_{0 \leq i, j \leq n-1} = \det\left(\mathcal{L}[u^{2i} \cdot u^j]\right)_{0 \leq i, j \leq n-1}.$$

Replace the column family $(u^j)_{j < n}$ by $(\text{He}_j(z))_{j < n}$, monic of degree j in u since $\text{He}_j(z) = \text{He}_j(u - c)$, and the row family $(u^{2i})_{i < n}$ by $(\Phi_i)_{i < n}$, monic even of degree $2i$ in u by Lemma 5.4. Both are unitriangular recombinations, so by Lemma 2.1

$$\ddot{H}_n = \det \left(\mathcal{L}[\Phi_i(z) \text{He}_j(z)] \right)_{0 \leq i, j \leq n-1}.$$

By Lemma 5.4 this last matrix is upper triangular (the (i, j) entry vanishes for $j < i$) with diagonal entries $i! (2c)^i$. Hence

$$\ddot{H}_n = \prod_{i=0}^{n-1} i! (2c)^i = (2c)^{\binom{n}{2}} \prod_{k=1}^{n-1} k!. \quad \square$$

Remark 5.6. (i) By the homogeneity $\ddot{H}_n(f(\sigma x)) = \sigma^{3\binom{n}{2}} \ddot{H}_n(f)$ (Lemma 3.7), Theorem 5.1 gives, for $f(x) = e^{bx+tx^2}$,

$$\ddot{H}_n(f) = (4bt)^{\binom{n}{2}} \prod_{k=1}^{n-1} k!.$$

(ii) For $c = 0$ (i.e. $f = e^{x^2/2}$, $a_{2k} = (2k-1)!!$, $a_{2k+1} = 0$) the theorem gives $\ddot{H}_n = 0$ for all $n \geq 2$.

Remark 5.7 (Gaussian rigidity among Appell families). The evaluation hinges on the shift identity of Lemma 5.3, $\mathcal{L}[\text{He}_p(c+z) \text{He}_q(z)] = q! \binom{p}{q} c^{p-q}$, which expresses that the Hermite polynomials form an *Appell* sequence, $\sum_{m \geq 0} \text{He}_m(z) t^m / m! = G(t) e^{tz}$ with $G(t) = e^{-t^2/2}$. The method therefore asks for a family (π_m) that is at once the sequence of monic orthogonal polynomials of $\mathcal{L}$ and an Appell sequence, and these two requirements already pin down the weight. Indeed, normalising $G(0) = 1$ and $\mathcal{L}[1] = 1$, orthogonality gives $\mathcal{L}[e^{\alpha z}] = 1/G(\alpha)$ and makes the mixed generating function

$$\sum_{p, q \geq 0} \mathcal{L}[\pi_p(z) \pi_q(z)] \frac{t^p}{p!} \frac{s^q}{q!} = \frac{G(t) G(s)}{G(t+s)}$$

a function of ts alone. With $\gamma = \log G$ this reads $\gamma(t) + \gamma(s) - \gamma(t+s) = \psi(ts)$, and matching the coefficient of $t^a s^b$ for $a \neq b$, $a, b \geq 1$, forces $\gamma_k = 0$ for $k \geq 3$, i.e. $\gamma(t) = \gamma_1 t + \gamma_2 t^2$. Hence $\mathcal{L}$ is Gaussian: up to a translation of z (the term γ_1) and a scaling (Remark 5.6(i)), the Hermite/Gaussian case is the only Appell family to which the one-functional triangularisation applies.

6. THE SINGLE SHIFT OF THE GAUSSIAN FAMILY

The same Hermite machinery (again the one-functional reduction **M3**) yields the derivative rule for this family. Since differentiation shifts the coefficient sequence, $a_n(f') = a_{n+1}(f)$, the dilated Hankel determinant of f' is the *shifted* dilated Hankel determinant of f : $\ddot{H}_n(f') = \det(a_{2i+j+1})_{0 \leq i, j \leq n-1}$. (For the rational chain the corresponding rule is the ratio formula in Corollary 4.4.)

Theorem 6.1 (Single shift of the Gaussian family). *Let $f(x) = e^{cx+x^2/2}$ as in Theorem 5.1. Then*

$$\ddot{H}_n(f') = c^n \ddot{H}_n(f) = c^n (2c)^{\binom{n}{2}} \prod_{k=1}^{n-1} k!.$$

In particular, for $c = 1$ the dilated Hankel determinant is invariant under differentiation: for the involution numbers a_n ,

$$\det(a_{2i+j+1})_{0 \leq i, j \leq n-1} = \det(a_{2i+j})_{0 \leq i, j \leq n-1} = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k!.$$

Proof. We keep the notation of the proof of Theorem 5.1 ($\mathcal{L}$, $u = c + z$, He_m , Φ_i). Since $a_{m+1} = \mathcal{L}[u^{m+1}]$,

$$\ddot{H}_n(f') = \det \left(\mathcal{L}[u^{2i} \cdot u^{j+1}] \right)_{0 \leq i, j \leq n-1}.$$

Replace the column family $(u^{j+1})_{j < n}$ by $(u \operatorname{He}_j(z))_{j < n}$ ($u \operatorname{He}_j(z) = u \operatorname{He}_j(u - c)$ is monic of degree $j + 1$ in u with no constant term) and the row family $(u^{2i})_{i < n}$ by

$$\Psi_i := \sum_{s=0}^i \frac{i!}{s!} (-2)^{i-s} \Phi_s,$$

monic and even of degree $2i$ in u ; both are unitriangular recombinations (Lemma 2.1). As in Remark 5.5, the weights $\frac{i!}{s!}(-2)^{i-s}$ are not guessed: they are exactly those that assemble the row generating functions into the truncated exponential T_i below, whose first-order differential equation delivers the required vanishing. We claim

$$\mathcal{L}[\Psi_i u \operatorname{He}_j] = \begin{cases} 0, & j < i, \\ c i! (2c)^i, & j = i, \end{cases} \quad (6.1)$$

which yields $\ddot{H}_n(f') = \prod_{i=0}^{n-1} c i! (2c)^i = c^n \ddot{H}_n(f)$ by triangularity.

To prove (6.1), first pair Φ_s with the new columns. By the recurrence $u \operatorname{He}_p(u) = \operatorname{He}_{p+1}(u) + p \operatorname{He}_{p-1}(u)$ applied to the argument $u = c + z$, and Lemma 5.3,

$$\mathcal{L}[\Phi_s u \operatorname{He}_j] = \sum_{k=0}^s \binom{s}{k} (-c^2)^k \left(j! \binom{2s-2k+1}{j} c^{2s-2k+1-j} + (2s-2k) j! \binom{2s-2k-1}{j} c^{2s-2k-1-j} \right).$$

Using $\sum_k (-1)^k \binom{s}{k} (1+y)^{2s-2k} = (y(2+y))^s$ and $m \binom{m-1}{j} = (j+1) \binom{m}{j+1}$, this becomes

$$\mathcal{L}[\Phi_s u \operatorname{He}_j] = j! c^{2s+1-j} [y^j] (1+y) (y(2+y))^s + (j+1)! c^{2s-1-j} [y^{j+1}] (y(2+y))^s.$$

Now put $w = w(y) := -\frac{c^2}{2} y(2+y)$, so that $c^{2s} (y(2+y))^s = (-2)^s w^s$ and, summing over s with the weights $\frac{i!}{s!} (-2)^{i-s}$,

$$\mathcal{L}[\Psi_i u \operatorname{He}_j] = i! (-2)^i j! c^{-1-j} \left(c^2 [y^j] (1+y) T_i + (j+1) [y^{j+1}] T_i \right), \quad T_i := \sum_{s=0}^i \frac{w(y)^s}{s!},$$

the truncated exponential. Since $\frac{dw}{dy} = -c^2(1+y)$ and $T'_i(w) = T_i(w) - \frac{w^i}{i!}$,

$$\frac{d}{dy} T_i + c^2(1+y) T_i = c^2(1+y) \frac{w^i}{i!},$$

whose right-hand side has y -valuation i (as w has valuation 1). Extracting the coefficient of y^j gives

$$(j+1) [y^{j+1}] T_i + c^2 [y^j] (1+y) T_i = \begin{cases} 0, & j < i, \\ \frac{(-1)^i c^{2i+2}}{i!}, & j = i, \end{cases}$$

using $[y^i] w^i = (-c^2)^i$. The case $j < i$ gives the vanishing in (6.1), and the case $j = i$ gives

$$\mathcal{L}[\Psi_i u \operatorname{He}_i] = i! (-2)^i \cdot i! c^{-1-i} \cdot \frac{(-1)^i c^{2i+2}}{i!} = i! 2^i c^{i+1} = c i! (2c)^i. \quad \square$$

7. THE EULER NUMBER FAMILY

The *Euler numbers* E_n are defined by [2, 43, 45, 18]

$$\tan x + \sec x = \sum_{n \geq 0} E_n \frac{x^n}{n!}, \quad (E_n)_{n \geq 0} = 1, 1, 1, 2, 5, 16, 61, 272, 1385, \dots \quad (7.1)$$

The even Euler numbers E_{2n} (resp. odd E_{2n+1}) are the secant (resp. tangent) numbers; by André's theorem E_n is the number of alternating permutations of $\{1, \dots, n\}$. The ordinary generating

functions of the secant and tangent numbers have the classical Stieltjes S -fractions

$$\sum_{k \geq 0} E_{2k} x^k = \frac{1}{1 - \frac{1 \cdot 1 x}{1 - \frac{2 \cdot 2 x}{1 - \ddots}}}, \quad \sum_{k \geq 0} E_{2k+1} x^k = \frac{1}{1 - \frac{1 \cdot 2 x}{1 - \frac{2 \cdot 3 x}{1 - \ddots}}},$$

with coefficients k^2 and $k(k+1)$ respectively.

We embed this pair into a two-parameter family, and specialise the general biorthogonal reduction **M2** of Section 3.2, in the S -fraction form of Section 3.2, to

$$u_j = j(j+s), \quad v_j = j(j+t) \quad (j \geq 1): \quad (7.2)$$

$\mathbf{a} = (a_n)_{n \geq 0}$ is the sequence whose even and odd parts have the Stieltjes continued fractions (3.10) with these coefficients. With coefficients $j(j+s)$, each series in (3.10) is a classical Stieltjes S -fraction [43, 47, 12]; at $s = 0, t = 1$ the two fractions reduce to the secant and tangent fractions above.

To describe the underlying function, we introduce the generalised secant numbers $E_{2k}^{(j)}$ by

$$\frac{1}{\cos(x)^j} = \sum_{k \geq 0} E_{2k}^{(j)} \frac{x^{2k}}{(2k)!}.$$

Their ordinary generating function has the Stieltjes S -fraction

$$\sum_{k \geq 0} E_{2k}^{(s+1)} x^k = \frac{1}{1 - \frac{1(1+s)x}{1 - \frac{2(2+s)x}{1 - \ddots}}},$$

whose coefficients are exactly $u_j = j(j+s)$ of (7.2). Thus, taking the even part of $\mathbf{a}$ from the secant fraction $u_j = j(j+s)$ and the odd part from $v_j = j(j+t)$, the exponential generating function of $\mathbf{a}$ is

$$\sum_{k \geq 0} a_k \frac{x^k}{k!} = \sum_{k \geq 0} a_{2k} \frac{x^{2k}}{(2k)!} + \sum_{k \geq 0} a_{2k+1} \frac{x^{2k+1}}{(2k+1)!} = \frac{1}{\cos(x)^{s+1}} + \int_0^x \frac{dy}{\cos(y)^{t+1}},$$

since $a_{2k} = E_{2k}^{(s+1)}$ and, differentiating the odd part, $a_{2k+1} = E_{2k}^{(t+1)}$. Accordingly we write

$$\ddot{H}_n = \ddot{H}_n \left(\frac{1}{\cos(x)^{s+1}} + \int_0^x \frac{dy}{\cos(y)^{t+1}} \right). \quad (7.3)$$

Throughout write $\bar{n} = \lceil n/2 \rceil$ and $\underline{n} = \lfloor n/2 \rfloor$, so $\bar{n} + \underline{n} = n$ and $\bar{n} \in \{\underline{n}, \underline{n} + 1\}$. The main result of this section is the following closed form, valid for *all* (s, t) ; the secant/tangent case $s = 0, t = 1$ and its relatives are collected in Section 27.

Proposition 7.1. *Let*

$$f(x) = \frac{1}{\cos(x)^{s+1}} + \int_0^x \frac{dy}{\cos(y)^{t+1}}.$$

Then,

$$\ddot{H}_n(f) = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{i=0}^{n-1} (2i)! \right) \left(\prod_{l=0}^{\bar{n}-1} (s+1)_{2l} \right) \left(\prod_{m=0}^{n-1} (t+1)_{2m} \right) \prod_{r=1}^{\bar{n}} \prod_{c=1}^{\underline{n}} \frac{(t-s)/2 + c - r}{n - r - c + 1}. \quad (7.4)$$

The sign $(-1)^{\binom{\bar{n}}{2}}$ together with the doubly-indexed product in (7.4) form the case-dependent *signed* factor

$$\Omega(\delta) := (-1)^{\binom{\bar{n}}{2}} \prod_{r=1}^{\bar{n}} \prod_{c=1}^{\underline{n}} \frac{\delta/2 + c - r}{n - r - c + 1} \quad (\delta = t - s),$$

so that (7.4) reads $\ddot{H}_n = \Omega(\delta) \prod_{i=0}^{n-1} (2i)! \prod_{l=0}^{\bar{n}-1} (s+1)_{2l} \prod_{m=0}^{\underline{n}-1} (t+1)_{2m}$.

The proof occupies Subsections 7.1–7.2 and is an instance of the collapse mechanism of Section 3.2: the orthogonal data of the family are explicit, the two functionals differ in the single parameter $(t-s)/2$, so the connection coefficients collapse to a single binomial term (Lemma 7.3), and the resulting binomial determinant is evaluated by the dual Jacobi–Trudi identity (Lemma 7.4). Subsection 7.3 then evaluates $\Omega(\delta)$ in closed form for odd δ (Lemma 7.5), the form in which (7.4) is specialised in Section 8 and in the corollaries of Section 27.

7.1. Orthogonal data and connection coefficients. Substituting (7.2) into Lemma 3.5, the recurrence coefficients of the $\mathcal{S}$ - and $\mathcal{T}$ -orthogonal polynomials P_i, Q_m become

$$\begin{aligned} c_i^S &= 2i(2i+s) + (2i+1)(2i+1+s), & \lambda_i^S &= (2i-1)(2i-1+s) 2i(2i+s), \\ c_m^T &= 2m(2m+t) + (2m+1)(2m+1+t), & \lambda_m^T &= (2m-1)(2m-1+t) 2m(2m+t) \end{aligned}$$

(so $c_0^S = 1+s$, $c_0^T = 1+t$), while the squared norms telescope into Pochhammer symbols,

$$h_i^S = \prod_{k=1}^{2i} u_k = \prod_{k=1}^{2i} k(k+s) = (2i)!(s+1)_{2i}, \quad h_m^T = \prod_{k=1}^{2m} v_k = (2m)!(t+1)_{2m}, \quad (7.5)$$

using $\prod_{k=1}^{2i} (k+s) = (s+1)(s+2) \cdots (s+2i) = (s+1)_{2i}$.

At the secant/tangent specialisation $s = 0$, $t = 1$ — needed repeatedly in later sections — the two families take the explicit form below; we write $\hat{P}_i := P_i|_{s=0}$ and $r_m := Q_m|_{t=1}$ for the resulting monic polynomials.

Lemma 7.2 (secant and tangent polynomials; classical). *The monic $\mathcal{S}$ -orthogonal polynomials $\hat{P}_i$ and the monic $\mathcal{T}$ -orthogonal polynomials r_m satisfy*

$$\begin{aligned} \hat{P}_{i+1} &= (y - (2i)^2 - (2i+1)^2) \hat{P}_i - ((2i-1)(2i))^2 \hat{P}_{i-1}, \\ r_{m+1} &= (y - 2(2m+1)^2) r_m - (2m-1)(2m)^2(2m+1) r_{m-1}, \end{aligned}$$

with $\mathcal{S}[\hat{P}_i \hat{P}_i] = \delta_{il} ((2i)!)^2$ and $\mathcal{T}[r_m r_{m'}] = \delta_{mm'} (2m)!(2m+1)!$.

Proof. This is the case $s = 0$ (resp. $t = 1$) of the present family. The secant and tangent numbers have the classical Stieltjes S -fractions [43, 47, 12] with coefficients $u_j = j^2$ and $v_j = j(j+1)$, and the recurrence coefficients and norms specialise to those displayed: $2m(2m+1) + (2m+1)(2m+2) = 2(2m+1)^2$, while (7.5) gives $h_i^S = (2i)!(1)_{2i} = ((2i)!)^2$ and $h_m^T = (2m)!(2)_{2m} = (2m)!(2m+1)!$. $\square$

For this family the connection-coefficient recurrence (3.9) of Lemma 3.3 has a closed-form solution — a single binomial term.

Lemma 7.3 (connection coefficients). *Write $P_i = \sum_{m=0}^i \kappa_{i,m} Q_m$ for the expansion of the $\mathcal{S}$ -orthogonal polynomials in the $\mathcal{T}$ -orthogonal basis. Then*

$$\kappa_{i,m} = \frac{(2i)!}{(2m)!} \binom{(t-s)/2}{i-m} \quad ((\binom{(t-s)/2}{d} = 0 \text{ for } d < 0),$$

and hence $\mathcal{T}[P_i Q_m] = \kappa_{i,m} (2m)!(t+1)_{2m} = (2i)!(t+1)_{2m} \binom{(t-s)/2}{i-m}$.

Proof. By Lemma 3.3 the $\kappa_{i,m}$ satisfy $\kappa_{i,i} = 1$, vanish for $m > i$ and $m < 0$, and obey the recurrence (3.9); together with $\kappa_{0,0} = 1$ these determine them. By (7.5), $\mathcal{T}[P_i Q_m] = \kappa_{i,m} \mathcal{T}[Q_m^2] = \kappa_{i,m} (2m)!(t+1)_{2m}$, so it remains to identify $\kappa_{i,m}$.

We verify that the candidate $\kappa_{i,m} = \frac{(2i)!}{(2m)!} \binom{(t-s)/2}{i-m}$, $d = i-m$, satisfies (3.9); for $i = 0, 1$ this holds by inspection. In general, divide (3.9) by $\kappa_{i,m}$ and use the neighbour ratios

$$\begin{aligned} \frac{\kappa_{i+1,m}}{\kappa_{i,m}} &= (2i+1)(2i+2) \frac{(t-s)/2 - d}{d+1}, & \frac{\kappa_{i,m-1}}{\kappa_{i,m}} &= (2m-1)(2m) \frac{(t-s)/2 - d}{d+1}, \\ \frac{\kappa_{i,m+1}}{\kappa_{i,m}} &= \frac{d}{((t-s)/2 - d + 1)(2m+1)(2m+2)}, & \frac{\kappa_{i-1,m}}{\kappa_{i,m}} &= \frac{d}{((t-s)/2 - d + 1)(2i-1)(2i)}, \end{aligned}$$

together with $\frac{\lambda_{m+1}^T}{(2m+1)(2m+2)} = (2m+1+t)(2m+2+t)$ and $\frac{\lambda_i^S}{(2i-1)(2i)} = (2i-1+s)(2i+s)$; the relation (3.9) becomes

$$\begin{aligned} & [(2i+1)(2i+2) - (2m-1)(2m)] \frac{(t-s)/2 - d}{d+1} \\ &= c_m^T - c_i^S + \frac{d}{(t-s)/2 - d + 1} [(2m+1+t)(2m+2+t) - (2i-1+s)(2i+s)]. \end{aligned}$$

With $i = m + d$ and the specialised values of c_i^S, c_m^T displayed above, both sides agree as polynomials in (m, d, s, t) — an elementary identity, routine to verify. Hence the candidate satisfies (3.9) and the initial conditions, so it equals $\kappa_{i,m}$. $\square$

7.2. A binomial determinant; proof of the closed form.

Lemma 7.4 (binomial determinant, general parameter). *Let $N \geq 1$, $q \in \{N, N+1\}$, and let a be arbitrary. Then*

$$\det \left(\binom{a}{q+r-m} \right)_{0 \leq r, m \leq N-1} = \prod_{r=1}^q \prod_{c=1}^N \frac{a+c-r}{(q-r) + (N-c) + 1}.$$

Proof. Recall the dual Jacobi–Trudi (Nägelsbach–Kostka) identity $s_\lambda = \det(e_{\lambda'_i - i + j})_{1 \leq i, j \leq \ell(\lambda')}$, where s_λ is the Schur symmetric function, e_k the elementary symmetric function, and λ' the conjugate partition of λ [31, I.3, (3.5)], [41, Cor. 7.16.2]. For integers $a \geq N$ this identity, applied to the rectangular partition (N^q) (whose conjugate is (q^N) , so $\lambda'_i = q$ and $e_{\lambda'_i - i + j} = e_{q-i+j}$) and specialised at a ones via $e_k(1^a) = \binom{a}{k}$, gives $\det(\binom{a}{q-i+j})$, which is the matrix below after transposing; hence

$$\det \left(\binom{a}{q+r-m} \right)_{0 \leq r, m \leq N-1} = s_{(N^q)}(\underbrace{1, \dots, 1}_a) = \prod_{r=1}^q \prod_{c=1}^N \frac{a+c-r}{(q-r) + (N-c) + 1}$$

by the hook content formula $s_\lambda(1^a) = \prod_{u \in \lambda} \frac{a+c(u)}{h(u)}$, where $c(u) = j-i$ is the content and $h(u)$ the hook length of the cell $u = (i, j)$ [41, Cor. 7.21.4], [31, I.3, Ex. 4]. Both sides are polynomials in a , so the identity holds for all a . $\square$

Proof of Proposition 7.1. By Lemma 7.3, $\kappa_{\bar{n}+r, m} = \frac{(2(\bar{n}+r))!}{(2m)!} \binom{(t-s)/2}{\bar{n}+r-m}$; pulling $(2(\bar{n}+r))!$ out of row r and $1/(2m)!$ out of column m and applying Lemma 7.4 with $a = (t-s)/2$,

$$\det(\kappa_{\bar{n}+r, m})_{0 \leq r, m \leq \bar{n}-1} = \left(\prod_{r=0}^{\bar{n}-1} (2(\bar{n}+r))! \right) \left(\prod_{m=0}^{\bar{n}-1} \frac{1}{(2m)!} \right) \times \prod_{r=1}^{\bar{n}} \prod_{c=1}^{\bar{n}} \frac{(t-s)/2 + c - r}{(\bar{n}-r) + (\bar{n}-c) + 1}.$$

Substituting into Lemma 3.4, with the norms $h_l^S = (2l)!(s+1)_{2l}$ and $h_m^T = (2m)!(t+1)_{2m}$ of (7.5), and cancelling the $(2m)!$ against the column norms, the surviving factors are

$$\prod_{m=0}^{\bar{n}-1} (t+1)_{2m} \quad \text{and} \quad \prod_{l=0}^{\bar{n}-1} (2l)! \cdot \prod_{r=0}^{\bar{n}-1} (2(\bar{n}+r))! = \prod_{i=0}^{\bar{n}-1} (2i)!,$$

which gives the stated formula. $\square$

7.3. The signed factor. We extend the double factorial to negative odd arguments by $(2k-1)!! = (2k+1)!!/(2k+1)$, so $(-1)!! = 1$ and $(-2k+1)!! = (-1)^{k-1}/(2k-3)!!$; with this convention the signed factor $\Omega(\delta)$ has the following closed evaluation.

Lemma 7.5. *With δ odd, we have*

$$\Omega(\delta) = (-1)^{\binom{\bar{n}}{2}} 2^{-\bar{n}\bar{n}} \frac{1}{\prod_{j=\bar{n}}^{n-1} j!} \prod_{j=0}^{\bar{n}-1} \frac{j!(\delta+2j)!!}{(\delta-2\bar{n}+2j)!!}. \quad (7.6)$$

In particular, for $\delta \in \{\pm 1, \pm 3\}$: when $n = 2m$ is even,

$$\begin{aligned}\Omega(1) &= 2^{-\binom{n}{2}}, & \Omega(-1) &= (-1)^m 2^{-\binom{n}{2}}, \\ \Omega(3) &= (-1)^{m-1} (n^2 - 1) 2^{-\binom{n}{2}}, & \Omega(-3) &= -(n^2 - 1) 2^{-\binom{n}{2}};\end{aligned}\tag{7.7}$$

when $n = 2m + 1$ is odd,

$$\Omega(1) = 2^{-\binom{n}{2}}, \quad \Omega(-1) = \Omega(3) = (-1)^m n 2^{-\binom{n}{2}}, \quad \Omega(-3) = -\frac{1}{3} n^2 (n^2 - 4) 2^{-\binom{n}{2}}.\tag{7.8}$$

Proof. Group the double product defining $\Omega(\delta)$ by rows. Since δ is odd, the numerator of row r clears to a double factorial,

$$\prod_{c=1}^{\underline{n}} \left(\frac{\delta}{2} + c - r \right) = 2^{-\underline{n}} \prod_{c=1}^{\underline{n}} (\delta - 2r + 2c) = 2^{-\underline{n}} \frac{(\delta + 2\underline{n} - 2r)!!}{(\delta - 2r)!!},$$

with the double-factorial convention above, while its denominator telescopes,

$$\prod_{c=1}^{\underline{n}} (n - r - c + 1) = \frac{(n - r)!}{(n - r - \underline{n})!} = \frac{(\bar{n} + \underline{n} - r)!}{(\bar{n} - r)!}$$

(recall $n - \underline{n} = \bar{n}$). Multiplying over $r = 1, \dots, \bar{n}$ produces the factor $2^{-\bar{n}\underline{n}}$ together with two telescoping ratios. The hook denominators give

$$\prod_{r=1}^{\bar{n}} \frac{(\bar{n} - r)!}{(\bar{n} + \underline{n} - r)!} = \frac{\prod_{j=0}^{\bar{n}-1} j!}{\prod_{j=\underline{n}}^{\bar{n}-1} j!} = \frac{\prod_{j=0}^{\underline{n}-1} j!}{\prod_{j=\bar{n}}^{\underline{n}-1} j!},$$

cancelling the common factorials $j = \underline{n}, \dots, \bar{n} - 1$; and, writing $\delta + 2\underline{n} - 2r = \delta - 2(r - \underline{n})$, the numerator double factorials give

$$\prod_{r=1}^{\bar{n}} \frac{(\delta + 2\underline{n} - 2r)!!}{(\delta - 2r)!!} = \frac{\prod_{k=1-\underline{n}}^{\bar{n}-\underline{n}} (\delta - 2k)!!}{\prod_{k=1}^{\bar{n}} (\delta - 2k)!!} = \frac{\prod_{i=0}^{\underline{n}-1} (\delta + 2i)!!}{\prod_{i=0}^{\underline{n}-1} (\delta - 2\bar{n} + 2i)!!},$$

cancelling the common arguments $k = 1, \dots, \bar{n} - \underline{n}$. Both ratios run over the same range $0 \leq j \leq \underline{n} - 1$, so the factorial numerator $\prod_{j=0}^{\underline{n}-1} j!$ merges into the double-factorial product; restoring the sign $(-1)^{\binom{\bar{n}}{2}}$ gives (7.6).

For $\delta = 1$ the two ratios above combine to a pure power of two. Writing $(1 - 2r)!! = (-1)^{r-1}/(2r - 3)!!$, the double-factorial ratio becomes $\prod_{r=1}^{\bar{n}} \frac{(2\underline{n}-2r+1)!!}{(1-2r)!!} = (-1)^{\binom{\bar{n}}{2}} \prod_{r=1}^{\bar{n}} (2r - 3)!! (2\underline{n} - 2r + 1)!!$; multiplied by the hook ratio $\prod_{r=1}^{\bar{n}} (r - 1)!/(\underline{n} + r - 1)!$ and the leading sign $(-1)^{\binom{\bar{n}}{2}}$, the two signs cancel and $\Omega(1) = 2^{-\bar{n}\underline{n}}Q$ with $Q = \prod_{r=1}^{\bar{n}} \frac{(2r-3)!! (2\underline{n}-2r+1)!! (r-1)!}{(\underline{n}+r-1)!}$. Along the ladder $(\underline{n}, \underline{n}) \rightarrow (\underline{n}+1, \underline{n}) \rightarrow (\underline{n}+1, \underline{n}+1)$ one has $Q = 1$ at $(0, 0)$ and each step multiplies Q by $2^{-\underline{n}}$ — via $(2\underline{n} - 1)!! \underline{n}!/(2\underline{n})!$ and $(2\underline{n} + 1)!! \underline{n}!/(2\underline{n} + 1)!$ respectively, both equal to $2^{-\underline{n}}$ — while $\bar{n}\underline{n} - \binom{n}{2}$ falls by $\underline{n}$; hence $Q = 2^{\bar{n}\underline{n} - \binom{n}{2}}$ and $\Omega(1) = 2^{-\bar{n}\underline{n}}Q = 2^{-\binom{n}{2}}$.

For $\delta \in \{-1, 3, -3\}$ we use a contiguous relation. Since $(x + 2)!! = (x + 2)x!!$ for every odd x (negative ones included, by the convention above), dividing (7.6) at $\delta + 2$ by the same at δ leaves only the top factor of each double factorial:

$$\frac{\Omega(\delta + 2)}{\Omega(\delta)} = \prod_{j=0}^{\underline{n}-1} \frac{\delta + 2 + 2j}{\delta + 2 - 2\bar{n} + 2j} = \begin{cases} \frac{(\delta + 2)(\delta + 4) \cdots (\delta + 2m)}{\delta(\delta - 2) \cdots (\delta - 2m + 2)}, & n = 2m, \\ \frac{(\delta + 2)(\delta + 4) \cdots (\delta + 2m)}{(\delta - 2m)(\delta - 2m + 2) \cdots (\delta - 2)}, & n = 2m + 1. \end{cases}$$

We chain outward from $\Omega(1) = 2^{-\binom{n}{2}}$.

$n = 2m$. At $\delta = 1$ the ratio is $\frac{(2m+1)!!}{(-1)^{m-1}(2m-3)!!} = (-1)^{m-1}(2m-1)(2m+1) = (-1)^{m-1}(n^2-1)$, so $\Omega(3) = (-1)^{m-1}(n^2-1)2^{-\binom{n}{2}}$. At $\delta = -1$ it is $\frac{(2m-1)!!}{(-1)^m(2m-1)!!} = (-1)^m$, so $\Omega(-1) = \Omega(1)/(-1)^m = (-1)^m 2^{-\binom{n}{2}}$. At $\delta = -3$ it is $\frac{-(2m-3)!!}{(-1)^m(2m+1)!!} = \frac{(-1)^{m+1}}{n^2-1}$, so $\Omega(-3) = (-1)^{m+1}(n^2-1)\Omega(-1) = -(n^2-1)2^{-\binom{n}{2}}$.

$n = 2m + 1$. The same three ratios are $\frac{(2m+1)!!}{(-1)^m(2m-1)!!} = (-1)^m n$ at $\delta = 1$, $\frac{(2m-1)!!}{(-1)^m(2m+1)!!} = \frac{(-1)^m}{n}$ at $\delta = -1$, and $\frac{-(2m-3)!!}{(-1)^m(2m+3)!!/3} = \frac{-3}{(-1)^m(n-2)n(n+2)}$ at $\delta = -3$ (using $(2m-1)(2m+1)(2m+3) = (n-2)n(n+2)$). Hence $\Omega(3) = (-1)^m n 2^{-\binom{n}{2}}$, $\Omega(-1) = (-1)^m n \Omega(1) = (-1)^m n 2^{-\binom{n}{2}}$ (so $\Omega(-1) = \Omega(3)$), and $\Omega(-3) = -\frac{(-1)^m(n-2)n(n+2)}{3} \Omega(-1) = -\frac{1}{3} n^2(n^2-4) 2^{-\binom{n}{2}}$. $\square$

8. THE DOUBLE SHIFT OF THE EULER NUMBER FAMILY

Alongside $\ddot{H}_n = \det(a_{2i+j})$ the most regular variant is the *double shift*

$$\ddot{H}_n^{(2)} := \det(a_{2i+j+2})_{0 \leq i, j \leq n-1}. \quad (8.1)$$

Shifting the column index by 2 *preserves* parity — in the moment picture (3.4) the even columns stay $\mathcal{S}$ -columns and the odd columns stay $\mathcal{T}$ -columns, each merely advanced by one power of y — so no even/odd swap occurs and the determinant stays a product for *all* (s, t) . (The single shift $\det(a_{2i+j+1})$, which swaps the two parities and is far less regular, is taken up in Section 9.)

Since $a_{2i+j+2} = \mathcal{E}[y^{i+1}\chi_j]$, the double shift is the dilated determinant of the *Christoffel transform* $y\mathcal{E}$ of $\mathcal{E}$ by $y = z^2$: its even and odd parts are $\mathcal{S}'[p] := \mathcal{S}[yp]$ and $\mathcal{T}'[p] := \mathcal{T}[yp]$; the biorthogonal reduction **M2** again applies. We first record what this transform does to the ingredients of Section 7 for *arbitrary* u_j, v_j , and express $\ddot{H}_n^{(2)}$ through the transformed connection determinant; only afterwards do we specialise to $u_j = j(j+s)$, $v_j = j(j+t)$, where that determinant collapses and the answer factors.

Lemma 8.1 (Christoffel transform by y , general u_j, v_j). *Let P_i, Q_m be the monic orthogonal polynomials of $\mathcal{S}, \mathcal{T}$ with squared norms h_i^S, h_m^T , and let P'_i, Q'_m be those of $\mathcal{S}', \mathcal{T}'$ with norms $h_i^{S'}, h_m^{T'}$ and connection coefficients $P'_i = \sum_m \kappa'_{i,m} Q'_m$. Put $D_n^S = (-1)^n P_n(0)$ and $D_m^T = (-1)^m Q_m(0)$. Then, for arbitrary u_j, v_j :*

- (i) $D_{n+1}^S = c_n^S D_n^S - \lambda_n^S D_{n-1}^S$ with $D_0^S = 1$, $D_1^S = c_0^S = u_1$, and likewise for D^T ;
- (ii) $\mathcal{S}', \mathcal{T}'$ are again moment functionals, with the J -fraction coefficients of the odd contraction (2.9), $\lambda_i^{S'} = u_{2i} u_{2i+1}$ and $c_i^{S'} = u_{2i+1} + u_{2i+2}$ (in particular $c_0^{S'} = u_1 + u_2$), and similarly for T' ;
- (iii) $h_i^{S'} = \frac{D_{i+1}^S}{D_i^S} h_i^S$ and $h_m^{T'} = \frac{D_{m+1}^T}{D_m^T} h_m^T$.

Proof. (i) Evaluating $P_{n+1} = (y - c_n^S)P_n - \lambda_n^S P_{n-1}$ at $y = 0$ gives $D_{n+1}^S = c_n^S D_n^S - \lambda_n^S D_{n-1}^S$, with $D_0^S = 1$ and (as $\lambda_0^S = 0$) $D_1^S = c_0^S = u_1$; likewise for D^T .

(ii) Multiplying $\mathcal{S}$ by its variable shifts the moments, $\mathcal{S}'[y^k] = \mathcal{S}[y^{k+1}] = \mu_{k+1}$, so $\mathcal{S}'$ is the Christoffel transform of $\mathcal{S}$ described in Section 2.6. Its J -fraction is the odd contraction (2.9) of the S -fraction $b_j = u_j$ of $\mathcal{S}$, which yields $c_i^{S'} = u_{2i+1} + u_{2i+2}$ and $\lambda_i^{S'} = u_{2i} u_{2i+1}$ at once [47, 12]. (Only the J -fraction of $\mathcal{S}'$ is needed below; $\mathcal{S}'$ itself need not have an S -fraction of the family.) The same computation with v_j gives $\mathcal{T}'$.

(iii) With $\mu_k = \mathcal{S}[y^k]$ and $H_i = \det(\mu_{a+b})_{a,b < i}$, the determinantal formula for P_i evaluated at $y = 0$ reads $(-1)^i P_i(0) H_i = \det(\mu_{a+b+1})_{a,b < i} = \det(\mathcal{S}'[y^{a+b}])_{a,b < i} =: H'_i$. Hence $h_i^{S'} = H'_{i+1}/H'_i = -\frac{P_{i+1}(0)}{P_i(0)} \frac{H_{i+1}}{H_i} = \frac{D_{i+1}^S}{D_i^S} h_i^S$; likewise for T' . $\square$

Proposition 8.2 (double shift, general form). *For arbitrary u_j, v_j ,*

$$\ddot{H}_n^{(2)} = (-1)^{\binom{n}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^{S'} \right) \left(\prod_{l=0}^{n-1} h_l^{T'} \right) \det(\kappa'_{\bar{n}+r,m})_{0 \leq r, m \leq n-1}.$$

Equivalently, telescoping the norms by Lemma 8.1(iii),

$$\ddot{H}_n^{(2)} = (-1)^n P_{\bar{n}}(0) Q_n(0) \frac{\det(\kappa'_{\bar{n}+r,m})}{\det(\kappa_{\bar{n}+r,m})} \ddot{H}_n, \quad (8.2)$$

where $\ddot{H}_n$ is the unshifted dilated determinant of Proposition 7.1.

Proof. $\ddot{H}_n^{(2)} = \ddot{H}_n(a_{\cdot+2})$ is the dilated determinant of the sequence with even part $\mathcal{S}'$ and odd part $\mathcal{T}'$, so the reduction of Lemma 3.4 applies verbatim with primes throughout, giving the first display. For the second, Lemma 8.1(iii) telescopes $\prod_{l=0}^{\bar{n}-1} h_l^{S'} = D_{\bar{n}}^S \prod_{l=0}^{\bar{n}-1} h_l^S$ and $\prod_{l=0}^{\bar{n}-1} h_l^{T'} = D_{\bar{n}}^T \prod_{l=0}^{\bar{n}-1} h_l^T$ (since $D_0^S = D_0^T = 1$); dividing by the same reduction (3.4) for $\ddot{H}_n$ and using $D_{\bar{n}}^S D_{\bar{n}}^T = (-1)^{\bar{n}+\underline{n}} P_{\bar{n}}(0) Q_{\underline{n}}(0) = (-1)^n P_{\bar{n}}(0) Q_{\underline{n}}(0)$ (here $\bar{n} + \underline{n} = n$) gives (8.2). $\square$

For general u_j, v_j the ratio $\det(\kappa')/\det(\kappa)$ in (8.2) is the one factor that need not simplify. We now specialise to the family $u_j = j(j+s)$, $v_j = j(j+t)$ of Section 7, where the ingredients of Lemma 8.1 acquire closed forms and—crucially—the connection determinant collapses.

Lemma 8.3 (Christoffel transform, special family). *For $u_j = j(j+s)$, $v_j = j(j+t)$, with $h_i^S = (2i)!(s+1)_{2i}$ and $h_m^T = (2m)!(t+1)_{2m}$:*

- (a) $D_n^S = (-1)^n P_n(0) = (2n-1)!! \prod_{k=0}^{n-1} (s+2k+1)$, and likewise $D_m^T = (2m-1)!! \prod_{k=0}^{m-1} (t+2k+1)$;
- (b) $h_i^{S'} = (2i+1)(s+2i+1) h_i^S$ and $h_m^{T'} = (2m+1)(t+2m+1) h_m^T$;
- (c) $\kappa'_{i,m} = \frac{(2i+1)!}{(2m+1)!} \binom{(t-s)/2}{i-m} = \frac{2i+1}{2m+1} \kappa_{i,m}$.

Proof. (a) By Lemma 8.1(i), $D_{n+1}^S = c_n^S D_n^S - \lambda_n^S D_{n-1}^S$ with $D_0^S = 1$, $D_1^S = s+1$. The displayed product satisfies it: $D_n^S/D_{n-1}^S = (2n-1)(s+2n-1)$, so $\lambda_n^S D_{n-1}^S/D_n^S = \lambda_n^S/[(2n-1)(s+2n-1)] = 2n(2n+s)$, and $c_n^S - 2n(2n+s) = (2n+1)(2n+1+s) = D_{n+1}^S/D_n^S$; the case of D^T is identical.

(b) By (a) $D_{i+1}^S/D_i^S = (2i+1)(s+2i+1)$, so Lemma 8.1(iii) gives the stated norms.

(c) By Lemma 8.1(ii) the pair $(\mathcal{S}', \mathcal{T}')$ is the construction of Section 7 with the index shift $j \mapsto j+1$ and the *same* parameter $(t-s)/2$; the connection coefficients therefore obey the recurrence (3.9) with $c^S, \lambda^S, c^T, \lambda^T$ replaced by their primed values from (ii). The displayed closed form satisfies it: the verification is that of Lemma 7.3 with $(2i)!, (2m)!$ advanced to $(2i+1)!, (2m+1)!$, again an identity of polynomials in (m, d, s, t) routine to check. $\square$

Proposition 8.4 (double shift, special family). *With $u_j = j(j+s)$, $v_j = j(j+t)$ as in Section 7,*

$$\ddot{H}_n^{(2)} := \det(a_{2i+j+2})_{0 \leq i, j \leq n-1} = (2n-1)!! \left(\prod_{k=0}^{\bar{n}-1} (s+2k+1) \right) \left(\prod_{k=0}^{\underline{n}-1} (t+2k+1) \right) \ddot{H}_n,$$

where $\ddot{H}_n$ is the unshifted dilated determinant of Proposition 7.1.

Proof. Start from the general form (8.2) of Proposition 8.2 and insert Lemma 8.3. By (c), pulling $2(\bar{n}+r)+1$ from row r and $1/(2m+1)$ from column m ,

$$\frac{\det(\kappa'_{\bar{n}+r, m})}{\det(\kappa_{\bar{n}+r, m})} = \frac{\prod_{r=0}^{\bar{n}-1} (2\bar{n}+2r+1)}{\prod_{m'=0}^{\bar{n}-1} (2m'+1)} = \frac{(2n-1)!!}{(2\bar{n}-1)!! (2\underline{n}-1)!!},$$

since $(2\bar{n}+1)(2\bar{n}+3) \cdots (2n-1) = (2n-1)!!/(2\bar{n}-1)!!$ (here $\bar{n} + \underline{n} = n$). By (a),

$$(-1)^n P_{\bar{n}}(0) Q_{\underline{n}}(0) = D_{\bar{n}}^S D_{\underline{n}}^T = (2\bar{n}-1)!! (2\underline{n}-1)!! \left(\prod_{k=0}^{\bar{n}-1} (s+2k+1) \right) \left(\prod_{k=0}^{\underline{n}-1} (t+2k+1) \right).$$

Multiplying, the $(2\bar{n}-1)!! (2\underline{n}-1)!!$ cancel, leaving the asserted factor times $\ddot{H}_n$. $\square$

The two new products are the odd-shift products $(s+1)(s+3) \cdots (s+2\bar{n}-1)$ and $(t+1)(t+3) \cdots (t+2\underline{n}-1)$. In particular $\ddot{H}_n^{(2)}$ factors into linear forms in (s, t) for every (s, t) : the $(t-s)/2$ part is inherited unchanged from $\ddot{H}_n$, and the only new factors are these odd-shift products and the scalar $(2n-1)!!$.

9. THE SINGLE SHIFT OF THE EULER NUMBER FAMILY ON THE LINE $t = 1$

After the double shift of Section 8 we turn to the *single shift*

$$\ddot{H}_n^{(1)} = \ddot{H}_n^{(1)}(\mathbf{a}) = \det(a_{2i+j+1})_{0 \leq i, j \leq n-1}, \quad (9.1)$$

the minor of the infinite Hankel matrix (a_{p+q}) that selects the *odd* rows $1, 3, \dots, 2n-1$ (rather than the even rows $0, 2, \dots, 2n-2$ of $\ddot{H}_n$). In the moment picture (3.4) the entry is $a_{2i+j+1} = \mathcal{E}[z^{2i} \cdot z^{j+1}]$, so the column family is shifted from $(\chi_0, \dots, \chi_{n-1})$ to $(\chi_1, \dots, \chi_n)$, and the parity of every column *swaps*: the z -type (odd) columns now form the full moment family attached to $\mathcal{T}$, while the even columns become the defective family $y^1, \dots, y^n$, missing y^0 . The biorthogonal reduction **M2** therefore applies with the two parities interchanged, and it is the *odd* part of $\mathbf{a}$ that governs the determinant.

The outcome is markedly less uniform than for $\ddot{H}_n$ (Proposition 7.1) and for the double shift (Section 8), both of which factor for *all* (s, t) . The single shift (9.1) factors into linear forms over $\mathbb{Q}$ on the line $t = 1$ (the canonical tangent fraction $v_j = j(j+1)$; Proposition 9.1) and—exceptionally—on the line $t = 3$ (Theorem 10.6); for the other integer values $t = 2, 4, 5, 6, \dots$ an irreducible quadratic (then cubic, $\dots$) factor appears already at $n = 4$. On the line $t = 1$ every factor is classical; at $t = 3$ the only departure from the $t = 1$ shape is a single *non-classical* linear factor with non-integer root.

Proposition 9.1 (shifted determinant, $t = 1$). *Let $u_j = j(j+s)$ and $v_j = j(j+1)$ (so the odd part of $\mathbf{a}$ is the tangent fraction). Then*

$$\ddot{H}_n^{(1)} = \tilde{K}_n \prod_{j=1}^n \left(\frac{s+1}{2}\right)_j^2 \prod_{j=1}^{n-1} \left(\frac{s}{2} + 1\right)_j \prod_{j=1}^{\bar{n}-1} \left(\frac{1-s}{2}\right)_j, \quad (9.2)$$

with a positive scalar $\tilde{K}_n$ independent of s . Equivalently, clearing the Pochhammer symbols, $\ddot{H}_n^{(1)} = K_n R_n(s)$ for another constant K_n independent of s , of sign $\text{sign } K_n = (-1)^{\binom{\bar{n}}{2}}$ (the sign contributed by the back-shift factors $(\frac{1-s}{2})_j$ on passing to the monic form; in particular $K_n > 0$ for $n = 1, 2$), where $R_n(s)$ is the monic product of linear forms

$$R_n(s) = \underbrace{\left(\prod_{j=1}^n \prod_{k=0}^{j-1} (s+2k+1) \right)^2}_{\text{odd shifts, doubled}} \underbrace{\prod_{j=1}^{n-1} \prod_{k=1}^j (s+2k)}_{\text{even shifts}} \underbrace{\prod_{j=1}^{\bar{n}-1} \prod_{k=1}^j (s-2k+1)}_{\text{back shifts}}. \quad (9.3)$$

The constant is fixed by the secant/tangent specialisation $s = 0$ (Corollary 27.9): since $\ddot{H}_n^{(1)}|_{s=0} = (2n-1)!! \ddot{H}_n|_{s=0} = (2n-1)!! 2^{-\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!$, one has

$$\tilde{K}_n = \frac{(2n-1)!! 2^{-\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!}{\prod_{j=1}^n \left(\frac{1}{2}\right)_j^2 \prod_{j=1}^{n-1} j! \prod_{j=1}^{\bar{n}-1} \left(\frac{1}{2}\right)_j}, \quad \left(\frac{1}{2}\right)_j = \frac{(2j-1)!!}{2^j} = \frac{(2j)!}{4^j j!}. \quad (9.4)$$

The first values are $\tilde{K}_n = 1, 12, 1440, 9676800, 7023034368000, 149513097738977280000, \dots$ for $n = 1, \dots, 6$.

The exponent of each linear form in (9.3) is transparent: $(s+2k+1)$ occurs to the power $2(\underline{n}-k)$, the even shift $(s+2k)$ to the power $\underline{n}-k$, and the back shift $(s-2k+1)$ to the power $\bar{n}-k$.

Like the double shift, the single shift is the dilated determinant of a shifted sequence, but now of $G_k := a_{k+1}$: indeed $\ddot{H}_n^{(1)} = \ddot{H}_n(G)$, and the even/odd split of G is

$$G_{2k} = a_{2k+1} = \mathcal{T}[y^k], \quad G_{2k+1} = a_{2k+2} = \mathcal{S}[y^{k+1}] = \mathcal{S}'[y^k],$$

so G has *even* functional $\mathcal{T}$ and *odd* functional the Christoffel transform $\mathcal{S}' = y\mathcal{S}$ of Lemma 8.1. The parities have swapped, and the relevant connection now links the orthogonal polynomials of the two *different* functionals $\mathcal{T}$ and $\mathcal{S}'$. On the line $t = 1$ it is still a single binomial.

Lemma 9.2 (mixed connection at $t = 1$). *Let $t = 1$, let Q_i be the $\mathcal{T}$ -orthogonal polynomials ($v_j = j(j+1)$), and let P'_m be the monic $\mathcal{S}'$ -orthogonal polynomials of Lemma 8.1. Writing $Q_i = \sum_m \tilde{\kappa}_{i,m} P'_m$,*

$$\tilde{\kappa}_{i,m} = \frac{(2i+1)!}{(2m+1)!} \binom{(s+1)/2}{i-m}.$$

Proof. By Lemma 8.1(ii) the P'_m obey $yP'_m = P'_{m+1} + c'_m P'_m + \lambda'_m P'_{m-1}$ with $c'_m = u_{2m+1} + u_{2m+2}$, $\lambda'_m = u_{2m} u_{2m+1}$, while at $t = 1$ the Q_i obey $Q_{i+1} = (y - c'_i)Q_i - \lambda'_i Q_{i-1}$ with c'_i, λ'_i from Lemma 3.5. Substituting the expansion $Q_i = \sum_m \tilde{\kappa}_{i,m} P'_m$ into the Q -recurrence and collecting the coefficient of P'_m yields a three-term recurrence in (i, m) of exactly the shape (3.9); the displayed closed form satisfies it, by the same elementary verification as in Lemma 7.3, now with $(2i)!, (2m)!$ advanced to $(2i+1)!, (2m+1)!$ and parameter $(s+1)/2$. $\square$

Proof of Proposition 9.1. Apply the reduction of Lemma 3.4 to G , whose even functional $\mathcal{T}$ has orthogonal polynomials Q and norms $h_l^T = (2l)!(t+1)_{2l}$, and whose odd functional $\mathcal{S}'$ has orthogonal polynomials P' , norms $h_c^{S'}$, and connection $Q_i = \sum_c \tilde{\kappa}_{i,c} P'_c$:

$$\ddot{H}_n^{(1)} = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^T \right) \left(\prod_{c=0}^{\bar{n}-1} h_c^{S'} \right) \det(\tilde{\kappa}_{\bar{n}+r,c})_{0 \leq r, c \leq \bar{n}-1}.$$

By Lemma 9.2, $\tilde{\kappa}_{\bar{n}+r,c} = \frac{(2(\bar{n}+r)+1)!}{(2c+1)!} \binom{(s+1)/2}{\bar{n}+r-c}$; pulling $(2(\bar{n}+r)+1)!$ out of row r and $1/(2c+1)!$ out of column c and applying the binomial determinant of Lemma 7.4 with $a = (s+1)/2$, $q = \bar{n}$, $N = \underline{n}$ (admissible since $\bar{n} \in \{\underline{n}, \underline{n}+1\}$),

$$\det(\tilde{\kappa}_{\bar{n}+r,c}) = \left(\prod_{r=0}^{\bar{n}-1} (2(\bar{n}+r)+1)! \right) \left(\prod_{c=0}^{\bar{n}-1} \frac{1}{(2c+1)!} \right) \prod_{r=1}^{\bar{n}} \prod_{c=1}^{\underline{n}} \frac{(s+1)/2 + c - r}{(\bar{n} - r) + (\underline{n} - c) + 1}.$$

With $h_l^T = (2l)!(2l+1)!$ and $h_c^{S'} = (2c+1)(s+2c+1)(2c)!(s+1)_{2c}$ (Lemma 8.3(b)), every factor is now linear in s ; an elementary rearrangement of Pochhammer symbols collects them into the closed forms (9.2)–(9.3). This is the parity-swapped analogue of Proposition 7.1, with $\mathcal{S}'$ in place of $\mathcal{S}$ and the parameter $(t-s)/2$ specialised through $t = 1$ to $(s+1)/2$. $\square$

10. THE SINGLE SHIFT OF THE EULER NUMBER FAMILY ON THE LINE $t = 3$

The single binomial of Lemma 9.2 is special to $t = 1$. Dividing the connection recurrence (3.9) (with the row family $\mathcal{T}$ and the column family $\mathcal{S}'$) by $\tilde{\kappa}_{i,m}$, the backward term carries the factor

$$\frac{\lambda_i^T}{2i(2i+1)} = \frac{(2i-1)(2i-1+t)(2i+t)}{2i+1},$$

which is a polynomial in i only when $(2i+t)$ cancels the denominator $(2i+1)$, i.e. when $t = 1$. Off that line no single binomial can solve the recurrence. In the first case beyond, $t = 3$, the mixed connection coefficient splits into a sum of *two* binomials (Lemma 10.1), and this is exactly as much structure as still closes: the evaluation is again the biorthogonal reduction **M2**, now supplemented by one application of the Cauchy–Binet formula.

Lemma 10.1 (mixed connection at $t = 3$). *Let $t = 3$, let Q_i be the $\mathcal{T}$ -orthogonal polynomials ($v_j = j(j+3)$), and let P'_m be the monic $\mathcal{S}'$ -orthogonal polynomials of Lemma 8.1. Writing $Q_i = \sum_m \tilde{\kappa}_{i,m} P'_m$,*

$$\tilde{\kappa}_{i,m} = \frac{(2i)!}{(2m)!} \binom{(s-1)/2}{i-m} + (s+1) \frac{(2i)!}{(2m+1)!} \binom{(s-3)/2}{i-m-1}.$$

Equivalently, with $a = (s+1)/2$ and $d = i - m$, $\tilde{\kappa}_{i,m} = \frac{(2i)!}{(2m+1)!} [(2m+1) \binom{a-1}{d} + 2a \binom{a-2}{d-1}]$.

Proof. Let $Q_i^{(t)}$ denote the $\mathcal{T}$ -orthogonal polynomials for parameter t . Both $Q^{(3)}$ and $Q^{(1)}$ are secant-family polynomials in the sense of Lemma 7.3, so expanding the first in the second (parameter $(1-3)/2 = -1$, and $\binom{-1}{d} = (-1)^d$ for $d \geq 0$),

$$Q_i^{(3)} = \sum_{l=m}^i \frac{(2i)!}{(2l)!} (-1)^{i-l} Q_l^{(1)}.$$

Substituting $Q_l^{(1)} = \sum_m \frac{(2l+1)!}{(2m+1)!} \binom{(s+1)/2}{l-m} P'_m$ from Lemma 9.2 and using $\frac{(2i)!}{(2l)!} (2l+1)! = (2i)! (2l+1)$,

$$\tilde{\kappa}_{i,m} = \frac{(2i)!}{(2m+1)!} \sum_{l=m}^i (-1)^{i-l} (2l+1) \binom{a}{l-m}, \quad a = \frac{s+1}{2}.$$

With $l = m + p$ the sum is $(-1)^d [(2m+1) \sum_{p=0}^d (-1)^p \binom{a}{p} + 2 \sum_{p=0}^d (-1)^p p \binom{a}{p}]$, $d = i - m$. The two alternating sums collapse by $\sum_{p=0}^d (-1)^p \binom{a}{p} = (-1)^d \binom{a-1}{d}$ and, via $p \binom{a}{p} = a \binom{a-1}{p-1}$, $\sum_{p=0}^d (-1)^p p \binom{a}{p} = a (-1)^d \binom{a-2}{d-1}$, giving $(2m+1) \binom{a-1}{d} + 2a \binom{a-2}{d-1}$. Since $a-1 = (s-1)/2$, $a-2 = (s-3)/2$ and $2a = s+1$, this is the stated form. $\square$

Remark 10.2. The second binomial is the “defect” term absent at $t = 1$; it vanishes only on the diagonal ($d = 0$, where $\tilde{\kappa}_{i,i} = 1$). It makes the reduction of Lemma 3.4 produce $\det(\tilde{\kappa}_{\bar{n}+r,m})$ as the determinant of a *sum* of two binomial matrices, which the single hook-content evaluation of Lemma 7.4 no longer closes by itself. Nonetheless the determinant *does* close: one application of the Cauchy–Binet formula (Lemma 10.3) turns it into a short sum of hook-content products (Proposition 10.5), and the outcome still factors completely into linear forms over $\mathbb{Q}$ (Theorem 10.6). The sole effect of the defect is to replace one classical linear factor by a single *non-classical* one (the carrier Γ_n of Theorem 10.6), with non-integer root. Thus $t = 3$ is a *second* factoring line beside $t = 1$; an irreducible higher-degree factor first occurs at the other values $t = 2, 4, 5, 6, \dots$. More generally for odd $t = 2r + 1$ one expands $Q^{(2r+1)}$ in $Q^{(1)}$ with parameter $(1 - (2r + 1))/2 = -r$, producing an $(r+1)$ -term sum of binomials, hence (after Cauchy–Binet) a sum of $\binom{n+r}{r}$ hook-content products; the present case is $r = 1$.

We now carry this out. Write $a = \frac{s+1}{2}$ and $b = a - 2 = \frac{s-3}{2}$ throughout, and recall $\bar{n} + \underline{n} = n$, $\bar{n} \in \{\underline{n}, \underline{n} + 1\}$.

Lemma 10.3 (Cauchy–Binet collapse at $t = 3$). *Let $t = 3$. Pulling $(2(\bar{n}+r))!$ from row r and $1/(2m+1)!$ from column m of $(\tilde{\kappa}_{\bar{n}+r,m})_{0 \leq r, m \leq \underline{n}-1}$ (Lemma 10.1),*

$$\det(\tilde{\kappa}_{\bar{n}+r,m}) = \left(\prod_{r=0}^{\underline{n}-1} (2\bar{n}+2r)! \right) \left(\prod_{m=0}^{\underline{n}-1} \frac{1}{(2m+1)!} \right) \det N, \quad N_{r,m} = (2m+1) \binom{b}{d} + (s+2m+2) \binom{b}{d-1}, \quad (10.1)$$

with $d = \bar{n} + r - m$. Let $V = ((\binom{b}{\bar{n}+r-k}))_{0 \leq r \leq \underline{n}-1, 0 \leq k \leq \underline{n}}$ be the $\underline{n} \times (\underline{n}+1)$ Pascal block, and let B be the $(\underline{n}+1) \times \underline{n}$ bidiagonal matrix with $B_{k,m} = (2m+1) [k=m] + (s+2m+2) [k=m+1]$. Then $N = VB$, and the Cauchy–Binet formula gives

$$\det N = \sum_{p=0}^{\underline{n}} c_p D_p, \quad c_p = \left(\prod_{m=0}^{p-1} (2m+1) \right) \left(\prod_{m=p}^{\underline{n}-1} (s+2m+2) \right), \quad (10.2)$$

$$D_p = \det \left(\binom{b}{\bar{n}+r-k} \right)_{\substack{0 \leq r \leq \underline{n}-1 \\ k \in \{0, \dots, \underline{n}\} \setminus \{p\}}}.$$

Proof. The row/column extraction (10.1) is immediate from the second display of Lemma 10.1, after rewriting the bracket by $\binom{a-1}{d} = \binom{a-2}{d} + \binom{a-2}{d-1}$ and $2a = s+1$ as $(2m+1) \binom{b}{d} + (s+2m+2) \binom{b}{d-1}$ (here $b = a - 2$). In this form column m of N is $(2m+1)V_{(m)} + (s+2m+2)V_{(m+1)}$, where $V_{(k)} = ((\binom{b}{\bar{n}+r-k}))_r$ is the k -th column of V ; this is exactly $N = VB$ with B as displayed. Since V is $\underline{n} \times (\underline{n}+1)$ and B is $(\underline{n}+1) \times \underline{n}$, the Cauchy–Binet formula reads $\det N = \det(VB) = \sum_{p=0}^{\underline{n}} \det V^{[\hat{p}]} \det B^{[\hat{p}]}$, the sum over the deleted common index p , where $[\hat{p}]$ omits column p of

V (resp. row p of B). By definition $\det V^{[\bar{p}]} = D_p$. Deleting row p of the bidiagonal B leaves a triangular matrix (rows $0, \dots, p-1$ carry the diagonal entries $2m+1$, rows $p+1, \dots, \underline{n}$ carry the subdiagonal entries $s+2m+2$), whence $\det B^{[\bar{p}]} = \prod_{m < p} (2m+1) \prod_{m \geq p} (s+2m+2) = c_p$. $\square$

Lemma 10.4 (hook-content value of D_p). *Each D_p in (10.2) is a determinant of binomial coefficients $\binom{b}{\beta_c+r}$ with distinct column shifts $\beta_c \in \{\bar{n}, \bar{n}-1, \dots, \bar{n}-\underline{n}\} \setminus \{\bar{n}-p\}$. By the dual Jacobi–Trudi identity (with $e_j = \binom{b}{j}$) it equals, up to sign, the Schur function $s_{\lambda^{(p)}}$ of the partition $\lambda^{(p)}$ read off from the strictly decreasing shifts β_c (for $p \in \{0, \underline{n}\}$ a rectangle, as in Lemma 7.4; otherwise a rectangle with one indentation), and the hook-content formula evaluates it as*

$$D_p = \prod_{u \in \lambda^{(p)}} \frac{b + c(u)}{h(u)},$$

a product of linear forms in $b = (s-3)/2$, hence in s . Since each content $c(u)$ is an integer, every root is an odd integer in s ; in particular each D_p factors completely into linear forms with integer roots. $\square$

Proof. This is the evaluation used already for the rectangular case in Lemma 7.4; deleting the single column $k = p$ from the contiguous block $\{0, \dots, \underline{n}\}$ replaces the rectangular partition there by the partition $\lambda^{(p)}$ with the stated one-row indentation. The dual Jacobi–Trudi identity $s_\lambda = \det(e_{\lambda'_i - i + j})$ applied with $e_j = \binom{b}{j}$ matches the matrix $((\binom{b}{\beta_c+r}))$ up to the reindexing $\beta_c \mapsto \lambda'$, and the hook-content formula [26, 27] closes it; both sides are polynomials in b , so the evaluation holds for symbolic b . $\square$

Combining Lemma 3.4 (in its parity-swapped form, as in the proof of Proposition 9.1) with Lemmas 10.3–10.4 gives a closed, fully proved evaluation of the single shift at $t = 3$.

Proposition 10.5 (single shift at $t = 3$: closed evaluation). *Let $u_j = j(j+s)$, $v_j = j(j+3)$. With $h_l^T = (2l)!(4)_{2l}$ and $h_c^{S'} = (2c+1)(s+2c+1)(2c)!(s+1)_{2c}$ (Lemma 8.3(b) at $t = 3$),*

$$\ddot{H}_n^{(1)}|_{t=3} = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^T \right) \left(\prod_{c=0}^{\bar{n}-1} \frac{h_c^{S'}}{(2c+1)!} \right) \left(\prod_{r=0}^{\bar{n}-1} (2\bar{n}+2r)! \right) \sum_{p=0}^{\bar{n}} c_p D_p, \quad (10.3)$$

with c_p, D_p from (10.2) and D_p evaluated by Lemma 10.4. Every factor on the right of (10.3), except possibly the sum $\sum_p c_p D_p$, is a product of linear forms in s ; whether $\ddot{H}_n^{(1)}|_{t=3}$ factors completely thus rests on that sum alone, and this is settled in Theorem 10.6.

Proof. Apply the reduction of Lemma 3.4 to $G_k = a_{k+1}$ exactly as in the proof of Proposition 9.1: the even functional is $\mathcal{T}$ (orthogonal polynomials Q , norms h_l^T) and the odd functional is the Christoffel transform $\mathcal{S}'$ (orthogonal polynomials P' , norms $h_c^{S'}$), with mixed connection $Q_i = \sum_c \tilde{\kappa}_{i,c} P'_c$. This gives $\ddot{H}_n^{(1)} = (-1)^{\binom{\bar{n}}{2}} (\prod_l h_l^T) (\prod_c h_c^{S'}) \det(\tilde{\kappa}_{\bar{n}+r,c})$. At $t = 3$ insert (10.1) and (10.2) of Lemma 10.3, cancelling one $(2c+1)!$ per column against the $h_c^{S'}$ factor as written. The norms $h_l^T, h_c^{S'}$ and the factorials are products of linear forms in s (the s -dependence sits in $(s+1)_{2c}$ and $s+2c+1$); each D_p is one by Lemma 10.4. $\square$

Theorem 10.6 (single shift at $t = 3$: factorisation). *Let $u_j = j(j+s)$, $v_j = j(j+3)$. Then $\ddot{H}_n^{(1)}|_{t=3} = \det(a_{2i+j+1})$ factors completely into linear forms over $\mathbb{Q}$. All its factors have integer roots except one, the unique non-classical factor (the carrier)*

$$\Gamma_n = \Gamma_n(s) := (2\bar{n}+1)s + (4\bar{n}\underline{n} - 2\bar{n} - 1) = (2\bar{n}+1)(s + 2\underline{n}) - (2n+1), \quad (10.4)$$

whose root $s = -2\underline{n} + \frac{2n+1}{2\bar{n}+1}$ is never an integer. Equivalently, in the collapse of Proposition 10.5,

$$\sum_{p=0}^{\bar{n}} c_p D_p = D_0 \cdot \frac{L_n}{2\bar{n}+1} \left(\prod_{j=0}^{\bar{n}-2} (s + 2j + 1) \right) \Gamma_n(s) \quad (10.5)$$

for a positive rational constant L_n , so the only factor of $\ddot{H}_n^{(1)}|_{t=3}$ that is not classical (i.e. not already present, with the same shape, on the line $t = 1$ of Proposition 9.1) is the carrier (10.4).

The first values illustrate the statement (an overall nonzero rational constant omitted):

$$\begin{aligned} n = 2 : & \quad (s+1)(3s+1), \\ n = 3 : & \quad (s-3)(s+1)(5s+3), \\ n = 4 : & \quad (s-3)(s-1)(s+1)^3(s+2)(s+3)(5s+11), \\ n = 5 : & \quad (s-5)(s-3)^2(s-1)(s+1)^3(s+2)(s+3)(7s+17), \end{aligned}$$

the rightmost factor in each line being the carrier Γ_n of (10.4).

The negative integer roots $s = -(2k+1)$ admit a clean structural explanation, which we record next; it rigorously accounts for a large part of the factorisation.

Lemma 10.7 (atomic degeneration at $s = -(2k+1)$). *Fix an integer $k \geq 0$. At $s = -(2k+1)$ the even part $1/\cos^{s+1} = \cos^{2k}$ is a trigonometric polynomial, so the even functional $\mathcal{S}$ becomes the atomic functional supported on the $k+1$ points $y_l = -(2l)^2$ ($l = 0, \dots, k$):*

$$a_{2j}|_{s=-(2k+1)} = \mathcal{S}[y^j] = \sum_{l=0}^k \omega_l (-4l^2)^j, \quad \omega_0 = \frac{1}{4^k} \binom{2k}{k}, \quad \omega_l = \frac{2}{4^k} \binom{2k}{k-l} \quad (l \geq 1). \quad (10.6)$$

Consequently, in $\ddot{H}_n^{(1)} = \det(a_{2i+j+1})$ at $s = -(2k+1)$ every odd column j (where a_{2i+j+1} is an even moment) is $(\sum_{l=1}^k \omega_l y_l^{(j+1)/2} y_l^i)_i$, a combination of the k geometric vectors $(y_l^i)_i$, $l = 1, \dots, k$ (the atom $y_0 = 0$ contributes nothing, since $y_0^{(j+1)/2} = 0$). Hence for $0 \leq k \leq \underline{n} - 1$ the $\underline{n}$ odd columns span a space of dimension $\leq k$, the corank of $\ddot{H}_n^{(1)}$ at $s = -(2k+1)$ is at least $\underline{n} - k$, and

$$(s + 2k + 1)^{\underline{n}-k} \mid \ddot{H}_n^{(1)}|_{t=3}. \quad (10.7)$$

Proof. Writing $\cos^{2k} x = 4^{-k} \sum_{l=-k}^k \binom{2k}{k+l} e^{2ilx} = 4^{-k} [\binom{2k}{k} + 2 \sum_{l=1}^k \binom{2k}{k-l} \cos 2lx]$ and reading off $a_{2j} = (2j)! [x^{2j}] \cos^{2k} x$ with $[x^{2j}] \cos 2lx = (-1)^j (2l)^{2j} / (2j)!$ gives (10.6). The odd-column statement follows since $a_{2i+j+1} = a_{2(i+(j+1)/2)} = \mathcal{S}[y^{i+(j+1)/2}]$, and an $n \times n$ matrix, $\underline{n}$ of whose columns lie in a k -dimensional space, has rank $\leq \bar{n} + k$, hence corank $\geq n - \bar{n} - k = \underline{n} - k$; a polynomial matrix of corank c at s_0 has det divisible by $(s - s_0)^c$, giving (10.7). $\square$

The bound (10.7) captures only part of the multiplicity; the rest comes from the reduced determinant $\det N$, whose rank at $s = -(2k+1)$ we now pin down exactly. This is the technical core.

Lemma 10.8 (rank of N at $s = -(2k+1)$). *For $0 \leq k \leq \underline{n} - 1$, the matrix N of (10.2) has rank $\leq k+1$ at $s = -(2k+1)$; equivalently corank $N \geq \underline{n} - 1 - k$, so $(s + 2k + 1)^{\underline{n}-1-k} \mid \det N$.*

Proof. At $s = -(2k+1)$ one has $b = -(k+2)$, so $\binom{b}{d} = (-1)^d \binom{k+1+d}{d}$; with $u = \bar{n} + r - m$ and the identity $\binom{k+u}{k+1} = \frac{u}{k+1} \binom{k+u}{k}$, the entry $N_{r,m} = (2m+1) \binom{b}{\bar{n}+r-m} + (s+2m+2) \binom{b}{\bar{n}+r-m-1}$ becomes $N_{r,m} = (-1)^u R(r, m)$ with

$$R(r, m) = (2m+1) \binom{k+u}{k} + 2k \binom{k+u}{k+1}.$$

The signs $(-1)^u = (-1)^{\bar{n}}(-1)^r(-1)^m$ split into nonzero row and column factors, so $\text{rank } N = \text{rank}(R(r, m))$. Put $X = \bar{n} + r$. By the Chu–Vandermonde identity $\binom{k+X-m}{i} = \sum_j \binom{k+X}{j} \binom{-m}{i-j}$,

$$R = \sum_{j=0}^{k+1} \binom{k+X}{j} G_j(m), \quad G_j(m) = (2m+1) \binom{-m}{k-j} + 2k \binom{-m}{k+1-j}. \quad (10.8)$$

The $k+2$ functions $\binom{k+X}{j}$, $j = 0, \dots, k+1$, are linearly independent (degrees $0, \dots, k+1$ in X), so $\text{rank } R = \dim \text{span}\{G_j\}_{j=0}^{k+1}$. Now both terms of G_j have degree $k+1-j$ in m , with combined leading coefficient

$$\frac{2(-1)^{k-j}}{(k-j)!} \left(1 - \frac{k}{k+1-j}\right) = \frac{2(-1)^{k-j}}{(k-j)!} \cdot \frac{1-j}{k+1-j},$$

which vanishes precisely at $j = 1$. Hence $\deg G_0 = k+1$, $\deg G_1 \leq k-1$, and $\deg G_j = k+1-j$ for $2 \leq j \leq k+1$ (degrees $k-1, k-2, \dots, 0$). The $k+2$ polynomials G_j therefore realise only the $k+1$ distinct degrees $\{0, 1, \dots, k-1\} \cup \{k+1\}$ (the value k is absent), so $\dim \text{span}\{G_j\} \leq k+1$. Thus $\text{rank } N \leq k+1$, and a polynomial matrix of corank $\geq \underline{n} - 1 - k$ has determinant divisible by $(s + 2k + 1)^{\underline{n}-1-k}$. $\square$

Remark 10.9. Lemmas 10.7 and 10.8 together pin the negative odd multiplicities exactly. In the two-sided form (10.3), $\ddot{H}_n^{(1)} = (s\text{-free constant}) \cdot \prod_c h_c^{S'} \cdot \det N$, the norm product $\prod_c h_c^{S'}$ contributes multiplicity $\underline{n} - k$ to $(s + 2k + 1)$ (from the single factor $h_k^{S'}$ and from $(s+1)_{2c}$ for $c \geq k+1$), while Lemma 10.8 contributes $\underline{n} - 1 - k$ from $\det N$; hence

$$(s + 2k + 1)^{2\underline{n}-1-2k} \mid \ddot{H}_n^{(1)}|_{t=3} \quad (0 \leq k \leq \underline{n} - 1), \quad (10.9)$$

which is the exact multiplicity of Theorem 10.6. The positive and non-classical factors are supplied by $\det N = D_0 \cdot \Sigma$, with D_0 the hook-content product of Lemma 10.4 and Σ of degree $\underline{n}$: that Σ vanishes at $s = -(2j+1)$ for $0 \leq j \leq \underline{n}-2$ follows from Lemma 10.8 ($\text{ord}_{-(2j+1)} \det N \geq \underline{n} - 1 - j$) together with $\text{ord}_{-(2j+1)} D_0 = \underline{n} - 2 - j$ (an explicit hook count), and the single remaining root is that of the carrier (10.4), determined in the proof of Theorem 10.6 below.

The last ingredient pins down the non-classical factor through the leading symbols of $\det N$; for this we need one trace identity for the binomial matrix $\bar{M} = \left(\binom{\bar{n}+r}{m}\right)_{0 \leq r, m \leq \underline{n}-1}$.

Lemma 10.10 (a trace identity). *Let $D_m = \text{diag}(0, 1, \dots, \underline{n} - 1)$. For every diagonal matrix $D_a = \text{diag}(a_0, \dots, a_{\underline{n}-1})$,*

$$\text{tr}(\bar{M}^{-1} D_a \bar{M} D_m) = \sum_{r=0}^{\underline{n}-1} a_r (\bar{n} + r) - \bar{n} a_0 \underline{n}. \quad (10.10)$$

Proof. From $m \binom{N}{m} = N \binom{N-1}{m-1}$ and Pascal's rule $\binom{N-1}{m-1} = \binom{N}{m} - \binom{N-1}{m}$ with $N = \bar{n} + r$, $m \binom{\bar{n}+r}{m} = (\bar{n} + r) \left[\binom{\bar{n}+r}{m} - \binom{\bar{n}+r-1}{m} \right]$, i.e. $\bar{M} D_m = D_{\bar{n}+r} (\bar{M} - \bar{M}^\downarrow)$, where $\bar{M}_{r,m}^\downarrow = \binom{\bar{n}+r-1}{m} = \bar{M}_{r-1,m}$ (the row $r = 0$ being the extension row $\eta_m = \binom{\bar{n}-1}{m}$). Thus $\bar{M}^\downarrow = Z^+ \bar{M} + e_0 \eta^\top$ with Z^+ the down-shift ($Z^+_{r,r'} = [r = r' + 1]$), and, using $D_{\bar{n}+r} e_0 = \bar{n} e_0$,

$$\bar{M} D_m \bar{M}^{-1} = D_{\bar{n}+r} - D_{\bar{n}+r} Z^+ - \bar{n} e_0 \eta^\top \bar{M}^{-1}.$$

Taking $\text{tr}(D_a \cdot)$: the first term gives $\sum_r a_r (\bar{n} + r)$, the second vanishes ($D_a D_{\bar{n}+r} Z^+$ is strictly lower-triangular), and the third gives $-\bar{n} a_0 \eta^\top \bar{M}^{-1} e_0$. Finally $v := \bar{M}^{-1} e_0$ satisfies $\sum_m \binom{\bar{n}+r}{m} v_m = \delta_{r,0}$, so the polynomial $f(x) := \sum_{m=0}^{\underline{n}-1} \binom{x}{m} v_m$ (of degree $\leq \underline{n} - 1$) has $f(\bar{n} + r) = \delta_{r,0}$, whence $f(x) = \prod_{j=1}^{\underline{n}-1} \frac{x - \bar{n} - j}{-j}$ and

$$\eta^\top \bar{M}^{-1} e_0 = f(\bar{n} - 1) = \prod_{j=1}^{\underline{n}-1} \frac{-(j+1)}{-j} = \prod_{j=1}^{\underline{n}-1} \frac{j+1}{j} = \underline{n}.$$

This gives (10.10). $\square$

Proof of Theorem 10.6. By (10.9) (Lemmas 10.7 and 10.8), $(s + 2k + 1)^{2\underline{n}-1-2k} \mid \ddot{H}_n^{(1)}$ for $0 \leq k \leq \underline{n} - 1$; this turns out to be the exact multiplicity of the negative odd roots. By Proposition 10.5 the rest of the statement is the collapse (10.5), i.e. that the polynomial $\Sigma(s) := \sum_p c_p D_p / D_0$, of degree $\underline{n}$, equals $A \prod_{j=0}^{\underline{n}-2} (s + 2j + 1) \cdot \Gamma_n(s)$ for a constant A . Its $\underline{n} - 1$ integer roots $s = -(2j + 1)$, $j = 0, \dots, \underline{n} - 2$, are already established (Remark 10.9: Lemma 10.8 gives $\text{ord}_{-(2j+1)} \det N \geq \underline{n} - 1 - j$ and the hook count gives $\text{ord}_{-(2j+1)} D_0 = \underline{n} - 2 - j$, so $\Sigma(-(2j + 1)) = 0$).

The last (non-classical) root is the sum of roots of Σ less the known ones: $\rho = \text{rootsum } \Sigma + (\underline{n} - 1)^2 = \text{rootsum}(\det N) - \text{rootsum}(D_0) + (\underline{n} - 1)^2$. The root-sum of D_0 is explicit: D_0 has the consecutive columns $\{1, \dots, \underline{n}\}$, hence is the rectangular case $p = 0$ of Lemma 10.4, the hook-content product over the rectangle with $\bar{n} - 1$ rows and $\underline{n}$ columns, whose b -factors

$b + c(u) = b + c - r$ have roots $s = 3 + 2(r - c)$; summing over $1 \leq r \leq \bar{n} - 1$, $1 \leq c \leq \underline{n}$,

$$\text{rootsum}(D_0) = \sum_{r=1}^{\bar{n}-1} \sum_{c=1}^{\underline{n}} (3 + 2(r - c)) = (\bar{n} - 1) \underline{n} (\bar{n} - \underline{n} + 2). \quad (10.11)$$

For $\text{rootsum}(\det N)$ we use the leading s -symbols. Writing $N_{r,m} = L_{r,m}s^u + L'_{r,m}s^{u-1} + \dots$ ($u = \bar{n} + r - m$) with $L_{r,m} = \frac{2\bar{n} + 2r + 1}{2^u u!}$, $L'_{r,m} = \frac{u(4 - u(2\bar{n} + 2r + 1))}{2^u u!}$, the factorisation $N = s^{\bar{n}} \text{diag}(s^r)(L + s^{-1}L' + \dots) \text{diag}(s^{-m})$ gives $\det N = s^{\bar{n}\underline{n}} \det L (1 + s^{-1} \text{tr}(L^{-1}L') + \dots)$ (here $\det L \neq 0$, since $L = \text{diag}(\frac{2\bar{n}+2r+1}{2^{\bar{n}+r}(\bar{n}+r)!}) \bar{M} \text{diag}(2^m m!)$ and $\det \bar{M} = 1$ by the Vandermonde of $\bar{n}, \dots, \bar{n} + \underline{n} - 1$), so $\text{rootsum}(\det N) = -\text{tr}(L^{-1}L')$.

To evaluate the trace, write $L' = W \odot L$ (Hadamard product) with $W_{r,m} = \frac{4u}{2\bar{n}+2r+1} - u^2$, $u = \bar{n} + r - m$, and split $W = W_r^{\text{row}} + W_m^{\text{col}} + W^{\text{mix}}$ into $W_r^{\text{row}} = \frac{4(\bar{n}+r)}{2\bar{n}+2r+1} - (\bar{n}+r)^2$, $W_m^{\text{col}} = -m^2$ and $W^{\text{mix}} = 2(\bar{n}+r)m - \frac{4m}{2\bar{n}+2r+1}$. By conjugation invariance of the trace the separable parts contribute $\text{tr}(L^{-1}D_{W^{\text{row}}}L) = \sum_r W_r^{\text{row}}$ and $\text{tr}(L^{-1}LD_{W^{\text{col}}}) = -\sum_m m^2$, while the mixed part gives $2\text{tr}(L^{-1}D_{\bar{n}+r}LD_m) - 4\text{tr}(L^{-1}PD_m)$, where $P_{r,m} = \frac{1}{2^u u!} = \frac{1}{2\bar{n}+2r+1} L_{r,m}$. Since $L = \text{diag}(\frac{2\bar{n}+2r+1}{2^{\bar{n}+r}(\bar{n}+r)!}) \bar{M} \text{diag}(2^m m!)$ and the two outer diagonals commute through the conjugation, both mixed traces collapse onto $\bar{M}$:

$$\text{tr}(L^{-1}D_{\bar{n}+r}LD_m) = \text{tr}(\bar{M}^{-1}D_{\bar{n}+r}\bar{M}D_m), \quad \text{tr}(L^{-1}PD_m) = \text{tr}(\bar{M}^{-1}D_{1/(2\bar{n}+2r+1)}\bar{M}D_m),$$

and Lemma 10.10 (with $a_r = \bar{n} + r$, resp. $a_r = \frac{1}{2\bar{n}+2r+1}$) evaluates them to $\sum_r (\bar{n} + r)^2 - \bar{n}^2 \underline{n}$, resp. $\sum_r \frac{\bar{n}+r}{2\bar{n}+2r+1} - \frac{\bar{n}\underline{n}}{2\bar{n}+1}$. Adding the four contributions, the $\frac{\bar{n}+r}{2\bar{n}+2r+1}$ terms cancel and, using $\sum_r (\bar{n} + r)^2 - \sum_m m^2 = \bar{n}\underline{n}(\bar{n} + \underline{n} - 1)$,

$$\text{tr}(L^{-1}L') = \bar{n}\underline{n}(\bar{n} + \underline{n} - 1) - 2\bar{n}^2 \underline{n} + \frac{4\bar{n}\underline{n}}{2\bar{n} + 1} = \bar{n}\underline{n}(\underline{n} - \bar{n} - 1) + \frac{4\bar{n}\underline{n}}{2\bar{n} + 1}. \quad (10.12)$$

Therefore $\rho = -\text{tr}(L^{-1}L') - (\bar{n} - 1)\underline{n}(\bar{n} - \underline{n} + 2) + (\underline{n} - 1)^2$; the $\bar{n}\underline{n}$ -terms collapse to $\underline{n}(2 - \underline{n})$ and the squares cancel, leaving $\rho = -2\underline{n} + 1 + \frac{2\underline{n}}{2\bar{n} + 1} = -2\underline{n} + \frac{2\underline{n} + 1}{2\bar{n} + 1}$ (using $2\bar{n} + 1 + 2\underline{n} = 2n + 1$), i.e. exactly the root of the carrier (10.4). Hence $\Sigma(s)$ has degree $\underline{n}$, the $\underline{n} - 1$ integer roots $-(2j + 1)$ and the root ρ , so it equals $A \prod_{j=0}^{\underline{n}-2} (s + 2j + 1) \cdot \Gamma_n(s)$ with $A = L_n / (2\bar{n} + 1)$. Finally $L_n > 0$: the quantity $A(2\bar{n} + 1)$ is the ratio of the leading coefficients of $\det N$ and D_0 , and both are positive — $\text{lc}(\det N) = \det L > 0$, as L is a product of two positive diagonal matrices and $\bar{M}$ with $\det \bar{M} = 1$, while D_0 is a hook-content product in $b = (s - 3)/2$ with positive leading coefficient. This proves the collapse (10.5) and with it Theorem 10.6. $\square$

We close this section with a complement that is not needed for the proof above: the moments and the determinant at $t = 3$ also satisfy an exact, though multi-term, *contiguous relation in s* .

Lemma 10.11 (contiguous relation at $t = 3$). *The moments $a_k = a_k(s)$ at $t = 3$ satisfy*

$$a_{2k}(s) = \frac{a_{2k+2}(s-2) + (s-1)^2 a_{2k}(s-2)}{s(s-1)}, \quad a_{2k+1}(s) = a_{2k+1}(s-2) \quad (10.13)$$

(the even functional $\mathcal{S}_s = \frac{1}{s(s-1)}(y + (s-1)^2)\mathcal{S}_{s-2}$ is a Christoffel transform of $\mathcal{S}_{s-2}$; the odd functional is s -independent). Consequently, since $a_{2k+2}(s-2)$ sits two columns to the right of $a_{2k}(s-2)$ in the Hankel array (a_{2i+j+1}) ,

$$(s(s-1))^{\underline{n}} \ddot{H}_n^{(1)}|_{t=3}(s) = \sum_{q=0}^{\underline{n}} (s-1)^{2q} G_q(s-2), \quad G_{\underline{n}}(s-2) = \ddot{H}_n^{(1)}|_{t=3}(s-2), \quad (10.14)$$

where G_q is the minor of the array at parameter $s - 2$ whose even columns are unchanged and whose $\underline{n}$ odd columns are the original ones for the first q and advanced by two indices for the remaining $\underline{n} - q$ (so G_q uses the odd columns $\{1, 3, \dots, 2\underline{n} + 1\} \setminus \{2q + 1\}$, one of them a border column).

Proof. With $f_w = \cos^{-w}$ one has $f_w'' = w(w+1)f_{w+2} - w^2 f_w$ (differentiate twice and use $\tan^2 = \sec^2 - 1$); reading off the coefficient of $x^{2k}/(2k)!$ with $w = s-1$ gives the even relation in (10.13), while $a_{2k+1} = E_{2k}^{(4)}$ is independent of s . In $M(s) = (a_{2i+j+1}(s))$, the even columns (j even, so $2i+j+1$ odd) are s -independent, while for odd j the relation reads $s(s-1)M(s)_{ij} = M(s-2)_{i,j+2} + (s-1)^2 M(s-2)_{ij}$ (here $M(s-2)_{i,j+2}$ is the column- $(j+2)$ entry by the Hankel index shift, or a border entry for the last odd column). Factoring $1/[s(s-1)]$ from each of the n odd columns and expanding each transformed odd column $M(s-2)_{i,j+2} + (s-1)^2 M(s-2)_{ij}$ by multilinearity, only the choices giving distinct column indices survive; these are indexed by a threshold q (the first q odd columns keep their index, the rest advance by two), with weight $(s-1)^{2q}$, yielding (10.14). $\square$

11. THE SECANT-NUMBER FAMILY $(1+x)/\cos(x)^{s+1}$: THE CASE $s=0$

Throughout this section and the next we study

$$f(x) = \frac{1+x}{\cos(x)^{s+1}},$$

splitting its exponential generating function into even and odd parts:

$$\sum_{k \geq 0} a_k \frac{x^k}{k!} = \sum_{k \geq 0} a_{2k} \frac{x^{2k}}{(2k)!} + \sum_{k \geq 0} a_{2k+1} \frac{x^{2k+1}}{(2k+1)!} = \frac{1}{\cos(x)^{s+1}} + \frac{x}{\cos(x)^{s+1}}.$$

The even part is the generating function of the generalised secant numbers $E_{2k}^{(s+1)}$ of Section 7, and in the odd part $x \cdot x^{2k}/(2k)! = (2k+1) x^{2k+1}/(2k+1)!$; hence

$$a_{2k} = E_{2k}^{(s+1)}, \quad a_{2k+1} = (2k+1) E_{2k}^{(s+1)}.$$

In the present section we evaluate the determinant at $s=0$, where both parity functionals turn out to be classical; Section 12 then treats general s , where the odd functional is not.

The case $s=0$. Here $a_{2k} = E_{2k}$ and $a_{2k+1} = (2k+1)E_{2k}$ are the secant numbers, so the even columns again see the secant functional $\mathcal{S}$ of Lemma 7.2, while the odd columns see the functional

$$\mathcal{T}^*[y^k] := (2k+1) E_{2k}.$$

Proposition 11.1. *Let $f(x) = \frac{1+x}{\cos(x)}$, so that $a_{2k} = E_{2k}$ and $a_{2k+1} = (2k+1)E_{2k}$ (secant numbers). Then for all $n \geq 1$,*

$$\ddot{H}_n(f) = 2^{\binom{n}{2}} ((n-1)!!)^2 \prod_{k=1}^{n-2} (k!!)^6.$$

The proof is the biorthogonal reduction **M2**: Lemma 11.2 identifies $\mathcal{T}^*$ as a classical (continuous dual Hahn) functional, Lemma 11.4 computes the connection coefficients between the two orthogonal families, and the determinant then collapses through Lemma 3.4.

Lemma 11.2 (the functional $\mathcal{T}^*$ at $s=0$ is classical). *The monic orthogonal polynomials ρ_m of $\mathcal{T}^*$ satisfy*

$$\rho_{m+1} = (y - 8m^2 - 8m - 3)\rho_m - (2m)^4 \rho_{m-1},$$

with orthogonality and norms $\mathcal{T}^[\rho_m \rho_{m'}] = \delta_{mm'} ((2m)!!)^4$.*

Proof. By the classical integral representation $E_{2k} = \int_{\mathbb{R}} u^{2k} \omega(u) du$ with $\omega(u) = \frac{1}{2} \operatorname{sech}(\pi u/2)$, an integration by parts gives

$$(2k+1)E_{2k} = \int_{\mathbb{R}} (u^{2k+1})' \omega(u) du = \int_{\mathbb{R}} u^{2k} \cdot u(-\omega'(u)) du,$$

so $\mathcal{T}^*$ is represented by the positive even weight $u(-\omega'(u)) = \frac{\pi}{4} \frac{u \sinh(\pi u/2)}{\cosh^2(\pi u/2)}$. Writing $u = 2w$

(so $y = 4w^2$), this weight is proportional to $w \sinh(2\pi w) \operatorname{sech}^3(\pi w) \propto |\Gamma(\frac{1}{2} + iw)^3 / \Gamma(2iw)|^2$, the continuous dual Hahn weight with parameters $(a, b, c) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. The standard continuous dual

Hahn data $A_m = (m + a + b)(m + a + c)$, $C_m = m(m + b + c - 1)$ then give, in the variable $y = 4w^2$,

$$c_m^* = 4(A_m + C_m - a^2) = 8m^2 + 8m + 3, \quad \lambda_m^* = 16 A_{m-1} C_m = 16m^4 = (2m)^4,$$

and norms $16^m m! (a + b)_m (a + c)_m (b + c)_m = 16^m (m!)^4 = ((2m)!!)^4$. (For comparison, $\mathcal{S}$ itself is the continuous dual Hahn measure with parameters $(\frac{1}{2}, \frac{1}{2}, 0)$, which gives back the secant data of Lemma 7.2.) $\square$

Remark 11.3 (J-fraction form). The ρ_m recurrence of Lemma 11.2 is equivalent to the Jacobi continued fraction for the generating function of $\mathcal{T}^*$: since $\mathcal{T}^*[y^k] = (2k + 1)E_{2k}$ with $\mathcal{T}^*[1] = 1$,

$$\sum_{k \geq 0} (2k + 1)E_{2k} t^k = \frac{1}{1 - c_0^* t - \frac{\lambda_1^* t^2}{1 - c_1^* t - \frac{\lambda_2^* t^2}{1 - c_2^* t - \frac{\lambda_3^* t^2}{1 - \ddots}}}}, \quad c_m^* = 8m^2 + 8m + 3, \quad \lambda_m^* = (2m)^4.$$

The first convergents reproduce $(2k + 1)E_{2k} = 1, 3, 25, 427, 12465, \dots$

Lemma 11.4 (connection formula). *For all $i, m \geq 0$,*

$$\mathcal{T}^*[\hat{P}_i \rho_m] = (i!)^2 (m!)^2 2^{2i+2m} \binom{1/2}{i-m},$$

$$\text{equivalently } \hat{P}_i = \sum_{m=0}^i \tilde{\kappa}_{i,m} \rho_m \text{ with } \tilde{\kappa}_{i,m} = \left(\frac{i!}{m!}\right)^2 4^{i-m} \binom{1/2}{i-m}.$$

Proof. As in the proof of Lemma 7.3: the recurrences of $\hat{P}_i$ (Lemma 7.2) and of ρ_m (Lemma 11.2) give

$$\tilde{\kappa}_{i+1,m} = \tilde{\kappa}_{i,m-1} + (c_m^* - c_i) \tilde{\kappa}_{i,m} + \lambda_{m+1}^* \tilde{\kappa}_{i,m+1} - \lambda_i \tilde{\kappa}_{i-1,m}.$$

The closed form holds for $i = 0, 1$ (indeed $\hat{P}_1 = y - 1 = \rho_1 + 2$); inserting it and dividing by $\tilde{\kappa}_{i,m}$ (with $d = i - m$) reduces the recurrence to the rational identity

$$4[(i+1)^2 - m^2] \frac{\frac{1}{2} - d}{d+1} = c_m^* - c_i + \frac{d}{\frac{3}{2} - d} [4(m+1)^2 - (2i-1)^2], \quad i = m + d,$$

again an elementary polynomial identity in (m, d) . $\square$

Proof of Proposition 11.1. Both parity functionals are classical: $\mathcal{S}$ has monic orthogonal polynomials $\hat{P}_l$ with norms $h_l^{\mathcal{S}} = ((2l)!)^2$ (Lemma 7.2), and $\mathcal{T}^*$ has monic orthogonal polynomials ρ_m with norms $h_m^{\mathcal{T}^*} = ((2m)!!)^4$ (Lemma 11.2). Lemma 3.4 therefore gives

$$\ddot{H}_n = (-1)^{\binom{n}{2}} \prod_{l=0}^{\bar{n}-1} ((2l)!)^2 \cdot \prod_{m=0}^{\bar{n}-1} ((2m)!!)^4 \cdot \det(\tilde{\kappa}_{\bar{n}+r,m})_{0 \leq r, m \leq \bar{n}-1},$$

with the connection coefficients $\tilde{\kappa}_{i,m} = (i!/m!)^2 4^{i-m} \binom{1/2}{i-m}$ of Lemma 11.4. Pulling $(i!)^2 4^i$ out of row r (where $i = \bar{n} + r$) and $(m!)^{-2} 4^{-m}$ out of column m leaves $\det\left(\binom{1/2}{\bar{n}+r-m}\right)_{0 \leq r, m \leq \bar{n}-1}$; together with the sign $(-1)^{\binom{\bar{n}}{2}}$, this determinant is the factor $\Omega(1) = 2^{-\binom{n}{2}}$ of Lemma 7.5 (Lemma 7.4 at $a = \frac{1}{2}$). Since $((2m)!!)^4 (m!)^{-2} 4^{-m} = (m!)^2 4^m$, this yields

$$\ddot{H}_n = 2^{-\binom{n}{2}} \prod_{l=0}^{\bar{n}-1} ((2l)!)^2 \cdot \prod_{i=\bar{n}}^{n-1} (i!)^2 4^i \cdot \prod_{m=0}^{\bar{n}-1} (m!)^2 4^m.$$

It remains to identify the right-hand side with $2^{\binom{n}{2}} ((n-1)!!)^2 \prod_{k=1}^{n-2} (k!!)^6$. Both sides equal 1 for $n = 1$, and when $n \rightarrow n + 1$ both sides are multiplied by $2^n (n!!)^2 ((n-1)!!)^4$: for the right-hand side this is immediate, and for the left-hand side one checks the two parities separately using $n! = n!! (n-1)!!$ (for n even, the ratio is $(n!)^4 / ((n/2)!)^2 = 2^n (n!!)^2 ((n-1)!!)^4$; for n odd it is $2^{2n-1} (\frac{n-1}{2}!)^2 (n!)^2 = 2^n (n!!)^2 ((n-1)!!)^4$). $\square$

12. THE SECANT-NUMBER FAMILY $(1+x)/\cos(x)^{s+1}$: GENERAL s

We now evaluate $\ddot{H}_n(f)$ for $f = \frac{1+x}{\cos(x)^{s+1}}$ in closed form, *simultaneously for all s* . Recall from Section 11 that $a_{2k} = E_{2k}^{(s+1)}$ and $a_{2k+1} = (2k+1)E_{2k}^{(s+1)}$; the corresponding even and odd functionals in the variable y are

$$\mathcal{S}[y^k] = E_{2k}^{(s+1)}, \quad \mathcal{T}^*[y^k] = (2k+1)E_{2k}^{(s+1)},$$

so that the matrix entries are $a_{2i+j} = \mathcal{S}[y^{i+l}]$ when $j = 2l$ is even and $a_{2i+j} = \mathcal{T}^*[y^{i+l}]$ when $j = 2l+1$ is odd.

Theorem 12.1. *Let $f(x) = \frac{1+x}{\cos(x)^{s+1}}$. Then for all $n \geq 1$,*

$$\ddot{H}_n(f) = c_n \prod_{i=1}^{n-1} (s+1)_i = c_n \prod_{j=1}^{n-1} (s+j)^{n-j}, \quad c_n = \frac{2^{\binom{n}{2}} ((n-1)!!)^2 \prod_{k=1}^{n-2} (k!!)^6}{\prod_{i=1}^{n-1} i!}, \quad (12.1)$$

where $(s+1)_i = (s+1)(s+2)\cdots(s+i)$. The constant c_n is exactly the value $\ddot{H}_n(f)|_{s=0}$ of Proposition 11.1 divided by $\prod_{i=1}^{n-1} i! = \prod_{i=1}^{n-1} (1)_i$.

The point of (12.1) is that, unlike the (s, t) family of Section 7, where *both* the even and the odd functional are classical, here the odd functional $\mathcal{T}^*$ is *not* a classical orthogonal-polynomial functional once $s \geq 1$: its monic three-term recurrence coefficients are non-polynomial rationals (its weight is $\propto u |\Gamma(\frac{s+1}{2} + iu)|^2 \Im \psi(\frac{s+1}{2} + iu)$, whose digamma factor degenerates to an elementary tanh only at $s = 0$). Nevertheless the determinant remains a smooth product for every s . The mechanism is a one-sided biorthogonalisation **M2** that uses *only* the classical even functional $\mathcal{S}$.

The proof plays a divisibility bound against a degree bound. Lemma 12.2 expresses $\ddot{H}_n(f)$ through a determinant $\det D$ built from $\mathcal{S}$ alone; Lemma 12.3 shows that the product $\prod_{j=1}^{n-1} (s+j)^{n-j}$, of degree $\binom{n}{2}$, divides $\ddot{H}_n(f)$; and Lemma 12.4 supplies the matching bound $\deg_s \ddot{H}_n(f) \leq \binom{n}{2}$. The quotient is therefore constant in s , and Proposition 11.1 evaluates it at $s = 0$.

Lemma 12.2 (one-sided reduction). *Let $P_0, P_1, \dots$ be the monic $\mathcal{S}$ -orthogonal polynomials, with $\mathcal{S}[P_i P_l] = \delta_{il} h_i^S$ and $h_i^S = \prod_{k=1}^{2l} k(k+s) = (2l)!(s+1)_{2l}$. Put $\bar{n} = \lceil n/2 \rceil$, $\underline{n} = \lfloor n/2 \rfloor$. Then*

$$\ddot{H}_n(f) = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^S \right) \det(\mathcal{T}^*[P_{\bar{n}+r} y^m])_{0 \leq r, m \leq \underline{n}-1}. \quad (12.2)$$

Moreover $\mathcal{T}^* = \mathcal{S} \circ (2y\partial_y + 1)$, so for $i > m$

$$\mathcal{T}^*[P_i y^m] = 2\mathcal{S}[y^{m+1} P'_i] \quad (\text{the term } (2m+1)\mathcal{S}[y^m P_i] \text{ vanishes since } \deg y^m < i). \quad (12.3)$$

Proof. Identity (12.2) is the general one-sided reduction (Lemma 3.2), applied with even functional $\mathcal{S}$ and odd functional $\mathcal{T}^*$: that lemma orthogonalises only the even columns and needs *only* that $\mathcal{S}$ be quasi-definite, holding for an arbitrary odd functional—here $\mathcal{T}^*$, whose orthogonal-polynomial structure is never used—with the explicit sign $(-1)^{\binom{\bar{n}}{2}}$. Finally $\mathcal{T}^*[y^k] = (2k+1)\mathcal{S}[y^k] = \mathcal{S}[(2y\partial_y + 1)y^k]$ gives $\mathcal{T}^*[P_i y^m] = \mathcal{S}[(2y\partial_y + 1)(P_i y^m)] = 2\mathcal{S}[y^{m+1} P'_i] + (2m+1)\mathcal{S}[y^m P_i]$, and the last term vanishes for $i > m$ by orthogonality, which holds throughout the block because $\bar{n} + r \geq \bar{n} \geq \underline{n} > m$. $\square$

Lemma 12.3 (rank drop at negative integers). *For each integer p with $1 \leq p \leq n-1$, the matrix $(a_{2i+j})_{0 \leq i, j < n}$ has rank at most p at $s = -p$; consequently $(s+p)^{n-p} \mid \ddot{H}_n(f)$ and, the factors being coprime, $\prod_{j=1}^{n-1} (s+j)^{n-j} \mid \ddot{H}_n(f)$.*

Proof. At $s = -p$ the even weight is the cosine polynomial

$$\cos(x)^{p-1} = 2^{-(p-1)} \sum_{t=0}^{p-1} \binom{p-1}{t} e^{i(p-1-2t)x},$$

so $a_{2k} = \sum_{t=0}^{p-1} w_t \theta_t^k$ with $w_t = 2^{-(p-1)} \binom{p-1}{t}$ and $\theta_t = -(p-1-2t)^2$, while $a_{2k+1} = (2k+1)a_{2k}$. Hence every column of (a_{2i+j}) lies in the span of the vectors $(\theta^i)_i$ and $(i\theta^i)_i$ taken over the $\lceil p/2 \rceil$ distinct values θ : indeed $a_{2i+2l} = \sum_{\theta} (\cdots) (\theta^i)_i$ and $a_{2i+2l+1} = \sum_{\theta} (\cdots) (2(i\theta^i)_i + (2l+1)(\theta^i)_i)$. A nonzero θ contributes two such vectors, while $\theta = 0$ (present iff p is odd) contributes only $(\theta^i)_i = e_0$ since $(i\theta^i)_i = 0$; so the column space has dimension $\leq p$. As $\ddot{H}_n(f) \neq 0$ (it is nonzero at $s = 0$ by Proposition 11.1) and the matrix has corank $\geq n - p$ at $s = -p$, a constant row change of basis makes $n - p$ rows divisible by $(s + p)$, so $(s + p)^{n-p} \mid \ddot{H}_n(f)$. $\square$

Lemma 12.4 (derivative connection coefficients). *Let $P_0, P_1, \dots$ be the monic $\mathcal{S}$ -orthogonal polynomials, with three-term recurrence $P_{i+1} = (y - b_i)P_i - \lambda_i P_{i-1}$, where $b_i = (4i+1)s + 8i^2 + 4i + 1$ and $\lambda_i = 2i(2i-1)(s+2i-1)(s+2i)$, and set*

$$T_i := yP'_i - iP_i = \sum_{a=0}^{i-1} \Delta_{i,a} P_a. \quad (12.4)$$

Then

$$\Delta_{i,a} = (-1)^{i-a-1} \frac{2^{i-a-1}}{2(i-a)-1} \frac{i!}{a!} \frac{(2i-1)!!}{(2a-1)!!} \left(s + \frac{2(i+a)+1}{2(i-a)+1} \right), \quad (12.5)$$

and in particular $\deg_s \Delta_{i,a} \leq 1$ for all $0 \leq a \leq i-1$.

Proof. Differentiating the recurrence gives $P'_{i+1} = (y - b_i)P'_i + P_i - \lambda_i P'_{i-1}$; multiplying by y and using (12.4) yields the structure recurrence

$$T_{i+1} = (y - b_i)T_i - \lambda_i T_{i-1} + b_i P_i + 2\lambda_i P_{i-1}.$$

Expanding in the P -basis with $(y - b_i)P_a = P_{a+1} + (b_a - b_i)P_a + \lambda_a P_{a-1}$ gives

$$\Delta_{i+1,a} = \Delta_{i,a-1} + (b_a - b_i)\Delta_{i,a} + \lambda_{a+1}\Delta_{i,a+1} - \lambda_i \Delta_{i-1,a} + b_i [a=i] + 2\lambda_i [a=i-1],$$

together with $T_0 = 0$ and $T_1 = yP'_1 - P_1 = s + 1$. The expression (12.5) satisfies this at every a — the homogeneous relation for $a \leq i-2$ and the two inhomogeneous boundary relations at $a = i-1$ and $a = i$ — and the base case; since the recurrence determines all $\Delta_{i,a}$ uniquely, (12.5) holds. Each check is a direct rational-function identity in i, a, s . (At $s \rightarrow \infty$, with $y = s\eta$, the functional $\mathcal{S}$ degenerates to the classical Laguerre ^{$(-1/2)$} functional and (12.5) reduces to its structure relation $\eta\pi'_i = i\pi_i + \rho_i\pi_{i-1}$; for finite s the lower coefficients are merely linear in s .) $\square$

Proof of Theorem 12.1. By Lemma 12.2, $\ddot{H}_n(f) = (-1)^{\binom{\bar{n}}{2}} (\prod_{l=0}^{\bar{n}-1} h_l^S) \det D$, whose entries are $D_{r,m} = 2\mathcal{S}[y^{m+1}P'_{\bar{n}+r}]$. Since $\mathcal{S}[y^{m+1}P'_i] = \mathcal{S}[y^m yP'_i] = \mathcal{S}[y^m(T_i + iP_i)] = \mathcal{S}[y^m T_i]$ (the term $i\mathcal{S}[y^m P_i]$ vanishes because $m < \bar{n} \leq i$), Lemma 12.4 gives

$$D_{r,m} = 2\mathcal{S}[y^m T_{\bar{n}+r}] = 2 \sum_{a=0}^m \Delta_{\bar{n}+r,a} \mathcal{S}[y^m P_a],$$

where only $a \leq m$ survive by orthogonality. Writing $P_a = \sum_t p_{a,t} y^t$, the recurrence gives $\deg_s p_{a,t} \leq a - t$, and $\deg_s E_{2k}^{(s+1)} = k$, so $\deg_s \mathcal{S}[y^m P_a] = \deg_s \sum_t p_{a,t} E_{2(m+t)}^{(s+1)} \leq m + a \leq 2m$. Combined with $\deg_s \Delta_{\bar{n}+r,a} \leq 1$ this yields $\deg_s D_{r,m} \leq 2m + 1$, hence

$$\deg_s \det D \leq \sum_{m=0}^{\bar{n}-1} (2m+1) = \underline{n}^2, \quad \deg_s \ddot{H}_n(f) \leq \underline{n}^2 + \bar{n}(\bar{n}-1) = \binom{\bar{n}}{2}.$$

On the other hand, Lemma 12.3 shows $\prod_{j=1}^{\bar{n}-1} (s+j)^{n-j}$, of degree $\binom{\bar{n}}{2}$, divides $\ddot{H}_n(f)$. The two facts force

$$\ddot{H}_n(f) = c_n \prod_{j=1}^{\bar{n}-1} (s+j)^{n-j} = c_n \prod_{i=1}^{\bar{n}-1} (s+1)_i$$

with c_n a constant, fixed by setting $s = 0$: there $\prod_{i=1}^{\bar{n}-1} (1)_i = \prod_{i=1}^{\bar{n}-1} i!$ and $\ddot{H}_n(f)|_{s=0}$ is the value of Proposition 11.1, giving c_n as in (12.1). $\square$

Remark 12.5 (alternative evaluation via Cauchy–Binet). The determinant $\det D$ can also be expanded directly. Writing $y^{m+1} = \sum_{a \leq m+1} \beta_a^{(m+1)} P_a$ (with $\beta_{m+1}^{(m+1)} = 1$) and $P'_i = \sum_{a \leq i-1} \gamma_a^{(i)} P_a$ and using $\mathcal{S}[P_a P_b] = \delta_{ab} h_a^S$,

$$D_{r,m} = 2 \sum_{a=0}^{m+1} \beta_a^{(m+1)} \gamma_a^{(\bar{n}+r)} h_a^S, \quad \text{i.e.} \quad D = 2 \Gamma \operatorname{diag}(h_a^S) B^\top,$$

with $\Gamma_{r,a} = \gamma_a^{(\bar{n}+r)}$ and $B_{m,a} = \beta_a^{(m+1)}$ ($0 \leq a \leq \underline{n}$). Here Cauchy–Binet is unavoidable: the inner index a ranges over the $\underline{n} + 1$ values $0, \dots, \underline{n}$ (it reaches $a = m + 1 = \underline{n}$ when $m = \underline{n} - 1$), one more than the $\underline{n}$ rows r and columns m , so Γ and $B^\top$ are *rectangular* and the naive factorisation $\det D = 2^{\underline{n}} \det \Gamma \det \operatorname{diag}(h_a^S) \det B^\top$ is meaningless. Instead, Cauchy–Binet expands $\det D$ as the sum over the $\underline{n}$ -subsets $S \subset \{0, \dots, \underline{n}\}$ of $2^{\underline{n}} \det(\Gamma_{\cdot, S}) (\prod_{a \in S} h_a^S) \det(B_{\cdot, S})$ — one term for each of the $\underline{n} + 1$ ways to drop a single inner index. Each entry $D_{r,m}$ carries a factor $(s+1)$ and column m has s -degree $2m+1$ (the per-column form of $\deg_s \Delta_{i,a} \leq 1$ in Lemma 12.4), recovering $\det D = K_n (s+1)^{\underline{n}} \prod_{j=1}^{\underline{n}-1} [(s+2j)(s+2j+1)]^{\underline{n}-j}$; multiplying by $\prod_{l=0}^{\bar{n}-1} (2l)! (s+1)_{2l}$ gives (12.1) again.

Remark 12.6 (first values of c_n). At $s = 0$ the formula recovers Proposition 11.1. The first values of the constant are

$$c_1, c_2, \dots = 1, 2, 2^4, 2^{10} 3, 2^{17} 3^4, 2^{31} 3^5 \cdot 5, \dots$$

13. THE SINGLE SHIFT OF THE SECANT-NUMBER FAMILY $(1+x)/\cos(x)^{s+1}$

Recall from Section 12 the family

$$f(x) = \frac{1+x}{\cos(x)^{s+1}}, \quad a_{2k} = E_{2k}^{(s+1)}, \quad a_{2k+1} = (2k+1)E_{2k}^{(s+1)},$$

with $\cos(x)^{-(s+1)} = \sum_{k \geq 0} E_{2k}^{(s+1)} x^{2k} / (2k)!$, and the two moment functionals on $\mathbb{Q}[y]$,

$$\mathcal{S}[y^k] = E_{2k}^{(s+1)}, \quad \mathcal{T}^*[y^k] = (2k+1)E_{2k}^{(s+1)}.$$

Here $\mathcal{S}$ is the classical secant functional, with monic orthogonal polynomials P_i , three-term recurrence $P_{i+1} = (y - c_i^S)P_i - \lambda_i^S P_{i-1}$,

$$c_i^S = (4i+1)s + 8i^2 + 4i + 1, \quad \lambda_i^S = 2i(2i-1)(s+2i-1)(s+2i), \quad (13.1)$$

squared norms $h_i^S = \mathcal{S}[P_i^2] = (2i)!(s+1)_{2i}$, and (Lemma 12.4 at $a = 0$)

$$\mathcal{T}^*[P_i] = (-1)^{i-1} 2^i i! (2i-3)!! (s+1) \quad (i \geq 1). \quad (13.2)$$

The dilated determinant of f is, for an explicit constant c_n (Theorem 12.1),

$$\ddot{H}_n(f) = \det(a_{2i+j})_{0 \leq i, j < n} = c_n \prod_{i=1}^{n-1} (s+1)_i. \quad (13.3)$$

Throughout $\bar{n} = \lceil n/2 \rceil$, $\underline{n} = \lfloor n/2 \rfloor$, and $\prod_{i=1}^{n-1} (s+1)_i = \prod_{j=1}^{n-1} (s+j)^{n-j}$ (degree $\binom{n}{2}$). The single shift (the dilated determinant of f') is

$$\ddot{H}_n^{(1)} := \det(a_{2i+j+1})_{0 \leq i, j < n}.$$

Theorem 13.1. *For all $n \geq 1$,*

$$\ddot{H}_n^{(1)} = ((n-1)!!)^2 \ddot{H}_n(f) = c_n^{(1)} \prod_{i=1}^{n-1} (s+1)_i, \quad c_n^{(1)} = ((n-1)!!)^2 c_n.$$

The proof has two parts. Part 1 shows that the ratio $\ddot{H}_n^{(1)} / \ddot{H}_n(f)$ is a constant, independent of s : the full product $\prod_{i=1}^{n-1} (s+1)_i$ divides $\ddot{H}_n^{(1)}$ (a rank-drop argument at the negative integers $s = -p$), while an odd-sided reduction bounds $\deg_s \ddot{H}_n^{(1)}$ by $\binom{n}{2}$, so the quotient is a scalar. Part 2 evaluates the ratio at $s = 0$, where the odd functional $\mathcal{T}^*$ also becomes classical: the biorthogonal reductions of $\ddot{H}_n^{(1)}$ and $\ddot{H}_n(f)$ then share the same connection determinant, which cancels upon dividing, leaving a ratio of norms.

13.1. Part 1: the ratio $\ddot{H}_n^{(1)}/\ddot{H}_n(f)$ is constant in s . Put $b_m = a_{m+1}$, so $\ddot{H}_n^{(1)} = \det(b_{2i+j})$. The even and odd parts of $\mathbf{b}$ are

$$b_{2k} = (2k+1)E_{2k}^{(s+1)} = \mathcal{T}^*[y^k], \quad b_{2k+1} = E_{2k+2}^{(s+1)} = \mathcal{S}_1[y^k],$$

where $\mathcal{S}_1[y^k] := \mathcal{S}[y^{k+1}]$ is the Christoffel transform $y\mathcal{S}$ of the secant functional.

Lemma 13.2 (kernel polynomial data). $\mathcal{S}_1$ is quasi-definite, with monic orthogonal polynomials $\hat{P}_m$ (the kernel polynomials of $\mathcal{S}$). Writing $\nu_m := P_{m+1}(0)/P_m(0)$, one has

$$\nu_m = -(2m+1)(s+2m+1), \quad (13.4)$$

and, by the Christoffel transform formulas,

$$c_m^{S_1} = c_{m+1}^S + \nu_{m+1} - \nu_m = (4m+3)s + 8m^2 + 12m + 5, \quad (13.5)$$

$$\lambda_m^{S_1} = \lambda_m^S \frac{\nu_m}{\nu_{m-1}} = 2m(2m+1)(s+2m)(s+2m+1), \quad (13.6)$$

$$h_m^{S_1} = -\nu_m h_m^S = (2m+1)!(s+1)_{2m+1} \quad (\deg_s h_m^{S_1} = 2m+1). \quad (13.7)$$

Proof. Evaluating $P_{m+1} = (y - c_m^S)P_m - \lambda_m^S P_{m-1}$ at $y = 0$ gives $\nu_m \nu_{m-1} + c_m^S \nu_{m-1} + \lambda_m^S = 0$; with (13.1) and $\nu_0 = -c_0^S = -(s+1)$ one checks that $-(2m+1)(s+2m+1)$ satisfies this recurrence, proving (13.4). The expressions for $c_m^{S_1}$, $\lambda_m^{S_1}$, $h_m^{S_1}$ are the standard Christoffel ($y\mathcal{S}$) formulas; substituting (13.1), (13.4) and $h_m^S = (2m)!(s+1)_{2m}$ and simplifying gives the closed forms (each is an elementary identity in (m, s)). $\square$

Lemma 13.3 (odd-sided reduction). $\ddot{H}_n^{(1)} = (-1)^{\binom{n+1}{2}} \left(\prod_{m=0}^{n-1} h_m^{S_1} \right) \det(\mathcal{T}^*[\hat{P}_{\underline{n}+r} y^l])_{0 \leq r, l \leq \bar{n}-1}$.

Proof. This is the one-sided reduction **M2** (Lemma 3.2) applied to $\mathbf{b}$, orthogonalising the odd columns against the quasi-definite $\mathcal{S}_1$: replacing the rows (y^i) and the odd-column family by the $\mathcal{S}_1$ -orthogonal $(\hat{P}_i)$ — both unitriangular recombinations — makes the odd-odd block diagonal, $\mathcal{S}_1[\hat{P}_i \hat{P}_m] = \delta_{im} h_m^{S_1}$, on the rows $i < \underline{n}$; Laplace expansion along those rows leaves the $\bar{n}$ even $(\mathcal{T}^*)$ columns on the rows $\underline{n}, \dots, n-1$. $\square$

Lemma 13.4 (divisibility). $\prod_{j=1}^{n-1} (s+j)^{n-j} \mid \ddot{H}_n^{(1)}$.

Proof. At $s = -p$ ($1 \leq p \leq n-1$) the even weight is the cosine polynomial $\cos(x)^{p-1}$, so $a_{2k} = \sum_t w_t \theta_t^k$, where $\theta_t = -(p-1-2t)^2$ runs over $\lceil p/2 \rceil$ distinct values, while $a_{2k+1} = (2k+1)a_{2k}$. This is the rank-drop argument of Lemma 12.3, unaffected by the shift: replacing a_{2i+j} by a_{2i+j+1} only relabels which columns are even and which odd, leaving the spanning vectors unchanged. Every column of $(a_{2i+j+1})_{0 \leq i, j < n}$ thus lies in the span of the $(\theta^i)_i$ and $(i\theta^i)_i$, a nonzero θ contributing two and $\theta = 0$ (present iff p is odd) only one, so the column space has dimension $\leq p$. Hence the rank at $s = -p$ is $\leq p$ and $(s+p)^{n-p} \mid \ddot{H}_n^{(1)}$; the factors $(s+1), \dots, (s+n-1)$ being coprime, their product divides $\ddot{H}_n^{(1)}$. $\square$

Lemma 13.5 (per-column degrees). For $0 \leq l < a$ one has $\deg_s \mathcal{T}^*[\hat{P}_a y^l] \leq 2l$. Moreover $\mathcal{T}^*[\hat{P}_a] = (-1)^a (2a)!$ (the case $l = 0$).

Proof. We use the operator identity $\mathcal{T}^* = \mathcal{S} \circ (2y\partial_y + 1)$ (valid because $a_{2k+1} = (2k+1)a_{2k}$) together with the secant evaluation $\mathcal{T}^*[P_i y^m] = 2\mathcal{S}[y^{m+1}P_i']$ for $i > m$ (Lemma 12.2), which has $\deg_s \mathcal{T}^*[P_i y^m] \leq 2m+1$ (proof of Theorem 12.1), and the kernel relation $y\hat{P}_a = P_{a+1} - \nu_a P_a$ (Lemma 13.2), which differentiates to

$$y\hat{P}_a' = P_{a+1}' - \nu_a P_a' - \hat{P}_a. \quad (13.8)$$

Case $1 \leq l < a$. From $\mathcal{T}^* = \mathcal{S} \circ (2y\partial_y + 1)$,

$$\mathcal{T}^*[\hat{P}_a y^l] = 2\mathcal{S}[y^{l+1}\hat{P}_a'] + (2l+1)\mathcal{S}[y^l\hat{P}_a].$$

Now $\mathcal{S}[y^l \hat{P}_a] = \mathcal{S}_1[y^{l-1} \hat{P}_a] = 0$ (since $\deg y^{l-1} = l-1 < a$). Multiplying (13.8) by y^l , applying $\mathcal{S}$, and using $\mathcal{S}[y^l \hat{P}_a] = 0$ again, $\mathcal{S}[y^{l+1} \hat{P}'_a] = \mathcal{S}[y^l P'_{a+1}] - \nu_a \mathcal{S}[y^l P'_a]$, whence

$$\mathcal{T}^*[\hat{P}_a y^l] = \mathcal{T}^*[P_{a+1} y^{l-1}] - \nu_a \mathcal{T}^*[P_a y^{l-1}]. \quad (13.9)$$

The two terms have $\deg_s \leq 2(l-1) + 1 = 2l-1$ and $\leq 1 + (2l-1) = 2l$ (as $\deg_s \nu_a = 1$); thus $\deg_s \mathcal{T}^*[\hat{P}_a y^l] \leq 2l$.

Case $l = 0$. Set $u_a := \mathcal{T}^*[\hat{P}_a]$. From the kernel three-term recurrence $\hat{P}_a = (y - c_{a-1}^{S_1})\hat{P}_{a-1} - \lambda_{a-1}^{S_1}\hat{P}_{a-2}$ and $\mathcal{T}^*$ -linearity,

$$u_a = \mathcal{T}^*[y\hat{P}_{a-1}] - c_{a-1}^{S_1}u_{a-1} - \lambda_{a-1}^{S_1}u_{a-2} \quad (a \geq 2).$$

By (13.9) applied to $\hat{P}_{a-1}$ with $l = 1$ (valid since $\mathcal{S}[y\hat{P}_{a-1}] = \mathcal{S}_1[\hat{P}_{a-1}] = 0$ for $a \geq 2$), $\mathcal{T}^*[y\hat{P}_{a-1}] = \mathcal{T}^*[\hat{P}_{a-1}y] = \mathcal{T}^*[P_a] - \nu_{a-1}\mathcal{T}^*[P_{a-1}]$, which is explicit by (13.2). Substituting the claimed value $u_a = (-1)^a(2a)!$ together with (13.2), (13.4), (13.5), (13.6) turns the recurrence into a polynomial identity in (a, s) , which one verifies directly. With the base cases $u_0 = \mathcal{T}^*[1] = E_0^{(s+1)} = 1$ and $u_1 = \mathcal{T}^*[y - (3s+5)] = 3E_2^{(s+1)} - (3s+5) = -2$ (using $E_2^{(s+1)} = s+1$), induction gives $u_a = (-1)^a(2a)!$. In particular $\deg_s \mathcal{T}^*[\hat{P}_a] = 0$. $\square$

Lemma 13.6 (degree bound). $\deg_s \ddot{H}_n^{(1)} \leq \binom{n}{2}$.

Proof. In the determinant of Lemma 13.3 the l -th column has all entries of s -degree $\leq 2l$ by Lemma 13.5, so $\deg_s \det \leq \sum_{l=0}^{\bar{n}-1} 2l = \bar{n}(\bar{n}-1)$; and $\deg_s \prod_{m=0}^{\bar{n}-1} h_m^{S_1} = \sum_{m=0}^{\bar{n}-1} (2m+1) = \underline{n}^2$ by (13.7). Hence $\deg_s \ddot{H}_n^{(1)} \leq \underline{n}^2 + \bar{n}(\bar{n}-1) = \binom{n}{2}$. $\square$

Proposition 13.7. $\ddot{H}_n^{(1)} = c_n^{(1)} \prod_{i=1}^{n-1} (s+1)_i$ for a constant $c_n^{(1)}$; equivalently $\ddot{H}_n^{(1)}/\ddot{H}_n(f)$ is independent of s .

Proof. By Lemma 13.4, $\prod_{j=1}^{n-1} (s+j)^{n-j} = \prod_{i=1}^{n-1} (s+1)_i$ (degree $\binom{n}{2}$) divides $\ddot{H}_n^{(1)}$; by Lemma 13.6 the degree is at most $\binom{n}{2}$. Hence $\ddot{H}_n^{(1)} = c_n^{(1)} \prod_{i=1}^{n-1} (s+1)_i$, and by (13.3) the ratio $\ddot{H}_n^{(1)}/\ddot{H}_n(f) = c_n^{(1)}/c_n$ is constant. $\square$

13.2. Part 2: evaluation at $s = 0$. By Proposition 13.7 it suffices to compute $\ddot{H}_n^{(1)}/\ddot{H}_n(f)$ at $s = 0$, where all three functionals are classical: $\mathcal{S}, \mathcal{T}^*, \mathcal{S}_1$ are the continuous dual Hahn functionals $\text{CDH}(\frac{1}{2}, \frac{1}{2}, c)$ for $c = 0, \frac{1}{2}, 1$, with recurrence coefficients (from (13.1), (13.5), (13.6) at $s = 0$, and Lemma 11.2 for $\mathcal{T}^*$)

$$\begin{aligned} c_i^S &= 8i^2 + 4i + 1, & \lambda_i^S &= ((2i)(2i-1))^2; \\ c_i^T &= 8i^2 + 8i + 3, & \lambda_i^T &= (2i)^4; \\ c_i^{S_1} &= 8i^2 + 12i + 5, & \lambda_i^{S_1} &= ((2i)(2i+1))^2, \end{aligned}$$

and squared norms (from (13.7), $h_l^S = (2l)!(s+1)_{2l}$ at $s = 0$, and $\mathcal{T}^*[\rho_m^2] = ((2m)!!)^4$ from Lemma 11.2)

$$h_l^S|_0 = ((2l)!)^2, \quad h_l^T|_0 = ((2l)!!)^4, \quad h_m^{S_1}|_0 = ((2m+1)!)^2. \quad (13.10)$$

Because both functionals are classical at $s = 0$, the full biorthogonal reduction (Lemma 3.4) applies to each determinant. For $\ddot{H}_n(f)$ this is the evaluation of Proposition 11.1:

$$\ddot{H}_n(f)|_0 = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^S \right) \left(\prod_{m=0}^{\bar{n}-1} h_m^T \right) \det(\tilde{\kappa}_{\bar{n}+r, m})_{0 \leq r, m \leq \bar{n}-1}, \quad (13.11)$$

where $\tilde{\kappa}$ expands the $\mathcal{S}$ -orthogonal polynomials in the $\mathcal{T}^*$ -orthogonal basis ρ_m , $P_i = \sum_{m \leq i} \tilde{\kappa}_{i, m} \rho_m$ (in Lemma 11.4 the polynomials $P_i|_{s=0}$ are written $\hat{P}_i$; here the hat is reserved for the kernel polynomials), and

$$\tilde{\kappa}_{i, m} = \left(\frac{i!}{m!} \right)^2 4^{i-m} \binom{1/2}{i-m}. \quad (13.12)$$

For the shifted determinant, the same reduction applied to $\mathbf{b}$ (even functional $\mathcal{T}^*$, odd functional $\mathcal{S}_1$) gives

$$\ddot{H}_n^{(1)}|_0 = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^T \right) \left(\prod_{m=0}^{\underline{n}-1} h_m^{S_1} \right) \det(\kappa_{\bar{n}+r, m}^b)_{0 \leq r, m \leq \underline{n}-1}, \quad (13.13)$$

where κ^b expands the $\mathcal{T}^*$ -orthogonal ρ_i in the $\mathcal{S}_1$ -orthogonal basis of kernel polynomials, $\rho_i = \sum_{m \leq i} \kappa_{i, m}^b \hat{P}_m$ (all at $s = 0$).

Lemma 13.8. $\kappa_{i, m}^b = \tilde{\kappa}_{i, m}$ for all $i \geq m \geq 0$.

Proof. By the connection-coefficient recurrence (Lemma 3.3) for the pair $(\mathcal{T}^*, \mathcal{S}_1)$,

$$\kappa_{i+1, m}^b = \kappa_{i, m-1}^b + (c_m^{S_1} - c_i^T) \kappa_{i, m}^b + \lambda_{m+1}^{S_1} \kappa_{i, m+1}^b - \lambda_i^T \kappa_{i-1, m}^b, \quad \kappa_{0, 0}^b = 1,$$

which determines κ^b uniquely from $\kappa_{0, 0}^b = 1$. Substituting the closed form (13.12) and the explicit coefficients $c^T, \lambda^T, c^{S_1}, \lambda^{S_1}$ above, and dividing through by $\tilde{\kappa}_{i, m}$ (using $\binom{1/2}{d+1} / \binom{1/2}{d} = (\frac{1}{2} - d)/(d+1)$, $d = i - m$), the recurrence reduces to an elementary rational identity in (i, m) , which one verifies directly — exactly as in the proof of the third connection formula (Lemma 11.4). Hence $\tilde{\kappa}$ satisfies the recurrence and the base case, so $\kappa_{i, m}^b = \tilde{\kappa}_{i, m}$. (The reduction reflects that $\mathcal{T}^* \rightarrow \mathcal{S}_1$ is the half-step $c \mapsto c + \frac{1}{2}$ in the continuous dual Hahn hierarchy, the same step as $\mathcal{S} \rightarrow \mathcal{T}^*$ governing $\tilde{\kappa}$.) $\square$

Proof of Theorem 13.1. By Lemma 13.8 the determinants in (13.11) and (13.13) coincide; dividing and using (13.10),

$$\frac{\ddot{H}_n^{(1)}|_0}{\ddot{H}_n(f)|_0} = \frac{\prod_{l=0}^{\bar{n}-1} h_l^T \prod_{m=0}^{\underline{n}-1} h_m^{S_1}}{\prod_{l=0}^{\bar{n}-1} h_l^S \prod_{m=0}^{\underline{n}-1} h_m^T} = \frac{\prod_{l=0}^{\bar{n}-1} ((2l)!!)^4 \prod_{m=0}^{\underline{n}-1} ((2m+1)!)^2}{\prod_{l=0}^{\bar{n}-1} ((2l)!)^2 \prod_{m=0}^{\underline{n}-1} ((2m)!!)^4}.$$

$n = 2p$ even ($\bar{n} = \underline{n} = p$): the $\mathcal{T}^*$ -norm factors cancel and, using $(2m+1)!/(2m)! = 2m+1$,

$$\frac{\ddot{H}_n^{(1)}|_0}{\ddot{H}_n|_0} = \prod_{m=0}^{p-1} \frac{((2m+1)!)^2}{((2m)!)^2} = \prod_{m=0}^{p-1} (2m+1)^2 = ((2p-1)!!)^2 = ((n-1)!!)^2.$$

$n = 2p+1$ odd ($\bar{n} = p+1$, $\underline{n} = p$): the $\mathcal{T}^*$ -norms leave $((2p)!!)^4$, while

$$\frac{\prod_{m=0}^{p-1} ((2m+1)!)^2}{\prod_{l=0}^p ((2l)!)^2} = \frac{1}{((2p)!)^2} \prod_{m=0}^{p-1} (2m+1)^2 = \frac{((2p-1)!!)^2}{((2p)!)^2},$$

so, with $(2p)! = (2p)!!(2p-1)!!$,

$$\frac{\ddot{H}_n^{(1)}|_0}{\ddot{H}_n|_0} = ((2p)!!)^4 \cdot \frac{((2p-1)!!)^2}{((2p)!)^2} = ((2p)!!)^2 = ((n-1)!!)^2.$$

Thus $\ddot{H}_n^{(1)}/\ddot{H}_n(f) = ((n-1)!!)^2$ at $s = 0$; by Proposition 13.7 it equals $((n-1)!!)^2$ for all s , which with (13.3) is Theorem 13.1. $\square$

Remark 13.9 (the continuous dual Hahn hierarchy). The mechanism is that $\mathcal{S}, \mathcal{T}^*, \mathcal{S}_1$ are consecutive members $\text{CDH}(\frac{1}{2}, \frac{1}{2}, c)$, $c = 0, \frac{1}{2}, 1$, of one continuous dual Hahn hierarchy; the connection coefficient for each half-step $c \mapsto c + \frac{1}{2}$ is the same binomial (13.12) (Lemma 13.8), so the common determinantal factor cancels and only the norms (13.10) survive.

14. THE DOUBLE SHIFT OF THE SECANT-NUMBER FAMILY $(1+x)/\cos(x)^{s+1}$ AT $s = 1$

For the coefficient sequence $\mathbf{a}$ of $f = (1+x)/\cos(x)^{s+1}$, alongside the single shift of Section 13 we form the *double shift*

$$\ddot{H}_n^{(2)} := \det(a_{2i+j+2})_{0 \leq i, j < n}.$$

For general s the ratio $\ddot{H}_n^{(2)}/\ddot{H}_n$ is no longer constant in s (Remark 14.6), but at $s = 1$ it collapses to a pure product.

Theorem 14.1. *For all $n \geq 1$,*

$$\ddot{H}_n^{(2)}|_{s=1} = 2^n (n!)^2 \ddot{H}_n|_{s=1} = 2^{n+\binom{n}{2}} (n!)^3 ((n-1)!!)^2 \prod_{k=1}^{n-2} (k!!)^6. \quad (14.1)$$

Equivalently, by the single-shift Theorem 13.1 ($\ddot{H}_n^{(1)} = ((n-1)!!)^2 \ddot{H}_n$) and $n! = n!! (n-1)!!$,

$$\frac{\ddot{H}_n^{(2)}}{\ddot{H}_n^{(1)}} \Big|_{s=1} = 2^n (n!!)^2.$$

The proof has three steps. A one-sided reduction **M2** against the even functional of the shifted sequence (§14.1) leaves a residual determinant R_n built from the derivatives $\hat{P}'_i$; a *structure relation* then replaces $y\hat{P}'_i$ by $T_i = y\hat{P}'_i - i\hat{P}_i$ and turns R_n into a *square* determinant of the structure coefficients $\Delta_{i,a}$ of the classical functional $\mathcal{S}_1$ (§14.2); finally, at $s = 1$ the closed form of $\Delta_{i,a}$ degenerates into a rank-one factor times a Cauchy kernel, which collapses the determinant (§14.3).

Recall from §12 and §13 the even and odd moment functionals attached to f ,

$$\mathcal{S}[y^k] = a_{2k}, \quad \mathcal{T}^*[y^k] = a_{2k+1} = (2k+1) a_{2k},$$

the classical secant data (13.1)–(13.2) for $\mathcal{S}$, and the Christoffel transform $\mathcal{S}_1 = y \mathcal{S}$ (Lemma 13.2), whose monic orthogonal polynomials $\hat{P}_m$ (the kernel polynomials of $\mathcal{S}$) have three-term recurrence $\hat{P}_{m+1} = (y - c_m^{S_1})\hat{P}_m - \lambda_m^{S_1} \hat{P}_{m-1}$ with

$$\begin{aligned} c_m^{S_1} &= (4m+3)s + 8m^2 + 12m + 5, & \lambda_m^{S_1} &= 2m(2m+1)(s+2m)(s+2m+1), \\ h_m^{S_1} &= (2m+1)!(s+1)_{2m+1}. \end{aligned} \quad (14.2)$$

Throughout $\bar{n} = \lceil n/2 \rceil$, $\underline{n} = \lfloor n/2 \rfloor$.

14.1. The functionals of the doubly shifted sequence. Put $d_m := a_{m+2}$, so $\ddot{H}_n^{(2)} = \det(d_{2i+j})$. The even and odd parts of $\mathbf{d}$ are

$$d_{2k} = a_{2k+2} = \mathcal{S}_1[y^k], \quad d_{2k+1} = a_{2k+3} = (2k+3) \mathcal{S}_1[y^k] =: \mathcal{T}_1^*[y^k],$$

where $\mathcal{T}_1^*$ is the Christoffel transform $y \mathcal{T}^*$ of the odd functional. The two are linked by the same first-order operator that links $\mathcal{S}$ and $\mathcal{T}^*$, shifted by 2:

$$\mathcal{T}_1^* = \mathcal{S}_1 \circ (2y\partial_y + 3), \quad \text{since} \quad (2k+3)y^k = (2y\partial_y + 3)y^k. \quad (14.3)$$

The decisive structural difference from the single-shift case (§13) is that here the *even* functional is the classical one — $\mathcal{S}_1$, the Christoffel transform of $\mathcal{S}$, with data (14.2) — while the odd functional $\mathcal{T}_1^*$ is non-classical for $s \geq 1$; in the single-shift case the parities were reversed. Consequently the one-sided reduction **M2** is performed against $\mathcal{S}_1$.

Lemma 14.2 (one-sided reduction against $\mathcal{S}_1$). *For all $n \geq 1$,*

$$\ddot{H}_n^{(2)} = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^{S_1} \right) R_n, \quad R_n := \det(\mathcal{T}_1^*[\hat{P}_{\bar{n}+r} y^m])_{0 \leq r, m \leq \underline{n}-1}, \quad (14.4)$$

and for $i > m$,

$$\mathcal{T}_1^*[\hat{P}_i y^m] = 2 \mathcal{S}_1[y^{m+1} \hat{P}'_i] \quad (\text{the term } (2m+3)\mathcal{S}_1[y^m \hat{P}_i] \text{ vanishes since } \deg y^m < i). \quad (14.5)$$

Proof. Identity (14.4) is the general one-sided reduction (Lemma 3.2) applied to $\mathbf{d}$ with even functional $\mathcal{S}_1$ and odd functional $\mathcal{T}_1^*$: it orthogonalises only the $\bar{n}$ even columns against the quasi-definite $\mathcal{S}_1$, leaving the $\underline{n}$ odd ($\mathcal{T}_1^*$) columns, with sign $(-1)^{\binom{\bar{n}}{2}}$ exactly as in Lemma 12.2; no orthogonal-polynomial structure of $\mathcal{T}_1^*$ is needed. Finally (14.3) gives $\mathcal{T}_1^*[\hat{P}_i y^m] = 2\mathcal{S}_1[y^{m+1} \hat{P}'_i] + (2m+3)\mathcal{S}_1[y^m \hat{P}_i]$, and the last term vanishes for $i > m$ by orthogonality, which holds throughout the block since $\bar{n} + r \geq \bar{n} > \underline{n} - 1 \geq m$. $\square$

14.2. Reduction of R_n to a structure determinant. Let $T_i := y\hat{P}'_i - i\hat{P}_i$. Since $\hat{P}_i$ is monic of degree i , the leading terms cancel and $\deg T_i \leq i - 1$; write

$$T_i = \sum_{a=0}^{i-1} \Delta_{i,a} \hat{P}_a. \quad (14.6)$$

Lemma 14.3 (structure reduction). *For all $n \geq 1$ and all s ,*

$$R_n = 2^{\underline{n}} \left(\prod_{a=0}^{\underline{n}-1} h_a^{S_1} \right) \det(\Delta_{\bar{n}+r,a})_{0 \leq r, a \leq \underline{n}-1}. \quad (14.7)$$

Proof. Fix $i = \bar{n} + r$ and m with $0 \leq r, m \leq \underline{n} - 1$, so $i > m$. By (14.5) and $y\hat{P}'_i = T_i + i\hat{P}_i$,

$$\mathcal{T}_1^*[\hat{P}_i y^m] = 2\mathcal{S}_1[y^m y\hat{P}'_i] = 2\mathcal{S}_1[y^m T_i] + 2i\mathcal{S}_1[y^m \hat{P}_i] = 2\mathcal{S}_1[y^m T_i],$$

the last term vanishing because $m < i$. Expand $y^m = \sum_{b=0}^m \beta_b^{(m)} \hat{P}_b$ in the orthogonal basis (a lower-unitriangular change of basis, $\beta_m^{(m)} = 1$); then $\mathcal{S}_1[y^m \hat{P}_a] = \beta_a^{(m)} h_a^{S_1}$, which is 0 for $a > m$. Hence by (14.6)

$$\mathcal{T}_1^*[\hat{P}_i y^m] = 2 \sum_{a=0}^{i-1} \Delta_{i,a} \mathcal{S}_1[y^m \hat{P}_a] = 2 \sum_{a=0}^{\underline{n}-1} \Delta_{i,a} h_a^{S_1} \beta_a^{(m)}.$$

In matrix form the residual matrix is $(\mathcal{T}_1^*[\hat{P}_{\bar{n}+r} y^m]) = 2 \Delta' \text{diag}(h_0^{S_1}, \dots, h_{\underline{n}-1}^{S_1}) B^\top$, where $\Delta'_{r,a} = \Delta_{\bar{n}+r,a}$ and $B_{m,a} = \beta_a^{(m)}$ is $\underline{n} \times \underline{n}$ lower-unitriangular, so $\det B = 1$. Taking determinants gives (14.7). $\square$

The point of (14.7) is that the rectangular Cauchy–Binet of Remark 12.5 has become a *square* $\underline{n} \times \underline{n}$ determinant: replacing the raw derivative $\hat{P}'_i$ by the structure form T_i lowers the summation range from $a \leq m + 1$ to $a \leq m$, and the power-expansion matrix B drops out as unitriangular.

Lemma 14.4 (structure coefficients of $\mathcal{S}_1$). *For $0 \leq a \leq i - 1$,*

$$\Delta_{i,a} = (-1)^{i-a-1} \frac{2^{i-a-1}}{2(i-a)-1} \frac{i!}{a!} \frac{(2i+1)!!}{(2a+1)!!} \left(s + \frac{2(i+a)+3}{2(i-a)+1} \right), \quad (14.8)$$

in particular $\deg_s \Delta_{i,a} = 1$.

Proof. This is the structure relation of Lemma 12.4 transported from $\mathcal{S}$ to its Christoffel transform $\mathcal{S}_1$. Differentiating the recurrence $\hat{P}_{i+1} = (y - c_i^{S_1})\hat{P}_i - \lambda_i^{S_1}\hat{P}_{i-1}$, multiplying by y and using $y\hat{P}'_j = T_j + j\hat{P}_j$ together with $(y - c_i^{S_1})\hat{P}_i = \hat{P}_{i+1} + \lambda_i^{S_1}\hat{P}_{i-1}$ yields the same structure recurrence as in Lemma 12.4,

$$T_{i+1} = (y - c_i^{S_1})T_i - \lambda_i^{S_1}T_{i-1} + c_i^{S_1}\hat{P}_i + 2\lambda_i^{S_1}\hat{P}_{i-1},$$

which in the $\hat{P}$ -basis reads

$$\Delta_{i+1,a} = \Delta_{i,a-1} + (c_a^{S_1} - c_i^{S_1})\Delta_{i,a} + \lambda_{a+1}^{S_1}\Delta_{i,a+1} - \lambda_i^{S_1}\Delta_{i-1,a} + c_i^{S_1}[a=i] + 2\lambda_i^{S_1}[a=i-1],$$

with $T_0 = 0$ and $T_1 = y\hat{P}'_1 - \hat{P}_1 = c_0^{S_1} = 3s + 5$. The expression (14.8) satisfies the base case $\Delta_{1,0} = 3s + 5$ and every instance of this recurrence — a direct rational-function identity in (i, a, s) after inserting (14.2), exactly as in Lemma 12.4; the only change from the secant case is $(2i-1)!!/(2a-1)!! \mapsto (2i+1)!!/(2a+1)!!$ and the root $\frac{2(i+a)+1}{2(i-a)+1} \mapsto \frac{2(i+a)+3}{2(i-a)+1}$, reflecting the index shift $\mathcal{S} \mapsto \mathcal{S}_1$. Since the recurrence determines all $\Delta_{i,a}$ uniquely, (14.8) holds. $\square$

14.3. The value at $s = 1$. At $s = 1$ the linear factor in (14.8) simplifies: for $i = \bar{n} + r$,

$$1 + \frac{2(i+a)+3}{2(i-a)+1} = \frac{2(i-a)+1+2(i+a)+3}{2(i-a)+1} = \frac{4(i+1)}{2(i-a)+1},$$

so, using $\frac{1}{2(i-a)-1} \cdot \frac{1}{2(i-a)+1} = \frac{1}{4(i-a)^2-1}$ and $i! \cdot 4(i+1) = 4(i+1)!$,

$$\Delta_{i,a}|_{s=1} = \underbrace{(-1)^i 2^{i+2} (i+1)! (2i+1)!!}_{\alpha_i} \cdot \underbrace{(-1)^{a+1} \frac{2^{-a}}{a! (2a+1)!!}}_{\beta_a} \cdot 2^{-1} \cdot \frac{1}{4(i-a)^2-1}. \quad (14.9)$$

Pulling the row factor $\alpha_{\bar{n}+r}$ from each row r , the column factor β_a from each column a , and the constant 2^{-1} from each of the $\underline{n}$ rows,

$$\det(\Delta_{\bar{n}+r,a})|_{s=1} = 2^{-\underline{n}} \left(\prod_{r=0}^{\underline{n}-1} \alpha_{\bar{n}+r} \right) \left(\prod_{a=0}^{\underline{n}-1} \beta_a \right) \det\left(\frac{1}{4(\bar{n}+r-a)^2-1}\right)_{0 \leq r, a \leq \underline{n}-1}. \quad (14.10)$$

Lemma 14.5 (Cauchy determinant). *For integers $M \geq 1$ and $N \geq 0$,*

$$\det\left(\frac{1}{4(M+r-a)^2-1}\right)_{0 \leq r, a \leq N-1} = (-1)^{\binom{N}{2}} 4^{-N} \frac{\prod_{k=0}^{N-1} k! \prod_{k=1}^N k!}{\prod_{k=1}^N \left(M^2 - \frac{(2k-1)^2}{4}\right)^{N+1-k}}. \quad (14.11)$$

Proof. The entry is $\frac{1}{4}(M+r-a-\frac{1}{2})^{-1}(M+r-a+\frac{1}{2})^{-1}$, so the matrix is $\frac{1}{4}(\frac{1}{(x_r+y_a)(x_r+y_a+1)})$ with $x_r = M+r-\frac{1}{2}$ and $y_a = -a$. This is the Cauchy double alternant with two consecutive denominators [26]; evaluating it and substituting the arithmetic progressions x_r, y_a (whose pairwise differences are $\prod_{r < r'}(r-r') = \pm \prod k!$ and likewise for y) gives (14.11). $\square$

Proof of Theorem 14.1. Combining Lemmas 14.2, 14.3, 14.4 and (14.10)–(14.11) expresses $\ddot{H}_n^{(2)}|_{s=1}$ as an explicit product of factorials and double factorials: with $M = \bar{n}$, $N = \underline{n}$ and $h_l^{S_1}|_{s=1} = (2l+1)!(2l+2)!$,

$$\ddot{H}_n^{(2)}|_{s=1} = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} (2l+1)!(2l+2)! \right) 2^{\underline{n}} \left(\prod_{a=0}^{\underline{n}-1} (2a+1)!(2a+2)! \right) \det(\Delta_{\bar{n}+r,a})|_{s=1},$$

the last factor given by (14.10)–(14.11). On the other side of (14.1), Theorem 12.1 gives $\ddot{H}_n = c_n \prod_{i=1}^{n-1} (s+1)_i$. Since $\prod_{i=1}^{n-1} (2)_i = \prod_{i=1}^{n-1} (i+1)! = n! \prod_{i=1}^{n-1} i!$ and

$$c_n = 2^{\binom{n}{2}} ((n-1)!!)^2 \prod_{k=1}^{n-2} (k!!)^6 \Big/ \prod_{i=1}^{n-1} i!,$$

setting $s = 1$ gives

$$\ddot{H}_n|_{s=1} = 2^{\binom{n}{2}} n! ((n-1)!!)^2 \prod_{k=1}^{n-2} (k!!)^6.$$

Both sides of (14.1) are thus explicit products; their equality is an elementary identity in n , verified at $n = 1$ and propagated by the ratio $n \mapsto n+1$ (the two parities treated separately, as in Proposition 11.1). Hence (14.1) holds for all n . $\square$

Remark 14.6 (why $s = 1$ is special). The mechanism contrasts sharply with the single-shift case. There the ratio $\ddot{H}_n^{(1)}/\ddot{H}_n$ is independent of s (Proposition 13.7), and its value is read off at $s = 0$, where both functionals are classical and the connection determinants cancel. Here the ratio $\ddot{H}_n^{(2)}/\ddot{H}_n$ genuinely depends on s — beyond linear factors $(s+j)$ it carries a factor $Q_n(s)$ of degree $\underline{n}$, irreducible over $\mathbb{Q}$ in the computed range (e.g. $Q_4 = s^2 + \frac{36}{7}s + \frac{169}{35}$, $Q_5 = s^2 + \frac{44}{9}s + \frac{269}{63}$) — and $\mathcal{T}_1^*$ remains non-classical at $s = 1$, so no cancellation of connection determinants is available. What saves the evaluation is that the *even* functional $\mathcal{S}_1$ is classical for *all* s , so its structure coefficients $\Delta_{i,a}$ have the closed form (14.8); and at the special value $s = 1$ the linear factor of $\Delta_{i,a}$ degenerates into a rank-one (row $\times$ column) factor times the Cauchy kernel $1/(4(i-a)^2-1)$, which collapses the determinant.

15. A RANK-ONE PERTURBATION OF THE EULER NUMBER FAMILY: $(\sin x + 1)/\cos^2 x + s \sin x$

We evaluate the dilated Hankel determinant of a one-parameter, rank-one perturbation of the shifted Euler sequence $a_n = E_{n+1}$, whose exponential generating function is $(\sin x + 1)/\cos^2 x$ and whose dilated determinant is the $s = 0$ value of Proposition 9.1, in the product form

$$\ddot{H}_n^E := \prod_{k=1}^{n-1} (k!)^2 (2k+1)!!.$$

We consider the family

$$f_s(x) = \frac{\sin x + 1}{\cos^2 x} + s \sin x \quad (s \in \mathbb{C}).$$

Its most striking member is $s = -1$: there $\sin x (1 - \sin^2 x)/\cos^2 x = \sin x$ collapses the perturbation, so that $f_{-1}(x) = (\sin^3 x + 1)/\cos^2 x$, and the determinant looks irregular, carrying “sporadic” prime factors 29, 37, 23 for $n = 8, 9, 10$. They are not sporadic. Since

$$29 = \binom{8}{2} + 1, \quad 37 = \binom{9}{2} + 1, \quad 46 = \binom{10}{2} + 1 = 2 \cdot 23,$$

the apparent prime 23 is merely half of $\binom{10}{2} + 1$; in fact the whole family has the closed form below, affine in s .

Proposition 15.1. *For every $s \in \mathbb{C}$ and all $n \geq 1$, the family $f_s(x) = \frac{\sin x + 1}{\cos^2 x} + s \sin x$ has*

$$\ddot{H}_n(f_s) = \left(1 - s \binom{n}{2}\right) \prod_{k=1}^{n-1} (k!)^2 (2k+1)!! = \left(1 - s \binom{n}{2}\right) \ddot{H}_n^E.$$

In particular $s = 0$ recovers Proposition 9.1, while $s = -1$ gives, for $f_{-1}(x) = (\sin^3 x + 1)/\cos^2 x$,

$$\ddot{H}_n(f_{-1}) = \left(\binom{n}{2} + 1\right) \ddot{H}_n^E.$$

The closed form follows from a *rank-one* reduction (M6, via the matrix determinant lemma, Lemma 15.2 below — not the divisor method M4, cf. Remark 3.12) to Proposition 9.1: the perturbation multiplies the determinant by the scalar $1 - s \binom{n}{2}$, so that the entire content of the proposition reduces to the value of a single bilinear form, (15.2) below — *independent of s* — which is then evaluated by transporting it through the same M2 biorthogonal reduction used for Proposition 9.1.

15.1. The rank-one structure.

Lemma 15.2 (rank-one structure). *Write $a_n = E_{n+1}$ for the shifted Euler sequence, $A = (a_{2i+j})_{0 \leq i, j \leq n-1}$, and let σ_n be the coefficient sequence of $\sin x$, i.e. $\sigma_{2k} = 0$, $\sigma_{2k+1} = (-1)^k$. Then the sequence of f_s is $\tilde{a}_n = a_n + s \sigma_n$, and*

$$\ddot{H}_n(f_s) = \det(A + s u w^\top) = \ddot{H}_n^E (1 + s w^\top A^{-1} u), \quad u_i = (-1)^i, \quad w_j = \sigma_j.$$

Proof. The perturbing term $s \sin x$ contributes $s \sigma_n$ to the coefficient sequence, so $\tilde{a}_n = a_n + s \sigma_n$; the member $s = -1$ is $(\sin^3 x + 1)/\cos^2 x$ because $\sin x (1 - \sin^2 x)/\cos^2 x = \sin x$. The shift of the index by $2i$ preserves parity and introduces a sign $(-1)^i$: for even j both σ_{2i+j} and σ_j vanish, while for $j = 2k + 1$,

$$\sigma_{2i+j} = \sigma_{2(i+k)+1} = (-1)^{i+k} = (-1)^i \sigma_j.$$

Hence $\sigma_{2i+j} = (-1)^i \sigma_j$ for all i, j , so the perturbation matrix $s(\sigma_{2i+j})_{i,j}$ equals $s u w^\top$ with $u_i = (-1)^i$ and $w_j = \sigma_j$. The matrix determinant lemma [19, Theorem 18.1.1]

$$\det(A + s u w^\top) = \det A (1 + s w^\top A^{-1} u) \tag{15.1}$$

now gives the claim, with $\det A = \ddot{H}_n^E$ the $s = 0$ value of Proposition 9.1. $\square$

By Lemma 15.2, Proposition 15.1 is equivalent to the single scalar identity

$$w^\top A^{-1}u = -\binom{n}{2}, \quad (15.2)$$

which does not involve s : once it is proved, $\ddot{H}_n(f_s) = \ddot{H}_n^E(1 - s\binom{n}{2})$ for every s . We prove (15.2) by carrying the pair (u, w) through the biorthogonal reduction already used for Proposition 9.1. Throughout put $\bar{n} = \lceil n/2 \rceil$ and $\underline{n} = \lfloor n/2 \rfloor$, so $\bar{n} + \underline{n} = n$.

15.2. Shifted secant data and evaluations at $v = -1$. The odd columns of the shifted family pair through the shifted-secant functional $\mathcal{S}_1[v^k] := E_{2k+2} = \mathcal{S}[v^{k+1}]$, written in the even variable $v = z^2$ (the letter y is reserved for the linear system below); we first record its monic orthogonal polynomials.

Lemma 15.3 (shifted secant family; classical). *The monic $\mathcal{S}_1$ -orthogonal polynomials $\hat{P}_l^{(1)}$ satisfy*

$$\hat{P}_{l+1}^{(1)} = (v - (2l+1)^2 - (2l+2)^2)\hat{P}_l^{(1)} - ((2l)(2l+1))^2 \hat{P}_{l-1}^{(1)},$$

with $\mathcal{S}_1[\hat{P}_l^{(1)}\hat{P}_{l'}^{(1)}] = \delta_{ll'}((2l+1)!)^2$.

Proof. This is the *odd* contraction of the secant S -fraction of Lemma 7.2 (the even contraction gave $\hat{P}_i$): by (2.9)–(2.10), an S -fraction with coefficients $(b_1, b_2, \dots)$ yields for the shifted moments μ_{k+1} the data $c'_l = b_{2l+1} + b_{2l+2}$, $\lambda'_l = b_{2l}b_{2l+1}$ and norms $h'_l = \prod_{j=1}^{2l+1} b_j$. Here $b_j = j^2$, so $c'_l = (2l+1)^2 + (2l+2)^2$, $\lambda'_l = ((2l)(2l+1))^2$ and $h'_l = ((2l+1)!)^2$. $\square$

We shall need the following two evaluations at $v = -1$.

Lemma 15.4. *The tangent polynomials r_m and the shifted secant polynomials $\hat{P}_l^{(1)}$ of Lemmas 7.2 and 15.3 satisfy*

$$r_m(-1) = (-1)^m(2m-1)!!(2m+1)!!, \quad \hat{P}_l^{(1)}(-1) = (-1)^l(2l+1)!,$$

and in particular $r_m(-1)/(2m+1)! = \binom{-1/2}{m}$.

Proof. Both follow by induction on the three-term recurrences. For r_m the recurrence of Lemma 7.2, written here in v , is $r_{m+1} = (v - c_m^T)r_m - \lambda_m^T r_{m-1}$ with $c_m^T = 2(2m+1)^2$, $\lambda_m^T = (2m-1)(2m)^2(2m+1)$. The claim is equivalent to $\theta_m := r_m(-1)/r_{m-1}(-1) = -(2m-1)(2m+1)$; assuming it at level m ,

$$\frac{r_{m+1}(-1)}{r_m(-1)} = -1 - c_m^T - \frac{\lambda_m^T}{\theta_m} = -1 - 2(2m+1)^2 + (2m)^2 = -(2m+1)(2m+3) = \theta_{m+1},$$

since $\lambda_m^T/\theta_m = -(2m)^2$, with base case $r_1(-1) = -1 - c_0^T = -3$. The same computation for $\hat{P}_l^{(1)}$, with $c'_l = (2l+1)^2 + (2l+2)^2$ and $\lambda'_l = ((2l)(2l+1))^2$ (Lemma 15.3), gives $\hat{P}_l^{(1)}(-1)/\hat{P}_{l-1}^{(1)}(-1) = -(2l)(2l+1)$, hence $\hat{P}_l^{(1)}(-1) = (-1)^l(2l+1)!$. Finally, with $(2m+1)! = (2m+1)!!2^m m!$ and $(2m-1)!!/(2^m m!) = (-1)^m \binom{-1/2}{m}$, the displayed quotient follows. $\square$

15.3. Transport through the biorthogonal reduction. The **M2** reduction behind Proposition 9.1, taken at $s = 0$, is a determinant-preserving change of the row and column families (Lemma 2.1, with $v = z^2$): the rows v^i are replaced by the tangent polynomials r_i , the even columns $z v^m$ by $z r_m$, and the odd columns v^{l+1} by $v \hat{P}_l^{(1)}$. Writing the resulting unitriangular factorisation as $A = P^{-1}KQ^{-\top}$ ($\det P = \det Q = 1$), the even and odd columns of A pair respectively through the tangent functional $\mathcal{T}$ and the shifted secant functional $\mathcal{S}_1$, so that

$$K_{i,2m} = \mathcal{T}[r_i r_m] = \delta_{im}(2m)!(2m+1)!, \quad K_{i,2l+1} = \mathcal{S}_1[r_i \hat{P}_l^{(1)}] = (2i+1)!(2l+1)!\binom{1/2}{i-l}$$

(the odd-column value is Lemma 9.2 at $s = 0$, multiplied by the norm $((2l+1)!)^2$ of Lemma 15.3), and $\det K = \ddot{H}_n^E$. Since P, Q are unitriangular,

$$w^\top A^{-1}u = (Qw)^\top K^{-1}(Pu) = c^\top K^{-1}b, \quad (15.3)$$

where the very same operations evaluate the two transported vectors. The row operations send $u_i = (-1)^i = \text{ev}_{-1}[v^i]$ to $b_i = r_i(-1)$; the column operations send w (supported on the odd columns, $w_{2l+1} = (-1)^l$) to the covector c with $c_{2m} = 0$ and

$$c_{2l+1} = \sum_{l'} (-1)^{l'} [\hat{P}_l^{(1)}]_{l'} = \hat{P}_l^{(1)}(-1) = (-1)^l (2l+1)!,$$

$[\hat{P}_l^{(1)}]_{l'}$ being the coefficient of $v^{l'}$ (Lemma 15.4).

It remains to evaluate $c^\top K^{-1}b$, i.e. to solve $Ky = b$ and form $c^\top y = \sum_l (-1)^l (2l+1)! y_{2l+1}$. Each even column $2m$ of K has its only nonzero entry in row m , so the rows $i = \bar{n}, \dots, n-1$ involve the odd unknowns alone:

$$\sum_{l=0}^{\bar{n}-1} (2i+1)! (2l+1)! \binom{1/2}{i-l} y_{2l+1} = r_i(-1), \quad i = \bar{n}, \dots, n-1.$$

Dividing by $(2i+1)!$, putting $i = \bar{n} + p$ and $\tilde{y}_l = (2l+1)! y_{2l+1}$, and using $r_i(-1)/(2i+1)! = \binom{-1/2}{i}$ (Lemma 15.4), this becomes

$$\sum_{l=0}^{\bar{n}-1} \binom{1/2}{\bar{n}+p-l} \tilde{y}_l = \binom{-1/2}{\bar{n}+p} \quad (p = 0, \dots, \bar{n}-1), \quad w^\top A^{-1}u = \sum_{l=0}^{\bar{n}-1} (-1)^l \tilde{y}_l. \quad (15.4)$$

Lemma 15.5. *The unique solution of the system in (15.4) satisfies $\sum_{l=0}^{\bar{n}-1} (-1)^l \tilde{y}_l = -\binom{n}{2}$.*

Proof. Let $Y(x) = \sum_{l=0}^{\bar{n}-1} \tilde{y}_l x^l$. Since $\binom{1/2}{\bar{n}+p-l} = [x^{\bar{n}+p-l}](1+x)^{1/2}$, the left side of (15.4) is $[x^{\bar{n}+p}]((1+x)^{1/2}Y(x))$, so the system reads

$$[x^k]((1+x)^{1/2}Y(x)) = \binom{-1/2}{k} = [x^k](1+x)^{-1/2}, \quad k = \bar{n}, \dots, n-1.$$

The system matrix $(\binom{1/2}{\bar{n}+p-l})_{p,l}$ is, up to nonzero row and column factors, the block of K that survives the Laplace expansion along the even columns, and $\det K = \ddot{H}_n^E \neq 0$; hence it is invertible, and it suffices to exhibit one polynomial Y of degree $\leq \bar{n}-1$ meeting these $\bar{n}$ equations, after which $\sum_l (-1)^l \tilde{y}_l = Y(-1)$. Take

$$Y(x) = -\sum_{j=1}^{\bar{n}} \binom{n}{2j} (1+x)^{j-1},$$

of degree $\bar{n}-1$. With $t = (1+x)^{1/2}$ (the letter s is the perturbation parameter),

$$(1+x)^{1/2}Y(x) = -\sum_{j=1}^{\bar{n}} \binom{n}{2j} (1+x)^{j-1/2} = (1+x)^{-1/2} - \Phi(x),$$

$$\Phi(x) := \sum_{j=0}^{\bar{n}} \binom{n}{2j} t^{2j-1} = \frac{(1+t)^n + (1-t)^n}{2t},$$

the last equality because $\sum_j \binom{n}{2j} t^{2j} = \frac{1}{2}((1+t)^n + (1-t)^n)$. It remains to show $[x^k]\Phi = 0$ for $\bar{n} \leq k \leq n-1$. Substituting $x = t^2 - 1$ (so a small loop of x about 0 becomes a small loop of t about 1, $dx = 2t dt$),

$$[x^k]\Phi = \frac{1}{2\pi i} \oint_{t=1} \frac{(1+t)^n + (1-t)^n}{(t-1)^{k+1}(t+1)^{k+1}} dt,$$

and writing $(1-t)^n = (-1)^n(t-1)^n$ splits the integrand as

$$\frac{(1+t)^{n-k-1}}{(t-1)^{k+1}} + (-1)^n \frac{(t-1)^{n-k-1}}{(t+1)^{k+1}}.$$

For $\bar{n} \leq k \leq n-1$ the second summand is regular at $t=1$ (exponent $n-k-1 \geq 0$); the first has residue $\binom{n-k-1}{k} 2^{n-2k-1}$, which vanishes because $n-k-1 \leq \bar{n}-1 < \bar{n} \leq k$ forces $\binom{n-k-1}{k} = 0$.

Hence $[x^k]\Phi = 0$ throughout, Y solves the system, and

$$\sum_{l=0}^{n-1} (-1)^l \tilde{y}_l = Y(-1) = - \sum_{j=1}^n \binom{n}{2j} [j=1] = - \binom{n}{2}. \quad \square$$

Proof of Proposition 15.1. Combining Lemma 15.5 with (15.4) and (15.3) proves (15.2); together with Lemma 15.2 this gives $\ddot{H}_n(f_s) = (1 - s \binom{n}{2}) \ddot{H}_n^E$ for all s and $n \geq 1$. $\square$

Remark 15.6 (affine dependence on s). Since the perturbation is rank one, $\ddot{H}_n(f_s)$ is *affine* in s : it equals $\ddot{H}_n^E$ at $s = 0$ and drops by $\binom{n}{2} \ddot{H}_n^E$ per unit of s , vanishing at the single value $s = 1/\binom{n}{2}$ (for $n \geq 2$). Corollaries 27.21–27.25 record the integer members $s = -2, \dots, 2$.

Remark 15.7 (a perturbation that is not rank one). By contrast, the perturbation $(\sin x + 1)/\cos^2 x - \sin 2x$ is not rank one — it mixes two conjugate frequency pairs — and has no product formula at all: its determinants carry irregular prime factors and change sign.

16. THE SPRINGER NUMBER FAMILY

The *Springer numbers* S_n are defined by

$$\frac{1}{\cos x - \sin x} = \sum_{n \geq 0} S_n \frac{x^n}{n!}, \quad (S_n)_{n \geq 0} = 1, 1, 3, 11, 57, 361, 2763, 24611, \dots$$

(OEIS A001586); they count the type- B snakes [40] and form a moment sequence [39]. Their dilated Hankel determinant, and that of a whole *Springer number family* containing them, is as simple as that of the factorials.

The Hankel determinants of the Springer numbers, and the equivalent continued-fraction expansions, have received considerable attention. Since (S_n) is a moment sequence [39], its ordinary Hankel determinant $\det(S_{i+j})$ is a classical positive quantity accessible through the Jacobi–Stieltjes theory of orthogonal polynomials; the associated continued fractions enumerating snakes and cycle-alternating permutations were obtained by Josuat-Vergès [22], and the closed-form Hankel determinants of the companion Euler numbers were established by Han [18], in parallel with the classical evaluations of the tangent and secant (Euler) numbers going back to André [2]. What follows goes beyond the ordinary determinant: we evaluate the *dilated* Hankel determinant $\det(a_{2i+j})$ — and, more than that, the determinant of an entire one-parameter Springer number family — in closed product form.

Theorem 16.1. *For every integer $r \geq 1$,*

$$\ddot{H}_n\left(\frac{1}{(\cos x - t \sin x)^r}\right) = (t(t^2 + 1))^{\binom{n}{2}} \ddot{H}_n\left(\frac{1}{(1 - x)^r}\right).$$

The right-hand sequence is $a_n = (r)_n$, whose dilated determinant is the Beta evaluation of §4; at $t = r = 1$ this is $\ddot{H}_n((\cos x - \sin x)^{-1}) = 4^{\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!$, the Springer evaluation. More generally $t = 1$ gives, for $r = 1, 2$,

$$\ddot{H}_n\left(\frac{1}{\cos x - \sin x}\right) = 4^{\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!, \quad \ddot{H}_n\left(\frac{1}{(\cos x - \sin x)^2}\right) = 4^{\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k + 1)!.$$

Idea of the proof. Write $f_t(x) = (\cos x - t \sin x)^{-r}$. We show that $\ddot{H}_n(f_t)$ is a *polynomial* in t of degree $\leq 3\binom{n}{2}$ that is divisible by $(t(t^2 + 1))^{\binom{n}{2}}$ — it vanishes to order $\binom{n}{2}$ at each of the three roots $t = 0, \pm i$ — and hence equals a constant multiple of it (Observation 3.8); a scaling limit identifies the constant. The orders at $t = 0$ and at $t = \pm i$ come from two elementary representations of f_t (a rotation to an even function; an exponential series), each turning the order into a one-line determinant expansion via the divisor method **M4** of §3.4.

We shall repeatedly use *Faà di Bruno's formula*, the chain rule for the n -th derivative of a composite $g(h(x))$: it expands $(g \circ h)^{(n)}$ as a sum of terms $g^{(k)}(h) \prod_j (h^{(j)})^{m_j}$ with $\sum_j m_j = k \leq n$ and $\sum_j j m_j = n$ [7, Ch. I].

Proof of Theorem 16.1. Let $a_n(t) = n! [x^n](\cos x - t \sin x)^{-r} = f_t^{(n)}(0)$, a composite with $h(x) = \cos x - t \sin x$ and $g(u) = u^{-r}$. The x -derivatives of all orders of h at $x = 0$ lie in $\{\pm 1, \pm t\}$, of degree ≤ 1 in t ; since $h(0) = 1 \neq 0$ the coefficients $g^{(k)}(h(0))$ are constants, so by Faà di Bruno each term is a product of at most n of these derivatives and $a_n \in \mathbb{Q}[t]$ with $\deg_t a_n \leq n$. Hence $\ddot{H}_n(f_t)$ is a polynomial in t with

$$\deg_t \ddot{H}_n(f_t) \leq \max_{\sigma} \sum_i \deg_t a_{2i+\sigma(i)} \leq \max_{\sigma} \sum_i (2i + \sigma(i)) = 2 \binom{n}{2} + \binom{n}{2} = 3 \binom{n}{2}.$$

Order at $t = 0$. From $\cos x - t \sin x = \sqrt{1+t^2} \cos(x + \phi)$, $\phi = \arctan t$, we get $f_t(x) = (1+t^2)^{-r/2} \sec^r(x + \phi)$, so by the homogeneity Lemma 3.7 $\ddot{H}_n(f_t) = (1+t^2)^{-rn/2} \ddot{H}_n(\sec^r(\cdot + \phi))$. The prefactor is a unit at $t = 0$ and $\phi = \arctan t$ is a local analytic isomorphism ($\phi'(0) = 1$); since $\sec^r$ is even, the confluent Lemma 3.11 gives $\text{ord}_{t=0} \ddot{H}_n(f_t) \geq \binom{n}{2}$.

Order at $t = \pm i$. As $\ddot{H}_n(f_t) \in \mathbb{Q}[t]$, $\text{ord}_{t=-i} = \text{ord}_{t=i}$. Put $u = \frac{t-i}{2i}$; then, using $\sin x = (e^{ix} - e^{-ix})/(2i)$,

$$\cos x - t \sin x = e^{-ix} - (t-i) \sin x = (1+u)e^{-ix} - u e^{ix} = (1+u)e^{-ix}(1 - q' e^{2ix}), \quad q' = \frac{u}{1+u},$$

whence $(\cos x - t \sin x)^{-r} = (1+u)^{-r} \sum_{k \geq 0} \binom{k+r-1}{k} q'^k e^{i(2k+r)x}$, so the moments

$$a_m(t) = (1+u)^{-r} \sum_{k \geq 0} \binom{k+r-1}{k} q'^k (i(2k+r))^m$$

form an exponential sum (Lemma 3.9) with distinct nodes $\zeta_k = i(2k+r)$ and coefficient orders $\text{ord}_{q'} c_k = k$. The factor $(1+u)^{-r}$ is a unit at $t = i$ and q' vanishes there to first order; by the order Corollary 3.10, with $o_k = k$, the unique minimal subset is $\{0, 1, \dots, n-1\}$, so

$$\text{ord}_{t=i} \ddot{H}_n(f_t) = \text{ord}_{q'=0} \ddot{H}_n = \sum_{k=0}^{n-1} k = \binom{n}{2},$$

in particular $\ddot{H}_n(f_t) \neq 0$.

Conclusion. $\ddot{H}_n(f_t)$ is a nonzero polynomial of degree $\leq 3 \binom{n}{2}$ vanishing to order $\geq \binom{n}{2}$ at each of $t = 0, i, -i$; the three orders already sum to $3 \binom{n}{2} \geq \deg_t \ddot{H}_n(f_t)$, so by the divisor principle (Observation 3.8) every inequality is an equality and

$$\ddot{H}_n(f_t) = c t^{\binom{n}{2}} (t^2 + 1)^{\binom{n}{2}}, \quad c \neq 0.$$

To find c , replace x by x/t and let $t \rightarrow \infty$: coefficientwise $\cos \frac{x}{t} - t \sin \frac{x}{t} \rightarrow 1 - x$, so the left side tends to $\ddot{H}_n((1-x)^{-r})$, while by Lemma 3.7 it equals $t^{-3 \binom{n}{2}} \ddot{H}_n(f_t) = c(1+t^{-2})^{\binom{n}{2}} \rightarrow c$. Hence $c = \ddot{H}_n((1-x)^{-r})$, which is the theorem. $\square$

17. A DERIVATIVE OF THE SPRINGER NUMBER FAMILY AT $t = 1$

Since $\frac{d}{dx}(\cos x - \sin x) = -(\cos x + \sin x)$, the numerator below is the derivative of the Springer base, and for $s \neq 1$

$$\frac{\cos x + \sin x}{(\cos x - \sin x)^s} = \frac{1}{s-1} \frac{d}{dx} (\cos x - \sin x)^{1-s}, \quad (17.1)$$

the derivative of the $t = 1$, exponent- $(s-1)$ member of the family of §16. Its dilated Hankel determinant again factors completely — now as a product of shifted factors in the exponent s .

Theorem 17.1. *For all $n \geq 1$ and all s ,*

$$\ddot{H}_n\left(\frac{\cos x + \sin x}{(\cos x - \sin x)^s}\right) = 4^{\binom{n}{2}} \left(\prod_{k=1}^{n-1} k!\right) \prod_{j=0}^{n-2} [(s+2j)(s+2j+1)]^{n-1-j}.$$

Equivalently the factor $(s+a)$ occurs with multiplicity $n-1 - \lfloor a/2 \rfloor$ for $a = 0, 1, \dots, 2n-3$ (highest on $s, s+1$, decreasing in consecutive pairs); at $s = 1$ this evaluates $\ddot{H}_n(\frac{\cos x + \sin x}{\cos x - \sin x})$. The constant $4^{\binom{n}{2}} \prod_{k < n} k!$ is the Springer constant of §16 (at $r = 1$) divided by $\prod_{k < n} (2k)!$.

Write $f_s(x) = \frac{\cos x + \sin x}{(\cos x - \sin x)^s}$ and $a_k(s) = k![x^k]f_s$. The proof is again the divisor method **M4** (Observation 3.8), but now with the *exponent* s as the parameter: the degenerations are the non-positive integers, where f_s becomes a trigonometric polynomial, together with $s = \infty$. Unlike §16 (and its elliptic deformation, Section 19), the parameter is not a coefficient t — indeed the coefficient deformation $\frac{\cos x + t \sin x}{(\cos x - t \sin x)^s}$ acquires extra s -dependent zeros in t and does *not* yield to the method. We first record the polynomial dependence on s and the structure of its high-order coefficients.

Lemma 17.2. $a_k(s)$ is a monic polynomial in s of degree k , and its coefficient

$$\pi_r(k) := [s^{k-r}]a_k(s)$$

is a polynomial in k of degree $\leq 2r$ with leading coefficient $1/r!$.

Proof. Put $w(x) := -\log(\cos x - \sin x) = x + x^2 + \frac{2}{3}x^3 + \cdots = x\omega(x)$, so $\omega(0) = 1$, $[x^1]\log \omega = 1$, and $(\cos x - \sin x)^{-s} = e^{sw}$. With $\phi := \cos x + \sin x$,

$$a_k(s) = k![x^k](\phi e^{sw}) = k! \sum_{d \geq 0} \frac{s^d}{d!} [x^k](\phi w^d).$$

As w^d has x -order d , $[x^k](\phi w^d) = 0$ for $d > k$, so $\deg_s a_k = k$; the top term is $k! \frac{s^k}{k!} [x^k]w^k = s^k$ (since $w^k = x^k + \cdots$), so a_k is monic. With $d = k - r$ and $w^{k-r} = x^{k-r}\omega^{k-r}$,

$$\pi_r(k) = \frac{k!}{(k-r)!} [x^k](\phi w^{k-r}) = \frac{k!}{(k-r)!} [x^r](\phi \omega^{k-r}).$$

Here $\frac{k!}{(k-r)!} = k(k-1)\cdots(k-r+1)$ has degree r in k ; and $\omega^{k-r} = \exp((k-r)\log \omega)$ gives $[x^r](\phi \omega^{k-r}) = \sum_{p=0}^r \frac{(k-r)^p}{p!} [x^r](\phi (\log \omega)^p)$, a polynomial in k of degree $\leq r$ with leading coefficient $\frac{1}{r!}([x^1]\log \omega)^r \phi(0) = \frac{1}{r!}$. Hence $\deg_k \pi_r \leq 2r$ and $[k^{2r}]\pi_r = \frac{1}{r!}$. $\square$

The high- s behaviour is thus governed by the *leading symbol* $\hat{a}_k(s) = \sum_{r \geq 0} \frac{k^{2r}}{r!} s^{k-r} = s^k e^{k^2/s}$, whose dilated determinant is a Gaussian kernel.

Lemma 17.3. $\det(e^{\sigma(2i+j)^2})_{0 \leq i,j < n} = 4^{\binom{n}{2}} \left(\prod_{k=1}^{n-1} k! \right) \sigma^{\binom{n}{2}} + O(\sigma^{\binom{n}{2}+1})$.

Proof. Factor $e^{4\sigma i^2}$ from row i and $e^{\sigma j^2}$ from column j ($(2i+j)^2 = 4i^2 + 4ij + j^2$); these are $1 + O(\sigma)$, so to leading order $\det(e^{\sigma(2i+j)^2}) = (1 + O(\sigma)) \det(e^{4\sigma ij})$. Expanding $e^{4\sigma ij} = \sum_{m \geq 0} \frac{(4\sigma)^m}{m!} i^m j^m$, Cauchy–Binet gives

$$\det(e^{4\sigma ij}) = \sum_{0 \leq m_0 < \cdots < m_{n-1}} \left(\prod_a \frac{(4\sigma)^{m_a}}{m_a!} \right) \det(i^{m_a})_{i,a} \det(j^{m_a})_{j,a},$$

of lowest order $\sum_a m_a = \binom{n}{2}$, attained only by $(m_a) = (0, 1, \dots, n-1)$, where each Vandermonde is $\prod_{0 \leq i < j < n} (j-i) = \prod_{k=1}^{n-1} k!$ and $\prod_a m_a! = \prod_{k=1}^{n-1} k!$. The leading term is

$$(4\sigma)^{\binom{n}{2}} \left(\prod_{k < n} k! \right)^2 / \prod_{k < n} k! = 4^{\binom{n}{2}} \left(\prod_{k < n} k! \right) \sigma^{\binom{n}{2}}.$$

$\square$

Proof of Theorem 17.1. Step 1: degree and leading coefficient (the order at $s = \infty$). $\ddot{H}_n = \det(a_{2i+j})$ is a polynomial in s of degree $\leq \max_{\tau} \sum_i (2i + \tau(i)) = 3\binom{n}{2}$ (Lemma 17.2), the maximum over permutations τ . Pulling s^{2i} from row i and s^j from column j ,

$$\ddot{H}_n(f_s) = s^{3\binom{n}{2}} \det(\gamma_{2i+j}(\sigma)), \quad \sigma := 1/s, \quad \gamma_k(\sigma) := \sum_{r \geq 0} \pi_r(k) \sigma^r.$$

Collecting $\gamma_k(\sigma) = \sum_{l \geq 0} E_l(\sigma) k^l$ by powers of k , Lemma 17.2 ($\deg_k \pi_r \leq 2r$) gives $\text{ord}_\sigma E_l \geq \lfloor l/2 \rfloor$. Writing $(2i+j)^l = \sum_b \binom{l}{b} (2i)^{l-b} j^b$ factors $\gamma_{2i+j} = \sum_b j^b F_b(2i, \sigma)$ with $F_b(u, \sigma) = \sum_{l \geq b} \binom{l}{b} E_l(\sigma) u^{l-b}$, so by Cauchy–Binet

$$\det(\gamma_{2i+j}) = \sum_B \det(F_b(2i, \sigma))_{i, b \in B} \det(j^b)_{j, b \in B},$$

over n -subsets $B \subset \mathbb{Z}_{\geq 0}$. In column b the coefficient of σ^r is a polynomial in i of degree $\leq 2r-b$, so a nonzero $\sigma^{\sum r_b}$ -term needs the degrees $\{2r_b - b\}_{b \in B}$ to dominate $\{0, \dots, n-1\}$; with $r_b \geq \lfloor b/2 \rfloor$ this forces $\sum_b r_b = \frac{1}{2} \sum_b ((2r_b - b) + b) \geq \frac{1}{2} (\binom{n}{2} + \binom{n}{2}) = \binom{n}{2}$. Hence $\text{ord}_\sigma \det(\gamma_{2i+j}) \geq \binom{n}{2}$, i.e. $\deg_s \ddot{H}_n \leq 2\binom{n}{2}$. Equality in the bound forces both $\{b\} = \{0, \dots, n-1\}$ and the multiset $\{2r_b - b\} = \{0, \dots, n-1\}$, so the $\sigma^{\binom{n}{2}}$ -coefficient uses, in each column, only the top-degree part of π_r — i.e. only $[k^{2r}] \pi_r = \frac{1}{r!} (k^{2r})$ (Lemma 17.2) — and is therefore unchanged when γ_k is replaced by its symbol $\sum_r \frac{k^{2r}}{r!} \sigma^r = e^{\sigma k^2}$. By Lemma 17.3 that coefficient is $4^{\binom{n}{2}} \prod_{k < n} k! \neq 0$; hence $\deg_s \ddot{H}_n = 2\binom{n}{2}$ with leading coefficient $c_n = 4^{\binom{n}{2}} \prod_{k=1}^{n-1} k!$.

Step 2: order at $s = -a$. For an integer $a \geq 0$, with $y = x + \frac{\pi}{4}$,

$$f_{-a}(x) = (\cos x + \sin x)(\cos x - \sin x)^a = 2^{(a+1)/2} \sin(y) \cos^a(y),$$

an *odd* trigonometric polynomial whose nonzero frequencies are $\{\pm 1, \pm 3, \dots, \pm(a+1)\}$ for a even and $\{\pm 2, \pm 4, \dots, \pm(a+1)\}$ for a odd — in either case $d_a := \lfloor a/2 \rfloor + 1$ distinct absolute values, none zero. Writing $f_{-a} = \sum_\nu c_\nu e^{i\nu x}$, its moment matrix $B^{(0)} = (b_{2i+j})$, $b_m = \sum_\nu c_\nu (i\nu)^m$, factors (using $(i\nu)^{2i} = (-i\nu)^{2i}$) as one rank-one term per *distinct* absolute value (Lemma 3.9), so $\text{rank } B^{(0)} \leq d_a$. Now $a_{2i+j}(s)$ is analytic in $\delta := s + a$ with $a_{2i+j}(-a) = b_{2i+j}$, so $\ddot{H}_n(f_s) = \det(B^{(0)} + \delta B^{(1)} + \dots)$; a matrix pencil whose constant term has rank ρ has determinant of order $\geq n - \rho$ at $\delta = 0$. Hence

$$\text{ord}_{s=-a} \ddot{H}_n(f_s) \geq n - \text{rank } B^{(0)} \geq n - d_a = n - 1 - \lfloor a/2 \rfloor.$$

Step 3: assembly. $\ddot{H}_n(f_s)$ has degree $2\binom{n}{2}$ and is divisible by $\prod_{a=0}^{2n-3} (s+a)^{n-1-\lfloor a/2 \rfloor} = \prod_{j=0}^{n-2} [(s+2j)(s+2j+1)]^{n-1-j}$, itself of degree $\sum_{j=0}^{n-2} 2(n-1-j) = 2\binom{n}{2}$. By Observation 3.8 they agree up to the leading constant $c_n = 4^{\binom{n}{2}} \prod_{k < n} k!$, which is Theorem 17.1. $\square$

Remark 17.4. By (17.1) this determinant is, up to the scalar $(s-1)^{-n}$, the *single shift* $\det(b_{2i+j+1})$ of the Springer sequence $b_m = m! [x^m] (\cos x - \sin x)^{-(s-1)}$ of §16. The two degenerations are of the kinds met earlier: $s = \infty$ reproduces, after rescaling, the rank-one geometric collapse met at $t = \pm i$ in the Springer proof (the limit being e^x , its symbol the Gaussian kernel of Lemma 17.3; the same collapse reappears at $m = 1$ in §19.2), while the integers $s = -a$ give an even/low-rank collapse as in §16 (here f_{-a} is an odd trigonometric polynomial).

18. THE RECIPROCAL-SINE CASE $(1+x)x/\sin x$

The secant family of Section 11 evaluates $(1+x)/\cos x$, and more generally $(1+x)/\cos(x)^{s+1}$, in closed product form. It is natural to try $\sin$ in place of $\cos$. Since $\sin x$ is odd and vanishes at the origin, the literal analogue $1/\sin x$ is not a power series; the even analytic correction $x/\sin x$ repairs this and leads to the object of this section,

$$f(x) = (1+x) \frac{x}{\sin x}.$$

Its two moment functionals turn out to be *Wilson* functionals sharing three of their four parameters (Propositions 18.3 and 18.5); this is what drives the evaluation, which we prove in detail below. Unlike the cosine case, the exponent deformation does *not* survive: numerically, $(1+x)(x/\sin x)^{s+1}$ shows no comparably nice dilated determinant once $s \neq 0$, so we study only the member $s = 0$ displayed above. Write $g(x) = x/\sin x$ and

$$b_k = (2k)! [x^{2k}] g \quad (b_0 = 1, b_1 = \frac{1}{3}, b_2 = \frac{7}{15}, b_3 = \frac{31}{21}, \dots).$$

Since g is even and xg is odd, the moments of f are

$$a_{2k} = b_k, \quad a_{2k+1} = (2k+1)b_k, \quad (18.1)$$

so the two functionals attached to the quadratic decomposition (the even $\mathcal{S}$ and the odd $\mathcal{T}$ of Section 3.2) are

$$\mathcal{S}[y^k] = b_k, \quad \mathcal{T}[y^k] = (2k+1)b_k = \mathcal{T}^*[y^k].$$

The situation is the mirror image of the cosine family of Section 11: here the *odd* functional $\mathcal{T}$ is the classical one — and, less obviously, the even functional $\mathcal{S}$ turns out to be classical as well, indeed itself a Wilson functional (Proposition 18.5). The purpose of this section is to prove the following evaluation.

Theorem 18.1 (closed form for $(1+x)x/\sin x$). *Set*

$$Q(k) = \frac{144 \cdot 64^k (k!)^5 ((k+1)!)^6 (2k)!(2k+1)!}{(3k)!(3k+3)!(3k+4)!}. \quad (18.2)$$

Then the determinants obey the two-step recurrence $\ddot{H}_{n+2} = \frac{2}{3} Q(n) \ddot{H}_n$ ($n \geq 0$), and hence are the products of factorials

$$\ddot{H}_{2N} = \left(\frac{2}{3}\right)^N \prod_{k=0}^{N-1} Q(2k), \quad \ddot{H}_{2N+1} = \left(\frac{2}{3}\right)^N \prod_{k=0}^{N-1} Q(2k+1). \quad (18.3)$$

The proof occupies the remainder of the section; its plan is as follows. Subsections 18.2 and 18.3 make the two functionals explicit: $\mathcal{T}$ is the Wilson functional with parameters $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, and — the decisive structural fact — $\mathcal{S}$ is itself Wilson, with parameters $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0)$ (the Wilson and continuous Hahn families used throughout are recalled in §18.1). Subsection 18.4 applies the biorthogonal reduction **M2**, which confines the whole evaluation to a determinant of connection coefficients $\kappa_{i,m}$ between the two families. Because the families differ in a *single* Wilson parameter, κ collapses to a single hypergeometric term (Subsection 18.5), and the connection determinant becomes, after removal of row and column factors, a determinant of products of two *Catalan numbers*; that Catalan determinant is evaluated in closed form by Desnanot–Jacobi condensation (Subsection 18.6). Subsection 18.7 assembles the pieces.

18.1. The Wilson and continuous Hahn functionals. Both functionals of this section belong to the top two levels of the *Askey scheme* of hypergeometric orthogonal polynomials [25, 21]; we recall the two families and fix the notation used below.

Wilson. For parameters (a, b, c, d) with positive real parts (or occurring in conjugate pairs), the *Wilson functional* $\mathcal{W}_{a,b,c,d}$ acts on polynomials in the variable $y = x^2$ by

$$\mathcal{W}_{a,b,c,d}[p] = \frac{1}{K} \int_0^\infty p(x^2) \left| \frac{\Gamma(a+ix)\Gamma(b+ix)\Gamma(c+ix)\Gamma(d+ix)}{\Gamma(2ix)} \right|^2 dx,$$

with K the constant normalising $\mathcal{W}_{a,b,c,d}[1] = 1$. Its monic orthogonal polynomials are the *Wilson polynomials*, symmetric in (a, b, c, d) and given by the terminating series

$$\frac{W_n(x^2; a, b, c, d)}{(a+b)_n(a+c)_n(a+d)_n} = {}_4F_3 \left(\begin{matrix} -n, n+a+b+c+d-1, a+ix, a-ix \\ a+b, a+c, a+d \end{matrix}; 1 \right).$$

In the variable $y = x^2$ they obey the monic three-term recurrence $P_{n+1} = (y - c_n)P_n - \lambda_n P_{n-1}$ with [25, §9.1]

$$\begin{aligned} c_n &= A_n + C_n - a^2, & \lambda_n &= A_{n-1} C_n, \\ A_n &= \frac{(n+a+b+c+d-1)(n+a+b)(n+a+c)(n+a+d)}{(2n+a+b+c+d-1)(2n+a+b+c+d)}, \\ C_n &= \frac{n(n+b+c-1)(n+b+d-1)(n+c+d-1)}{(2n+a+b+c+d-2)(2n+a+b+c+d-1)}, \end{aligned}$$

and squared norms $h_n = \mathcal{W}[P_n^2] = \prod_{k=1}^n \lambda_k$ (the expression for c_n is symmetric in a, b, c, d despite its form). These stand at the very top of the Askey scheme: every other classical family is a limit or a specialisation of them. In this section the two functionals are Wilson functionals sharing

$a = b = c = \frac{1}{2}$ and differing *only* in the fourth parameter d — $d = \frac{1}{2}$ for the odd functional $\mathcal{T}$, $d = 0$ for the even functional $\mathcal{S}$; substituting these into the formulas above (in the rescaled variable $y = 4x^2$ of Proposition 18.3) reproduces the recurrence data of §18.2–18.3. This single-parameter difference is what collapses the connection coefficients in §18.5.

Continuous Hahn. One level below sit the *continuous Hahn polynomials* [25, §9.4], orthogonal on $\mathbb{R}$ in the variable x with weight $\Gamma(a+ix)\Gamma(b+ix)\Gamma(c-ix)\Gamma(d-ix)$ (parameters of positive real part, with $c = \bar{a}$, $d = \bar{b}$), and given by

$$p_n(x; a, b, c, d) = i^n \frac{(a+c)_n (a+d)_n}{n!} {}_3F_2 \left(\begin{matrix} -n, n+a+b+c+d-1, a+ix \\ a+c, a+d \end{matrix} ; 1 \right).$$

When the weight is *even* — as for $a = b = c = d = \frac{1}{2}$, where it equals $|\Gamma(\frac{1}{2}+ix)|^4 = \pi^2 \operatorname{sech}^2 \pi x$ — the monic polynomials are symmetric, $p_{n+1} = x p_n - \beta_n p_{n-1}$ with β_n rational in n , so odd and even degrees decouple. Pushing the measure forward under the quadratic map $y = x^2$ then turns them into a Wilson family: this is precisely the quadratic decomposition of §3.2. It is why the even functional $\mathcal{S}$ wears two faces, each used at a different point below: continuous Hahn in x , which yields its clean S -fraction $u_n = n^4/(4n^2 - 1)$ (Proposition 18.4), and Wilson in $y = x^2$, which yields its connection coefficients against $\mathcal{T}$ (Proposition 18.5).

18.2. The classical odd functional $\mathcal{T}$.

Proposition 18.2 (integral representation of $\mathcal{T}$). *For all $k \geq 0$,*

$$\mathcal{T}[y^k] = (2k+1) b_k = (2k+1)! \frac{2\eta(2k)}{\pi^{2k}}, \quad \eta(s) = \sum_{j \geq 1} \frac{(-1)^{j-1}}{j^s},$$

and $\mathcal{T}$ is the moment functional of the weight

$$\mathcal{T}[y^k] = \int_0^\infty z^k w(z) dz, \quad w(z) = \pi^2 \sum_{j \geq 1} (-1)^{j-1} j^2 e^{-j\pi\sqrt{z}} = \frac{\pi^2}{4} \frac{\sinh(\frac{\pi}{2}\sqrt{z})}{\cosh^3(\frac{\pi}{2}\sqrt{z})} \quad (z > 0). \quad (18.4)$$

Proof. From the Mittag-Leffler expansion [49, Ch. VII]

$$\frac{1}{\sin x} = \frac{1}{x} + \sum_{j \geq 1} (-1)^j \frac{2x}{x^2 - j^2 \pi^2},$$

we get $\frac{x}{\sin x} = 1 + \sum_{j \geq 1} (-1)^j \frac{2x^2}{x^2 - j^2 \pi^2}$, whence for $k \geq 1$

$$[x^{2k}] \frac{x}{\sin x} = 2 \sum_{j \geq 1} (-1)^j [x^{2k-2}] \frac{1}{x^2 - j^2 \pi^2} = 2 \sum_{j \geq 1} (-1)^{j-1} \frac{1}{(j\pi)^{2k}} = \frac{2\eta(2k)}{\pi^{2k}}.$$

Hence $b_k = (2k)! \frac{2\eta(2k)}{\pi^{2k}}$ for $k \geq 1$; with $\eta(0) = \frac{1}{2}$ the same formula gives $b_0 = 1$, and multiplying by $(2k+1)$ yields the first display. For the second, the substitution $z = u^2$ gives $\int_0^\infty z^k e^{-a\sqrt{z}} dz = 2(2k+1)!/a^{2k+2}$, so

$$\frac{(2k+1)!}{(j\pi)^{2k}} = \frac{(j\pi)^2}{2} \int_0^\infty z^k e^{-j\pi\sqrt{z}} dz.$$

Summing $2(-1)^{j-1}$ times this over $j \geq 1$ and interchanging sum and integral — legitimate because $\sum_{j \geq 1} j^2 e^{-j\pi\sqrt{z}}$ converges locally uniformly on $(0, \infty)$ and w is bounded near 0 — gives the series form of (18.4). The closed form, which also exhibits the boundedness just used, follows from $\sum_{j \geq 1} (-1)^{j-1} j^2 x^j = x(1-x)/(1+x)^3$ at $x = e^{-\pi\sqrt{z}}$. $\square$

The next proposition identifies $\mathcal{T}$ as a Wilson functional and makes its monic orthogonal polynomials ρ_m and their data explicit.

Proposition 18.3 ($\mathcal{T}$ is Wilson; the odd J -fraction). *The monic $\mathcal{T}$ -orthogonal polynomials satisfy $\rho_{m+1} = (y - c_m^\mathcal{T})\rho_m - \lambda_m^\mathcal{T}\rho_{m-1}$ with*

$$c_m^\mathcal{T} = 2m^2 + 2m + 1, \quad \lambda_m^\mathcal{T} = \frac{4m^6}{(2m-1)(2m+1)},$$

and squared norms

$$h_m^\mathcal{T} = \mathcal{T}[\rho_m^2] = \prod_{j=1}^m \lambda_j^\mathcal{T} = \frac{16^m (m!)^8}{(2m)!(2m+1)!}. \quad (18.5)$$

Equivalently, the generating function of the moments $\mathcal{T}[y^k] = (2k+1)b_k$ has the J -fraction

$$\sum_{k \geq 0} \mathcal{T}[y^k] t^k = \frac{1}{1 - c_0^\mathcal{T} t - \frac{\lambda_1^\mathcal{T} t^2}{1 - c_1^\mathcal{T} t - \frac{\lambda_2^\mathcal{T} t^2}{1 - \ddots}}} = \frac{1}{1 - t - \frac{\frac{4}{3} t^2}{1 - 5t - \frac{\frac{256}{15} t^2}{1 - 13t - \dots}}}. \quad (18.6)$$

Proof. By the reflection formula $|\Gamma(\frac{1}{2} + ix)|^2 = \pi \operatorname{sech}(\pi x)$ and $|\Gamma(2ix)|^2 = \pi/(2x \sinh 2\pi x)$, the Wilson weight with $a = b = c = d = \frac{1}{2}$ is

$$W(x) = \left| \frac{\Gamma(\frac{1}{2} + ix)^4}{\Gamma(2ix)} \right|^2 = 4\pi^3 \frac{x \sinh \pi x}{\cosh^3 \pi x}.$$

On the other hand, the closed form of the weight in (18.4), written in the variable $s = \sqrt{z}$ (so $dz = 2s ds$) and rescaled by $s = 2x$, gives

$$\mathcal{T}[y^k] = \frac{\pi^2}{2} \int_0^\infty s^{2k+1} \frac{\sinh(\frac{\pi}{2}s)}{\cosh^3(\frac{\pi}{2}s)} ds = 2\pi^2 4^k \int_0^\infty x^{2k} \frac{x \sinh \pi x}{\cosh^3 \pi x} dx = \frac{4^k}{2\pi} \int_0^\infty x^{2k} W(x) dx;$$

hence, in the Wilson variable $z = x^2$ (so $y = 4z$), $\mathcal{T}$ is the Wilson functional with parameters $a = b = c = d = \frac{1}{2}$, normalised by $\mathcal{T}[1] = 1$. The monic Wilson recurrence coefficients [25, §9.1] are $\alpha_m^W = A_m + C_m - a^2$ and $\lambda_m^W = A_{m-1}C_m$, where

$$A_m = \frac{(m+a+b+c+d-1)(m+a+b)(m+a+c)(m+a+d)}{(2m+a+b+c+d-1)(2m+a+b+c+d)},$$

$$C_m = \frac{m(m+b+c-1)(m+b+d-1)(m+c+d-1)}{(2m+a+b+c+d-2)(2m+a+b+c+d-1)}.$$

At $a = b = c = d = \frac{1}{2}$ these reduce to $A_m = \frac{(m+1)^3}{2(2m+1)}$, $C_m = \frac{m^3}{2(2m+1)}$, so that, using $(m+1)^3 + m^3 = (2m+1)(m^2+m+1)$,

$$\alpha_m^W = \frac{m^2 + m + 1}{2} - \frac{1}{4} = \frac{2m^2 + 2m + 1}{4}, \quad \lambda_m^W = \frac{m^6}{4(4m^2 - 1)}.$$

The substitution $y = 4z$ multiplies α by 4 and λ by 16, giving $c_m^\mathcal{T} = 2m^2 + 2m + 1$ and $\lambda_m^\mathcal{T} = \frac{4m^6}{4m^2 - 1}$. Finally $h_m^\mathcal{T} = h_0^\mathcal{T} \prod_{j=1}^m \lambda_j^\mathcal{T} = \prod_{j=1}^m \frac{4j^6}{4j^2 - 1} = \frac{16^m (m!)^8}{(2m)!(2m+1)!}$, since $h_0^\mathcal{T} = \mathcal{T}[1] = b_0 = 1$. $\square$

Of this data, the sequel uses the recurrence coefficients $c_m^\mathcal{T}, \lambda_m^\mathcal{T}$ (in the connection recurrence (18.13) below) and the norms (18.5).

18.3. The even functional $\mathcal{S}$. Contrary to a first impression, the even functional $\mathcal{S}$ is also elementary: its Stieltjes S -fraction is as clean as can be, the single rational coefficient $u_n = n^4/(4n^2 - 1)$. The orthogonal-polynomial recurrence (the J -fraction) is then fixed by the even contraction (§2.5) as $c_i^S = u_{2i} + u_{2i+1}$, $\lambda_i^S = u_{2i-1}u_{2i}$ — explicit, but bulkier than u_n itself, which is why we work from the S -fraction.

Proposition 18.4 (the even functional). *The functional $\mathcal{S}[y^k] = b_k$ is the moment functional of the weight*

$$\mathcal{S}[y^k] = \int_0^\infty z^k w_S(z) dz, \quad w_S(z) = \frac{\pi}{4\sqrt{z} \cosh^2(\frac{\pi}{2}\sqrt{z})} \quad (z > 0), \quad (18.7)$$

equivalently, in the variable $s = \sqrt{z}$, the even weight $\frac{\pi}{2} \operatorname{sech}^2(\frac{\pi}{2}s)$ on $\mathbb{R}$. Its generating function $\sum_{k \geq 0} b_k t^k$ has the Stieltjes continued fraction with coefficients

$$u_n = \frac{n^4}{(2n-1)(2n+1)} = \frac{n^4}{4n^2-1} \quad (n \geq 1), \quad (18.8)$$

so that the squared norms of the monic $\mathcal{S}$ -orthogonal polynomials P_l are the explicit products

$$h_l^{\mathcal{S}} = \mathcal{S}[P_l^2] = \prod_{j=1}^{2l} u_j = \frac{16^l ((2l)!)^6}{(4l)!(4l+1)!}. \quad (18.9)$$

Proof. The representation (18.7) follows as in Proposition 18.2: from $b_k = (2k)! 2\eta(2k)/\pi^{2k}$ and $\int_0^\infty z^k e^{-a\sqrt{z}} dz/\sqrt{z} = 2(2k)!/a^{2k+1}$ one gets

$$b_k = \int_0^\infty z^k \frac{\pi}{\sqrt{z}} \sum_{j \geq 1} (-1)^{j-1} j e^{-j\pi\sqrt{z}} dz,$$

and $\sum_{j \geq 1} (-1)^{j-1} j x^j = x/(1+x)^2$ at $x = e^{-\pi\sqrt{z}}$ gives the closed form $\frac{\pi}{4\sqrt{z}} \operatorname{sech}^2(\frac{\pi}{2}\sqrt{z})$. For the continued fraction (18.8): with $s = 2x$, $|\Gamma(\frac{1}{2} + ix)|^2 = \pi \operatorname{sech}(\pi x)$ gives $\frac{1}{\pi^2} |\Gamma(\frac{1}{2} + ix)|^4 = \operatorname{sech}^2(\pi x)$, so ν is (the $s = 2x$ rescaling of) the continuous Hahn weight with $a = b = c = d = \frac{1}{2}$. This weight being even, the monic continuous Hahn polynomials are symmetric, $p_{n+1}(s) = s p_n(s) - \beta_n p_{n-1}(s)$, and their recurrence coefficients [25, §9.4] specialise to $\beta_n = \frac{n^4}{4n^2-1}$ (the value $\frac{n^4}{4(4n^2-1)}$ in the variable x , multiplied by 4 for $s = 2x$). For a symmetric measure the Stieltjes S -fraction of the even-moment series $\sum_k b_k t^k$ has coefficients equal to these recurrence coefficients (the even contraction, §2.5), so $u_n = \beta_n = \frac{n^4}{4n^2-1}$. Finally (18.9) is the elementary product $\prod_{j=1}^{2l} \frac{j^4}{(2j-1)(2j+1)} = \frac{((2l)!)^4}{(4l-1)!(4l+1)!} = \frac{16^l ((2l)!)^6}{(4l)!(4l+1)!}$. $\square$

The even functional carries one more piece of structure, which will turn out to be the key to the closed-form evaluation: it is itself a *Wilson* functional, in the same variable as $\mathcal{T}$.

Proposition 18.5 (the even functional is also Wilson). *In the variable $y = 4x^2$ of Proposition 18.3, $\mathcal{S}$ is the Wilson functional with parameters $(a, b, c, d) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0)$.*

Proof. By the duplication formula $\Gamma(2ix) = 2^{2ix-1} \pi^{-1/2} \Gamma(ix) \Gamma(\frac{1}{2} + ix)$, the Wilson weight with parameters $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0)$ is

$$\left| \frac{\Gamma(\frac{1}{2} + ix)^3 \Gamma(ix)}{\Gamma(2ix)} \right|^2 = \left| 2^{1-2ix} \sqrt{\pi} \Gamma(\frac{1}{2} + ix)^2 \right|^2 = 4\pi \left| \Gamma(\frac{1}{2} + ix) \right|^4 = 4\pi^3 \operatorname{sech}^2(\pi x),$$

which is proportional to the weight $\frac{\pi}{2} \operatorname{sech}^2(\frac{\pi}{2}s)$ of (18.7) under $s = 2x$ (so $z = s^2 = 4x^2 = y$). Both functionals being normalised by $\mathcal{S}[1] = 1$, they coincide. $\square$

Thus the two monic families P_i and ρ_m attached to f are *Wilson families differing in a single parameter*: $d = 0$ for the even functional, $d = \frac{1}{2}$ for the odd one. This is the structural fact behind the closed form of the connection coefficients in Subsection 18.5 below.

18.4. Reduction to a connection-coefficient determinant. We now reduce Theorem 18.1 to a single finite determinant identity. Since *both* moment functionals of f are classical, both families of monic orthogonal polynomials are explicit: the Stieltjes S -fraction (18.8) of $\mathcal{S}$ gives, through the even contraction (Lemma 3.5),

$$c_i^{\mathcal{S}} = u_{2i} + u_{2i+1}, \quad \lambda_i^{\mathcal{S}} = u_{2i-1} u_{2i}, \quad u_n = \frac{n^4}{4n^2-1}. \quad (18.10)$$

Since $\mathcal{T}$ is also quasi-definite, with monic orthogonal polynomials ρ_m (the Wilson polynomials of Proposition 18.3, $c_m^{\mathcal{T}} = 2m^2 + 2m + 1$, $\lambda_m^{\mathcal{T}} = \frac{4m^6}{(2m-1)(2m+1)}$, J -fraction (18.6)), the full biorthogonal

reduction **M2** (the determinant reduction Lemma 3.4) applies verbatim and gives, for all $n \geq 1$,

$$\ddot{H}_n = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^{\mathcal{S}} \right) \left(\prod_{m=0}^{n-1} h_m^{\mathcal{T}} \right) \det(\kappa_{\bar{n}+r, m})_{0 \leq r, m \leq \underline{n}-1}, \quad (18.11)$$

where $h_l^{\mathcal{S}}$ and $h_m^{\mathcal{T}}$ are the explicit factorial products (18.9) and (18.5), and the $\kappa_{i, m}$ are the connection coefficients expressing the $\mathcal{S}$ -orthogonal P_i in the $\mathcal{T}$ -orthogonal basis,

$$P_i = \sum_{m=0}^i \kappa_{i, m} \rho_m. \quad (18.12)$$

By the connection recurrence (Lemma 3.3) these are determined, with no free parameter, by $\kappa_{0,0} = 1$ (and $\kappa_{i, m} = 0$ for $m < 0$ or $m > i$) and

$$\kappa_{i+1, m} = \kappa_{i, m-1} + (c_m^{\mathcal{T}} - c_i^{\mathcal{S}}) \kappa_{i, m} + \lambda_{m+1}^{\mathcal{T}} \kappa_{i, m+1} - \lambda_i^{\mathcal{S}} \kappa_{i-1, m}. \quad (18.13)$$

The first connection coefficients are $\kappa_{i, i} = 1$, $\kappa_{i, i-1} = \frac{2i^3}{4i-1}$ and $\kappa_{i, i-2} = -\frac{8(i-1)^3 i^4}{(4i-3)(4i-2)(4i-1)}$.

The prefactor in (18.11) being a known factorial product, Theorem 18.1 is now reduced to the evaluation of the connection determinant $\det(\kappa_{\bar{n}+r, m})$ — a purely algebraic identity, in which every entry is an explicitly generated rational number and all of the analytic data (η -values, the sech^2 and Wilson weights) has been discharged.

At first sight this determinant looks intractable: a connection array between two *distinct* classical families is in general a balanced ${}_3F_2$ -type double sum, the matrix in (18.11) is neither of Hankel nor of Toeplitz type, and — unlike the double shift of the cosine family (Section 14) — no free parameter is in sight. The present pair is closer than it looks: by Propositions 18.3 and 18.5 the two families are Wilson families differing in a *single* parameter ($d = 0$ versus $d = \frac{1}{2}$), and for a one-parameter change the connection coefficients collapse to a single hypergeometric term (Lemma 18.6 below). After removing row and column factors, the connection determinant becomes the Hadamard product of a Toeplitz and a Hankel matrix of *Catalan numbers*, and that determinant is closed under Desnanot–Jacobi condensation once its two shifts are treated as free parameters (Theorem 18.8).

18.5. The connection coefficients in closed form.

Lemma 18.6 (closed form of κ). *For $0 \leq m \leq i$, with $d = i - m$,*

$$\kappa_{i, m} = \frac{\left(\frac{3}{2} - d\right)_d (i + m + 2)_d \left((m + 1)_d\right)^3}{d! \left(m + \frac{3}{2}\right)_d \left(i + m + \frac{1}{2}\right)_d}. \quad (18.14)$$

Proof. Write $k_{i, m}$ for the right-hand side of (18.14), extended by $k_{i, m} = 0$ for $m < 0$ or $m > i$. By Lemma 3.3 the array κ is the unique solution of the recurrence (18.13) whose row $i = 0$ is $\kappa_{0,0} = 1$, $\kappa_{0, m} = 0$ ($m \neq 0$). The candidate has the same row $i = 0$ (at $d = 0$ every product in (18.14) is empty), so it suffices to show that k satisfies (18.13) for all $i \geq 0$ and all m . Every factor of (18.14) is nonzero on the triangle $0 \leq m \leq i$, and telescoping each Pochhammer symbol gives the four neighbour ratios as rational functions of (i, m) (recall $d = i - m$):

$$\begin{aligned} \frac{k_{i+1, m}}{k_{i, m}} &= -\frac{4(2d-1)(i+1)^4(2i+2m+1)}{(d+1)(i+m+2)(4i+1)(4i+3)}, & \frac{k_{i, m-1}}{k_{i, m}} &= -\frac{2(2d-1)(i+m+1)m^3}{(d+1)(2m+1)(2i+2m-1)}, \\ \frac{k_{i, m+1}}{k_{i, m}} &= \frac{d(2m+3)(2i+2m+1)}{2(3-2d)(i+m+2)(m+1)^3}, & \frac{k_{i-1, m}}{k_{i, m}} &= -\frac{d(i+m+1)(4i-3)(4i-1)}{4(2d-3)i^4(2i+2m-1)}. \end{aligned}$$

Dividing (18.13) by $k_{i, m}$ and inserting these ratios together with the data $c_i^{\mathcal{S}}, \lambda_i^{\mathcal{S}}$ of (18.10) and $c_m^{\mathcal{T}} = 2m^2 + 2m + 1$, $\lambda_{m+1}^{\mathcal{T}} = \frac{4(m+1)^6}{(2m+1)(2m+3)}$ turns the recurrence into a rational-function identity in (i, m) , verified by clearing denominators. The boundary columns take care of themselves: at $m = 0$ the second ratio carries the factor m^3 and vanishes, as required by $\kappa_{i, -1} = 0$; at $m = i$ the third ratio carries the factor d and vanishes, as required by $\kappa_{i, i+1} = 0$; and at $m = i + 1$ the recurrence reduces to $k_{i+1, i+1} = k_{i, i} = 1$. $\square$

The diagonals $\kappa_{i,i-1}$, $\kappa_{i,i-2}$ recorded after (18.13) are the cases $d = 1, 2$ of (18.14). The single-term form (18.14) now splits into row factors, column factors, and a kernel of *Catalan numbers*.

Lemma 18.7 (Catalan splitting). *Let $C_k = \frac{1}{k+1} \binom{2k}{k}$ denote the Catalan numbers and put $\gamma(k) = (2k)!/(4^k k!)$, so that $\Gamma(k + \frac{1}{2}) = \sqrt{\pi} \gamma(k)$. For $i > m$,*

$$\kappa_{i,m} = \alpha_i \beta_m C_{i-m-1} C_{i+m}, \quad \alpha_i = \frac{2(-1)^i}{16^i} \cdot \frac{(2i+1)! (i!)^3}{\gamma(i+1) \gamma(2i)}, \quad \beta_m = (-1)^{m+1} \frac{\gamma(m+1)}{(m!)^3}. \quad (18.15)$$

Proof. Each Pochhammer symbol in (18.14) telescopes into factorials and half-integer Gamma values:

$$\begin{aligned} (i+m+2)_d &= \frac{(2i+1)!}{(i+m+1)!}, & ((m+1)_d)^3 &= \left(\frac{i!}{m!}\right)^3, \\ (m+\frac{3}{2})_d &= \frac{\gamma(i+1)}{\gamma(m+1)}, & (i+m+\frac{1}{2})_d &= \frac{\gamma(2i)}{\gamma(i+m)}, \end{aligned}$$

while $(\frac{3}{2} - d)_d = \frac{1}{2} \prod_{k=2}^d (\frac{3}{2} - k) = (-1)^{d-1} \frac{1}{2} (\frac{1}{2})_{d-1} = (-1)^{d-1} \frac{1}{2} \gamma(d-1)$. Hence

$$\kappa_{i,m} = (-1)^{d-1} \frac{\gamma(d-1)}{2d!} \cdot \frac{(2i+1)!}{(i+m+1)!} \cdot \frac{(i!)^3}{(m!)^3} \cdot \frac{\gamma(m+1)}{\gamma(i+1)} \cdot \frac{\gamma(i+m)}{\gamma(2i)},$$

and the two Catalan evaluations $\gamma(d-1)/d! = C_{d-1}/4^{d-1}$ and $\gamma(i+m)/(i+m+1)! = C_{i+m}/4^{i+m}$ turn this into (18.15): the powers of 4 combine to $4^{-(d-1)-(i+m)} = 4 \cdot 16^{-i}$ and the sign to $(-1)^{d-1} = (-1)^i (-1)^{m+1}$. $\square$

In the residual block of (18.11) every row index $i = \bar{n} + r \geq \bar{n}$ exceeds every column index $m \leq \underline{n} - 1 < \bar{n}$, so Lemma 18.7 applies to every entry: pulling $\alpha_{\bar{n}+r}$ out of row r and β_m out of column m ,

$$\det(\kappa_{\bar{n}+r,m})_{0 \leq r, m \leq \underline{n}-1} = \left(\prod_{r=0}^{\underline{n}-1} \alpha_{\bar{n}+r} \right) \left(\prod_{m=0}^{\underline{n}-1} \beta_m \right) F_{\underline{n}}(\bar{n}-1, \bar{n}), \quad (18.16)$$

where, for integers $N \geq 0$ and $a, b \geq 0$,

$$F_N(a, b) := \det(C_{a+r-m} C_{b+r+m})_{0 \leq r, m \leq N-1}. \quad (18.17)$$

All the analytic content is now discharged: (18.17) is the determinant of the Hadamard product of a Toeplitz and a Hankel matrix of Catalan numbers, and it evaluates in closed form.

18.6. A Catalan determinant.

Theorem 18.8 (Catalan determinant). *Let $N \geq 0$ and let a, b be integers with $a \geq N-1$ and $b \geq a+1$. Then, with $\gamma(k) = (2k)!/(4^k k!)$ as above and the staircase Pochhammer products*

$$\Lambda_N(t) := \prod_{j=1}^{N-1} (t+j)_j \quad (\Lambda_0 = \Lambda_1 = 1), \quad (18.18)$$

the determinant (18.17) equals

$$\begin{aligned} F_N(a, b) &= (-4)^{\binom{N}{2}} 4^{N(a+b)} \prod_{k=1}^{N-1} (2k+1)! \\ &\quad \times \frac{\prod_{k=0}^{N-1} \gamma(a-k) \gamma(b+k)}{\prod_{k=2}^{N+1} (a+k-1)! \prod_{k=N+1}^{2N} (b+k-1)!} \Lambda_N(b-a) \Lambda_N(a+b+\frac{3}{2}). \end{aligned} \quad (18.19)$$

Proof. Write $G_N(a, b)$ for the right-hand side of (18.19) and

$$G_N(a, b) = c_N 4^{N(a+b)} A_N(a) B_N(b) \Lambda_N(b-a) \Lambda_N(a+b+\frac{3}{2}), \quad c_N = (-4)^{\binom{N}{2}} \prod_{k=1}^{N-1} (2k+1)!,$$

with $A_N(a) = \prod_{k=0}^{N-1} \gamma(a-k) / \prod_{k=2}^{N+1} (a+k-1)!$ and $B_N(b) = \prod_{k=0}^{N-1} \gamma(b+k) / \prod_{k=N+1}^{2N} (b+k-1)!$. The proof is by induction on N through the Desnanot–Jacobi identity, in the manner of the condensation proofs of the determinant calculus [26].

For $N = 0$ both sides equal 1; for $N = 1$, $G_1(a, b) = 4^{a+b} \gamma(a) \gamma(b) / ((a+1)! (b+1)!) = C_a C_b = F_1(a, b)$. Let $N \geq 2$. In the region $a \geq N-1$, $b \geq a+1$ every Catalan index in (18.17) is nonnegative, and inspection of the entries $C_{a+r-m} C_{b+r+m}$ identifies the four corner minors of the $N \times N$ matrix as F_{N-1} at (a, b) , $(a, b+2)$, $(a+1, b+1)$, $(a-1, b+1)$ and its central minor as $F_{N-2}(a, b+2)$, so that the Desnanot–Jacobi identity reads

$$F_N(a, b) F_{N-2}(a, b+2) = F_{N-1}(a, b) F_{N-1}(a, b+2) - F_{N-1}(a+1, b+1) F_{N-1}(a-1, b+1). \quad (18.20)$$

All five smaller determinants stay in the region $(a-1 \geq N-2)$, and the gaps $b-a$, $b-a+2$ remain ≥ 1 , so they equal the corresponding G 's by the induction hypothesis; and $G_{N-2}(a, b+2) \neq 0$, every factor of (18.19) being finite and nonzero there. Hence (18.20) determines $F_N(a, b)$, and it remains to check that G satisfies the same relation, i.e. that $R_1 - R_2 = 1$ for

$$R_1 := \frac{G_{N-1}(a, b) G_{N-1}(a, b+2)}{G_N(a, b) G_{N-2}(a, b+2)}, \quad R_2 := \frac{G_{N-1}(a+1, b+1) G_{N-1}(a-1, b+1)}{G_N(a, b) G_{N-2}(a, b+2)}.$$

Both quotients collapse block by block. The one-step ratios

$$\frac{A_N(a)}{A_{N-1}(a)} = \frac{\gamma(a-N+1)}{(a+N)!}, \quad \frac{B_N(b)}{B_{N-1}(b)} = \frac{\gamma(b+N-1)(b+N-1)!}{(b+2N-2)!(b+2N-1)!},$$

$$\frac{\Lambda_N(t)}{\Lambda_{N-1}(t)} = (t+N-1)_{N-1},$$

together with $c_{N-1}^2 / (c_N c_{N-2}) = -\frac{1}{4} / ((2N-1)(2N-2))$ and $\gamma(x+1)/\gamma(x) = x + \frac{1}{2}$, give: the powers of 4 contribute 4^2 to each quotient; the Λ -blocks contribute, at $t = b-a$ and at $t = a+b+\frac{3}{2}$ alike (the four Λ -arguments are the same in R_1 and R_2),

$$\frac{\Lambda_{N-1}(t) \Lambda_{N-1}(t+2)}{\Lambda_N(t) \Lambda_{N-2}(t+2)} = \frac{(t+N)_{N-2}}{(t+N-1)_{N-1}} = \frac{1}{t+N-1}; \quad (18.21)$$

and telescoping the (finitely overlapping) γ - and factorial windows of the A - and B -blocks leaves $(a-N+\frac{3}{2})(a+N)$ and $(b+N-\frac{1}{2})(b+N)$ in R_1 , respectively $(a+\frac{1}{2})(a+1)$ and $(b+\frac{1}{2})(b+2N-1)$ in R_2 . Altogether

$$R_1 = \frac{-4(a-N+\frac{3}{2})(a+N)(b+N-\frac{1}{2})(b+N)}{(2N-1)(2N-2)(b-a+N-1)(a+b+N+\frac{1}{2})},$$

$$R_2 = \frac{-4(a+\frac{1}{2})(a+1)(b+\frac{1}{2})(b+2N-1)}{(2N-1)(2N-2)(b-a+N-1)(a+b+N+\frac{1}{2})},$$

and $R_1 - R_2 = 1$ amounts to the polynomial identity

$$(a+\frac{1}{2})(a+1)(b+\frac{1}{2})(b+2N-1) - (a-N+\frac{3}{2})(a+N)(b+N-\frac{1}{2})(b+N) = \frac{(N-1)(2N-1)}{2} (b-a+N-1)(a+b+N+\frac{1}{2}), \quad (18.22)$$

checked by expanding both sides. This completes the induction. $\square$

Remark 18.9 (the two-parameter family is essential). The condensation closes only because the *two-parameter* family (18.17) is considered: the four corner minors of $F_N(a, b)$ shift (a, b) in four different directions, and no one-parameter subfamily (in particular not the diagonal $b = a+1$ that the application requires) is stable under (18.20). The hypotheses $a \geq N-1$ and $b \geq a+1$ are what the condensation argument needs, keeping all five smaller determinants of (18.20) within the range of the induction; the application uses only the diagonal $b = a+1$.

18.7. Proof of the closed form.

Proof of Theorem 18.1. Combining the biorthogonal reduction (18.11) with the splitting (18.16) and with Theorem 18.8 — applicable to $F_{\underline{n}}(\bar{n}-1, \bar{n})$ since $\bar{n}-1 \geq \underline{n}-1$ and the gap is $b-a=1$ — expresses $\ddot{H}_n$ as the explicit product

$$\ddot{H}_n = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^{\mathcal{S}} \right) \left(\prod_{m=0}^{\underline{n}-1} h_m^{\mathcal{T}} \right) \left(\prod_{r=0}^{\bar{n}-1} \alpha_{\bar{n}+r} \right) \left(\prod_{m=0}^{\underline{n}-1} \beta_m \right) G_{\underline{n}}(\bar{n}-1, \bar{n}) =: \Pi_n, \quad (18.23)$$

with $h^{\mathcal{S}}, h^{\mathcal{T}}$ the factorial products (18.9), (18.5), α, β as in (18.15), and G the right-hand side of (18.19). Direct evaluation gives $\Pi_0 = \Pi_1 = 1$. The two-step quotient Π_{n+2}/Π_n is an elementary telescoping: the step $n \mapsto n+2$ replaces $(\bar{n}, \underline{n})$ by $(\bar{n}+1, \underline{n}+1)$, so the quotient collects the factors $h_{\bar{n}}^{\mathcal{S}}, h_{\underline{n}}^{\mathcal{T}}, \alpha_n \alpha_{n+1}/\alpha_{\bar{n}}, \beta_{\underline{n}}$ and the ratio $G_{\underline{n}+1}(\bar{n}, \bar{n}+1)/G_{\underline{n}}(\bar{n}-1, \bar{n})$, the latter an explicit ratio of factorials by the one-step ratios in the proof of Theorem 18.8 together with

$$\frac{\Lambda_{q+1}(t)}{\Lambda_q(t)} = (t+q)_q, \quad \frac{\Lambda_q(t+2)}{\Lambda_q(t)} = \frac{\Gamma(t+1)\Gamma(t+2q)}{\Gamma(t+q)\Gamma(t+q+1)}.$$

Simplifying the resulting factorial products (the two parities of n treated separately, as in Proposition 11.1) yields

$$\frac{\Pi_{n+2}}{\Pi_n} = \frac{2}{3} Q(n) \quad (n \geq 0),$$

with Q as in (18.2). Hence $\ddot{H}_n = \Pi_n$ satisfies the two-step recurrence $\ddot{H}_{n+2} = \frac{2}{3} Q(n) \ddot{H}_n$ with $\ddot{H}_0 = \ddot{H}_1 = 1$, and iterating it gives the products (18.3). $\square$

Remark 18.10 (the collapse mechanism). The proof is an instance of the two-stage collapse mechanism of the reduction **M2** described at the end of Section 3.2. The single-term connection (Lemma 18.6) is the analogue, on the Wilson level, of the classical fact that connection coefficients between two Jacobi families sharing one parameter factor completely; and the kernel $C_{i-m-1}C_{i+m}$ it leaves behind generates the two-parameter Catalan family (18.17), which is stable under Desnanot–Jacobi condensation. In particular, no deformation parameter and no recourse to the divisor method **M4** is needed.

19. AN ELLIPTIC DEFORMATION OF THE EULER NUMBERS

The Hankel determinants attached to elliptic functions have been studied by Thomas Jan Stieltjes [42], Philippe Flajolet [12], Hubert Prodinger [37] and Dominique Dumont [9]. Stieltjes computed the continued fractions — equivalently the Hankel determinants — of many special series; Flajolet gave the combinatorial theory relating Hankel determinants, J -fractions and weighted lattice paths; Dumont interpreted the Taylor coefficients of the Jacobi elliptic functions combinatorially; and Prodinger obtained the continued fractions of several tangent- and secant-type deformations.

Recall the *Jacobi elliptic functions* $\operatorname{sn}(x, m)$, $\operatorname{cn}(x, m)$, $\operatorname{dn}(x, m)$, of modulus parameter $m = k^2$. Writing the incomplete elliptic integral of the first kind $u = \int_0^\phi (1 - m \sin^2 \theta)^{-1/2} d\theta$, they are defined by inverting $u \mapsto \phi$ and setting $\operatorname{sn}(u, m) = \sin \phi$, $\operatorname{cn}(u, m) = \cos \phi$, $\operatorname{dn}(u, m) = (1 - m \sin^2 \phi)^{1/2}$; they are doubly periodic meromorphic functions satisfying $\operatorname{sn}^2 + \operatorname{cn}^2 = 1$ and $\operatorname{dn}^2 + m \operatorname{sn}^2 = 1$, and reduce to $\sin x, \cos x, 1$ at $m = 0$ (Whittaker–Watson [49, Ch. 22]).

Replacing $\sin, \cos$ by the Jacobi elliptic functions $\operatorname{sn}(x, m)$, $\operatorname{cn}(x, m)$ deforms the Euler generating function $\sec x + \tan x = (1 + \sin x)/\cos x$ into

$$g_m(x) := \frac{1 + \operatorname{sn}(x, m)}{\operatorname{cn}(x, m)} = \sum_{n \geq 0} E_n(m) \frac{x^n}{n!}, \quad E_n(0) = E_n, \quad (19.1)$$

since $\operatorname{sn}(x, 0) = \sin x$, $\operatorname{cn}(x, 0) = \cos x$. The $E_n(m)$ are polynomials in m ($E_2 = 1$, $E_3 = 2 - m$, $E_4 = 5 - 4m$, $E_5 = 16 - 16m + m^2$).

Theorem 19.1. *For every m and $n \geq 1$,*

$$\ddot{H}_n(g_m) = \ddot{H}_n\left(\frac{1 + \operatorname{sn}(x, m)}{\operatorname{cn}(x, m)}\right) = \left(\frac{1-m}{2}\right)^{\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!.$$

Equivalently, since $\ddot{H}_n(g_0) = 2^{-\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!$ (Corollary 27.9), $\ddot{H}_n(g_m) = (1-m)^{\binom{n}{2}} \ddot{H}_n(g_0)$: the determinant is the Euler value times a clean $\binom{n}{2}$ -power of the *complementary* modulus. This is the exact analogue of the Springer ladder (Theorem 16.1); the proof follows the same divisor strategy **M4**, and is in fact simpler — only one special value of the parameter occurs. As there, the orders are read off *exponential expansions* of the generating function, through the dilated Cauchy–Binet Lemma 3.9; here the new feature is that the frequencies are *two-sided*, and only those of distinct absolute value contribute, because the even rows z^{2i} cannot separate ν from $-\nu$.

19.1. Reduction to two orders. $\ddot{H}_n(g_m)$ is a polynomial in m , being a determinant with entries $E_k(m) \in \mathbb{Q}[m]$. Theorem 19.1 follows from:

- (A) $\deg_m \ddot{H}_n(g_m) \leq \binom{n}{2}$;
- (B) $(1-m)^{\binom{n}{2}}$ divides $\ddot{H}_n(g_m)$.

Proof of Theorem 19.1, assuming (A) and (B). A polynomial of degree $\leq \binom{n}{2}$ divisible by the degree- $\binom{n}{2}$ polynomial $(1-m)^{\binom{n}{2}}$ equals $c_n(1-m)^{\binom{n}{2}}$ (Observation 3.8); evaluating at $m = 0$ gives $c_n = \ddot{H}_n(g_0) = 2^{-\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!$. $\square$

Both (A) and (B) are instances of the dilated Cauchy–Binet Lemma 3.9, reached through two classical transformations of the Jacobi functions; they occupy the next two subsections.

19.2. The degree bound (A), via the reciprocal modulus. By Jacobi’s reciprocal-modulus transformation, $\operatorname{sn}(x, m) = m^{-1/2} \operatorname{sn}(m^{1/2}x, 1/m)$, $\operatorname{cn}(x, m) = \operatorname{dn}(m^{1/2}x, 1/m)$ [49]. Substituting $x \mapsto x/m^{1/2}$ and applying the homogeneity Lemma 3.7 ($\beta = m^{-1/2}$),

$$\ddot{H}_n(g_m) = m^{\frac{3}{2}\binom{n}{2}} \ddot{H}_n(G), \quad G(x) := g_m(x/m^{1/2}) = \frac{1 + \epsilon \operatorname{sn}(x, \epsilon^2)}{\operatorname{dn}(x, \epsilon^2)}, \quad \epsilon := m^{-1/2}. \quad (19.2)$$

Thus $\deg_m \ddot{H}_n(g_m) \leq \binom{n}{2}$ is equivalent to $\operatorname{ord}_{\epsilon=0} \ddot{H}_n(G) \geq \binom{n}{2}$ (a polynomial of m -degree d becomes $m^{\frac{3}{2}\binom{n}{2}} \times$ a series of ϵ -order $3\binom{n}{2} - 2d$).

We obtain this lower bound from Lemma 3.9 applied to G itself, after producing its exponential expansion.

From $\operatorname{sn}' = \operatorname{cn} \operatorname{dn}$, $\operatorname{dn}' = -\epsilon^2 \operatorname{sn} \operatorname{cn}$ and $\operatorname{dn}^2 = 1 - \epsilon^2 \operatorname{sn}^2$,

$$\frac{G'}{G} = \frac{\epsilon \operatorname{sn}' \operatorname{dn} - (1 + \epsilon \operatorname{sn}) \operatorname{dn}'}{\operatorname{dn}(1 + \epsilon \operatorname{sn})} = \frac{\epsilon \operatorname{cn} [\operatorname{dn}^2 + \epsilon \operatorname{sn}(1 + \epsilon \operatorname{sn})]}{\operatorname{dn}(1 + \epsilon \operatorname{sn})} = \epsilon \frac{\operatorname{cn}(x, \epsilon^2)}{\operatorname{dn}(x, \epsilon^2)}, \quad (19.3)$$

so $G = \exp(\epsilon\Psi)$ with $\Psi(x) = \int_0^x \frac{\operatorname{cn}}{\operatorname{dn}} dt$. Now $\operatorname{cn}/\operatorname{dn}$ is even and, since $\operatorname{cn}(x + 2K) = -\operatorname{cn}(x)$ and $\operatorname{dn}(x + 2K) = \operatorname{dn}(x)$, anti-periodic with period $2K$; hence it has *zero mean* and only odd harmonics. Therefore Ψ is an *odd*, $4K$ -periodic function (no secular term), and $G = e^{\epsilon\Psi}$ is $4K$ -periodic and real-analytic ($\operatorname{dn} > 0$ on $\mathbb{R}$). With $\rho := \pi/2K$ ($\rho \rightarrow 1$ as $\epsilon \rightarrow 0$), the Fourier expansion of G and term-by-term differentiation at $x = 0$ give the moment form

$$G(x) = \sum_{\nu \in \mathbb{Z}} c_\nu(\epsilon) e^{i\nu\rho x}, \quad a_m = m![x^m]G = \sum_{\nu \in \mathbb{Z}} c_\nu (i\nu\rho)^m, \quad (19.4)$$

an exponential sum with symmetric nodes $\zeta_\nu = i\nu\rho$ (both series converge, the c_ν decaying geometrically in the nome).

It remains to bound $\operatorname{ord}_\epsilon c_\nu$. Write $\operatorname{cn}/\operatorname{dn} = \sum_p \gamma_p e^{ip\rho x}$; integrating (19.3) term by term,

$$\Psi(x) = \sum_{p \neq 0} \psi_p e^{ip\rho x} + \text{const}, \quad \psi_p = \frac{\gamma_p}{ip\rho}, \quad (19.5)$$

the $p = 0$ term being absent because $\operatorname{cn}/\operatorname{dn}$ has zero mean. The Fourier coefficients of $\operatorname{cn}/\operatorname{dn}$ involve only odd p , the $(2k+1)$ -th entering at nome order k ; thus $\operatorname{ord}_\epsilon \gamma_p \geq |p| - 1$, and hence

$\text{ord}_\epsilon \psi_p \geq |p| - 1$ as well ($\rho = 1 + O(\epsilon^2)$ is a unit) [49, §22.6]. Expanding the exponential $G = e^{\epsilon\Psi}$ and collecting the $e^{i\nu\rho x}$ -coefficient,

$$c_\nu = (\text{unit}) \sum_{N \geq 0} \frac{\epsilon^N}{N!} \sum_{\substack{p_1, \dots, p_N \neq 0 \\ p_1 + \dots + p_N = \nu}} \psi_{p_1} \cdots \psi_{p_N},$$

where the global unit is $e^{-\epsilon \cdot \text{const}}$. Each monomial has ϵ -order $\geq N + \sum_j (|p_j| - 1) = \sum_j |p_j| \geq |\sum_j p_j| = |\nu|$, so

$$\text{ord}_\epsilon c_\nu \geq |\nu| \quad (\nu \in \mathbb{Z}). \quad (19.6)$$

(Equivalently, substituting (19.4) into (19.3) and comparing $e^{i\nu\rho x}$ -coefficients yields the recurrence

$$i\nu\rho c_\nu = \epsilon \sum_p \gamma_p c_{\nu-p} \quad (\nu \neq 0), \quad (19.7)$$

with which (19.6) is consistent; but (19.7) alone is not triangular in $|\nu|$, so the order bound is read off the exponential (19.5) as above.) Applying the order Corollary 3.10 with $o_\nu = |\nu|$, and noting that surviving subsets have distinct $|\nu|$,

$$\text{ord}_{\epsilon=0} \ddot{H}_n(G) \geq \min_T \sum_{\nu \in T} |\nu| = 0 + 1 + \cdots + (n-1) = \binom{n}{2},$$

proving (A). (Only the lower bound is needed; that the constant c_n above is nonzero comes from the value $\ddot{H}_n(g_0) \neq 0$ at $m = 0$.)

Remark 19.2 (why the leading symbol is not enough). One might hope to replace G by $e^{\epsilon \sin x}$ (its $\epsilon \rightarrow 0$ symbol) and argue that a_m agrees with $m![x^m]e^{\epsilon \sin x}$ to relative order 2. This is *false* from the per-entry orders alone: for $n \geq 6$ a perturbation of each a_m by a term of order $\text{ord}_\epsilon a_m + 2$ can *lower* $\text{ord}_\epsilon \ddot{H}_n$ below $\binom{n}{2}$ (numerically, to 13 at $n = 6$ and 17 at $n = 7$). The argument above instead uses the genuine moment structure (19.4)–(19.7) of G , which forces $\text{ord}_\epsilon c_\nu \geq |\nu|$; that is what makes it correct.

19.3. Vanishing at $m = 1$ (B), via the imaginary transformation. At $m = 1$, $\text{sn}(x, 1) = \tanh x$, $\text{cn}(x, 1) = \text{sech } x$, so $g_1(x) = (1 + \tanh x) \cosh x = e^x$: a rank-one geometric sequence. We show the vanishing of $\ddot{H}_n(g_m)$ there has order $\binom{n}{2}$. Set $\mu := 1 - m$; by Jacobi's imaginary transformation $\text{sn}(ix, m) = i \text{sn}(x, 1 - m) / \text{cn}(x, 1 - m)$, $\text{cn}(ix, m) = 1 / \text{cn}(x, 1 - m)$ [49],

$$g_m(ix) = \frac{1 + \text{sn}(ix, m)}{\text{cn}(ix, m)} = \text{cn}(x, \mu) + i \text{sn}(x, \mu) =: h_\mu(x),$$

the elliptic analogue of $e^{ix} = \cos x + i \sin x$. By the homogeneity Lemma 3.7 ($\beta = i$), $\ddot{H}_n(g_m) = i^{-3} \binom{n}{2} \ddot{H}_n(h_\mu)$, and $\mu = 1 - m$ vanishes to first order at $m = 1$, so it suffices that $\text{ord}_{\mu=0} \ddot{H}_n(h_\mu) = \binom{n}{2}$.

For modulus $\mu \rightarrow 0$ the nome $\mathfrak{q} = \mathfrak{q}(\mu)$ is analytic with $\mathfrak{q} = \mu/16 + O(\mu^2)$, $K(\mu) \rightarrow \pi/2$, $\rho := \pi/2K(\mu) \rightarrow 1$ [49]. The Fourier expansions of sn, cn [49, §22.6] combine to

$$h_\mu(x) = \frac{2\pi}{K\sqrt{\mu}} \sum_{k \geq 0} \frac{\mathfrak{q}^{k+1/2}}{1 - \mathfrak{q}^{2(2k+1)}} \left(e^{i(2k+1)\rho x} - \mathfrak{q}^{2k+1} e^{-i(2k+1)\rho x} \right),$$

an exponential sum with nodes $\zeta_{\pm(2k+1)} = \pm i(2k+1)\rho$ (odd multiples of ρ , symmetric) and coefficients of order, in $\mathfrak{q}$,

$$\text{ord}_{\mathfrak{q}} c_{2k+1} = k, \quad \text{ord}_{\mathfrak{q}} c_{-(2k+1)} = 3k + 1$$

(the prefactor $\frac{2\pi}{K\sqrt{\mu}} \mathfrak{q}^{1/2} \rightarrow 1$ is a unit). By the order Corollary 3.10, the surviving subsets have distinct $|\nu| \in \{1, 3, 5, \dots\}$ and the unique minimal one is $T^* = \{1, 3, \dots, 2n-1\}$ taken with the cheaper $+$ signs (orders $0, 1, \dots, n-1$, summing to $\binom{n}{2}$); its leading coefficient is a product of two Vandermondes in the distinct nodes $i(2k+1)\rho$ ($\rho \rightarrow 1$), hence nonzero. Therefore $\text{ord}_{\mathfrak{q}=0} \ddot{H}_n(h_\mu) = \binom{n}{2}$, and since $\mathfrak{q} \sim \mu/16$, $\text{ord}_{\mu=0} \ddot{H}_n(h_\mu) = \binom{n}{2}$, proving (B). Together with (A) and the reduction of Subsection 19.1, this completes the proof of Theorem 19.1.

Remark 19.3. The elliptic proof draws on classical Jacobi-function theory — the reciprocal-modulus and imaginary transformations, the Fourier/nome expansions of sn , cn , and the nome order $\text{ord}_\epsilon \gamma_p \geq |p| - 1$ of the coefficients of cn / dn [49, §22.6] — as cited inputs; everything else (the ODE (19.3) and recurrence (19.7), the order counts) is elementary. Its skeleton is identical to the Springer proof: $m = 1$ (where $g_1 = e^x$) plays the role of $t = \pm i$; the imaginary transformation, turning m into $1 - m$, plays the role of the twist $t \mapsto i$ that produced $t^2 + 1$; the reciprocal modulus ($m \rightarrow \infty$) plays the role of $t \rightarrow \infty$; and $m = 0$ fixes the constant. The structural novelty is that the exponential expansions are two-sided (Lemma 3.9) and that, for the degree bound, the lower order $\text{ord}_\epsilon c_\nu \geq |\nu|$ must be read from G 's moment structure, not from per-entry orders. The further deformation $(1 + t \text{sn}) / \text{cn}$ retains this ladder form when $t \neq 1$.

20. AN ALGEBRAIC FAMILY: $(1 + x)/(1 - x^2)^{s/2}$

Replacing $\cos(x)^{-(s+1)}$ by the algebraic even series $(1 - x^2)^{-s/2} = \sum_{k \geq 0} \binom{-s/2}{k} (-1)^k x^{2k}$ gives another one-parameter family with the same closed-product behaviour. A scale parameter t , i.e. $(1 - tx^2)^{-s/2}$, would be inessential: it multiplies the moment a_{2i+j} by $t^{i+\lfloor j/2 \rfloor}$ and hence $\ddot{H}_n$ by the constant $t^{\binom{n}{2} + \lfloor (n-1)^2/4 \rfloor}$, so we set $t = 1$.

Notably, *neither* of the two moment functionals $\mathcal{S}[y^k] = a_{2k}$ and $\mathcal{T}^*[y^k] = a_{2k+1}$ is classical here: the even moments $\mu_k = 4^k \binom{1}{2}_k \binom{s}{2}_k$ form a Hadamard product of two Laguerre moment sequences, and the three-term recurrence coefficients of both $\mathcal{S}$ and $\mathcal{T}^*$ are rational, but not polynomial, functions of s . Accordingly the proof below uses no orthogonal polynomials at all: it rests on an elementary contiguous relation in s (M5), which relates $\ddot{H}_n(s)$ to $\ddot{H}_n(s-2)$ by column operations and is solved by a periodicity argument, anchored by a Vandermonde evaluation M1 at the single point $s = 2$ (Step 3 of the proof) — M1 is thus only the base case, not the mechanism that carries the result across all s .

Theorem 20.1. *Let $f(x) = \frac{1+x}{(1-x^2)^{s/2}}$, so that $a_{2k} = \frac{(2k)!}{k!} (s/2)_k$ and $a_{2k+1} = (2k+1)a_{2k}$. Then for all $n \geq 1$,*

$$\ddot{H}_n(f) = \left(\prod_{k=1}^{n-1} (2k)! \right) \prod_{j=1}^{n-1} (s+2j-2)^{n-j} = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} (2k)! \prod_{i=1}^{n-1} (s/2)_i.$$

Proof. Write $\mu_k = \mu_k(s) = \frac{(2k)!}{k!} (s/2)_k = 4^k \binom{1}{2}_k \binom{s}{2}_k$, so that $a_{2k} = \mu_k$ and $a_{2k+1} = (2k+1)\mu_k$. The moments satisfy the first-order recurrence

$$\mu_{k+1}(s) = (2k+1)(2k+s)\mu_k(s), \quad (20.1)$$

since $\mu_{k+1}/\mu_k = 4(k + \frac{1}{2})(k + \frac{s}{2})$.

Abbreviate $\ddot{H}_n(s) := \ddot{H}_n(f) = \det(a_{2i+j}(s))$. We prove $\ddot{H}_n(s) = c_n D(s)$ with $D(s) = \prod_{j=1}^{n-1} (s+2j-2)^{n-j}$ and $c_n = \prod_{k=1}^{n-1} (2k)!$.

Step 1 (a contiguous relation). For $m = 2k$ or $m = 2k+1$ one has $\mu_k(s)/\mu_k(s-2) = (s/2)_k/(s/2-1)_k = (s+2k-2)/(s-2)$, so with $\lfloor m/2 \rfloor = k$,

$$a_m(s) = \frac{s-2+2\lfloor m/2 \rfloor}{s-2} a_m(s-2) \quad (m \geq 0). \quad (20.2)$$

Put $\alpha_i = s-2+2i$ and $C_{ij} = a_{2i+j}(s-2)$, so $\det C = \ddot{H}_n(s-2)$. By (20.2) with $m = 2i+j$ (whence $\lfloor m/2 \rfloor = i + \lfloor j/2 \rfloor$), $(s-2)a_{2i+j}(s) = (\alpha_i + 2\lfloor j/2 \rfloor)C_{ij}$, so

$$(s-2)^n \ddot{H}_n(s) = \det \tilde{M}, \quad \tilde{M}_{ij} = (\alpha_i + 2\lfloor j/2 \rfloor)C_{ij}.$$

Let $r_i = \mu_i(s-2) = C_{i0}$. Iterating (20.1) at $s-2$, namely $\mu_{k+1}(s-2) = (2k+1)\alpha_k \mu_k(s-2)$, and using $2(i+p)+1 = \alpha_i + 2p+3-s$ (as $2i = \alpha_i - s + 2$), the columns of C factor as

$$C_{ij} = r_i \psi_j(\alpha_i), \quad \psi_{2l}(\alpha) = \prod_{p=0}^{l-1} (\alpha+2p)(\alpha+2p+3-s), \quad \psi_{2l+1}(\alpha) = (\alpha+2l+3-s) \psi_{2l}(\alpha), \quad (20.3)$$

where ψ_j is *monic of degree j* in α . For $j \geq 2$ the product defining ψ_j contains the factor $(\alpha + 0) = \alpha$, so $\alpha \mid \psi_j$. As $\psi_0, \dots, \psi_{j-1}$ are monic of degrees $0, \dots, j-1$, they form a basis of the polynomials of degree $< j$; expanding ψ_j/α in it gives $\psi_j(\alpha) = \alpha \sum_{j' < j} \xi_{j',j} \psi_{j'}(\alpha)$ with $\xi_{j',j} \in \mathbb{Q}(s)$, that is, by (20.3),

$$C_{\bullet,j} = \alpha \odot \sum_{j' < j} \xi_{j',j} C_{\bullet,j'} \quad (j \geq 2), \quad (20.4)$$

where $\alpha \odot (\cdot)$ is entrywise multiplication by $(\alpha_i)_i$. Now reduce $\tilde{M}$ by column operations, which leave the determinant unchanged. Its j -th column is $\tilde{M}_{\bullet,j} = \alpha \odot C_{\bullet,j} + 2\lfloor j/2 \rfloor C_{\bullet,j}$; for $j = 0, 1$ it already equals $\alpha \odot C_{\bullet,j}$. Inductively, once columns $0, \dots, j-1$ have been turned into $\alpha \odot C_{\bullet,0}, \dots, \alpha \odot C_{\bullet,j-1}$, relation (20.4) expresses $2\lfloor j/2 \rfloor C_{\bullet,j}$ as a combination of them, and subtracting it turns column j into $\alpha \odot C_{\bullet,j}$. Hence $\tilde{M}$ reduces to $\text{diag}(\alpha) C$, and

$$(s-2)^n \ddot{H}_n(s) = \left(\prod_{i=0}^{n-1} (s-2+2i) \right) \ddot{H}_n(s-2) = (s-2) \Pi_{n-1}(s) \ddot{H}_n(s-2), \quad \Pi_{n-1}(s) = \prod_{r=0}^{n-2} (s+2r).$$

Cancelling one factor $s-2$,

$$(s-2)^{n-1} \ddot{H}_n(s) = \Pi_{n-1}(s) \ddot{H}_n(s-2). \quad (20.5)$$

Step 2 (solving the relation). Collecting exponents base by base gives $D(s)/D(s-2) = \Pi_{n-1}(s)/(s-2)^{n-1}$, so D satisfies (20.5) as well. Dividing, the rational function $g(s) = \ddot{H}_n(s)/D(s)$ satisfies $g(s) = g(s-2)$; a rational function of period 2 is constant (for a avoiding the finitely many poles of g , the function $g - g(a)$ vanishes at the infinitely many points of $a + 2\mathbb{Z}$ that also avoid them, hence identically). Hence $\ddot{H}_n(s) = c_n D(s)$ for a constant c_n .

Step 3 (the constant). At $s = 2$ we have $(s/2)_k = (1)_k = k!$, so $a_m = m!$ for every m : the family degenerates to the factorial member of the Beta chain, whose dilated determinant was evaluated by the Vandermonde reduction **M1** in Corollary 4.4 (case $r = 0$):

$$\ddot{H}_n(2) = \det((2i+j)!) = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k! \prod_{k=1}^{n-1} (2k)!.$$

As $D(2) = \prod_{j=1}^{n-1} (2j)^{n-j} = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k!$, we obtain $c_n = \ddot{H}_n(2)/D(2) = \prod_{k=1}^{n-1} (2k)!$, the first displayed form. The equivalent form follows from $\prod_{i=1}^{n-1} (s/2)_i = 2^{-\binom{n}{2}} \prod_{j=1}^{n-1} (s+2j-2)^{n-j}$. $\square$

21. THE SQUARED ALGEBRAIC FAMILY $(1+x)^2/(1-x^2)^{s/2}$

The prefactor $1+x$ of Theorem 20.1 can be squared. The resulting family is *not* a reparametrisation of the previous one: writing $(1+x)^2(1-x^2)^{-s/2} = (1+x)^{2-s/2}(1-x)^{-s/2}$, a rescaling $x \mapsto \beta x$ must preserve the singular pair $\{\pm 1\}$, hence $\beta = \pm 1$, and matching the exponents at $\mp \beta$ against $(1+x)^{1-\sigma/2}(1-x)^{-\sigma/2}$ forces two incompatible values of σ for either sign. The determinant confirms that the family is new: the product form survives except in its topmost factor, where $(s/2)_{n-1}$ deforms into a *binomial* expression carrying the denominator $s-3$. The excluded value $s = 3$ is precisely the point where this binomial degenerates into odd harmonic numbers (Remark 21.3). The proof below runs the contiguous step of **M5** *once*, from s to $s-2$, and closes it by a Cauchy–Binet expansion instead of the periodicity argument; no anchoring special value is needed. The same machinery evaluates all the shifted determinants $\ddot{H}_n^{(r)} = \det(a_{2i+j+r})$ of both families; this is the subject of the next section (Section 22), where those of the algebraic family turn out to be pure products for *every* shift r , while those of the squared algebraic family split by the parity of r — odd shifts erase the defect entirely, even shifts carry the same $s-3$ binomial.

Throughout the section, $a_m = a_m(s)$ and $\mu_k = \mu_k(s) = \frac{(2k)!}{k!} (s/2)_k$ are the moments of $(1+x)(1-x^2)^{-s/2}$ as in Theorem 20.1.

Theorem 21.1. *Let $f(x) = \frac{(1+x)^2}{(1-x^2)^{s/2}}$, with coefficients $b_m = m! [x^m]f$, that is,*

$$b_{2k} = \mu_k + 2k(2k-1) \mu_{k-1}, \quad b_{2k+1} = 2(2k+1) \mu_k.$$

Then for all $n \geq 1$ and $s \neq 3$,

$$\ddot{H}_n(f) = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} (2k)! \prod_{i=1}^{n-2} (s/2)_i \cdot \frac{(2n-1)!! + (s-4) 2^{n-1} (s/2)_{n-1}}{s-3}. \quad (21.1)$$

The last factor is a monic polynomial in s of degree $n-1$ (its numerator vanishes at $s=3$), so that, once cleared of the denominator, (21.1) is a polynomial identity valid for every s .

Proof. All identities below are between rational functions of s . We keep the notation of the proof of Theorem 20.1: $\alpha_i = s-2+2i$, $r_i = \mu_i(s-2)$, $C_{ij} = a_{2i+j}(s-2)$, and ψ_j is the monic degree- j polynomial of (20.3). Write also $\varphi_n(\sigma) := \sigma(\sigma+2) \cdots (\sigma+2n-2) = 2^n (\sigma/2)_n$.

Step 1 (one contiguous step). From $(1+x)^2 = 2(1+x) - (1-x^2)$ we get $f = 2(1+x)(1-x^2)^{-s/2} - (1-x^2)^{-(s-2)/2}$, hence $b_m(s) = 2a_m(s) - e_m(s-2)$, where $e_{2k}(\sigma) = \mu_k(\sigma)$ and $e_{2k+1} = 0$ are the moments of $(1-x^2)^{-\sigma/2}$. For j even, $e_{2i+j}(s-2) = \mu_{i+j/2}(s-2) = C_{ij}$; for j odd it vanishes. Combining with (20.2), $(s-2)a_{2i+j}(s) = (\alpha_i + 2\lfloor j/2 \rfloor) C_{ij}$, we obtain the entrywise multiplier form

$$(s-2)b_{2i+j}(s) = (2\alpha_i + \tau_j) C_{ij}, \quad \tau_{2l} = 4l - (s-2), \quad \tau_{2l+1} = 4l. \quad (21.2)$$

Step 2 (bidiagonality in the ψ -basis). The polynomials (20.3) form a multiplicative chain:

$$\psi_j(\alpha) = \prod_{m=0}^{j-1} (\alpha + \delta_m), \quad \delta_{2p} = 2p + 3 - s, \quad \delta_{2p+1} = 2p,$$

(indeed $\psi_{2l+1}/\psi_{2l} = \alpha + 2l + 3 - s$ and $\psi_{2l+2}/\psi_{2l+1} = \alpha + 2l$), whence the two-term relation $\alpha \psi_j = \psi_{j+1} - \delta_j \psi_j$. Therefore

$$(2\alpha + \tau_j) \psi_j = 2\psi_{j+1} + (\tau_j - 2\delta_j) \psi_j = 2\psi_{j+1} + \epsilon_j \psi_j, \quad \epsilon_{2l} = s-4, \quad \epsilon_{2l+1} = 0,$$

since $\tau_{2l} - 2\delta_{2l} = 4l - s + 2 - 2(2l + 3 - s) = s-4$ and $\tau_{2l+1} - 2\delta_{2l+1} = 4l - 4l = 0$. With $C_{ij} = r_i \psi_j(\alpha_i)$, equation (21.2) becomes $(s-2)b_{2i+j}(s) = r_i(2\psi_{j+1}(\alpha_i) + \epsilon_j \psi_j(\alpha_i))$, i.e.

$$(s-2)^n \ddot{H}_n(f) = \left(\prod_{i=0}^{n-1} r_i \right) \det(GT), \quad G = (\psi_k(\alpha_i))_{\substack{0 \leq i \leq n-1 \\ 0 \leq k \leq n}}, \quad T = \begin{pmatrix} \epsilon_0 & & & \\ 2 & \epsilon_1 & & \\ & 2 & \ddots & \\ & & & \epsilon_{n-1} \\ & & & 2 \end{pmatrix}, \quad (21.3)$$

G of size $n \times (n+1)$ and T of size $(n+1) \times n$, lower bidiagonal.

Step 3 (Cauchy–Binet). By the Cauchy–Binet formula, $\det(GT) = \sum_{u=0}^n \det G_{\hat{u}} \det T_{\hat{u}}$, where $G_{\hat{u}}$ deletes the column ψ_u of G and $T_{\hat{u}}$ the row u of T . Deleting row u makes $T_{\hat{u}}$ block-triangular, so $\det T_{\hat{u}} = \prod_{j < u} \epsilon_j \cdot 2^{n-u}$, which vanishes as soon as $u \geq 2$ because $\epsilon_1 = 0$. Only two terms survive:

$$\det(GT) = 2^n \det G_{\hat{0}} + (s-4) 2^{n-1} \det G_{\hat{1}}. \quad (21.4)$$

Step 4 (the two gap alternants). Set $V := \prod_{0 \leq i < j \leq n-1} (\alpha_j - \alpha_i) = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k!$ and $\Lambda(\beta) := \prod_{i=0}^{n-1} (\beta - \alpha_i)$. Border G with the row $(\psi_0(\beta), \dots, \psi_n(\beta))$: since the ψ_k are monic of degree k , the $(n+1) \times (n+1)$ alternant equals the Vandermonde determinant of $(\alpha_0, \dots, \alpha_{n-1}, \beta)$, namely $V\Lambda(\beta)$. Laplace expansion along the added row gives

$$\sum_{u=0}^n (-1)^{n+u} \psi_u(\beta) \det G_{\hat{u}} = V \Lambda(\beta).$$

Expanding Λ in the chain basis (a Newton expansion), $\Lambda(\beta) = \sum_{u=0}^n \gamma_u \psi_u(\beta)$, and comparing coefficients, $\det G_{\hat{u}} = (-1)^{n+u} \gamma_u V$. The first two coefficients are read off by evaluation. At $\beta = s-3$ every ψ_u with $u \geq 1$ vanishes (factor $\beta + \delta_0$), so $\gamma_0 = \Lambda(s-3) = \prod_{i=0}^{n-1} (-(2i+1)) = (-1)^n (2n-1)!!$. At $\beta = 0$ every ψ_u with $u \geq 2$ vanishes (factor $\beta + \delta_1$) and $\psi_1(0) = \delta_0 = 3-s$, so $(-1)^n \varphi_n(s-2) = \Lambda(0) = \gamma_0 + (3-s)\gamma_1$. Hence

$$\det G_{\hat{0}} = (2n-1)!! V, \quad \det G_{\hat{1}} = \frac{\varphi_n(s-2) - (2n-1)!!}{s-3} V. \quad (21.5)$$

Step 5 (assembly). Substituting (21.5) into (21.4) and using $2(s-3) - (s-4) = s-2$ together with $\varphi_n(s-2) = (s-2)\varphi_{n-1}(s)$,

$$\begin{aligned}\det(GT) &= 2^{n-1} \frac{(s-2)(2n-1)!! + (s-4)\varphi_n(s-2)}{s-3} V \\ &= 2^{n-1}(s-2) \frac{(2n-1)!! + (s-4)\varphi_{n-1}(s)}{s-3} V.\end{aligned}$$

Moreover $\mu_i(s-2) = \mu_i(s)(s-2)/(s+2i-2)$ (as in Step 1 of Theorem 20.1) and $\prod_{i=0}^{n-1}(s-2+2i) = (s-2)\varphi_{n-1}(s)$ give $\prod_i r_i = (\prod_i \mu_i(s))(s-2)^{n-1}/\varphi_{n-1}(s)$. Inserting both into (21.3) and cancelling $(s-2)^n$,

$$\ddot{H}_n(f) = \left(\prod_{i=0}^{n-1} \mu_i(s) \right) V \cdot \frac{2^{n-1}((2n-1)!! + (s-4)\varphi_{n-1}(s))}{(s-3)\varphi_{n-1}(s)}.$$

Finally $\prod_{i=0}^{n-1} \mu_i(s) \cdot V = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} (2k)! \prod_{i=1}^{n-1} (s/2)_i$ and $2^{n-1} \prod_{i=1}^{n-1} (s/2)_i / \varphi_{n-1}(s) = \prod_{i=1}^{n-2} (s/2)_i$, which is (21.1) after writing $\varphi_{n-1}(s) = 2^{n-1}(s/2)_{n-1}$ in the numerator. $\square$

Remark 21.2 (the same computation re-proves Theorem 20.1). For $(1+x)(1-x^2)^{-s/2}$ the multiplier in (21.2) is $\alpha_i + 2\lfloor j/2 \rfloor$, and the same two-term reduction gives $(\alpha + 2l)\psi_{2l} = \psi_{2l+1} + (s-3)\psi_{2l}$ and $(\alpha + 2l)\psi_{2l+1} = \psi_{2l+2}$: the diagonal of T becomes $\epsilon_{2l} = s-3$, $\epsilon_{2l+1} = 0$. Steps 3–5 then yield $\det(GT) = ((2n-1)!! + (s-3) \cdot \frac{\varphi_n(s-2) - (2n-1)!!}{s-3}) V = \varphi_n(s-2) V$: the double factorials cancel *exactly*, the denominator $s-3$ disappears, and the pure product of Theorem 20.1 drops out — with no periodicity argument and no anchoring value $s=2$. Squaring the prefactor shifts the diagonal from $s-3$ to $s-4$: the mismatch with the denominator $s-3$ of $\det G_1$ ruins the cancellation, and the binomial numerator of (21.1) is the trace it leaves.

Remark 21.3 (the excluded value $s=3$). The last factor of (21.1) is also the terminating sum

$$\frac{(2n-1)!! + (s-4)2^{n-1}(s/2)_{n-1}}{s-3} = (2n-1)!! \left(1 + (s-4) \sum_{k=0}^{n-2} \frac{2^k (s/2)_k}{(2k+3)!!} \right),$$

valid for every s : with $t_k = 2^k (s/2)_k / (2k+1)!!$ one has $t_{k+1} - t_k = (s-3)2^k (s/2)_k / (2k+3)!!$, and the sum telescopes to $(2^{n-1}(s/2)_{n-1} / (2n-1)!! - 1) / (s-3)$. At $s=3$ the summand is $(2k+1)!! / (2k+3)!! = 1/(2k+3)$, and the factor degenerates into $(2n-1)!!(2 - O_n)$ with $O_n = 1 + \frac{1}{3} + \dots + \frac{1}{2n-1}$:

$$\ddot{H}_n(f)|_{s=3} = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} (2k)! \prod_{i=1}^{n-2} (3/2)_i \cdot (2n-1)!! (2 - O_n).$$

The odd harmonic number O_n is why $s=3$ must be excluded from any product-type statement; it is the value of s at which the telescoping ratio $2^k (s/2)_k / (2k+1)!!$ becomes constant in k .

Remark 21.4 (degenerate values). At $s=4$ one has $f = (1-x)^{-2}$ and $b_m = (m+1)!$; the binomial term drops out ($s-4=0$) and (21.1) reduces to $2^{\binom{n}{2}} \prod_{k=1}^{n-1} (2k)! \prod_{i=1}^{n-2} (i+1)! (2n-1)!! = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k! \prod_{i=0}^{n-1} (2i+1)!$, the case $r=1$ of Corollary 4.4. At $s=0$ one has $f = (1+x)^2$ and (21.1) correctly gives $\ddot{H}_1 = 1$, $\ddot{H}_2 = -4$ and $\ddot{H}_n = 0$ for $n \geq 3$.

22. SHIFTED DETERMINANTS OF THE TWO ALGEBRAIC FAMILIES

Continuing with the two algebraic families of Sections 20–21, we now evaluate all their shifted dilated determinants

$$\ddot{H}_n^{(r)}(f) := \det(a_{2i+j+r})_{0 \leq i, j \leq n-1} \quad (r \geq 0).$$

For the algebraic family the answer is a pure product for *every* r , and the proof is a direct Vandermonde reduction **M1**; for the squared algebraic family the parity of r decides whether the $s-3$ binomial of Theorem 21.1 survives. Throughout, $\rho := \lfloor r/2 \rfloor$.

Proposition 22.1 (all shifts of the algebraic family). *Let $f_1(x) = (1+x)(1-x^2)^{-s/2}$, with moments a_m as in Theorem 20.1. Then for all $n \geq 1$ and $r \geq 0$,*

$$\ddot{H}_n^{(r)}(f_1) = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} k! \prod_{i=0}^{n-1} a_{2i+r}(s), \quad (22.1)$$

that is, $\prod_i a_{2i+r} = \prod_{i=0}^{n-1} \mu_{i+\rho}$ for r even and $\prod_{i=0}^{n-1} (2i+r) \mu_{i+\rho}$ for r odd.

Proof. By (20.1) the term ratio of $\mathbf{a}$ has two linear branches,

$$\frac{a_{m+1}}{a_m} = \begin{cases} m+1, & m \text{ even}, \\ m+s-1, & m \text{ odd}. \end{cases}$$

The row indices $x_i = 2i$ are all even, so which branch is taken at offset p from the anchor depends only on the parity of $r+p$, not on the row:

$$a_{x+r+j} = a_{x+r} Q_j(x), \quad Q_j(x) = \prod_{p=0}^{j-1} \begin{cases} x+r+p+1, & r+p \text{ even}, \\ x+r+p+s-1, & r+p \text{ odd}, \end{cases}$$

a monic polynomial of degree j in x , the same for every row. Lemma 3.1 with $w(x) = a_{x+r}$ applies, the coefficient matrix is unit triangular, and $\ddot{H}_n^{(r)}(f_1) = \prod_{i < j} (2j-2i) \cdot \prod_i a_{2i+r}$, which is (22.1). $\square$

Remark 22.2 (a third proof of Theorem 20.1). The case $r=0$ of Proposition 22.1 is Theorem 20.1 itself: the algebraic family is, after all, within reach of **M1**. Its term ratio has degree one in each parity class — which places it outside the Beta family of Section 4 and defeats the classical Hankel treatment — but the dilated rows sit on even indices and never mix the two classes, so the row factorisation (3.1) holds with monic Q_j of degree j . The three proofs illuminate different structure: **M5** the contiguous relation in s , Remark 21.2 the exact double-factorial cancellation, and **M1** the two-branch Beta pattern.

Theorem 22.3 (all shifts of the squared algebraic family). *Let $f(x) = (1+x)^2(1-x^2)^{-s/2}$, with moments b_m as in Theorem 21.1, and let $n \geq 1$.*

(i) *For odd r the defect disappears entirely:*

$$\ddot{H}_n^{(r)}(f) = 2^n \ddot{H}_n^{(r)}(f_1) = 2^{\binom{n}{2}+n} \prod_{k=1}^{n-1} k! \prod_{i=0}^{n-1} (2i+r) \mu_{i+\rho}(s).$$

(ii) *For even $r = 2\rho$ and $s \neq 3$,*

$$\ddot{H}_n^{(r)}(f) = 2^{n-1} \ddot{H}_n^{(r)}(f_1) \cdot \frac{P_{n,\rho}(s)}{\Theta_{n,\rho}(s)}, \quad \Theta_{n,\rho}(s) := \prod_{i=0}^{n-1} (s-2+2\rho+2i), \quad (22.2)$$

where, with the convention $(-1)!! = 1$,

$$P_{n,\rho}(s) := \frac{(s-2) \frac{(2n+2\rho-1)!!}{(2\rho-1)!!} + (s-4) \Theta_{n,\rho}(s)}{s-3}$$

is a monic polynomial in s of degree n (the numerator vanishes at $s=3$, since $\Theta_{n,\rho}(3) = (2n+2\rho-1)!!/(2\rho-1)!!$). For $\rho \geq 1$ the quotient in (22.2) clears entrywise, $\mu_{i+\rho}(s)/(s-2+2\rho+2i) = \frac{1}{2} \frac{(2i+2\rho)!}{(i+\rho)!} (s/2)_{i+\rho-1}$, giving the fully explicit form

$$\ddot{H}_n^{(2\rho)}(f) = 2^{\binom{n}{2}-1} \prod_{k=1}^{n-1} k! \prod_{i=0}^{n-1} \frac{(2i+2\rho)!}{(i+\rho)!} (s/2)_{i+\rho-1} \cdot P_{n,\rho}(s) \quad (\rho \geq 1);$$

the case $\rho = 0$ is Theorem 21.1.

Proof. The five steps of Theorem 21.1 go through with the chain shifted by r ; we record the changes.

Steps 1–2. With $m = 2i + j + r$, the identities of Step 1 read $(s-2) b_{2i+j+r}(s) = (2\alpha_i + \tau_{j+r}) C_{ij}^{(r)}$, where $C_{ij}^{(r)} = a_{2i+j+r}(s-2) = r_i \psi_{j+r}(\alpha_i)$ (the factorisation (20.3) holds for every column index) and τ_m is as in (21.2), indexed by the parity of $m = j+r$. Factor $\psi_r(\alpha_i)$ — that is, $a_{2i+r}(s-2)/r_i$ — out of row i ; the remaining columns form the shifted chain $\chi_j := \psi_{j+r}/\psi_r = \prod_{m=0}^{j-1} (\alpha + \delta_{r+m})$, and the computation of Step 2, applied verbatim to the parity of $j+r$, gives

$$(2\alpha + \tau_{j+r}) \chi_j = 2 \chi_{j+1} + \epsilon_j \chi_j, \quad \epsilon_j = \begin{cases} s-4, & j+r \text{ even}, \\ 0, & j+r \text{ odd}. \end{cases}$$

Hence $(s-2)^n \ddot{H}_n^{(r)}(f) = \prod_i a_{2i+r}(s-2) \cdot \det(G^{(r)} T^{(r)})$ with $G^{(r)} = (\chi_k(\alpha_i))$ of size $n \times (n+1)$ and $T^{(r)}$ lower bidiagonal as in (21.3), its diagonal now being ϵ_j .

Steps 3–4, r odd. Here $\epsilon_0 = 0$, so in the Cauchy–Binet expansion only $u = 0$ survives and $\det(G^{(r)} T^{(r)}) = 2^n \det G_0^{(r)}$. Factoring $\chi_1(\alpha_i) = \alpha_i + \delta_r = \alpha_i + 2\rho$ out of each row of $G_0^{(r)}$ leaves the initial segment of a monic chain, whose alternant is V ; hence $\det(G^{(r)} T^{(r)}) = 2^n \Theta_{n,\rho}(s) V$, since $\prod_i (\alpha_i + 2\rho) = \Theta_{n,\rho}(s)$.

Steps 3–4, $r = 2\rho$ even. As in Theorem 21.1, $u = 0, 1$ survive. The Newton argument of Step 4 now runs at the nodes $-\delta_r = s-3-2\rho$ and $-\delta_{r+1} = -2\rho$, and again $\chi_1(-\delta_{r+1}) = \delta_r - \delta_{r+1} = 3-s$:

$$\gamma_0 = \Lambda(s-3-2\rho) = (-1)^n \frac{(2n+2\rho-1)!!}{(2\rho-1)!!}, \quad \Lambda(-2\rho) = (-1)^n \Theta_{n,\rho}(s) = \gamma_0 + (3-s)\gamma_1,$$

so that, exactly as in (21.5)–(21.4),

$$\det(G^{(r)} T^{(r)}) = 2^{n-1} \frac{(s-2) \frac{(2n+2\rho-1)!!}{(2\rho-1)!!} + (s-4) \Theta_{n,\rho}(s)}{s-3} V = 2^{n-1} P_{n,\rho}(s) V.$$

Step 5. By (20.2), $a_{2i+r}(s-2) = a_{2i+r}(s)(s-2)/(s-2+2(i+\rho))$, so $\prod_i a_{2i+r}(s-2) = \prod_i a_{2i+r}(s) \cdot (s-2)^n / \Theta_{n,\rho}(s)$. For r odd the factor $\Theta_{n,\rho}(s)$ cancels against Step 3, and $\ddot{H}_n^{(r)}(f) = 2^n \prod_i a_{2i+r}(s) \cdot V = 2^n \ddot{H}_n^{(r)}(f_1)$ by Proposition 22.1. For r even it survives in the denominator, giving (22.2). The explicit form for $\rho \geq 1$ follows from $(s/2)_{i+\rho} = (s/2)_{i+\rho-1} (s+2i+2\rho-2)/2$. $\square$

Remark 22.4 (the contrast, and $s = 3$ again). An odd shift moves the even-column defect $\epsilon_j = s-4$ off the column $j = 0$; the zero ϵ_0 then truncates the Cauchy–Binet expansion to its first term, and the squared algebraic family costs only the overall factor 2^n relative to the algebraic family. For even shifts the binomial survives, and the odd-harmonic degeneration of Remark 21.3 persists: the same telescoping gives

$$P_{n,\rho}(3) = \frac{(2n+2\rho-1)!!}{(2\rho-1)!!} \left(2 - \sum_{j=\rho+1}^{n+\rho} \frac{1}{2j-1} \right),$$

so no even shift admits a product form at $s = 3$.

23. A BESSEL ANALOGUE OF THE EULER NUMBER FAMILY

The Bessel cosine cosb_s . For $s > -1$ define the *Bessel cosine* of order s as the normalised Bessel function

$$\text{cosb}_s(x) := \Gamma(s+1) \left(\frac{2}{x} \right)^s J_s(x) = \sum_{k \geq 0} \frac{(-1)^k}{4^k k! (s+1)_k} x^{2k}, \quad \text{cosb}_s(0) = 1. \quad (23.1)$$

(The normalisation is classical — it is the function written Λ_s in the characteristic-function literature — but the name and symbol cosb are ours.) The name is earned by the exact identities

$$\text{cosb}_{-1/2}(x) = \cos x, \quad \text{cosb}_{1/2}(x) = \frac{\sin x}{x}, \quad \text{cosb}'_s(x) = -\frac{x}{2s+2} \text{cosb}_{s+1}(x),$$

so that with $\text{sinb}_s := -\text{cosb}'_s$ one has $\text{sinb}_{-1/2} = \sin x$ and the derivative ladder generalises $\cos' = -\sin$. At half-integer orders cosb_s is elementary (trigonometric); at all other orders it is a genuine Bessel function. By the Poisson integral representation, for $s > -\frac{1}{2}$,

$$\text{cosb}_s(x) = \frac{\Gamma(s+1)}{\Gamma(s+\frac{1}{2})\Gamma(\frac{1}{2})} \int_{-1}^1 e^{ixt} (1-t^2)^{s-1/2} dt,$$

i.e. cosb_s is the characteristic function of the symmetric Beta (Gegenbauer) distribution on $[-1, 1]$: arcsine at $s = 0$, uniform at $s = \frac{1}{2}$, Wigner semicircle at $s = 1$. Note where the parameter lives: in the trigonometric family the power sits on the *function* side ($1/\cos(x)^{s+1}$), here it sits on the *weight* side ($(1-t^2)^{s-1/2}$). A literal power form $\text{cosb}_s = h(x)^{-(s+1)}$ with one fixed h is impossible: it would force $\text{cosb}_0 \text{cosb}_2 = \text{cosb}_1^2$, and in fact $\text{cosb}_0 \text{cosb}_2 - \text{cosb}_1^2 = -\frac{x^2}{12} + \frac{5x^4}{384} - \dots \neq 0$.

The family and the closed form. The trigonometric (s, t) family of Section 7 is $1/\cos(x)^{s+1} + \int_0^x \cos(y)^{-(t+1)} dy$, evaluated for all (s, t) by Proposition 7.1. Its direct Bessel analogue, with the exponent ladder replaced by the order ladder, is

$$f_{s,t}(x) = \text{cosb}_s(x) + \int_0^x \text{cosb}_t(y) dy. \quad (23.2)$$

Writing $a_k = k! [x^k] f_{s,t}$ and using $(2k)! = 4^k k! (\frac{1}{2})_k$, the coefficients (23.1) give

$$a_{2k} = (-1)^k \frac{(\frac{1}{2})_k}{(s+1)_k}, \quad a_{2k+1} = (-1)^k \frac{(\frac{1}{2})_k}{(t+1)_k} \quad (k \geq 0), \quad (23.3)$$

the odd case because integration divides the coefficient of x^{2k+1} by $2k+1$ and $(2k+1)!/(2k+1) = (2k)!$. Thus the even and odd moment generating functions are one and the same Gauss series at the two parameters,

$$\sum_{k \geq 0} a_{2k} x^k = {}_2F_1\left(\frac{1}{2}, 1; s+1; -x\right), \quad \sum_{k \geq 0} a_{2k+1} x^k = {}_2F_1\left(\frac{1}{2}, 1; t+1; -x\right) :$$

the even and odd moment functionals belong to a *single* classical hierarchy (they are Jacobi–Gegenbauer functionals — in $y = z^2$, the moments of the weight $y^{-1/2}(1-y)^{s-1/2}$ carried to $[-1, 0]$) and differ in the single parameter $s \leftrightarrow t$. This is precisely the collapse hypothesis of the biorthogonal reduction **M2** (Section 3.2), and the whole family evaluates in closed form.

Theorem 23.1 (Bessel (s, t) family; **M2**). *Let $f_{s,t}$ be as in (23.2). Then, as an identity of rational functions of (s, t) ,*

$$\ddot{H}_n(f_{s,t}) = (-1)^{\binom{n}{2}} \frac{\prod_{i=0}^{n-1} (2i)!}{2^{n(n-1)}} \left(\prod_{l=0}^{\bar{n}-1} \frac{(s+\frac{1}{2})_l}{(s+1)_{n-1+l}} \right) \left(\prod_{m=0}^{\bar{n}-1} \frac{(t+\frac{1}{2})_m}{(t+1)_{n-1+m}} \right) \prod_{r=1}^{\bar{n}} \prod_{c=1}^{\bar{n}} \frac{t-s+c-r}{n-r-c+1}. \quad (23.4)$$

Both sides of (23.4) lie in $\mathbb{Q}(s, t)$, so the identity may be specialised at every (s, t) with $s, t \notin \{-1, -2, \dots\}$ — in particular throughout the Poisson range $s, t > -\frac{1}{2}$. The sign together with the double product is exactly the signed factor of the trigonometric family at *doubled* argument, $\Omega(2(t-s))$ in the notation of Section 7.3; we return to this parallel in Remark 23.8. The contrast observed numerically — smooth at half-integer offset, degenerate at integer offset — is now a one-line consequence.

Corollary 23.2 (smooth and degenerate offsets). *Let $s, t > -\frac{1}{2}$.*

- (i) *If $t-s \in \mathbb{Z}$, then $\ddot{H}_n(f_{s,t}) = 0$ precisely for $n \geq 2(t-s)+1$ when $t > s$, for $n \geq 2$ when $t = s$, and for $n \geq 2(s-t)+2$ when $t < s$.*
- (ii) *If $t-s \notin \mathbb{Z}$, then $\ddot{H}_n(f_{s,t}) \neq 0$ for every n , and (23.4) is a closed product of linear forms. For half-integer offset $t-s \in \frac{1}{2} + \mathbb{Z}$ the signed factor moreover evaluates in double factorials by Lemma 7.5, applied with the odd argument $\delta = 2(t-s)$.*

Proof. For $s, t > -\frac{1}{2}$ every Pochhammer symbol in (23.4) is a product of positive reals, so $\ddot{H}_n = 0$ if and only if the double product vanishes, i.e. if and only if $t - s = r - c$ for some $1 \leq r \leq \bar{n}$, $1 \leq c \leq \underline{n}$. This forces $t - s \in \mathbb{Z}$, and conversely, for $t - s \in \mathbb{Z}$, such (r, c) exist if and only if $\bar{n} \geq t - s + 1$ and $\underline{n} \geq s - t + 1$, i.e. $n \geq \max(2(t - s) + 1, 2(s - t) + 2, 2)$, which is the stated threshold in each of the three cases. $\square$

Example 23.3 (J_0 plus the sine integral). At $(s, t) = (0, \frac{1}{2})$ the member (23.2) is $J_0(x) + \text{Si}(x)$, since $\cos b_0 = J_0$ and $\int_0^x \cos b_{1/2}(y) dy = \int_0^x \sin(y)/y dy$. Here $2(t - s) = 1$, $\Omega(1) = 2^{-\binom{n}{2}}$ by Lemma 7.5, and (23.4) specialises to the factorial product

$$\ddot{H}_n(J_0 + \text{Si}) = \frac{\prod_{i=0}^{n-1} (2i)!}{2^{n(n-1)} 2^{\binom{n}{2}}} \prod_{l=0}^{\bar{n}-1} \frac{(\frac{1}{2})_l}{(n-1+l)!} \prod_{m=0}^{\underline{n}-1} \frac{m!}{(\frac{3}{2})_{n-1+m}} = 1, \frac{1}{6}, \frac{1}{960}, \frac{1}{1693440}, \dots$$

($n = 1, 2, 3, 4$) — the numerators are 1, which is the arithmetic signature that first singled the half-integer lines out.

Remark 23.4 (the boundary of the ladder). The closed form also quantifies the collapse at the edge of the order ladder. At $s = -\frac{1}{2}$ the numerators $(s + \frac{1}{2})_l$ vanish for every $l \geq 1$, so $\ddot{H}_n = 0$ as soon as $\bar{n} \geq 2$, i.e. for all $n \geq 3$; at $t = -\frac{1}{2}$ likewise $\ddot{H}_n = 0$ for all $n \geq 4$ ($\underline{n} \geq 2$). Thus $\cos x + \int_0^x J_0$ degenerates from $n = 3$ and $J_0(x) + \sin x$ from $n = 4$, although both pairs have half-integer offset: rule (ii) of Corollary 23.2 genuinely needs $s, t > -\frac{1}{2}$. More generally $(s + \frac{1}{2})_l = 0$ for $l > k$ at $s = -\frac{1}{2} - k$, giving $\ddot{H}_n = 0$ for $n \geq 2k + 3$. The explanation is visible in the Poisson formula: at $s = -\frac{1}{2}$ the weight $(1 - t^2)^{s-1/2}$ ceases to be a measure and $\cos x$ is the characteristic function of the *atomic* two-point measure at ± 1 — a rank collapse, the same degeneracy as the $(1 + x)$ cos-type atomic cases elsewhere in the paper.

The proof of Theorem 23.1 occupies the next three subsections: the orthogonal data of the family are explicit (Lemma 23.5), the connection coefficients collapse to a single term because the two functionals differ in one parameter (Lemma 23.6), and the residual connection determinant — which, unlike the binomial determinant of the trigonometric family, carries the parameters s, t in its entries — is evaluated by Desnanot–Jacobi condensation (Theorem 23.7), in the manner of the Catalan determinant of Section 18.6.

23.1. Orthogonal data. By (23.3) the even part of $\mathbf{a}$ involves only s and the odd part only t , through one and the same one-parameter moment sequence; we record the data for the even part, the odd part being identical with t in place of s . All computations take place over the field $\mathbb{Q}(s, t)$.

Lemma 23.5 (orthogonal data). *The even and odd parts of $\mathbf{a}$ admit the Stieltjes S -fractions (3.10) with coefficients*

$$u_j = -\frac{j(j+2s-1)}{4(s+j-1)(s+j)}, \quad v_j = -\frac{j(j+2t-1)}{4(t+j-1)(t+j)} \quad (j \geq 1) \quad (23.5)$$

(for $j = 1$ the factor $s + j - 1 = s$ cancels: $u_1 = -\frac{1}{2(s+1)}$). Consequently both functionals are quasi-definite over $\mathbb{Q}(s, t)$, with squared norms

$$h_i^S = \frac{(2i)!(s + \frac{1}{2})_i}{4^i (s+i)_i (s+1)_{2i}}, \quad h_m^T = \frac{(2m)!(t + \frac{1}{2})_m}{4^m (t+m)_m (t+1)_{2m}}. \quad (23.6)$$

Proof. The even generating function is ${}_2F_1(\frac{1}{2}, 1; s+1; -x)$. Gauss's continued fraction [47, 8] expands the ratio $F(a, b+1; c+1; z)/F(a, b; c; z)$ of contiguous hypergeometric series as the formal S -fraction $1/(1 - \alpha_1 z/(1 - \alpha_2 z/(1 - \dots)))$ with

$$\alpha_{2k+1} = \frac{(a+k)(c-b+k)}{(c+2k)(c+2k+1)}, \quad \alpha_{2k} = \frac{(b+k)(c-a+k)}{(c+2k-1)(c+2k)}.$$

Taking $(a, b, c) = (\frac{1}{2}, 0, s)$, where $F(\frac{1}{2}, 0; s; z) = 1$, both parities merge into the single expression $\alpha_j = \frac{j(j+2s-1)}{4(s+j-1)(s+j)}$, and the substitution $z = -x$ turns $1/(1 - \alpha_1 z/\dots)$ into the S -fraction

in x with coefficients $u_j = -\alpha_j$, which is (23.5). By Lemma 3.5 the squared norms telescope,

$$h_i^S = \prod_{j=1}^{2i} u_j = \frac{(2i)! (2s)_{2i}}{16^i (s)_{2i} (s+1)_{2i}}$$

(the $2i$ signs cancel), and the duplication formula in Pochhammer form, $(2s)_{2i} = 4^i (s)_i (s + \frac{1}{2})_i$, together with $(s)_{2i} = (s)_i (s+i)_i$, gives (23.6). Each h_i^S is a nonzero element of $\mathbb{Q}(s, t)$, whence quasi-definiteness. $\square$

23.2. Connection coefficients. As in Section 3.2 let P_i and Q_m be the monic orthogonal polynomials of $\mathcal{S}$ and $\mathcal{T}$, and $P_i = \sum_{m=0}^i \kappa_{i,m} Q_m$ the connection expansion. Because the two functionals differ in a single parameter, the $\kappa_{i,m}$ collapse to a single term. Throughout put

$$\delta := t - s, \quad d := i - m.$$

Lemma 23.6 (connection coefficients). *For $0 \leq m \leq i$,*

$$\kappa_{i,m} = \frac{(2i)!}{(2m)! 4^d d!} \frac{\delta(\delta-1) \cdots (\delta-d+1)}{(t+2m+1)_d (s+i+m)_d} = \frac{(2i)!}{(2m)! 4^d} \binom{\delta}{d} \frac{1}{(t+2m+1)_d (s+i+m)_d}. \quad (23.7)$$

Proof. Write $k_{i,m}$ for the right-hand side, extended by $k_{i,m} = 0$ for $m < 0$ or $m > i$. By Lemma 3.3 the array κ is the unique solution of the recurrence (3.9) with row $\kappa_{0,0} = 1$, $\kappa_{0,m} = 0$ ($m \neq 0$); the candidate has the same row $i = 0$ (at $d = 0$ every product in (23.7) is empty), so it suffices to show that k satisfies (3.9). Every factor of (23.7) is nonzero on the triangle $0 \leq m \leq i$ over $\mathbb{Q}(s, t)$, and telescoping each block gives the four neighbour ratios as rational functions of (m, d, s, t) (recall $i = m + d$):

$$\begin{aligned} \frac{k_{i+1,m}}{k_{i,m}} &= \frac{(2i+1)(2i+2)(\delta-d)(s+i+m)}{4(d+1)(t+2m+d+1)(s+2i)(s+2i+1)}, \\ \frac{k_{i,m-1}}{k_{i,m}} &= \frac{(2m-1)(2m)(\delta-d)(t+2m+d)}{4(d+1)(t+2m-1)(t+2m)(s+i+m-1)}, \\ \frac{k_{i,m+1}}{k_{i,m}} &= \frac{4d(t+2m+1)(t+2m+2)(s+i+m)}{(2m+1)(2m+2)(\delta-d+1)(t+2m+d+1)}, \\ \frac{k_{i-1,m}}{k_{i,m}} &= \frac{4d(t+2m+d)(s+2i-2)(s+2i-1)}{(2i-1)(2i)(\delta-d+1)(s+i+m-1)}. \end{aligned}$$

Dividing (3.9) by $k_{i,m}$ and inserting these ratios together with the recurrence data of Lemma 23.5 — via Lemma 3.5, $c_i^S = u_{2i} + u_{2i+1}$, $\lambda_i^S = u_{2i-1}u_{2i}$, $c_m^T = v_{2m} + v_{2m+1}$, $\lambda_{m+1}^T = v_{2m+1}v_{2m+2}$ — turns the recurrence into a rational-function identity in (m, d, s, t) , verified by clearing denominators. The boundary columns take care of themselves: at $m = 0$ the second ratio carries the factor $2m$ and vanishes, as required by $\kappa_{i,-1} = 0$; at $m = i$ the third and fourth ratios carry the factor d and vanish, as required by $\kappa_{i,i+1} = 0$ and $\kappa_{i-1,i} = 0$; and at $m = i + 1$ the recurrence reduces to $k_{i+1,i+1} = k_{i,i} = 1$. $\square$

The binomial $\binom{\delta}{d}$ in (23.7) is the source of the integer-offset degeneracy: for $\delta = t - s \in \mathbb{Z}_{\geq 0}$ the connection array is *banded*, $\kappa_{i,m} = 0$ for $i - m > \delta$, and the residual block of Lemma 3.4 eventually has a zero diagonal. The next theorem makes this quantitative — and evaluates the determinant for every (s, t) at once.

23.3. The connection determinant. Unlike its trigonometric counterpart (Lemma 7.4), the kernel left over after removing row and column factors from (23.7) depends on $i + m$ as well as on $i - m$, so the dual Jacobi–Trudi identity does not apply. Instead we evaluate, more generally, *every* minor of the connection array on consecutive rows and consecutive columns: the two extra shift parameters make the family stable under Desnanot–Jacobi condensation, exactly as for the Catalan determinant of the reciprocal-sine function (Section 18.6).

Theorem 23.7 (connection determinant). *Let $p \geq a \geq 0$ and $N \geq 0$ be integers and put $q = p - a$. Then*

$$\det(\kappa_{p+r, a+m})_{0 \leq r, m \leq N-1} = \left(\prod_{r=0}^{N-1} \frac{(2p+2r)!}{(2a+2r)!} \right) 4^{-qN} \times \prod_{r=1}^q \prod_{c=1}^N \frac{\delta + c - r}{[(q-r) + (N-c) + 1] (s+2p-1+c-r) (t+2a+2N-1+r-c)}. \quad (23.8)$$

Proof. Write $D(p, a, N)$ for the determinant and $G(p, a, N)$ for the right-hand side, $D(p, a, 0) = G(p, a, 0) = 1$. Telescoping the products over r or over c turns G into Pochhammer blocks,

$$G(p, a, N) = \left(\prod_{r=0}^{N-1} \frac{(2p+2r)!}{(2a+2r)!} \right) \frac{4^{-qN} \prod_{r=1}^q (\delta + 1 - r)_N}{\prod_{j=1}^q (j)_N \prod_{c=1}^N (s+p+a+c-1)_q \prod_{r=1}^q (t+2a+N+r-1)_N} \quad (23.9)$$

(for the hook lengths, $\prod_{c=1}^N [(q-r) + (N-c) + 1] = (q-r+1)_N$; for the s -block, $2p-q = p+a$). The proof is by induction on N through the Desnanot–Jacobi identity, in the manner of the condensation proofs of the determinant calculus [26].

For $N = 0$ both sides equal 1. For $q = 0$ (any N) the matrix $(\kappa_{p+r, p+m})$ is unitriangular ($\kappa_{i,m} = 0$ for $m > i$, $\kappa_{i,i} = 1$), so $D = 1 = G$, every product in (23.8) being empty. For $N = 1$,

$$G(p, a, 1) = \frac{(2p)!}{(2a)!} \frac{4^{-q} \delta(\delta-1) \cdots (\delta-q+1)}{q! (s+p+a)_q (t+2a+1)_q} = \kappa_{p,a}$$

by (23.7). Let $N \geq 2$ and $q \geq 1$. Deleting the first or last row and the first or last column of the matrix of $D(p, a, N)$ shifts (p, a) and N inside the same family, so its four corner minors are $D(p+1, a+1, N-1)$, $D(p, a, N-1)$, $D(p+1, a, N-1)$, $D(p, a+1, N-1)$, and its central minor is $D(p+1, a+1, N-2)$; the Desnanot–Jacobi identity reads

$$D(p, a, N) D(p+1, a+1, N-2) = D(p+1, a+1, N-1) D(p, a, N-1) - D(p+1, a, N-1) D(p, a+1, N-1). \quad (23.10)$$

The five smaller determinants have offsets $q, q, q+1, q-1 \geq 0$ and sizes $N-1, N-2$, so they equal the corresponding G 's by the induction hypothesis, and $G(p+1, a+1, N-2) \neq 0$, every factor of (23.9) being a nonzero element of $\mathbb{Q}(s, t)$. Hence (23.10) determines $D(p, a, N)$, and it remains to check that G satisfies the same relation, i.e. that $R_1 - R_2 = 1$ for

$$R_1 := \frac{G(p+1, a+1, N-1) G(p, a, N-1)}{G(p, a, N) G(p+1, a+1, N-2)}, \quad R_2 := \frac{G(p+1, a, N-1) G(p, a+1, N-1)}{G(p, a, N) G(p+1, a+1, N-2)}.$$

Both quotients collapse block by block. First, $R_1 = \rho(p+1, a+1, N-1)/\rho(p, a, N)$ for the one-step ratio $\rho(p, a, N) := G(p, a, N)/G(p, a, N-1)$, which (23.9) telescopes to

$$\rho(p, a, N) = \frac{(2p+2N-2)!}{(2a+2N-2)!} \frac{(\delta+N-q)_q (t+2a+N-1)_q}{4^q (N)_q (s+p+a+N-1)_q (t+2a+2N-2)_q (t+2a+2N-1)_q}$$

(the t -block by $(t+2a+N+r-1)_N/(t+2a+N+r-2)_{N-1} = \frac{(t+2a+2N+r-3)(t+2a+2N+r-2)}{t+2a+N+r-2}$, whose three factors telescope over $r = 1, \dots, q$). In R_1 the factorial and 4-blocks cancel outright, the two long t -blocks coincide between the two levels, and each remaining Pochhammer quotient leaves a single linear factor — e.g. $(\delta+N-1-q)_q/(\delta+N-q)_q = (\delta+N-1-q)/(\delta+N-1)$ — giving

$$R_1 = \frac{(q+N-1)(\delta-q+N-1)(s+p+a+N-1)(t+2a+q+N-1)}{(N-1)(\delta+N-1)(s+2p+N-1)(t+2a+N-1)}.$$

Second, $R_2 = R_1 \cdot X$ with $X = \mu_a/\mu_{a+1}$, where $\mu_a := G(p+1, a, N-1)/G(p, a, N-1)$ is the offset step $q \mapsto q+1$ at size $N-1$. Blockwise,

$$\mu_a = \left(\prod_{r=0}^{N-2} (2p+2r+1)(2p+2r+2) \right)$$

$$\times \frac{4^{-(N-1)} (\delta - q)_{N-1}}{(q+1)_{N-1} (t+2a+N+q-1)_{N-1}} \prod_{c=1}^{N-1} \frac{s+p+a+c-1}{(s+2p+c-1)(s+2p+c)},$$

and in the quotient X everything depending on p alone cancels, leaving

$$X = \frac{q(\delta - q)(s+p+a)(t+2a+2N+q-2)}{(q+N-1)(\delta - q+N-1)(s+p+a+N-1)(t+2a+N+q-1)},$$

so that

$$R_2 = \frac{q(\delta - q)(s+p+a)(t+2a+q+2N-2)}{(N-1)(\delta + N-1)(s+2p+N-1)(t+2a+N-1)}.$$

Finally, in the variables $\sigma = s+p+a$, $\tau = t+2a$ and $M = N-1$ — for which $\delta - q = \tau - \sigma$, $\delta + M = \tau - \sigma + q + M$, $s+2p+N-1 = \sigma + q + M$ and $t+2a+N-1 = \tau + M$ — the required identity $R_1 - R_2 = 1$ becomes

$$\begin{aligned} (q+M)(\tau - \sigma + M)(\sigma + M)(\tau + q + M) - q(\tau - \sigma)\sigma(\tau + q + 2M) \\ = M(\tau - \sigma + q + M)(\sigma + q + M)(\tau + M), \end{aligned} \quad (23.11)$$

checked by expanding both sides. This completes the induction. $\square$

23.4. Proof of the closed form.

Proof of Theorem 23.1. By Lemma 23.5 the two functionals are quasi-definite over $\mathbb{Q}(s, t)$, and Lemma 3.4 applies. The residual block is the case $(p, a, N) = (\bar{n}, 0, \underline{n})$ of Theorem 23.7, in which $q = \bar{n}$ and the hook length $(q-r) + (N-c) + 1$ reads $n-r-c+1$; thus

$$\ddot{H}_n = (-1)^{\binom{\bar{n}}{2}} \left(\prod_{l=0}^{\bar{n}-1} h_l^S \right) \left(\prod_{m=0}^{\underline{n}-1} h_m^T \right) \det(\kappa_{\bar{n}+r, m})_{0 \leq r, m \leq \underline{n}-1}.$$

Three groups of factors merge. The factorials:

$$\prod_{l=0}^{\bar{n}-1} (2l)! \cdot \prod_{m=0}^{\underline{n}-1} (2m)! \cdot \prod_{r=0}^{\underline{n}-1} \frac{(2\bar{n}+2r)!}{(2r)!} = \prod_{i=0}^{\underline{n}-1} (2i)!.$$

The powers of two: $4^{-\binom{\bar{n}}{2} - \binom{\underline{n}}{2} - \bar{n}\underline{n}} = 4^{-\binom{n}{2}} = 2^{-n(n-1)}$. The s -dependent denominators: writing each Pochhammer symbol as a quotient of Gamma factors (all of which cancel in the end), $(s+l)_l (s+1)_{2l} = \frac{\Gamma(s+2l)\Gamma(s+2l+1)}{\Gamma(s+l)\Gamma(s+1)}$ and $(s+\bar{n}+c-1)_{\bar{n}} = \frac{\Gamma(s+2\bar{n}+c-1)}{\Gamma(s+\bar{n}+c-1)}$, so

$$\begin{aligned} \prod_{l=0}^{\bar{n}-1} (s+l)_l (s+1)_{2l} \cdot \prod_{c=1}^{\underline{n}} (s+\bar{n}+c-1)_{\bar{n}} &= \frac{\prod_{j=0}^{2\bar{n}-1} \Gamma(s+j)}{\Gamma(s+1)^{\bar{n}} \prod_{j=0}^{\bar{n}-1} \Gamma(s+j)} \cdot \frac{\prod_{j=2\bar{n}}^{2\bar{n}+\underline{n}-1} \Gamma(s+j)}{\prod_{j=\bar{n}}^{\bar{n}+\underline{n}-1} \Gamma(s+j)} \\ &= \frac{\prod_{j=n}^{n+\bar{n}-1} \Gamma(s+j)}{\Gamma(s+1)^{\bar{n}}} = \prod_{l=0}^{\bar{n}-1} (s+1)_{n-1+l}, \end{aligned}$$

and the same computation with $(t, \underline{n}, \bar{n})$ in place of $(s, \bar{n}, \underline{n})$ gives $\prod_{m < \underline{n}} (t+m)_m (t+1)_{2m} \cdot \prod_{r=1}^{\bar{n}} (t+\underline{n}+r-1)_{\underline{n}} = \prod_{m=0}^{\underline{n}-1} (t+1)_{n-1+m}$. Inverting these two products (they stand in the denominators of (23.6) and (23.8)) and restoring the numerators $(s+\frac{1}{2})_l, (t+\frac{1}{2})_m$ from (23.6) yields (23.4). $\square$

Remark 23.8 (the parallel with the trigonometric family). The parallel with Proposition 7.1 is exact, with one systematic change of scale. Both formulas share the skeleton $(-1)^{\binom{\bar{n}}{2}} \prod_{i < n} (2i)!$ times a signed double product; but the trigonometric signed factor is $\Omega(t-s)$ while the Bessel one is $\Omega(2(t-s))$: *the Bessel ladder walks in half steps*. The closed double-factorial evaluations of Lemma 7.5 therefore apply on the lines $t-s \in \frac{1}{2} + \mathbb{Z}$ (where $2(t-s)$ is odd), exactly as they apply to the trigonometric family on the lines $t-s$ odd; and the forced roots of Ω produce the degeneration on $t-s \in \mathbb{Z}$, as they do there on $t-s$ even. The parameter blocks, by contrast, are reciprocal: $(s+1)_{n-1+l}$ in the *denominator* here versus $(s+1)_{2l}$ in the numerator there, reflecting moments that are bounded (characteristic functions of probability measures) rather than factorially growing. Both proofs are instances of the collapse mechanism of **M2**: two functionals of a single classical hierarchy differing in one parameter — Jacobi–Gegenbauer here,

secant-type there, Wilson for the reciprocal-sine function. The kernel $\binom{\delta}{d} / ((t+2m+1)_d (s+i+m)_d)$ of (23.7) sits between the two known extremes: like the pure binomial $\binom{(t-s)/2}{d}$ of Lemma 7.3 it is binomial in $i - m$, but like the Catalan kernel of the reciprocal-sine family it also depends on $i + m$ — which is why the finishing step is condensation rather than Jacobi–Trudi. Finally, for half-integer s, t the members of (23.2) are elementary — e.g. $(s, t) = (\frac{1}{2}, 0)$ gives $\sin(x)/x + \int_0^x J_0$ — but for general order they are genuine Bessel functions, so the family lies strictly outside the scope of Section 7: this is not a reparametrisation of a known family. Consistently with the collapse mechanism, the crossbreeds mixing the trigonometric and Bessel hierarchies — one functional secant-type, the other Gegenbauer — were tested numerically and are never smooth: the single-hierarchy hypothesis of **M2** is not ornamental.

24. THE DOUBLE SHIFT OF THE BESSEL (s, t) FAMILY

As for the trigonometric family, alongside $\ddot{H}_n$ the most regular variant is the double shift $\ddot{H}_n^{(2)} = \det(a_{2i+j+2})$ of (8.1): shifting the column index by 2 preserves parity, and $\ddot{H}_n^{(2)}$ is the dilated determinant of the Christoffel transform $y\mathcal{E}$ of $\mathcal{E}$, with even and odd parts $\mathcal{S}'[p] = \mathcal{S}[yp]$, $\mathcal{T}'[p] = \mathcal{T}[yp]$. The general reduction of Proposition 8.2 and the transform data of Lemma 8.1 hold for arbitrary u_j, v_j ; what follows are the Bessel counterparts of the special ingredients of Lemma 8.3.

Lemma 24.1 (Christoffel transform, Bessel family). *For the coefficients u_j, v_j of (23.5), with $P_i, Q_m, \kappa_{i,m}$ as above:*

- (a) $D_n^S = (-1)^n P_n(0) = (-1)^n \frac{(2n-1)!!}{2^n (s+n)_n}$, and likewise $D_m^T = (-1)^m \frac{(2m-1)!!}{2^m (t+m)_m}$;
- (b) the connection coefficients $P'_i = \sum_m \kappa'_{i,m} Q'_m$ of the primed pair are

$$\kappa'_{i,m} = \frac{2i+1}{2m+1} \kappa_{i,m} \Big|_{(s,t) \rightarrow (s+1,t+1)} = \frac{(2i+1)!}{(2m+1)! 4^d} \binom{\delta}{d} \frac{1}{(t+2m+2)_d (s+i+m+1)_d} \quad (24.1)$$

(the parameter $\delta = t - s$ is invariant under the shift $(s, t) \rightarrow (s+1, t+1)$).

Proof. (a) By Lemma 8.1(i), $D_{n+1}^S = c_n^S D_n^S - \lambda_n^S D_{n-1}^S$ with $D_0^S = 1$, $D_1^S = u_1 = -\frac{1}{2(s+1)}$. The displayed product has

$$\frac{D_{n+1}^S}{D_n^S} = -\frac{2n+1}{2} \frac{(s+n)_n}{(s+n+1)_{n+1}} = -\frac{(2n+1)(s+n)}{2(s+2n)(s+2n+1)},$$

which equals u_1 at $n = 0$; inserting this ratio and $c_n^S = u_{2n} + u_{2n+1}$, $\lambda_n^S = u_{2n-1}u_{2n}$ from (23.5), the recurrence divided by D_n^S becomes a rational-function identity in (n, s) , routine to verify. The case of D^T is identical.

(b) By (a) and Lemma 8.1(iii) the primed norms $h_i^{S'} = (D_{i+1}^S/D_i^S) h_i^S$ and likewise $h_m^{T'}$ are nonzero in $\mathbb{Q}(s, t)$, so $\mathcal{S}', \mathcal{T}'$ are quasi-definite and, by Lemma 3.3, κ' is the unique solution of (3.9) with the primed recurrence data of Lemma 8.1(ii), $c_i^{S'} = u_{2i+1} + u_{2i+2}$, $\lambda_i^{S'} = u_{2i}u_{2i+1}$, and likewise for T' . The candidate (24.1) has $\kappa'_{i,i} = 1$, and its four neighbour ratios are those displayed in the proof of Lemma 23.6, taken at $(s+1, t+1)$ and corrected by the factors $\frac{2i+3}{2i+1}$, $\frac{2m+1}{2m-1}$, $\frac{2m+1}{2m+3}$, $\frac{2i-1}{2i+1}$ respectively. Dividing (3.9) by $\kappa'_{i,m}$ turns it again into a rational-function identity in (m, d, s, t) , verified by clearing denominators; the boundary columns take care of themselves exactly as in Lemma 23.6. $\square$

Proposition 24.2 (double shift, Bessel family; **M2**). *Let $f_{s,t}$ be as in (23.2). Then, as an identity in $\mathbb{Q}(s, t)$,*

$$\ddot{H}_n^{(2)}(f_{s,t}) = \det(a_{2i+j+2})_{0 \leq i,j \leq n-1} = (-1)^n \frac{(2n-1)!!}{2^n (s+n)_{\bar{n}} (t+n)_{\underline{n}}} \ddot{H}_n(f_{s,t}), \quad (24.2)$$

where $\ddot{H}_n(f_{s,t})$ is the unshifted determinant of Theorem 23.1.

Proof. By (8.2), writing $(-1)^n P_{\bar{n}}(0) Q_{\underline{n}}(0) = D_{\bar{n}}^S D_{\underline{n}}^T$,

$$\ddot{H}_n^{(2)} = D_{\bar{n}}^S D_{\underline{n}}^T \frac{\det(\kappa'_{\bar{n}+r,m})}{\det(\kappa_{\bar{n}+r,m})} \ddot{H}_n \quad (0 \leq r, m \leq \underline{n} - 1).$$

By Lemma 24.1(b), pulling $2(\bar{n}+r)+1$ out of row r and $1/(2m+1)$ out of column m ,

$$\det(\kappa'_{\bar{n}+r,m}) = \frac{(2n-1)!!}{(2\bar{n}-1)!!(2\underline{n}-1)!!} \det(\kappa_{\bar{n}+r,m}) \Big|_{(s,t) \rightarrow (s+1,t+1)}.$$

Both connection determinants are the case $(p, a, N) = (\bar{n}, 0, \underline{n})$ of Theorem 23.7, and in the quotient of (23.8) at $(s+1, t+1)$ by (23.8) at (s, t) everything cancels except the two parameter blocks — the factor $\delta + c - r$ is invariant — and these telescope: for fixed r , $\prod_{c=1}^{\underline{n}} \frac{s+2\bar{n}-1+c-r}{s+2\bar{n}+c-r} = \frac{s+2\bar{n}-r}{s+2\bar{n}+\underline{n}-r}$, and the product over $r = 1, \dots, \bar{n}$ gives, using $2\bar{n} + \underline{n} = n + \bar{n}$,

$$\frac{\det(\kappa_{\bar{n}+r,m}) \Big|_{(s+1,t+1)}}{\det(\kappa_{\bar{n}+r,m})} = \frac{(s+\bar{n})_{\bar{n}}}{(s+n)_{\bar{n}}} \cdot \frac{(t+\underline{n})_{\underline{n}}}{(t+n)_{\underline{n}}},$$

the t -block telescoping in the same way, over r first and then over c . By Lemma 24.1(a),

$$D_{\bar{n}}^S D_{\underline{n}}^T = (-1)^{\bar{n}+\underline{n}} \frac{(2\bar{n}-1)!!(2\underline{n}-1)!!}{2^{\bar{n}+\underline{n}}(s+\bar{n})_{\bar{n}}(t+\underline{n})_{\underline{n}}} = (-1)^n \frac{(2\bar{n}-1)!!(2\underline{n}-1)!!}{2^n(s+\bar{n})_{\bar{n}}(t+\underline{n})_{\underline{n}}}.$$

Multiplying the three displays, the double factorials and the products $(s+\bar{n})_{\bar{n}}$, $(t+\underline{n})_{\underline{n}}$ cancel, leaving (24.2). $\square$

Remark 24.3 (the double-shift parallel). The parallel with the double shift of the trigonometric family (Proposition 8.4) is again exact, with the change of scale of Remark 23.8. There the double shift *multiplies* $\ddot{H}_n$ by $(2n-1)!! \prod_{k < \bar{n}} (s+2k+1) \prod_{k < \underline{n}} (t+2k+1)$; here it *divides* by $(-1)^n 2^n (s+n)_{\bar{n}}(t+n)_{\underline{n}}/(2n-1)!!$: the universal factor $(2n-1)!!$ is one and the same, while the parameter blocks are again reciprocal — bounded moments — and the odd-step products $(s+1)(s+3)\dots$ become unit-step Pochhammer blocks, the half-step ladder once more. The transform data compare in the same way: $D_n^S = (2n-1)!! \prod_{k < n} (s+2k+1)$ there, $D_n^S = (-1)^n (2n-1)!!/(2^n(s+n)_{\bar{n}})$ here. Since the new factor in (24.2) has constant numerator and its poles lie at $s+n+l=0$ or $t+n+m=0$ only, for $s, t > -\frac{1}{2}$ it is finite and nonzero: the vanishing locus is untouched, the contrast of Corollary 23.2 holds verbatim for $\ddot{H}_n^{(2)}$, and on the offsets $t-s \notin \mathbb{Z}$ the double shift remains a closed product of linear forms. (The *single* shift $\det(a_{2i+j+1})$, which swaps the two parities, behaves differently: already at $n=4$ it carries an irreducible quadratic factor over $\mathbb{Q}(s, t)$, and we do not pursue it.)

25. A MULTIPLICATIVE BESSEL FAMILY: $(1+x) \cos b_\nu^2$ AT EVEN ORDERS

The additive family $\cos b_s + \int_0^x \cos b_t$ of Section 23 is the Bessel analogue of the Euler (s, t) family. The Bessel cosine introduced there,

$$\cos b_\nu(x) = \Gamma(\nu+1) \left(\frac{2}{x}\right)^\nu J_\nu(x) = \sum_{k \geq 0} \frac{(-1)^k}{4^k k! (\nu+1)_k} x^{2k}, \quad \cos b_\nu(0) = 1, \quad (25.1)$$

also produces an analogue of the *secant* family $(1+x)/\cos(x)^{s+1}$ of Section 11, with the exponent ladder replaced once more by the order ladder: the multiplicative family

$$f_\nu(x) = (1+x) \cos b_\nu(x)^2. \quad (25.2)$$

(The parameter cannot sit as a power on the function side: as recorded in Section 23, no fixed h gives $\cos b_\nu = h^{-(\nu+1)}$; and it is the *square* of the even factor that carries the structure, through the product formula of Lemma 25.1 below.) At $\nu = \frac{1}{2}$ the member (25.2) is $(1+x)(\sin x/x)^2$; at the boundary order $\nu = -\frac{1}{2}$ it is $(1+x) \cos^2 x$.

The determinants $\ddot{H}_n(f_\nu)$ display a sharp parity contrast: at even orders $n = 2N$ they factor completely into linear forms in ν , while at odd orders an irreducible factor of growing degree appears. This section proves the even half in closed form (Theorem 25.4); the odd orders, which remain open, are discussed in Remark 25.7.

The method deserves a word in advance. The even moment functional of f_ν is *not* classical — it is the self-convolution of the symmetric Beta law, not an orthogonality weight — so the biorthogonal collapse **M2** that evaluates the secant family and the additive Bessel family is out of reach. What replaces it is the most elementary reduction of the paper: the Vandermonde method **M1** of Section 3.1, applied here for the only time beyond the Beta family. Watson's classical product formula makes the even moments a *single* hypergeometric term; the $(1+x)$ -coupling preserves this; and for even n — and only for even n — the resulting row factorisation lands exactly on the Vandermonde degree bound. No orthogonal polynomials appear, and quasi-definiteness is never invoked.

25.1. The moments. Write $a_m = m! [x^m] f_\nu$ as usual.

Lemma 25.1 (the moments are a single hypergeometric term). *For all $k \geq 0$,*

$$a_{2k} = (-4)^k \frac{(\frac{1}{2})_k (\nu + \frac{1}{2})_k}{(\nu + 1)_k (2\nu + 1)_k}, \quad a_{2k+1} = (2k+1) a_{2k} = (-4)^k \frac{(\frac{3}{2})_k (\nu + \frac{1}{2})_k}{(\nu + 1)_k (2\nu + 1)_k}. \quad (25.3)$$

Proof. The odd–even coupling is the general $(1+x)$ -mechanism of Section 11: since $x \cdot x^{2k}/(2k)! = (2k+1) x^{2k+1}/(2k+1)!$, the odd part of f_ν contributes $a_{2k+1} = (2k+1)a_{2k}$, and the second expression follows from the first by $(\frac{3}{2})_k/(\frac{1}{2})_k = 2k+1$.

For the even part, square the series (25.1):

$$[x^{2k}] \text{cosb}_\nu^2 = \sum_{j=0}^k \frac{(-1)^j}{4^j j! (\nu + 1)_j} \cdot \frac{(-1)^{k-j}}{4^{k-j} (k-j)! (\nu + 1)_{k-j}} = \frac{(-1)^k}{4^k k!} \sum_{j=0}^k \binom{k}{j} \frac{1}{(\nu + 1)_j (\nu + 1)_{k-j}}.$$

Put $c = \nu + 1$. From $\binom{k}{j} = (-1)^j (-k)_j / j!$ and $(c)_{k-j} = (c)_k / (c+k-j)_j$ with $(c+k-j)_j = (-1)^j (1-c-k)_j$, the sum is a terminating Gauss series at 1,

$$\sum_{j=0}^k \binom{k}{j} \frac{1}{(c)_j (c)_{k-j}} = \frac{1}{(c)_k} \sum_{j=0}^k \frac{(-k)_j (1-c-k)_j}{j! (c)_j} = \frac{{}_2F_1(-k, 1-c-k; c; 1)}{(c)_k} = \frac{(2c+k-1)_k}{(c)_k^2}$$

by the Chu–Vandermonde summation ${}_2F_1(-k, b; c; 1) = (c-b)_k / (c)_k$. Hence, with $(2k)! = 4^k k! (\frac{1}{2})_k$,

$$a_{2k} = (2k)! [x^{2k}] \text{cosb}_\nu^2 = (-1)^k (\frac{1}{2})_k \frac{(2\nu+k+1)_k}{(\nu+1)_k^2},$$

and the duplication formula in Pochhammer form, $(2\nu+1)_{2k} = 4^k (\nu + \frac{1}{2})_k (\nu+1)_k$, turns $(2\nu+k+1)_k = (2\nu+1)_{2k} / (2\nu+1)_k$ into $4^k (\nu + \frac{1}{2})_k (\nu+1)_k / (2\nu+1)_k$, which is (25.3). $\square$

25.2. The Vandermonde reduction at even orders. Throughout fix $N \geq 1$ and put $n = 2N$. Define

$$w_i = \frac{a_{2i}}{(\nu+1+i)_{N-1} (2\nu+1+i)_{N-1}} \quad (0 \leq i \leq 2N-1) \quad (25.4)$$

and, for $0 \leq l, m \leq N-1$, the polynomials

$$\begin{aligned} Q_{2l}(x) &= (-4)^l (x + \frac{1}{2})_l (x + \nu + \frac{1}{2})_l (x + \nu + 1 + l)_{N-1-l} (x + 2\nu + 1 + l)_{N-1-l}, \\ Q_{2m+1}(x) &= (2x + 2m + 1) Q_{2m}(x), \end{aligned} \quad (25.5)$$

of degrees $\deg Q_{2l} = 2N-2$ and $\deg Q_{2m+1} = 2N-1$, with coefficients in $\mathbb{Q}[\nu]$ (the Pochhammer symbols are taken in the variable x).

Proposition 25.2 (row factorisation and reduction). *For all $0 \leq i, j \leq 2N-1$,*

$$a_{2i+j} = w_i Q_j(i), \quad (25.6)$$

and consequently

$$\ddot{H}_{2N}(f_\nu) = \det M_N \prod_{k=1}^{2N-1} k! \prod_{i=0}^{2N-1} w_i, \quad (25.7)$$

where M_N is the coefficient matrix of the family (25.5), that is, $Q_j(x) = \sum_{k=0}^{2N-1} m_{kj} x^k$ with $0 \leq k, j \leq 2N-1$.

Proof. By Lemma 25.1 the even moments have the term ratio

$$\frac{a_{2i+2}}{a_{2i}} = -4 \frac{(i + \frac{1}{2})(i + \nu + \frac{1}{2})}{(i + \nu + 1)(i + 2\nu + 1)},$$

which telescopes to

$$a_{2i+2l} = a_{2i} (-4)^l \frac{(i + \frac{1}{2})_l (i + \nu + \frac{1}{2})_l}{(i + \nu + 1)_l (i + 2\nu + 1)_l} \quad (l \geq 0).$$

Every even column index $j = 2l$ of $\ddot{H}_{2N}$ has $l \leq N - 1$, so both denominators divide the fixed padding of (25.4): $(i + \nu + 1)_l (i + \nu + 1 + l)_{N-1-l} = (i + \nu + 1)_{N-1}$ and likewise for the second block. Hence $a_{2i+2l} = w_i Q_{2l}(i)$; and $a_{2i+2m+1} = (2i + 2m + 1) a_{2i+2m} = w_i (2i + 2m + 1) Q_{2m}(i) = w_i Q_{2m+1}(i)$, which is (25.6). The degrees in (25.5) are $l + l + (N - 1 - l) + (N - 1 - l) = 2N - 2$ and $2N - 1$, both $\leq n - 1$; so the matrix factors,

$$(a_{2i+j})_{0 \leq i, j \leq 2N-1} = \text{diag}(w_i) \cdot V \cdot M_N, \quad V = (i^k)_{0 \leq i, k \leq 2N-1},$$

exactly as in the proof of Lemma 3.1 (the Vandermonde reduction **M1**, applied to the array a_{2i+j} in the row variable i , with nodes $x_i = i$). Taking determinants, and evaluating $\det V = \prod_{0 \leq i < i' \leq 2N-1} (i' - i) = \prod_{k=1}^{2N-1} k!$, gives (25.7). $\square$

It remains to evaluate $\det M_N$, a polynomial in ν .

25.3. The coefficient determinant.

Lemma 25.3 (coefficient determinant).

$$\det M_N = 2^{N(2N-1)} \prod_{k=0}^{N-1} \left(\frac{1}{2}\right)_k (\nu + \frac{1}{2})_k^2 (2\nu + \frac{1}{2})_k.$$

Proof. All computations take place in $\mathbb{Q}(\nu)$. For *any* choice of nodes $u_0, \dots, u_{2N-1} \in \mathbb{Q}(\nu)$ the same factorisation as above gives $(Q_j(u_p))_{p,j} = V(u) M_N$ with $V(u) = (u_p^k)_{p,k}$, hence

$$\det(Q_j(u_p))_{0 \leq p, j \leq 2N-1} = \det M_N \prod_{0 \leq p < p' \leq 2N-1} (u_{p'} - u_p). \quad (25.8)$$

Choose the two interlaced arithmetic strings

$$u_{2r} = -\frac{1}{2} - r, \quad u_{2r+1} = -\nu - \frac{1}{2} - r \quad (0 \leq r \leq N-1), \quad (25.9)$$

whose $\binom{2N}{2}$ pairwise differences are all nonzero in $\mathbb{Q}(\nu)$. The point of this choice is that the matrix $T = (Q_j(u_p))$ becomes *lower triangular*.

Triangularity. At u_{2r} the block $(x + \frac{1}{2})_l$ of (25.5) evaluates to $(-r)_l$, which vanishes exactly for $l > r$; so $Q_{2l}(u_{2r}) = 0$ for $2l > 2r$. For the odd columns, $Q_{2m+1}(u_{2r}) = (2u_{2r} + 2m + 1) Q_{2m}(u_{2r}) = 2(m - r) Q_{2m}(u_{2r})$ vanishes for $m > r$ (through Q_{2m}) and at $m = r$ (through the linear factor); so $Q_{2m+1}(u_{2r}) = 0$ for $2m + 1 > 2r$. At u_{2r+1} the block $(x + \nu + \frac{1}{2})_l$ evaluates to $(-r)_l$, which vanishes for $l > r$; so $Q_{2l}(u_{2r+1}) = 0$ for $2l > 2r + 1$, and $Q_{2m+1}(u_{2r+1}) = 0$ for $2m + 1 > 2r + 1$ through the factor Q_{2m} . In all cases $T_{p,j} = 0$ for $j > p$.

The diagonal. At $x = u_{2r}$ the four blocks of Q_{2r} evaluate to $(-r)_r = (-1)^r r!$, $(\nu - r)_r$, $(\nu + \frac{1}{2})_{N-1-r}$ and $(2\nu + \frac{1}{2})_{N-1-r}$, so

$$T_{2r,2r} = Q_{2r}(u_{2r}) = 4^r r! (\nu - r)_r (\nu + \frac{1}{2})_{N-1-r} (2\nu + \frac{1}{2})_{N-1-r}.$$

At $x = u_{2r+1}$ the blocks of Q_{2r} evaluate to $(-\nu - r)_r = (-1)^r (\nu + 1)_r$, $(-r)_r = (-1)^r r!$, $(\frac{1}{2})_{N-1-r}$ and $(\nu + \frac{1}{2})_{N-1-r}$, while the linear factor is $2u_{2r+1} + 2r + 1 = -2\nu$; with $\nu(\nu + 1)_r = (\nu)_{r+1}$,

$$T_{2r+1,2r+1} = Q_{2r+1}(u_{2r+1}) = (-1)^{r+1} 2 \cdot 4^r r! (\nu)_{r+1} (\frac{1}{2})_{N-1-r} (\nu + \frac{1}{2})_{N-1-r}.$$

Multiplying down the diagonal ($\sum_r (r+1) = \binom{N+1}{2}$), and the tail blocks reindexed by $k = N-1-r$,

$$\det T = (-1)^{\binom{N+1}{2}} 2^{N(2N-1)} \left(\prod_{k=1}^{N-1} k! \right)^2 \prod_{r=0}^{N-1} (\nu - r)_r (\nu)_{r+1} \prod_{k=0}^{N-1} \left(\frac{1}{2}\right)_k (\nu + \frac{1}{2})_k^2 (2\nu + \frac{1}{2})_k, \quad (25.10)$$

the power of two being $2^N \cdot 4^{2\binom{N}{2}} = 2^{N(2N-1)}$.

The Vandermonde. The pairs $p < p'$ of (25.9) fall into four families: two pure strings, $u_{2r'} - u_{2r} = -(r' - r)$ and $u_{2r'+1} - u_{2r+1} = -(r' - r)$ for $r < r'$, contributing $\prod_{r < r'} (r' - r)^2 = (\prod_{k=1}^{N-1} k!)^2$; the mixed pairs $u_{2r'+1} - u_{2r} = -(\nu + r' - r)$ for $r \leq r'$ (that is, $2r < 2r' + 1$), where the difference $d = r' - r \in \{0, \dots, N-1\}$ occurs $N-d$ times; and the mixed pairs $u_{2r'} - u_{2r+1} = \nu - (r' - r)$ for $r < r'$, with $d = r' - r \in \{1, \dots, N-1\}$ occurring $N-d$ times. Hence

$$\prod_{p < p'} (u_{p'} - u_p) = (-1)^{\binom{N+1}{2}} \left(\prod_{k=1}^{N-1} k! \right)^2 \prod_{d=0}^{N-1} (\nu + d)^{N-d} \prod_{d=1}^{N-1} (\nu - d)^{N-d}. \quad (25.11)$$

Division. Since $(\nu)_{r+1} = \prod_{d=0}^r (\nu + d)$ and $(\nu - r)_r = \prod_{d=1}^r (\nu - d)$, one has $\prod_{r=0}^{N-1} (\nu)_{r+1} = \prod_{d=0}^{N-1} (\nu + d)^{N-d}$ and $\prod_{r=0}^{N-1} (\nu - r)_r = \prod_{d=1}^{N-1} (\nu - d)^{N-d}$: the two ν -blocks of (25.10) are exactly the two mixed blocks of (25.11), and the signs and squared superfactorials also match. Dividing (25.10) by (25.11) in (25.8) leaves the stated product. $\square$

25.4. The closed form.

Theorem 25.4 (even orders of $(1+x) \operatorname{cosh}_\nu^2$; M1). *Let $f_\nu = (1+x) \operatorname{cosh}_\nu^2$. For every $N \geq 1$, as an identity in $\mathbb{Q}(\nu)$,*

$$\ddot{H}_{2N}(f_\nu) = (-1)^N 2^{\binom{2N}{2}} \prod_{i=0}^{2N-1} (2i)! \prod_{k=0}^{N-1} \left(\frac{1}{2}\right)_k (\nu + \frac{1}{2})_k^2 (2\nu + \frac{1}{2})_k \prod_{i=0}^{2N-1} \frac{(\nu + \frac{1}{2})_i}{(\nu + 1)_{N-1+i} (2\nu + 1)_{N-1+i}}. \quad (25.12)$$

The left side is regular at every $\nu \notin \{-1, -2, \dots\}$, and (25.12) may be specialised there after the finitely many cancellations between the $(2\nu + 1)$ -blocks and the half-integer numerator factors.

Proof. Combine Proposition 25.2 and Lemma 25.3. In the product of the row factors (25.4), insert the moments (25.3) and telescope the padding, $(\nu + 1)_i (\nu + 1 + i)_{N-1} = (\nu + 1)_{N-1+i}$ and likewise for $(2\nu + 1)$:

$$\prod_{i=0}^{2N-1} w_i = \prod_{i=0}^{2N-1} \frac{(-4)^i \left(\frac{1}{2}\right)_i (\nu + \frac{1}{2})_i}{(\nu + 1)_{N-1+i} (2\nu + 1)_{N-1+i}}.$$

Here $\prod_i (-4)^i = (-1)^N 4^{\binom{2N}{2}}$, since $\sum_{i < 2N} i = \binom{2N}{2} = N(2N-1) \equiv N \pmod{2}$; and $\left(\frac{1}{2}\right)_i = (2i)! / (4^i i!)$ gives

$$\prod_{k=1}^{2N-1} k! \prod_{i=0}^{2N-1} \left(\frac{1}{2}\right)_i = 4^{-\binom{2N}{2}} \prod_{i=0}^{2N-1} (2i)!,$$

the two superfactorials cancelling. The powers of four cancel each other, and with the factor $2^{N(2N-1)} = 2^{\binom{2N}{2}}$ of Lemma 25.3 the product $\det M_N \prod_k k! \prod_i w_i$ collapses to (25.12).

For the regularity statement: each a_m is a rational function of ν with poles only at negative integers (visible in the Cauchy square in the proof of Lemma 25.1, whose denominators are products of $(\nu + 1)_j$), and $\ddot{H}_{2N}$ is a polynomial in the a_m . $\square$

The first values are

$$\ddot{H}_2(f_\nu) = (-1) \cdot 2 \cdot 2! \cdot \frac{\nu + \frac{1}{2}}{(\nu + 1)(2\nu + 1)} = -\frac{2}{\nu + 1}, \quad \ddot{H}_4(f_\nu) = \frac{2160(\nu + \frac{1}{4})(\nu + \frac{1}{2})(\nu + \frac{5}{2})}{(\nu + 1)^7(\nu + 2)^4(\nu + 3)^2(\nu + 4)},$$

the second after cancelling the half-integer factors of the $(2\nu + 1)$ -blocks against the numerator.

Corollary 25.5 (complete factorisation and the zero locus). *For every $N \geq 1$ the determinant $\ddot{H}_{2N}(f_\nu)$ is a nonzero rational function of ν that factors completely into linear forms over $\mathbb{Q}$: a closed product form at every even order. Its numerator zeros lie at negative half-integers (from $(\nu + \frac{1}{2})_k$) and at the points $\nu = -\frac{1}{4} - \frac{j}{2}$, $j \geq 0$ (from $(2\nu + \frac{1}{2})_k$); its poles at negative integers. In the Poisson range $\nu > -\frac{1}{2}$:*

$$\ddot{H}_{2N}(f_\nu) = 0 \iff \nu = -\frac{1}{4} \text{ and } N \geq 2,$$

and $\operatorname{sgn} \ddot{H}_{2N}(f_\nu) = (-1)^N$ for $\nu > -\frac{1}{4}$, while $\ddot{H}_{2N}(f_\nu) < 0$ throughout $-\frac{1}{2} < \nu < -\frac{1}{4}$.

Proof. Inspect (25.12). For $\nu > -\frac{1}{2}$ every Pochhammer factor is a product of positive reals except the blocks $(2\nu + \frac{1}{2})_k = \prod_{j=0}^{k-1} (2\nu + \frac{1}{2} + j)$, whose only possibly nonpositive factor is $2\nu + \frac{1}{2}$: these blocks vanish simultaneously at $\nu = -\frac{1}{4}$ (for every $k \geq 1$, hence for $N \geq 2$; for $N = 1$ the block range is empty), are positive for $\nu > -\frac{1}{4}$, and each block with $k \geq 1$ is negative for $-\frac{1}{2} < \nu < -\frac{1}{4}$, giving the extra sign $(-1)^{N-1}$ there. $\square$

Corollary 25.6 ($\nu = \frac{1}{2}$: the squared sinc).

$$\ddot{H}_{2N} \left((1+x) \left(\frac{\sin x}{x} \right)^2 \right) = (-1)^N 2^{8N^2-5N} \left(\prod_{k=1}^{2N-1} k! \right)^2 \frac{(N-1)!}{(3N-1)!} \prod_{i=0}^{2N-1} \frac{(2i)!}{(2N+2i-1)!},$$

with the values $-\frac{2^2}{3}, \frac{2^{16}}{3^4 5^3 7^2}, -\frac{2^{47}}{3^9 5^5 7^5 11^3 13^2}, \frac{2^{88}}{3^{14} 5^7 7^7 11^5 13^4 17^3 19^2}$ at $n = 2, 4, 6, 8$.

Proof. Specialise (25.12) at $\nu = \frac{1}{2}$, where $(\nu + \frac{1}{2})_k = k!$, $(2\nu + \frac{1}{2})_k = (\frac{3}{2})_k$ and $(2\nu + 1)_{N-1+i} = (2)_{N-1+i} = (N+i)!$. From $(\frac{1}{2})_k = (2k)!/(4^k k!)$ and $(\frac{3}{2})_k = (2k+1)!/(4^k k!)$ the middle block is $\prod_{k=0}^{N-1} (2k)! (2k+1)! 16^{-k} = 16^{-\binom{N}{2}} \prod_{k=1}^{2N-1} k!$, while $(\frac{3}{2})_{N-1+i} = (2N+2i-1)!/(4^{N-1+i} (N-1+i)!)$ turns the tail into

$$\prod_{i=0}^{2N-1} \frac{4^{N-1+i} i! (N-1+i)!}{(2N+2i-1)! (N+i)!} = 4^{2N(N-1) + \binom{2N}{2}} \left(\prod_{k=1}^{2N-1} k! \right) \frac{(N-1)!}{(3N-1)!} \prod_{i=0}^{2N-1} \frac{1}{(2N+2i-1)!},$$

using $\prod_i (N-1+i)!/(N+i)! = \prod_i (N+i)^{-1} = (N-1)!/(3N-1)!$. Collecting the powers of two, $N(2N-1) - 2N(N-1) + 4N(N-1) + 2N(2N-1) = 8N^2 - 5N$, and absorbing $\prod_i (2i)!$ into the last product gives the stated form. $\square$

The values in Corollary 25.6 have numerators that are pure powers of two and odd denominators. The member $(1+x)(\sin x/x)^2$ sits one step from the reciprocal-sine function of Section 18: there the sine is downstairs, $(1+x)x/\sin x$, the functionals are Wilson, and the evaluation runs through **M2** and condensation; here the sine is upstairs and squared, the functional is not classical at all, and the evaluation is pure Vandermonde — yet only the even orders survive.

Remark 25.7 (odd orders: where **M1** stops). At an odd order $n = 2N+1$ the even column indices reach $l = N$, so the padding (25.4) must be lengthened to $(\nu+1+i)_N(2\nu+1+i)_N$, and then

$$\deg Q_{2l} = 2N = n-1, \quad \deg Q_{2m+1} = 2N+1 = n > n-1:$$

the odd columns overflow the Vandermonde degree bound of Lemma 3.1 by exactly one, and **M1** yields no evaluation — for the dilated determinant of (25.2) the method works precisely at even orders. This failure is structural, not merely technical. The even half of the parity contrast is Theorem 25.4; the odd half remains open.

26. CONNECTION WITH A DETERMINANT OF CHAPOTON–HAN

The dilated Hankel determinant of this paper was first met by the author while studying a conjecture from his joint work with Chapoton on the roots of the Poupard and Kreweras polynomials [4]. This section closes the circle: after a change of basis, that conjecture *is* the dilated Hankel determinant of $(1+x)/\cos x$ evaluated in Proposition 11.1.

26.1. The evaluation ρ and the conjecture. Call the *index* of a nonzero polynomial its degree plus its valuation, and call P *palindromic* of index d if $x^d P(1/x) = P(x)$ — for example $P = x^i(1+x)^j$, of index $2i+j$. For such P the polynomial $(x^d+1)P(1) - 2P(x)$ vanishes to second order at $x = 1$ (palindromy gives $P'(1) = \frac{d}{2}P(1)$), so the *index-lowering operator* of [4, §5],

$$\mathcal{N}_0(P) = \frac{(x^d+1)P(1) - 2P(x)}{(x-1)^2},$$

is a polynomial, palindromic of index $d-2$ when nonzero. Iterating $\lfloor d/2 \rfloor$ times lands in index at most 1, and, following [4], the evaluation ρ is the final value of this iteration:

$$\rho(P) := \text{constant term of } \mathcal{N}_0^{\lfloor d/2 \rfloor}(P);$$

in particular $\rho(1) = 1$ and $\rho(c(1+x)) = c$. Since $P = (x-1)^2Q$ has $P(1) = 0$, giving $\mathcal{N}_0((x-1)^2Q) = -2Q$, and since $\rho(P) = \rho(\mathcal{N}_0P)$ by definition, the map ρ obeys, for palindromic Q ,

$$\rho((x-1)^2Q(x)) = -2\rho(Q(x)). \quad (26.1)$$

Note that ρ is linear on each fixed index but is *not* a linear functional on all of $\mathbb{Q}[x]$: for example $\rho((1+x)x^k) \neq \rho(x^k) + \rho(x^{k+1})$, the three indices $2k+1, 2k, 2k+2$ being distinct. Every manipulation below stays within a single index, on which ρ is linear. Put

$$A_k := \rho(x^k), \quad B_k := \rho((1+x)x^k) \quad (k \geq 0),$$

so that $(A_k) = 1, 1, 2, 10, 104, 1816, \dots$ are the generalised Euler numbers of type 2^n (OEIS A005799) and $(B_k/2^k) = 1, 1, 2, 8, 56, 608, \dots$ are the Genocchi medians (OEIS A005439). Following [4], let $\mathbf{M}_N$ be the $N \times N$ matrix

$$\mathbf{M}_N(i, j) = \rho(x^i(1+x)^j) 2^{-\lfloor j/2 \rfloor}, \quad 0 \leq i, j \leq N-1. \quad (26.2)$$

The conjecture in question [4, Conj. 5.4] asserts that

$$\det \mathbf{M}_N = \prod_{k=1}^{N-1} ((N-k)!)^{\epsilon(k)}, \quad \epsilon(k) = \begin{cases} 2 & k \text{ odd}, \\ 4 & k \text{ even}. \end{cases} \quad (26.3)$$

(Its first row, $j \mapsto \mathbf{M}_N(0, j)$, is exactly the sequence $1, 1, 1, 3, 5, 25, 61, 427, \dots$ of $f = (1+x)/\cos x$, which is what first suggested the connection.)

Theorem 26.1. *For all $N \geq 1$,*

$$\det \mathbf{M}_N = 2^{-\binom{N}{2}} \ddot{H}_N((1+x)/\cos x) = ((N-1)!!)^2 \prod_{k=1}^{N-2} (k!!)^6 = \prod_{k=1}^{N-1} ((N-k)!)^{\epsilon(k)}. \quad (26.4)$$

In particular [4, Conj. 5.4] holds.

The proof has three steps. First, a triangular change of column basis, powered by the reduction (26.1), strips the powers of $1+x$: $\det \mathbf{M}_N$ is an explicit power of 2 times the dilated Hankel determinant of the sequence interleaving (A_k) and (B_k) (Theorem 26.3). Second — the key step — we prove a moment representation of ρ (Lemma 26.5): on each index, ρ is the secant functional $\mathcal{S}$ (even index) or the functional $\mathcal{T}^*$ (odd index) of Section 11, composed with an explicit change of variable; in particular $A_k = \mathcal{S}([(1+y)/2]^k)$ and $B_k = \mathcal{T}^*([(1+y)/2]^k)$. This is where the secant numbers enter; the identification was observed, conjecturally, in [4, §5]. Third, the affine substitution $y = 2v - 1$ transports the orthogonal data of Section 11 (Lemma 26.6), and the biorthogonal reduction **M2** runs exactly as in the proof of Proposition 11.1. In passing, the first step also proves Conjecture 5.3 of [4] (Remark 26.4).

26.2. From $\mathbf{M}_N$ to a dilated Hankel determinant.

Lemma 26.2. *For all $i \geq 0$ and $m \geq 0$,*

$$\rho(x^i(1+x)^{2m}) = (-2)^m \sum_{l=0}^m \binom{m}{l} (-2)^l \rho(x^{i+l}) = (-2)^m \sum_{l=0}^m \binom{m}{l} (-2)^l A_{i+l}, \quad (26.5)$$

$$\rho(x^i(1+x)^{2m+1}) = (-2)^m \sum_{l=0}^m \binom{m}{l} (-2)^l \rho((1+x)x^{i+l}) = (-2)^m \sum_{l=0}^m \binom{m}{l} (-2)^l B_{i+l}. \quad (26.6)$$

Proof. Since $(1+x)^2 = (x-1)^2 + 4x$, the binomial theorem gives

$$x^i(1+x)^{2m} = \sum_{l=0}^m \binom{m}{l} 4^l x^{i+l} (x-1)^{2m-2l},$$

all of whose terms have index $2i+2m$, so ρ may be applied term by term. Iterating (26.1) yields $\rho(x^{i+l}(x-1)^{2m-2l}) = (-2)^{m-l} \rho(x^{i+l})$, so

$$\rho(x^i(1+x)^{2m}) = \sum_{l=0}^m \binom{m}{l} 4^l (-2)^{m-l} \rho(x^{i+l}) = (-2)^m \sum_{l=0}^m \binom{m}{l} (-2)^l \rho(x^{i+l}),$$

because $4^l(-2)^{m-l} = (-2)^m(-2)^l$. Multiplying the same expansion by $(1+x)$ and using $\rho((1+x)x^{i+l}(x-1)^{2m-2l}) = (-2)^{m-l}\rho((1+x)x^{i+l})$ gives (26.6). $\square$

Theorem 26.3. *Let a be the sequence interleaving A and B , namely $a_{2k} = A_k$ and $a_{2k+1} = B_k$. Then*

$$\det \mathbf{M}_N = 2^{\tau_N} \ddot{H}_N(a), \quad \tau_N = \binom{\bar{N}}{2} + \binom{\underline{N}}{2}, \quad (26.7)$$

where $\bar{N} = \lceil N/2 \rceil$ and $\underline{N} = \lfloor N/2 \rfloor$, as throughout. In other words, the Chapoton–Han determinant (26.3) is, up to an explicit power of two, the dilated Hankel determinant of the sequence $(1, 1, 1, 2, 2, 8, 10, 64, 104, \dots)$.

Proof. By (26.2) and Lemma 26.2 (with $\lfloor (2m+1)/2 \rfloor = m$), the columns of $\mathbf{M}_N$ are

$$\mathbf{M}_N(\cdot, 2m) = (-1)^m \sum_{l=0}^m \binom{m}{l} (-2)^l \alpha_l, \quad \mathbf{M}_N(\cdot, 2m+1) = (-1)^m \sum_{l=0}^m \binom{m}{l} (-2)^l \beta_l,$$

where $\alpha_l = (A_{i+l})_i$ and $\beta_l = (B_{i+l})_i$. Thus $\mathbf{M}_N$ arises from the matrix W with ordered columns $\alpha_0, \beta_0, \alpha_1, \beta_1, \dots$ by a triangular column operation: column $2m$ (resp. $2m+1$) involves only the α_l (resp. β_l) with $l \leq m$, with coefficient $(-1)^m \binom{m}{m} (-2)^m = 2^m$ on α_m (resp. β_m). Hence

$$\det \mathbf{M}_N = \left(\prod_{0 \leq 2m \leq N-1} 2^m \right) \left(\prod_{0 \leq 2m+1 \leq N-1} 2^m \right) \det W = 2^{\tau_N} \det W,$$

since $\sum_{2m < N} m + \sum_{2m+1 < N} m = \binom{\bar{N}}{2} + \binom{\underline{N}}{2} = \tau_N$. Finally $W(i, 2m) = A_{i+m} = a_{2i+2m}$ and $W(i, 2m+1) = B_{i+m} = a_{2i+2m+1}$, that is $W = (a_{2i+j})$, so $\det W = \ddot{H}_N(a)$. $\square$

Remark 26.4 (Conjecture 5.3 of Chapoton–Han). Lemma 26.2 also settles [4, Conj. 5.3], which asserts that $2^{\lfloor j/2 \rfloor}$ divides $\rho(x^i(1+x)^j)$ (so that the entries (26.2) are integers; the case of odd j was proved in [4]). Indeed $\mathcal{N}_0$ preserves integer coefficients — the division by the monic $(x-1)^2$ is exact — so $A_k, B_k \in \mathbb{Z}$, and (26.5)–(26.6) display $\rho(x^i(1+x)^{2m})$ and $\rho(x^i(1+x)^{2m+1})$ as $(-2)^m$ times an integer.

26.3. The evaluation ρ as secant moments. It remains to evaluate $\ddot{H}_N(a)$, and for that we must first identify A_k and B_k . The following moment representation is the bridge; it proves, in particular, the identification of the first row of $\mathbf{M}_N$ with the coefficients of $(1+x)/\cos x$ observed in [4, §5].

Lemma 26.5 (moment representation of ρ). *Let $\mathcal{S}[y^j] = E_{2j}$ and $\mathcal{T}^*[y^j] = (2j+1)E_{2j}$ be the functionals of Section 11, and set $\zeta := x + x^{-1} - 2 = (x-1)^2/x$. Every palindromic polynomial of even index $2e$ has the form $P = x^e \pi(\zeta)$, and every one of odd index $2e+1$ the form $P = (1+x)x^e \sigma(\zeta)$, with $\deg \pi, \deg \sigma \leq e$; and*

$$\rho(x^e \pi(\zeta)) = \mathcal{S}\left[\left(\frac{1+y}{2}\right)^e \pi\left(\frac{-4}{1+y}\right)\right], \quad \rho((1+x)x^e \sigma(\zeta)) = \mathcal{T}^*\left[\left(\frac{1+y}{2}\right)^e \sigma\left(\frac{-4}{1+y}\right)\right], \quad (26.8)$$

the arguments on the right being polynomials in y of degree $\leq e$. In particular ($\pi = \sigma = 1$, $e = k$),

$$A_k = \mathcal{S}\left[\left(\frac{1+y}{2}\right)^k\right] = 2^{-k} \sum_{j=0}^k \binom{k}{j} E_{2j}, \quad B_k = \mathcal{T}^*\left[\left(\frac{1+y}{2}\right)^k\right] = 2^{-k} \sum_{j=0}^k \binom{k}{j} (2j+1) E_{2j}. \quad (26.9)$$

Proof. If P is palindromic of index $2e$, the Laurent polynomial $x^{-e}P$ is invariant under $x \mapsto 1/x$, hence a polynomial in $x + x^{-1} = \zeta + 2$: this gives $P = x^e \pi(\zeta)$ with $\deg \pi \leq e$. If the index is odd, then $P(-1) = -P(-1)$, so $(1+x) \mid P$, and $P/(1+x)$ is palindromic of even index. At fixed e both sides of (26.8) are linear in π (resp. σ).

Introduce

$$G_e(\zeta) := x^e + x^{-e}, \quad H_e(\zeta) := \sum_{j=-e}^e (-1)^j x^j,$$

both invariant under $x \mapsto 1/x$, hence polynomials in ζ of degree e , with $G_e(0) = 2$ and $H_e(0) = (-1)^e$ (value at $x = 1$). Using $(x - 1)^2 = x\zeta$, $x^{2e} + 1 = x^e G_e(\zeta)$ and $x^{2e+1} + 1 = (-1)^e(1 + x)x^e H_e(\zeta)$, the operator $\mathcal{N}_0$ reads, in these coordinates,

$$\begin{aligned}\mathcal{N}_0(x^e \pi(\zeta)) &= x^{e-1} \tilde{\pi}(\zeta), & \tilde{\pi} &= \frac{G_e \pi(0) - 2\pi}{\zeta}, \\ \mathcal{N}_0((1+x)x^e \sigma(\zeta)) &= (1+x)x^{e-1} \tilde{\sigma}(\zeta), & \tilde{\sigma} &= \frac{2((-1)^e \sigma(0) H_e - \sigma)}{\zeta}\end{aligned}$$

(both numerators vanish at $\zeta = 0$, so the divisions are exact).

We prove (26.8) by induction on e . For $e = 0$ both sides equal the constant π (resp. σ , since $\rho(c(1+x)) = c$ and $\mathcal{T}^*[1] = 1$). For $e \geq 1$, since $\rho(P) = \rho(\mathcal{N}_0 P)$, it suffices to show that the right-hand sides are unchanged when (e, π) is replaced by $(e-1, \tilde{\pi})$, and (e, σ) by $(e-1, \tilde{\sigma})$. Writing $\zeta^* := -4/(1+y)$, so that $1/\zeta^* = -(1+y)/4$,

$$\begin{aligned}\left(\frac{1+y}{2}\right)^{e-1} \tilde{\pi}(\zeta^*) &= \left(\frac{1+y}{2}\right)^e \left(\pi(\zeta^*) - \frac{\pi(0)}{2} G_e(\zeta^*)\right), \\ \left(\frac{1+y}{2}\right)^{e-1} \tilde{\sigma}(\zeta^*) &= \left(\frac{1+y}{2}\right)^e \left(\sigma(\zeta^*) - (-1)^e \sigma(0) H_e(\zeta^*)\right),\end{aligned}$$

so the induction step amounts to the two vanishing statements

$$\mathcal{S}\left[\left(\frac{1+y}{2}\right)^e G_e(\zeta^*)\right] = 0, \quad \mathcal{T}^*\left[\left(\frac{1+y}{2}\right)^e H_e(\zeta^*)\right] = 0 \quad (e \geq 1). \quad (26.10)$$

These follow from the explicit expansions

$$\left(\frac{1+y}{2}\right)^e G_e(\zeta^*) = (-1)^e 2^{1-e} \sum_{j=0}^e (-1)^j \binom{2e}{2j} y^j, \quad \left(\frac{1+y}{2}\right)^e H_e(\zeta^*) = 2^{-e} \sum_{j=0}^e (-1)^j \binom{2e+1}{2j+1} y^j. \quad (26.11)$$

To verify (26.11), parametrise $y = \tan^2 \varphi$; then $\zeta^* = -4 \cos^2 \varphi$ corresponds to $x = -e^{-2i\varphi}$ (as $\zeta = x + x^{-1} - 2$), whence $G_e(\zeta^*) = (-1)^e 2 \cos 2e\varphi$ and $H_e(\zeta^*) = \sum_{j=-e}^e e^{-2ij\varphi} = \sin((2e+1)\varphi)/\sin \varphi$, while $1+y = \sec^2 \varphi$; since $(\cos \varphi + i \sin \varphi)^n = \cos^n \varphi (1 + i \tan \varphi)^n$ and the right-hand sides of (26.11) are $(-1)^e 2^{1-e} \operatorname{Re}(1 + i \tan \varphi)^{2e}$ and $2^{-e} \operatorname{Im}((1 + i \tan \varphi)^{2e+1})/\tan \varphi$, both equalities reduce to $\cos 2e\varphi = \cos 2e\varphi$ and $\sin(2e+1)\varphi = \sin(2e+1)\varphi$. Two polynomials in y agreeing for all $y = \tan^2 \varphi \geq 0$ agree identically. Applying the functionals to (26.11) now gives

$$\sum_{j=0}^e (-1)^j \binom{2e}{2j} E_{2j} = (-1)^e (2e)! [x^{2e}] (\sec x \cdot \cos x) = 0 \quad (e \geq 1),$$

since $\sec x \cdot \cos x = 1$, and, using $(2j+1) \binom{2e+1}{2j+1} = (2e+1) \binom{2e}{2j}$,

$$\sum_{j=0}^e (-1)^j \binom{2e+1}{2j+1} (2j+1) E_{2j} = (2e+1) \sum_{j=0}^e (-1)^j \binom{2e}{2j} E_{2j} = 0.$$

This proves (26.10), hence (26.8); and (26.9) follows by the binomial theorem. $\square$

By (26.9) and linearity, the moment functionals $\mathcal{A}[v^k] := A_k$ and $\mathcal{B}[v^k] := B_k$ satisfy $\mathcal{A}[F] = \mathcal{S}[F(\frac{1+y}{2})]$ and $\mathcal{B}[F] = \mathcal{T}^*[F(\frac{1+y}{2})]$ for every polynomial F : the even and odd parts of a are the secant pair of Section 11, transported by the affine substitution $y = 2v - 1$. All the orthogonal data transport along.

Lemma 26.6. *The functionals $\mathcal{A}$ and $\mathcal{B}$ are quasi-definite, with monic orthogonal polynomials*

$$p_i(v) = 2^{-i} \hat{P}_i(2v-1), \quad q_m(v) = 2^{-m} \varrho_m(2v-1),$$

where $\hat{P}_i$ and ϱ_m are the orthogonal polynomials of $\mathcal{S}$ and $\mathcal{T}^*$ (Lemmas 7.2 and 11.2; we write ϱ to keep the latter apart from the evaluation ρ). Their recurrence data and norms are

$$\begin{aligned}c_i^{\mathcal{A}} &= 4i^2 + 2i + 1, & \lambda_i^{\mathcal{A}} &= (i(2i-1))^2, & h_i^{\mathcal{A}} &= \mathcal{A}[p_i^2] = (i!)^2 ((2i-1)!!)^2 = 4^{-i} ((2i)!)^2; \\ c_m^{\mathcal{B}} &= 4m^2 + 4m + 2, & \lambda_m^{\mathcal{B}} &= 4m^4, & h_m^{\mathcal{B}} &= \mathcal{B}[q_m^2] = 4^m (m!)^4 = 4^{-m} ((2m)!!)^4,\end{aligned}$$

and the connection coefficients are

$$\mathcal{B}[p_i q_m] = 2^{i+m} (i!)^2 (m!)^2 \binom{1/2}{i-m}, \quad i.e. \quad p_i = \sum_{m=0}^i 2^{i-m} \left(\frac{i!}{m!}\right)^2 \binom{1/2}{i-m} q_m. \quad (26.12)$$

Proof. The polynomials p_i, q_m are monic of degrees i, m in v , and orthogonality and norms transport directly: $\mathcal{A}[p_i p_l] = \mathcal{S}[2^{-i-l} \hat{P}_i \hat{P}_l] = \delta_{il} 4^{-i} ((2i)!)^2$ by Lemma 7.2, and $\mathcal{B}[q_m q_{m'}] = \delta_{mm'} 4^{-m} ((2m)!)^4$ by Lemma 11.2. The affine substitution transforms the three-term recurrence data by $c \mapsto (c+1)/2$ and $\lambda \mapsto \lambda/4$; applied to $c_i^S = (2i)^2 + (2i+1)^2$, $\lambda_i^S = ((2i-1)(2i))^2$ and $c_m^* = 8m^2 + 8m + 3$, $\lambda_m^* = (2m)^4$, this yields the displayed values. Finally $\mathcal{B}[p_i q_m] = 2^{-i-m} \mathcal{T}^*[\hat{P}_i \varrho_m]$, so (26.12) is the image of the connection formula $\mathcal{T}^*[\hat{P}_i \varrho_m] = (i!)^2 (m!)^2 2^{2i+2m} \binom{1/2}{i-m}$ of Lemma 11.4. $\square$

Proof of Theorem 26.1. By Theorem 26.3 it suffices to evaluate $\ddot{H}_N(a)$. Both parity functionals of a are classical (Lemma 26.6), so, exactly as in the proof of Proposition 11.1, Lemma 3.4 gives

$$\ddot{H}_N(a) = (-1)^{\binom{\bar{N}}{2}} \prod_{l=0}^{\bar{N}-1} h_l^A \cdot \prod_{m=0}^{\frac{N-1}{2}} h_m^B \cdot \det(\kappa_{\bar{N}+r, m})_{0 \leq r, m \leq \frac{N-1}{2}}, \quad \kappa_{i, m} = 2^{i-m} \left(\frac{i!}{m!}\right)^2 \binom{1/2}{i-m},$$

with the connection coefficients of (26.12) — those of that proof with the weight 4^{i-m} replaced by 2^{i-m} . Pulling $(i!)^2 2^i$ out of row r (where $i = \bar{N} + r$) and $(m!)^{-2} 2^{-m}$ out of column m leaves the same binomial determinant, worth $\Omega(1) = 2^{-\binom{N}{2}}$ together with the sign (Lemma 7.5), so

$$\ddot{H}_N(a) = 2^{-\binom{N}{2}} \prod_{l=0}^{\bar{N}-1} 4^{-l} ((2l)!)^2 \cdot \prod_{i=\bar{N}}^{N-1} (i!)^2 2^i \cdot \prod_{m=0}^{\frac{N-1}{2}} (m!)^2 2^m.$$

Comparing factor by factor with the corresponding display in the proof of Proposition 11.1, which carries 4^i and 4^m in place of 2^i and 2^m and no 4^{-l} ,

$$\frac{\ddot{H}_N(a)}{\ddot{H}_N((1+x)/\cos x)} = \prod_{l=0}^{\bar{N}-1} 4^{-l} \cdot \prod_{i=\bar{N}}^{N-1} 2^{-i} \cdot \prod_{m=0}^{\frac{N-1}{2}} 2^{-m} = 2^{-\tau_N - \binom{N}{2}},$$

since $2\binom{\bar{N}}{2} + ((\binom{N}{2} - \binom{\bar{N}}{2})) + \binom{N}{2} = \tau_N + \binom{N}{2}$, using $\sum_{i=\bar{N}}^{N-1} i = \binom{N}{2} - \binom{\bar{N}}{2}$. Multiplying by 2^{τ_N} and using Proposition 11.1,

$$\det \mathbf{M}_N = 2^{\tau_N} \ddot{H}_N(a) = 2^{-\binom{N}{2}} \ddot{H}_N((1+x)/\cos x) = ((N-1)!)^2 \prod_{k=1}^{N-2} (k!!)^6.$$

It remains to identify this with the product of (26.3). Both equal 1 for $N = 1$, and both are multiplied by $(N!!)^2 ((N-1)!!)^4$ when $N \rightarrow N+1$: for the double-factorial product this is immediate, and for $\prod_{k=1}^{N-1} ((N-k)!)^{\epsilon(k)} = \prod_{j=1}^{N-1} (j!)^{\epsilon(N-j)}$ the shift $N \rightarrow N+1$ flips every exponent (as $\epsilon(k) + \epsilon(k+1) = 6$), so the ratio is

$$(N!)^2 \prod_{\substack{1 \leq j \leq N-1 \\ N-j \text{ odd}}} (j!)^2 \bigg/ \prod_{\substack{1 \leq j \leq N-1 \\ N-j \text{ even}}} (j!)^2 = (N!)^2 ((N-1)!!)^2 = (N!!)^2 ((N-1)!!)^4,$$

the middle equality because the quotient telescopes over consecutive factorials: $\frac{(N-1)!}{(N-2)!} \cdot \frac{(N-3)!}{(N-4)!} \cdots = (N-1)(N-3) \cdots = (N-1)!!$. $\square$

Thus the root conjecture of Chapoton–Han and the Euler dilated Hankel determinant are one and the same evaluation: the change of column basis of Theorem 26.3 strips the powers of $1+x$, and the substitution $y = 2v - 1$ of Lemma 26.6 carries what remains onto Proposition 11.1.

27. COROLLARIES: EXPLICIT EVALUATIONS

This section collects the explicit members of the families evaluated in this paper. No new proofs are needed: each corollary is a substitution of parameters into the theorem quoted at the head of its subsection, recording the generating function f , the determinant $\ddot{H}_n(f)$ and, where a closed form is available, the single and double shifts $\ddot{H}_n^{(1)} = \ddot{H}_n(f')$ and $\ddot{H}_n^{(2)} = \ddot{H}_n(f'')$.

27.1. The Beta family $a_m = \rho^m(\alpha)_m/(\beta)_m$. Every member of the Beta family (4.1) is a plain sequence $a_m = \rho^m(\alpha)_m/(\beta)_m$ (we normalise $a_0 = 1$), with term ratio $a_{m+1}/a_m = \rho(m+\alpha)/(m+\beta)$; when the denominator parameter β is absent the sequence is $\rho^m(\alpha)_m$. By Corollary 4.3 (resp. Proposition 4.2 when β is absent) the dilated Hankel determinant $\ddot{H}_n = \det(a_{2i+j})$ is a closed product of factorials. The shifted sequence a_{m+1} is again a Beta member, with $(\rho, \alpha, \beta) \mapsto (\rho, \alpha+1, \beta+1)$, so the single shift $\ddot{H}_n^{(1)} := \det(a_{2i+j+1})$ is likewise closed. We record several well-known members with their (ρ, α, β) , giving both $\ddot{H}_n$ and $\ddot{H}_n^{(1)}$.

Corollary 27.1 (Catalan numbers). $C_m = \frac{1}{m+1} \binom{2m}{m} = 4^m(\frac{1}{2})_m/(2)_m$, so $(\rho, \alpha, \beta) = (4, \frac{1}{2}, 2)$; the shift C_{m+1} is the member $(4, \frac{3}{2}, 3)$, and

$$\begin{aligned}\ddot{H}_n(C) &= 2^{\binom{n}{2}} \prod_{i=0}^{n-1} \binom{4i}{2i} \frac{i!(n+i)!}{(n+2i)!} = 1, 3, 32, 1232, \dots, \\ \ddot{H}_n^{(1)} &= 2^{\binom{n}{2}} \prod_{i=0}^{n-1} \frac{i!(4i+2)!(n+i)!}{(2i)!(2i+1)!(n+2i+1)!}.\end{aligned}$$

Corollary 27.2 (central binomial coefficients). $\binom{2m}{m} = 4^m(\frac{1}{2})_m/(1)_m$, so $(\rho, \alpha, \beta) = (4, \frac{1}{2}, 1)$; the shift $\binom{2m+2}{m+1}$ is the member $(4, \frac{3}{2}, 2)$, and

$$\begin{aligned}\ddot{H}_n &= 2^{\binom{n}{2}+n-1} \prod_{i=0}^{n-1} \binom{4i}{2i} \frac{i!(n-1+i)!}{(n-1+2i)!} = 1, 8, 224, \dots, \\ \ddot{H}_n^{(1)} &= 2^{\binom{n}{2}+n-1} \prod_{i=0}^{n-1} \frac{i!(4i+2)!(n-1+i)!}{(2i)!(2i+1)!(n+2i)!}.\end{aligned}$$

Corollary 27.3 ($\binom{2m+1}{m}$). $\binom{2m+1}{m} = 4^m(\frac{3}{2})_m/(2)_m$, so $(\rho, \alpha, \beta) = (4, \frac{3}{2}, 2)$; the shift $\binom{2m+3}{m+1}$ is the member $(4, \frac{5}{2}, 3)$, and

$$\ddot{H}_n = 2^{\binom{n}{2}} \prod_{i=0}^{n-1} \frac{(4i+1)!}{(2i+n)!} = 1, 5, 96, 6864, \dots, \quad \ddot{H}_n^{(1)} = 2^{\binom{n}{2}} \prod_{i=0}^{n-1} \frac{(4i+3)!}{(2i+1)(2i+n+1)!}.$$

Corollary 27.4 (factorials). $m! = (1)_m$, so $(\rho, \alpha, \beta) = (1, 1, -)$ (β absent); the shift is $(m+1)! = (2)_m$, and

$$\ddot{H}_n(m!) = 2^{\binom{n}{2}} \prod_{i=0}^{n-1} i!(2i)!, \quad \ddot{H}_n^{(1)} = 2^{\binom{n}{2}} \prod_{i=0}^{n-1} i!(2i+1)!.$$

Corollary 27.5 (double factorials). $(2m-1)!! = 2^m(\frac{1}{2})_m$ and $(2m+1)!! = 2^m(\frac{3}{2})_m$, so $(\rho, \alpha, \beta) = (2, \frac{1}{2}, -)$ and $(2, \frac{3}{2}, -)$; here the shift sends $(2m-1)!! \mapsto (2m+1)!! \mapsto (2m+3)!!$, so

$$\begin{aligned}\ddot{H}_n((2m-1)!!) &= 4^{\binom{n}{2}} \prod_{i=0}^{n-1} i!(4i-1)!!, \\ \ddot{H}_n((2m+1)!!) &= 4^{\binom{n}{2}} \prod_{i=0}^{n-1} i!(4i+1)!!, \\ \ddot{H}_n^{(1)}((2m+1)!!) &= 4^{\binom{n}{2}} \prod_{i=0}^{n-1} i!(4i+3)!!.\end{aligned}$$

Corollary 27.6 $((2m)!/m!). \frac{(2m)!}{m!} = 4^m(\frac{1}{2})_m$, so $(\rho, \alpha, \beta) = (4, \frac{1}{2}, -)$ (β absent); the shift is $\frac{(2m+2)!}{(m+1)!} = 2 \cdot 4^m(\frac{3}{2})_m$, and

$$\ddot{H}_n = 8^{(n)} \prod_{i=0}^{n-1} \frac{i! (4i)!}{(2i)!}, \quad \ddot{H}_n^{(1)} = 8^{(n)} \prod_{i=0}^{n-1} \frac{i! (4i+2)!}{(2i+1)!}.$$

27.2. The Gaussian family. By Theorem 5.1 the Gaussian generating function $f(x) = e^{cx+x^2/2}$ has a closed product $\ddot{H}_n(f)$, and by Theorem 6.1 its derivative $f' = (c+x)e^{cx+x^2/2}$, whose dilated Hankel determinant is the single shift $\ddot{H}_n(f') = \det(a_{2i+j+1})$, satisfies $\ddot{H}_n(f') = c^n \ddot{H}_n(f)$. We record the specialisations $c \in \{1, -1\}$, in each case giving f and its derivative f' .

Corollary 27.7 ($c = 1$). The moments a_n are the involution numbers of $\{1, \dots, n\}$, and

$$f = e^{x+x^2/2}, \quad \ddot{H}_n(f) = 2^{(n)} \prod_{k=1}^{n-1} k!.$$

For the derivative,

$$f' = (1+x)e^{x+x^2/2}, \quad \ddot{H}_n(f') = \ddot{H}_n(f) = 2^{(n)} \prod_{k=1}^{n-1} k!,$$

so the determinant is invariant under differentiation.

Corollary 27.8 ($c = -1$). The moments are the signed involution numbers $a_n = (-1)^n I_n$ (I_n the involution count), and

$$f = e^{-x+x^2/2}, \quad \ddot{H}_n(f) = (-2)^{(n)} \prod_{k=1}^{n-1} k!.$$

For the derivative,

$$f' = (x-1)e^{-x+x^2/2}, \quad \ddot{H}_n(f') = (-1)^n (-2)^{(n)} \prod_{k=1}^{n-1} k!.$$

27.3. The Euler number family. The cases $s < 0$, $t < 0$, or $t - s$ even are trivial; assume henceforth $s \geq 0$, $t \geq 0$ and $\delta = t - s$ odd. Throughout $\bar{n} = \lceil n/2 \rceil$, $\underline{n} = \lfloor n/2 \rfloor$. With these, Proposition 7.1 evaluates $\ddot{H}_n$ in closed form for every (s, t) ; its signed factor $\Omega(\delta)$ ($\delta = t - s$) is evaluated for $\delta \in \{\pm 1, \pm 3\}$ by Lemma 7.5.

Each corollary below records both f and, for the *double shift*, its second derivative f'' : the doubly shifted sequence $(F_{k+2})_{k \geq 0}$ has exponential generating function f'' , and by Proposition 8.4 the determinant $\ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'')$ is the unshifted one times the explicit positive scalar $(2n-1)!! \sigma_{\bar{n}}(s) \sigma_{\underline{n}}(t)$, where $\sigma_K(c) := \prod_{k=0}^{K-1} (c+2k+1)$; here

$$\sigma_K(0) = (2K-1)!!, \quad \sigma_K(1) = 2^K K!, \quad \sigma_K(2) = (2K+1)!!, \quad \sigma_K(3) = 2^K (K+1)!,$$

and explicitly $f'' = (s+1) \frac{(s+1) \sin^2 x + 1}{\cos^{s+3} x} + (t+1) \frac{\sin x}{\cos^{t+2} x}$. All the closed forms below were verified for $n \leq 12$.

Four of the cases ($s \in \{0, 2\}$, $t \in \{1, 3\}$) additionally carry a *single shift*: the shifted sequence $(F_{k+1})_{k \geq 0}$ has exponential generating function f' , and $\ddot{H}_n^{(1)} = \ddot{H}_n(f')$. On the line $t = 1$ this is a scalar multiple of $\ddot{H}_n(f)$ (Proposition 9.1); on $t = 3$ it carries the extra linear factor Γ_n of Theorem 10.6. The single shift is recorded in the four corresponding corollaries below.

We list the eight cases with $s+t$ odd and $s, t \in \{0, 1, 2, 3\}$, beginning with the secant/tangent (Euler) case $(s, t) = (0, 1)$.

Corollary 27.9 ($s = 0, t = 1$). For $u_j = j^2, v_j = j(j+1)$ the moments are the Euler numbers $a_n = E_n$ of (7.1) (the alternating permutations of $\{1, \dots, n\}$), and

$$f = \frac{1 + \sin x}{\cos x} = \sec x + \tan x, \quad \ddot{H}_n(f) = 2^{-\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)! = \prod_{k=1}^{n-1} (k!)^2 (2k-1)!!.$$

For the single shift,

$$f' = \frac{1 + \sin x}{\cos^2 x}, \quad \ddot{H}_n^{(1)}(f) = \ddot{H}_n(f') = (2n-1)!! \ddot{H}_n(f) = (2n-1)!! \prod_{k=1}^{n-1} (k!)^2 (2k-1)!!.$$

For the double shift,

$$f'' = \frac{(1 + \sin x)^2}{\cos^3 x}, \quad \ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (2n-1)!! (2\bar{n}-1)!! 2^{\underline{n}} \ddot{H}_n(f).$$

Corollary 27.10 ($s = 1, t = 2$). For $u_j = j(j+1), v_j = j(j+2)$,

$$f = \frac{\sin x + 2}{2 \cos^2 x} + \frac{1}{2} \log(\sec x + \tan x),$$

$$\ddot{H}_n(f) = 2^{-\binom{n}{2} - \underline{n}} \prod_{k=1}^{n-1} (2k)! \prod_{j=1}^n j! > 0.$$

For the double shift,

$$f'' = \frac{4 \sin^2 x + 3 \sin x + 2}{\cos^4 x}, \quad \ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (2n-1)!! 2^{\bar{n}} \bar{n}! (2\underline{n}+1)!! \ddot{H}_n(f).$$

Corollary 27.11 ($s = 2, t = 3$). For $u_j = j(j+2), v_j = j(j+3)$,

$$f = \frac{3 + \sin x (2 \cos^2 x + 1)}{3 \cos^3 x},$$

$$\ddot{H}_n(f) = 2^{-\binom{n+1}{2}} 3^{-\underline{n}} \prod_{k=1}^{n-1} (2k)! \prod_{j=1}^{n+1} j! > 0.$$

For the single shift,

$$f' = \frac{1 + 3 \sin x}{\cos^4 x}, \quad \ddot{H}_n^{(1)}(f) = \ddot{H}_n(f') = \Lambda_n(2) (n(n+1) + 1),$$

the carrier of Theorem 10.6 at $s = 2$ being $n(n+1) + 1$, with parity factor $\rho_n = \frac{1}{2\bar{n}+1}$ (here the same for both parities) and smooth part

$$\Lambda_n(2) = \frac{2^{-\binom{n}{2}}}{2\bar{n}+1} 6^{-\bar{n}} \prod_{l=0}^{\bar{n}-1} (2l)! (2l+3)! 2^{-\underline{n}} \prod_{c=0}^{n-1} (2c+3)(2c+2)! \prod_{r=0}^{\bar{n}-1} (2\bar{n}+2r+1)!. \quad (27.1)$$

For the double shift,

$$f'' = \frac{9 \sin^2 x + 4 \sin x + 3}{\cos^5 x}, \quad \ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (2n-1)!! (2\bar{n}+1)!! 2^{\underline{n}} (\underline{n}+1)! \ddot{H}_n(f).$$

Corollary 27.12 ($s = 1, t = 0$). For $u_j = j(j+1), v_j = j^2$,

$$f = \frac{1}{\cos^2 x} + \log(\sec x + \tan x),$$

$$\ddot{H}_n(f) = \Omega(-1) \prod_{i=0}^{n-1} (2i)! \prod_{l=0}^{\bar{n}-1} (2l+1)! \prod_{m=0}^{\bar{n}-1} (2m)!,$$

with $\Omega(-1)$ as in Lemma 7.5. For the double shift,

$$f'' = \frac{-\sin^3 x + 4 \sin^2 x + \sin x + 2}{\cos^4 x}, \quad \ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (2n-1)!! 2^{\bar{n}} \bar{n}! (2\underline{n}-1)!! \ddot{H}_n(f).$$

Corollary 27.13 ($s = 2, t = 1$). For $u_j = j(j+2), v_j = j(j+1)$,

$$f = \frac{1 + \sin x \cos^2 x}{\cos^3 x},$$

$$\ddot{H}_n(f) = 2^{-\bar{n}} \Omega(-1) \prod_{i=0}^{n-1} (2i)! \prod_{l=0}^{\bar{n}-1} (2l+2)! \prod_{m=0}^{\underline{n}-1} (2m+1)!.$$

For the single shift, $f' = \frac{\cos^2 x + 3 \sin x}{\cos^4 x}$, and specialising Proposition 9.1 at $s = 2$ (no carrier factor arises on the line $t = 1$),

$$\ddot{H}_n^{(1)}(f) = \ddot{H}_n(f') = (-1)^{\bar{n}-1} \frac{\underline{n}! ((2\underline{n}+1)!!)^2}{(2\bar{n}-3)!!} (2n-1)!! \prod_{k=1}^{n-1} (k!)^2 (2k-1)!!, \quad (27.2)$$

with the convention $(-1)!! = 1$; equivalently the secant/tangent single shift of Corollary 27.9 times $(-1)^{\bar{n}-1} \underline{n}! ((2\underline{n}+1)!!)^2 / (2\bar{n}-3)!!$. For the double shift,

$$f'' = \frac{-2 \sin^3 x + 9 \sin^2 x + 2 \sin x + 3}{\cos^5 x}, \quad \ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (2n-1)!! (2\bar{n}+1)!! 2^{\underline{n}} \underline{n}! \ddot{H}_n(f).$$

Corollary 27.14 ($s = 3, t = 2$). For $u_j = j(j+3), v_j = j(j+2)$,

$$f = \frac{2 + \sin x \cos^2 x}{2 \cos^4 x} + \frac{1}{2} \log(\sec x + \tan x),$$

$$\ddot{H}_n(f) = 2^{-\underline{n}} 6^{-\bar{n}} \Omega(-1) \prod_{i=0}^{n-1} (2i)! \prod_{l=0}^{\bar{n}-1} (2l+3)! \prod_{m=0}^{\underline{n}-1} (2m+2)!.$$

For the double shift,

$$f'' = \frac{-3 \sin^3 x + 16 \sin^2 x + 3 \sin x + 4}{\cos^6 x},$$

$$\ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (2n-1)!! 2^{\bar{n}} (\bar{n}+1)! (2\underline{n}+1)!! \ddot{H}_n(f).$$

Corollary 27.15 ($s = 0, t = 3$). For $u_j = j^2, v_j = j(j+3)$,

$$f = \frac{3 \cos^2 x + \sin x (2 \cos^2 x + 1)}{3 \cos^3 x},$$

$$\ddot{H}_n(f) = 6^{-\underline{n}} \Omega(3) \prod_{i=0}^{n-1} (2i)! \prod_{l=0}^{\bar{n}-1} (2l)! \prod_{m=0}^{\underline{n}-1} (2m+3)!,$$

with $\Omega(3)$ as in Lemma 7.5. For the single shift,

$$f' = \frac{1 + \sin x \cos^2 x}{\cos^4 x}, \quad \ddot{H}_n^{(1)}(f) = \ddot{H}_n(f') = \Lambda_n(0) \Gamma_n(0), \quad \Gamma_n(0) = n(n+1) - 1 - 4\bar{n},$$

the carrier of Theorem 10.6 at $s = 0$, with smooth part

$$\Lambda_n(0) = (-1)^{\underline{n}+1} \rho_n 2^{-\binom{n}{2}} 6^{-\bar{n}} \prod_{l=0}^{\bar{n}-1} (2l)! (2l+3)! \prod_{c=0}^{\underline{n}-1} (2c+1)! \prod_{r=0}^{\underline{n}-1} (2\bar{n}+2r+1)!, \quad (27.3)$$

with parity factor $\rho_n = \frac{1}{n+1}$ for n even and $\rho_n = \frac{n}{n+2}$ for n odd. For the double shift,

$$f'' = \frac{-\sin^4 x + 4 \sin x + 1}{\cos^5 x}, \quad \ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (2n-1)!! (2\bar{n}-1)!! 2^{\underline{n}} (\underline{n}+1)! \ddot{H}_n(f).$$

Corollary 27.16 ($s = 3, t = 0$). For $u_j = j(j+3), v_j = j^2$,

$$f = \frac{1}{\cos^4 x} + \log(\sec x + \tan x),$$

$$\ddot{H}_n(f) = \Omega(-3) 6^{-\bar{n}} \prod_{i=0}^{n-1} (2i)! \prod_{l=0}^{\bar{n}-1} (2l+3)! \prod_{m=0}^{\underline{n}-1} (2m)!,$$

with $\Omega(-3)$ as in Lemma 7.5. For the double shift,

$$f'' = \frac{\sin^5 x - 2 \sin^3 x + 16 \sin^2 x + \sin x + 4}{\cos^6 x},$$

$$\ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (2n-1)!! 2^{\bar{n}} (\bar{n}+1)! (2\underline{n}-1)!! \ddot{H}_n(f).$$

27.4. The secant-number family $(1+x)/\cos^{s+1}x$. By Theorem 12.1 the family $f(x) = (1+x)/\cos^{s+1}x$ has $\ddot{H}_n(f) = c_n \prod_{i=1}^{n-1} (s+1)_i$, with the constant c_n (independent of s) given there. The single shift is $((n-1)!!)^2$ times the unshifted determinant for every s (Theorem 13.1), while the double shift admits such a closed multiple only at $s = 1$ (Theorem 14.1). We record the cases $s = 0, 1, 2, 3$.

Corollary 27.17 ($s = 0$). For $f = \frac{1+x}{\cos x}$,

$$\ddot{H}_n(f) = c_n \prod_{i=1}^{n-1} i! = 2^{\binom{n}{2}} ((n-1)!!)^2 \prod_{k=1}^{n-2} (k!!)^6.$$

For the single shift, with $f' = \frac{\cos x + (1+x)\sin x}{\cos^2 x}$,

$$\ddot{H}_n^{(1)}(f) = \ddot{H}_n(f') = ((n-1)!!)^2 \ddot{H}_n(f).$$

Corollary 27.18 ($s = 1$). For $f = \frac{1+x}{\cos^2 x}$,

$$\ddot{H}_n(f) = c_n \prod_{i=1}^{n-1} (i+1)!.$$

For the single shift, with $f' = \frac{\cos x + 2(1+x)\sin x}{\cos^3 x}$,

$$\ddot{H}_n^{(1)}(f) = \ddot{H}_n(f') = ((n-1)!!)^2 \ddot{H}_n(f);$$

and for the double shift (Theorem 14.1), with $f'' = \frac{2(1+x)(1+2\sin^2 x) + 4\sin x \cos x}{\cos^4 x}$,

$$\ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = 2^n (n!)^2 \ddot{H}_n(f).$$

Corollary 27.19 ($s = 2$). For $f = \frac{1+x}{\cos^3 x}$,

$$\ddot{H}_n(f) = c_n \prod_{i=1}^{n-1} \frac{(i+2)!}{2}.$$

For the single shift, with $f' = \frac{\cos x + 3(1+x)\sin x}{\cos^4 x}$,

$$\ddot{H}_n^{(1)}(f) = \ddot{H}_n(f') = ((n-1)!!)^2 \ddot{H}_n(f).$$

Corollary 27.20 ($s = 3$). For $f = \frac{1+x}{\cos^4 x}$,

$$\ddot{H}_n(f) = c_n \prod_{i=1}^{n-1} \frac{(i+3)!}{6}.$$

For the single shift, with $f' = \frac{\cos x + 4(1+x)\sin x}{\cos^5 x}$,

$$\ddot{H}_n^{(1)}(f) = \ddot{H}_n(f') = ((n-1)!!)^2 \ddot{H}_n(f).$$

27.5. A rank-one perturbation of the Euler number family. By Proposition 15.1 the one-parameter family

$$f_s(x) = \frac{\sin x + 1}{\cos^2 x} + s \sin x$$

is a rank-one perturbation of the secant/tangent single shift (Proposition 9.1, generating function f_0), with $\ddot{H}_n(f_s) = (1 - s \binom{n}{2}) \ddot{H}_n^E$, where $\ddot{H}_n^E := \prod_{k=1}^{n-1} (k!)^2 (2k+1)!!$. The determinant is affine in s ; it is the unperturbed value $\ddot{H}_n^E > 0$ at $s = 0$, and at $s = -1$ the perturbation collapses to $f_{-1} = (\sin^3 x + 1)/\cos^2 x$. We record $s = -2, -1, 0, 1, 2$, writing each f_s as a single fraction (polynomial in $\sin x$)/ $\cos^2 x$.

Corollary 27.21 ($s = -2$). For $f_{-2} = \frac{2 \sin^3 x - \sin x + 1}{\cos^2 x}$,

$$\ddot{H}_n(f_{-2}) = (n^2 - n + 1) \prod_{k=1}^{n-1} (k!)^2 (2k+1)!! > 0.$$

Corollary 27.22 ($s = -1$). For $f_{-1} = \frac{\sin^3 x + 1}{\cos^2 x}$,

$$\ddot{H}_n(f_{-1}) = \left(\binom{n}{2} + 1 \right) \prod_{k=1}^{n-1} (k!)^2 (2k+1)!! > 0.$$

Corollary 27.23 ($s = 0$). For $f_0 = \frac{\sin x + 1}{\cos^2 x}$, the secant/tangent single shift of Corollary 27.9 (Proposition 9.1),

$$\ddot{H}_n(f_0) = \ddot{H}_n^E = \prod_{k=1}^{n-1} (k!)^2 (2k+1)!!.$$

Corollary 27.24 ($s = 1$). For $f_1 = \frac{-\sin^3 x + 2 \sin x + 1}{\cos^2 x}$,

$$\ddot{H}_n(f_1) = \left(1 - \binom{n}{2} \right) \prod_{k=1}^{n-1} (k!)^2 (2k+1)!!.$$

Here $\ddot{H}_1 = 1$, $\ddot{H}_2 = 0$, and $\ddot{H}_n < 0$ for $n \geq 3$: the perturbation exactly cancels the determinant at $n = 2$.

Corollary 27.25 ($s = 2$). For $f_2 = \frac{-2 \sin^3 x + 3 \sin x + 1}{\cos^2 x}$,

$$\ddot{H}_n(f_2) = (1 - n(n-1)) \prod_{k=1}^{n-1} (k!)^2 (2k+1)!!,$$

of sign (-1) for all $n \geq 2$.

27.6. The Springer number family $1/(\cos x - t \sin x)^r$. By Theorem 16.1 the determinant is, for every integer $r \geq 1$, the power $(t(t^2 + 1))^{\binom{n}{2}}$ times the Beta value $\ddot{H}_n((1-x)^{-r})$ for $a_n = (r)_n$ (§4). The factor $t(t^2 + 1)$ shows $t = 0$ is degenerate, so we take $t = 1$, the Springer line.

Corollary 27.26 ($t = 1$, $r = 1$; Springer numbers). For $f = \frac{1}{\cos x - \sin x}$, whose moments are the Springer numbers,

$$\ddot{H}_n(f) = 4^{\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!.$$

Corollary 27.27 ($t = 1$, $r = 2$). For $f = \frac{1}{(\cos x - \sin x)^2}$,

$$\ddot{H}_n(f) = 4^{\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k+1)!.$$

27.7. A derivative of the Springer number family $(\cos x + \sin x)/(\cos x - \sin x)^s$. By Theorem 17.1 the determinant factors completely into linear forms in s for every s ; a factor vanishes when s is a non-positive integer (degenerate), so we take $s = 1, 2, 3$.

Corollary 27.28 ($s = 1$).

$$\ddot{H}_n\left(\frac{\cos x + \sin x}{\cos x - \sin x}\right) = 4^{\binom{n}{2}} \left(\prod_{k=1}^{n-1} k!\right) \prod_{j=0}^{n-2} [(2j+1)(2j+2)]^{n-1-j}.$$

Corollary 27.29 ($s = 2$).

$$\ddot{H}_n\left(\frac{\cos x + \sin x}{(\cos x - \sin x)^2}\right) = 4^{\binom{n}{2}} \left(\prod_{k=1}^{n-1} k!\right) \prod_{j=0}^{n-2} [(2j+2)(2j+3)]^{n-1-j}.$$

Corollary 27.30 ($s = 3$).

$$\ddot{H}_n\left(\frac{\cos x + \sin x}{(\cos x - \sin x)^3}\right) = 4^{\binom{n}{2}} \left(\prod_{k=1}^{n-1} k!\right) \prod_{j=0}^{n-2} [(2j+3)(2j+4)]^{n-1-j}.$$

27.8. The reciprocal-sine case $(1+x)x/\sin x$. The function $f = (1+x)x/\sin x$ carries no free parameter to specialise, so we record its evaluation directly. By Theorem 18.1, with $Q(k)$ of (18.2),

$$\ddot{H}_{2m}(f) = \left(\frac{2}{3}\right)^m \prod_{k=0}^{m-1} Q(2k), \quad \ddot{H}_{2m+1}(f) = \left(\frac{2}{3}\right)^m \prod_{k=0}^{m-1} Q(2k+1),$$

a product of factorials — the deepest evaluation of the paper.

27.9. An elliptic deformation of the Euler numbers. The Jacobi-elliptic deformation $g_m = \frac{1 + \operatorname{sn}(x, m)}{\operatorname{cn}(x, m)}$ of the Euler generating function (with $g_0 = (1 + \sin x)/\cos x$) satisfies, by Theorem 19.1, $\ddot{H}_n(g_m) = (1-m)^{\binom{n}{2}} \ddot{H}_n(g_0)$ — the Euler value times a $\binom{n}{2}$ -power of the complementary modulus; the modulus $m = 1$ is degenerate ($\ddot{H}_n = 0$ for $n \geq 2$). We record $m = -1, 2, \frac{1}{2}$.

Corollary 27.31 ($m = -1$). For $g_{-1} = \frac{1 + \operatorname{sn}(x, -1)}{\operatorname{cn}(x, -1)}$,

$$\ddot{H}_n(g_{-1}) = \prod_{k=1}^{n-1} k! (2k)!.$$

Corollary 27.32 ($m = 2$). For $g_2 = \frac{1 + \operatorname{sn}(x, 2)}{\operatorname{cn}(x, 2)}$,

$$\ddot{H}_n(g_2) = \left(-\frac{1}{2}\right)^{\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!.$$

Corollary 27.33 ($m = \frac{1}{2}$). For $g_{1/2} = \frac{1 + \operatorname{sn}(x, \frac{1}{2})}{\operatorname{cn}(x, \frac{1}{2})}$,

$$\ddot{H}_n(g_{1/2}) = \left(\frac{1}{4}\right)^{\binom{n}{2}} \prod_{k=1}^{n-1} k! (2k)!.$$

27.10. An algebraic family $(1+x)/(1-x^2)^{s/2}$. By Theorem 20.1 the family $f(x) = (1+x)/(1-x^2)^{s/2}$, with moments $a_{2k} = \frac{(2k)!}{k!} (s/2)_k$, $a_{2k+1} = (2k+1)a_{2k}$, has a closed product evaluation with linear factors $(s+2j-2)^{n-j}$. (A scale parameter t , i.e. $(1-tx^2)^{-s/2}$, only multiplies $\ddot{H}_n$ by $t^{\binom{n}{2} + \lfloor (n-1)^2/4 \rfloor}$, so we take $t = 1$.)

The determinant is nonzero for all odd s — including negative ones, where $f = (1+x)(1-x^2)^{-s/2}$ is a genuine square-root series. For even $s \leq 0$, by contrast, f is a polynomial, the

moments terminate, and $\ddot{H}_n$ vanishes for $n \geq 2 - \frac{s}{2}$ (the factor $s + 2j - 2$ being zero at $j = 1 - \frac{s}{2}$; thus $\ddot{H}_n = 0$ for $n \geq 2$ at $s = 0$ and for $n \geq 3$ at $s = -2$). We record the cases $s = -3, -1, 1, 2, 3$.

Corollary 27.34 ($s = -3$). For $f = (1+x)(1-x^2)^{3/2}$,

$$\ddot{H}_n(f) = \left(\prod_{k=1}^{n-1} (2k)! \right) \prod_{j=1}^{n-1} (2j-5)^{n-j},$$

nonzero, of sign -1 for all $n \geq 2$.

Corollary 27.35 ($s = -1$). For $f = (1+x)\sqrt{1-x^2}$,

$$\ddot{H}_n(f) = \left(\prod_{k=1}^{n-1} (2k)! \right) \prod_{j=1}^{n-1} (2j-3)^{n-j},$$

nonzero, of sign $(-1)^{n-1}$.

Corollary 27.36 ($s = 1$). For $f = \frac{1+x}{\sqrt{1-x^2}}$,

$$\ddot{H}_n(f) = \left(\prod_{k=1}^{n-1} (2k)! \right) \prod_{j=1}^{n-1} (2j-1)^{n-j}.$$

Corollary 27.37 ($s = 2$). For $f = \frac{1+x}{1-x^2}$ (so that $a_n = n!$),

$$\ddot{H}_n(f) = \left(\prod_{k=1}^{n-1} (2k)! \right) \prod_{j=1}^{n-1} (2j)^{n-j} = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} (2k)! k!.$$

Corollary 27.38 ($s = 3$). For $f = \frac{1+x}{(1-x^2)^{3/2}}$,

$$\ddot{H}_n(f) = \left(\prod_{k=1}^{n-1} (2k)! \right) \prod_{j=1}^{n-1} (2j+1)^{n-j}.$$

27.11. The squared algebraic family $(1+x)^2/(1-x^2)^{s/2}$. By Theorem 21.1 the family $f(x) = (1+x)^2/(1-x^2)^{s/2}$ has, for $s \neq 3$, a closed product evaluation whose last factor is a monic polynomial in s of degree $n-1$ carrying the denominator $s-3$; the excluded value $s=3$ is where its numerator vanishes (Remark 21.3). We record $s=2, 4$.

Corollary 27.39 ($s = 2$). For $f = \frac{(1+x)^2}{1-x^2} = \frac{1+x}{1-x}$ (so $a_0 = 1$ and $a_m = 2m!$ for $m \geq 1$), the last factor is $2^n(n-1)! - (2n-1)!!$ and

$$\ddot{H}_n(f) = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} (2k)! \prod_{i=1}^{n-2} i! (2^n(n-1)! - (2n-1)!!) = 1, 4, 384, -39813120, \dots$$

Corollary 27.40 ($s = 4$). For $f = \frac{(1+x)^2}{(1-x^2)^2}$ the factor $(s-4)$ kills the deformation, the last factor reducing to $(2n-1)!!$, and

$$\ddot{H}_n(f) = 2^{\binom{n}{2}} \prod_{k=1}^{n-1} (2k)! \prod_{i=1}^{n-2} (2)_i (2n-1)!! = 1, 12, 11520, 2786918400, \dots$$

27.12. The Bessel (s, t) family $\text{cosb}_s + \int_0^x \text{cosb}_t$. By Theorem 23.1 the family $f_{s,t} = \text{cosb}_s(x) + \int_0^x \text{cosb}_t(y) dy$ has the closed evaluation (23.4), whose sign and double product together form the signed factor $\Omega(2(t-s))$ of the trigonometric family (Section 7.3). The determinant is a nonzero product exactly on the half-integer offsets $t-s \in \frac{1}{2} + \mathbb{Z}$ (Corollary 23.2); its double shift $\ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'')$ is a closed multiple of $\ddot{H}_n(f)$ (Proposition 24.2), where f'' is computed from the derivative ladder $\text{cosb}'_\nu = -\frac{x}{2\nu+2} \text{cosb}_{\nu+1}$. We record the two half-integer members with $s, t \in \{0, \frac{1}{2}\}$.

Corollary 27.41 ($s = 0, t = \frac{1}{2}$). Here $\text{cosb}_0 = J_0$ and $\int_0^x \text{cosb}_{1/2}(y) dy = \text{Si}(x)$, the sine integral, so $f = J_0(x) + \text{Si}(x)$. Specialising (23.4) and converting the half-integer Pochhammer symbols into factorials, every block merges into a single product per parity:

$$\begin{aligned}\ddot{H}_{2m}(f) &= \frac{1}{2^{\binom{2m}{2}}} \prod_{k=0}^{m-1} \frac{((2k)!)^2 (2m+2k)!}{(4m+2k-1)!}, \\ \ddot{H}_{2m+1}(f) &= \frac{(2m)!(4m)!}{2^{m(2m+5)} m! (3m)!} \prod_{k=0}^{m-1} \frac{((2k)!)^2 (2m+2k)!}{(4m+2k+1)!}.\end{aligned}$$

For the double shift,

$$f'' = \frac{x^2}{8} \text{cosb}_2 - \frac{1}{2} \text{cosb}_1 - \frac{x}{3} \text{cosb}_{3/2}, \quad \ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (-1)^n \frac{3((2n)!)^2}{2^{n+1} n! (3n)!} \ddot{H}_n(f).$$

Corollary 27.42 ($s = \frac{1}{2}, t = 0$). Here $f = \frac{\sin x}{x} + \int_0^x J_0(y) dy$. The even-order determinants coincide with those of Corollary 27.41 up to sign,

$$\ddot{H}_{2m}(f) = (-1)^m \ddot{H}_{2m}(J_0 + \text{Si}) = \frac{(-1)^m}{2^{\binom{2m}{2}}} \prod_{k=0}^{m-1} \frac{((2k)!)^2 (2m+2k)!}{(4m+2k-1)!},$$

while at odd order $n = 2m+1$

$$\ddot{H}_{2m+1}(f) = (-1)^m (2m+1) 2^{m(3-2m)} \frac{m! (3m)! (4m)!}{(6m+1)!} \prod_{k=0}^{m-1} \frac{((2k)!)^2 (2m+2k)!}{(4m+2k+1)!}.$$

For the double shift,

$$f'' = \frac{x^2}{15} \text{cosb}_{5/2} - \frac{1}{3} \text{cosb}_{3/2} - \frac{x}{2} \text{cosb}_1, \quad \ddot{H}_n^{(2)}(f) = \ddot{H}_n(f'') = (-1)^n \frac{\lfloor 3n/2 \rfloor ((2n)!)^2}{2^n n! (3n)!} \ddot{H}_n(f).$$

27.13. The multiplicative Bessel family $(1+x) \text{cosb}_\nu^2$. By Theorem 25.4 the even-order determinants of $f_\nu = (1+x) \text{cosb}_\nu^2$ factor completely into linear forms over $\mathbb{Q}$, by the closed evaluation (25.12); at odd orders no product form exists. We record $\nu = 0, \frac{1}{2}, 1$, converting all half-integer Pochhammer symbols into factorials.

Corollary 27.43 ($\nu = 0$). Here $\text{cosb}_0 = J_0$, and

$$\begin{aligned}f &= (1+x) J_0(x)^2, \\ \ddot{H}_{2m}(f) &= \frac{(-1)^m}{2^{m(6m-5)}} \prod_{k=0}^{m-1} \left(\frac{(2k)!}{k!} \right)^4 \prod_{i=0}^{2m-1} \frac{((2i)!)^2}{i! ((m-1+i)!)^2}.\end{aligned}$$

Corollary 27.44 ($\nu = \frac{1}{2}$). Here $\text{cosb}_{1/2} = \sin x/x$, and

$$\begin{aligned}f &= (1+x) \frac{\sin^2 x}{x^2}, \\ \ddot{H}_{2m}(f) &= (-1)^m 2^{m(8m-5)} \prod_{k=0}^{m-1} (2k)! (2k+1)! \prod_{i=0}^{2m-1} \frac{(2i)! i! (m-1+i)!}{(2m+2i-1)! (m+i)!}.\end{aligned}$$

Corollary 27.45 ($\nu = 1$). Here $\cos b_1 = 2J_1(x)/x$, and

$$f = 4(1+x) \frac{J_1(x)^2}{x^2},$$

$$\ddot{H}_{2m}(f) = \frac{(-1)^m}{2^{m(6m-5)} 3^m} \prod_{k=0}^{m-1} \frac{(2k)! ((2k+1)!)^2 (2k+4)!}{(k!)^3 (k+2)!} \prod_{i=0}^{2m-1} \frac{(2i)!(2i+1)!}{i!(m+i)!(m+i+1)!}.$$

28. THE LINDSTRÖM–GESSEL–VIENNOT APPROACH

The determinant $\ddot{H}_n$ counts something, and the natural first attempt at a combinatorial evaluation is the Lindström–Gessel–Viennot (LGV) lemma [30, 13, 23], which turns a determinant of path counts into a signed sum over non-intersecting path families. We recall the lemma, use it to see *why* the classical Hankel determinant is a clean product, and then read off the single change that the dilation makes. In the end this route did not deliver our evaluations — the algebraic reductions **M1–M6** did — but it locates precisely where the difficulty enters, and we record it for orientation.

28.1. Hankel minors as path systems. It is convenient to write, for finite index sets $R = \{r_0 < \dots < r_{n-1}\}$ and $C = \{c_0 < \dots < c_{n-1}\}$,

$$H(R; C) := \det(a_{r_i+c_j})_{0 \leq i, j \leq n-1},$$

the corresponding minor of the infinite Hankel matrix (a_{p+q}) . In this notation the classical and dilated determinants are

$$H_n = H(\{0, 1, \dots, n-1\}; \{0, 1, \dots, n-1\}), \quad \ddot{H}_n = H(\{0, 2, \dots, 2n-2\}; \{0, 1, \dots, n-1\}) :$$

they share the column set $\{0, 1, \dots, n-1\}$ and differ only in that the row (“top”) indices are *dilated* from $0, 1, \dots, n-1$ to $0, 2, \dots, 2n-2$ — exactly the even-row selection of Definition 1.1.

Lemma 28.1 (Lindström–Gessel–Viennot [30, 13]). *Let $\mathcal{D}$ be a locally finite acyclic directed graph with edge weights in a commutative ring, and for vertices u, v let $e(u, v) = \sum_{P: u \rightarrow v} \prod_{\text{edges of } P} w$ be the weight generating function of directed paths. For source and sink tuples $A = (A_0, \dots, A_{n-1})$ and $B = (B_0, \dots, B_{n-1})$,*

$$\det(e(A_i, B_j))_{0 \leq i, j \leq n-1} = \sum_{(P_0, \dots, P_{n-1})} \text{sgn}(\sigma) \prod_{k=0}^{n-1} w(P_k),$$

the sum over families in which P_k runs from A_k to $B_{\sigma(k)}$ for some permutation σ and the P_k are pairwise vertex-disjoint. If the endpoints are nonpermutable — every vertex-disjoint family has $\sigma = \text{id}$ — the determinant is the subtraction-free weight of the non-intersecting families joining A_k to B_k .

Now realise the moments as paths. If (a_k) are the moments of a quasi-definite functional with J -fraction coefficients (b_h, λ_h) — the (c_n, λ_n) of Section 2.4, renamed here because c denotes column indices — then by Flajolet’s theory [12, 46] a_k is the weight of all Motzkin paths of length k from height 0 to height 0, a level step at height h carrying weight b_h and a down-step from height h weight λ_h . Place a source A_i on the horizontal axis at abscissa $-r_i$ and a sink B_j at abscissa c_j ; a directed path $A_i \rightarrow B_j$ is then a Motzkin path of length $r_i + c_j$ that stays weakly above the axis, so $e(A_i, B_j) = a_{r_i+c_j}$ and

$$H(R; C) = \det(e(A_i, B_j))_{0 \leq i, j \leq n-1}.$$

Since all sources lie at abscissa ≤ 0 and all sinks at ≥ 0 , the endpoints are nonpermutable: the only vertex-disjoint families are the nested ones joining A_i to B_i (Figures 1 and 2). Hence *every* such minor $H(R; C)$ is a subtraction-free count of non-intersecting Motzkin paths.

28.2. What the dilation changes. For the classical H_n (Figure 1) the two index sets coincide, $R = C = \{0, 1, \dots, n-1\}$: the sources and sinks are equally spaced and symmetric about the origin, the nested family fills every level in turn, and its weight telescopes into the Heilmann product $H_n = \lambda_1^{n-1} \lambda_2^{n-2} \cdots \lambda_{n-1}$ [20, 46] (here $a_0 = 1$). This is the combinatorial face of the J -fraction, and it is what makes H_n “simple”.

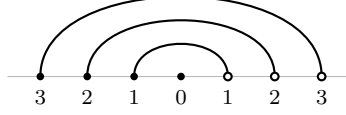


FIGURE 1. The classical $H_n = H(\{0, 1, 2, 3\}; \{0, 1, 2, 3\})$: equally spaced, symmetric nested arches.

The dilated determinant is the *same* picture with the sources pulled apart. In $\ddot{H}_n$ the top starting points move from $0, 1, \dots, n-1$ to $0, 2, \dots, 2n-2$ while the sinks are unchanged (Figure 2). The family is still forced and non-intersecting, so $\ddot{H}_n$ too is a subtraction-free count; but the source and sink spacings no longer match (2 against 1). The nested paths must now span a growing number of levels between consecutive sinks, the clean level-by-level telescoping that produced the J -fraction breaks down, and the lemma delivers only an unwieldy positive sum — not a product. This is the combinatorial shadow of the analytic obstruction of Section 1: interleaving the even and odd moments destroys the three-term recurrence on which the J -fraction rests. It is exactly here that the single-graph LGV picture stalls.

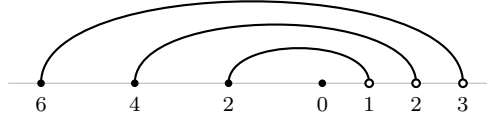


FIGURE 2. The dilated $\ddot{H}_n = H(\{0, 2, 4, 6\}; \{0, 1, 2, 3\})$: rows dilated, arches skewed, telescoping lost.

28.3. The even–odd splitting: the biorthogonal picture. There is nonetheless a way to keep $\ddot{H}_n$ subtraction-free, by splitting it into *two* path systems — one carried by the even moments, one by the odd moments. This is the combinatorial face of the biorthogonal reduction **M2**. Write $p = \lceil n/2 \rceil$ and $q = \lfloor n/2 \rfloor$. In $\ddot{H}_n = (a_{2i+j})_{0 \leq i, j \leq n-1}$ an even column $j = 2c$ has entries $a_{2i+2c} = a_{2(i+c)}$, involving only the even moments a_{2k} (the moments of the even functional $\mathcal{S}$); an odd column $j = 2c + 1$ has entries $a_{2i+2c+1} = a_{2(i+c)+1}$, involving only the odd moments a_{2k+1} (the moments of the odd functional $\mathcal{T}$). Move the p even columns to the front and the q odd columns to the back; the shuffle has sign $(-1)^{\binom{p}{2}}$ and leaves the block matrix $[M_{\mathcal{S}} \mid M_{\mathcal{T}}]$ with

$$M_{\mathcal{S}} = (a_{2(i+c)})_{0 \leq i \leq n-1, 0 \leq c \leq p-1} \quad (n \times p), \quad M_{\mathcal{T}} = (a_{2(i+c)+1})_{0 \leq i \leq n-1, 0 \leq c \leq q-1} \quad (n \times q).$$

Laplace expansion along the first p columns carries the sign $(-1)^{\sum_{i \in I} i + \binom{p}{2}}$ on the row subset I ; the $\binom{p}{2}$ cancels against the shuffle, leaving the subtraction-free-looking

$$\ddot{H}_n = \sum_{\substack{I \subseteq \{0, \dots, n-1\} \\ |I|=p}} (-1)^{\sum_{i \in I} i} \det(a_{2(i+c)})_{i \in I, 0 \leq c \leq p-1} \cdot \det(a_{2(i+c)+1})_{i \in I^c, 0 \leq c \leq q-1},$$

the sum over all p -subsets I of the rows, with complementary q -subset I^c . Each summand is a product of a $p \times p$ minor of the even Hankel matrix $(a_{2k+2l})_{k,l}$ and a $q \times q$ minor of the odd Hankel matrix $(a_{2k+2l+1})_{k,l}$; in the $H(R; C)$ notation above, and writing $H_{\mathcal{S}}, H_{\mathcal{T}}$ for Hankel minors of the even and odd moment sequences,

$$\ddot{H}_n = \sum_I (-1)^{\sum_{i \in I} i} H_{\mathcal{S}}(I; \{0, \dots, p-1\}) H_{\mathcal{T}}(I^c; \{0, \dots, q-1\}).$$

Combinatorially the two factors are non-intersecting Motzkin-path families in the two *separate* graphs carrying the J -fractions of $\mathcal{S}$ and of $\mathcal{T}$ — each of $(a_{2k})_k$ and $(a_{2k+1})_k$ being again a moment sequence. Drawing the even family above the axis and the odd family mirrored below it exhibits a whole term at once: the rows are two-coloured, one colour arching up and the complementary colour arching down, sharing the horizontal axis (Figure 3). The dilated determinant is thus a signed sum of products of two classical Hankel minors, one from each parity. The price is that the sum ranges over all $\binom{n}{p}$ splittings of the rows between the two systems, and nothing in the picture explains why it should collapse to a product. That collapse is captured algebraically by the biorthogonal reduction **M2**, which biorthogonalises against $\mathcal{S}$ and $\mathcal{T}$ simultaneously; it is exactly here that we leave the LGV picture for the reductions **M1**–**M6**.

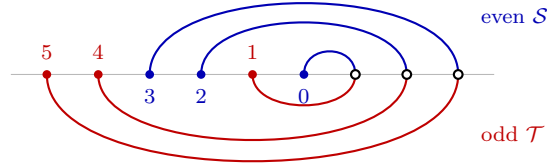


FIGURE 3. The even-odd splitting of $\ddot{H}_n$ ($n = 6$, generic): even system $I = \{0, 2, 3\}$ above, odd system $I^c = \{1, 4, 5\}$ mirrored below.

29. CLASSICAL HANKEL DETERMINANTS

Alongside the dilated determinant $\ddot{H}_n = \det(a_{2i+j})_{0 \leq i, j \leq n-1}$, each moment sequence produced in this paper also has a *classical* Hankel determinant

$$H_n(\mu) = \det(\mu_{i+j})_{0 \leq i, j \leq n-1}.$$

The evaluation of such determinants is a classical and much-studied subject; for surveys and general techniques see [26, 27], and for recent product and near-product evaluations of Hankel determinants of combinatorial moment sequences see [6, 29, 11, 10]. Whenever the Jacobi continued fraction of $\sum_k \mu_k t^k$ has been exhibited, this determinant is immediate: by Heilermann's formula (2.4) [20, 47],

$$H_n(\mu) = \mu_0^n \prod_{k=1}^{n-1} \lambda_k^{n-k}, \quad (29.1)$$

the λ_k being the denominator coefficients of the J -fraction. We record here, without proof, the resulting products. Several are classical or already in the literature, and are included only for completeness; we flag these below with their sources. The Bessel determinant is the classical Hankel determinant of Jacobi (Beta) moments [26], and the Springer determinant (the case $t = r = 1$) is known [3, 39] (sequence **A091804** in [33]). The remaining ones — for $(2k+1)E_{2k}$, for the two reciprocal-sine sequences, and for the two-parameter family $1/(\cos x - t \sin x)^r$ — appear not to have been recorded in this form. We begin with the sequence of Remark 11.3.

The sequence $(2k+1)E_{2k}$. Recall the secant numbers E_{2k} , the Taylor coefficients of the secant,

$$\frac{1}{\cos x} = \sum_{k \geq 0} E_{2k} \frac{x^{2k}}{(2k)!}, \quad E_0, E_2, E_4, E_6, \dots = 1, 1, 5, 61, 1385, \dots,$$

and form the sequence

$$\mu_k := (2k+1) E_{2k}, \quad \mu_0, \mu_1, \mu_2, \mu_3, \mu_4, \dots = 1, 3, 25, 427, 12465, \dots$$

By Remark 11.3 its generating function has the Jacobi continued fraction

$$\sum_{k \geq 0} (2k+1)E_{2k} t^k = \frac{1}{1 - c_0^* t - \frac{\lambda_1^* t^2}{1 - c_1^* t - \frac{\lambda_2^* t^2}{1 - c_2^* t - \frac{\lambda_3^* t^2}{1 - \ddots}}}}, \quad c_m^* = 8m^2 + 8m + 3, \quad \lambda_m^* = (2m)^4,$$

so that $\mu_0 = 1$ and, explicitly,

$$c_0^* = 3, \quad c_1^* = 19, \quad c_2^* = 51, \dots, \quad \lambda_1^* = 16, \quad \lambda_2^* = 256, \quad \lambda_3^* = 1296, \dots$$

Proposition 29.1. *For the sequence $\mu_k = (2k+1)E_{2k}$ the classical Hankel determinant is, for all $n \geq 1$,*

$$H_n(\mu) = \det((2(i+j)+1)E_{2(i+j)})_{0 \leq i,j \leq n-1} = \prod_{k=1}^{n-1} (2k)^{4(n-k)} = 2^{4\binom{n}{2}} \prod_{k=1}^{n-1} (k!)^4.$$

This is (29.1) with $\mu_0 = 1$ and $\lambda_k^* = (2k)^4$. The first values are

$$H_1, H_2, H_3, H_4, \dots = 1, 2^4, 2^{16}, 2^{32} 3^4, \dots$$

The sequence μ_k is A009843 in [33], where this continued fraction is also recorded; the Hankel determinants of the Euler numbers themselves are evaluated in [18], but the determinant of $(2k+1)E_{2k}$ above appears to be new.

The sequence $(2k+1)b_k$ of the reciprocal sine. Recall from Section 18 the even part $g(x) = x/\sin x$ of the reciprocal-sine function and its scaled Taylor coefficients b_k ,

$$\frac{x}{\sin x} = \sum_{k \geq 0} b_k \frac{x^{2k}}{(2k)!}, \quad b_0, b_1, b_2, b_3, \dots = 1, \frac{1}{3}, \frac{7}{15}, \frac{31}{21}, \dots,$$

and form the sequence

$$\mu_k := (2k+1)b_k, \quad \mu_0, \mu_1, \mu_2, \mu_3, \dots = 1, 1, \frac{7}{3}, \frac{31}{3}, \dots$$

This is the odd moment functional $\mathcal{T}[y^k] = (2k+1)b_k$ of that section. By Proposition 18.3 its generating function has the Jacobi continued fraction

$$\sum_{k \geq 0} (2k+1)b_k t^k = \frac{1}{1 - c_0^{\mathcal{T}} t - \frac{\lambda_1^{\mathcal{T}} t^2}{1 - c_1^{\mathcal{T}} t - \frac{\lambda_2^{\mathcal{T}} t^2}{1 - \ddots}}}, \quad c_m^{\mathcal{T}} = 2m^2 + 2m + 1, \quad \lambda_m^{\mathcal{T}} = \frac{4m^6}{(2m-1)(2m+1)},$$

so that $\mu_0 = 1$ and, explicitly,

$$c_0^{\mathcal{T}} = 1, \quad c_1^{\mathcal{T}} = 5, \quad c_2^{\mathcal{T}} = 13, \dots, \quad \lambda_1^{\mathcal{T}} = \frac{4}{3}, \quad \lambda_2^{\mathcal{T}} = \frac{256}{15}, \quad \lambda_3^{\mathcal{T}} = \frac{2916}{35}, \dots$$

Proposition 29.2. *For the sequence $\mu_k = (2k+1)b_k$ the classical Hankel determinant is, for all $n \geq 1$,*

$$H_n(\mu) = \det((2(i+j)+1)b_{i+j})_{0 \leq i,j \leq n-1} = \prod_{k=1}^{n-1} \left(\frac{4k^6}{(2k-1)(2k+1)} \right)^{n-k} = \prod_{k=0}^{n-1} \frac{16^k (k!)^8}{(2k)!(2k+1)!}.$$

This is (29.1) with $\mu_0 = 1$ and $\lambda_k^{\mathcal{T}} = \frac{4k^6}{(2k-1)(2k+1)}$; the second form is the product $\prod_{k=0}^{n-1} h_k^{\mathcal{T}}$ of the norms (18.5). The first values are

$$H_1, H_2, H_3, H_4, \dots = 1, \frac{4}{3}, \frac{4096}{135}, \frac{50331648}{875}, \dots$$

The reciprocal sine b_k itself. One may equally take the coefficients b_k as the moment sequence — this is the even functional $\mathcal{S}[y^k] = b_k$ of Section 18. Unlike the sequences above, $\sum_k b_k t^k$ is given not by a Jacobi but by a Stieltjes continued fraction: by Proposition 18.4,

$$\sum_{k \geq 0} b_k t^k = \frac{1}{1 - \frac{u_1 t}{1 - \frac{u_2 t}{1 - \frac{u_3 t}{1 - \ddots}}}}, \quad u_n = \frac{n^4}{(2n-1)(2n+1)},$$

with $\mu_0 = b_0 = 1$ and, explicitly, $u_1 = \frac{1}{3}$, $u_2 = \frac{16}{15}$, $u_3 = \frac{81}{35}$, $\dots$. For an S -fraction Heilermann's formula takes the telescoped form (2.8), $H_n = \prod_{i=0}^{n-1} \prod_{j=1}^{2i} u_j$.

Proposition 29.3. *For the sequence $b_k = (2k)! [x^{2k}](x/\sin x)$ the classical Hankel determinant is, for all $n \geq 1$,*

$$H_n(b) = \det(b_{i+j})_{0 \leq i, j \leq n-1} = \prod_{i=0}^{n-1} \prod_{j=1}^{2i} \frac{j^4}{(2j-1)(2j+1)} = \prod_{l=0}^{n-1} \frac{16^l ((2l)!)^6}{(4l)!(4l+1)!}.$$

The second form is the product $\prod_{l=0}^{n-1} h_l^S$ of the norms (18.9). The first values are

$$H_1, H_2, H_3, H_4, \dots = 1, \frac{16}{45}, \frac{65536}{55125}, \frac{4294967296}{18883865}, \dots$$

The shifted sequence $\mu'_k = b_{k+1}$ has, by the odd companion of (2.8),

$$\det(b_{i+j+1})_{0 \leq i, j \leq n-1} = \prod_{l=0}^{n-1} \prod_{j=1}^{2l+1} u_j = \prod_{l=0}^{n-1} h_l^S u_{2l+1}, \quad u_{2l+1} = \frac{(2l+1)^4}{(4l+1)(4l+3)},$$

with first values $\frac{1}{3}, \frac{48}{175}, \frac{65536}{11319}, \dots$

The Bessel-cosine moments. For $s > -1$ recall from Section 23 the Bessel cosine $\text{cosb}_s(x) = \Gamma(s+1)(2/x)^s J_s(x)$ and its scaled Taylor coefficients μ_k ,

$$\text{cosb}_s(x) = \sum_{k \geq 0} \mu_k \frac{x^{2k}}{(2k)!}, \quad \mu_k = (-1)^k \frac{(\frac{1}{2})_k}{(s+1)_k},$$

the even moment sequence $\mu_k = a_{2k}$ of the family (23.2). Its ordinary generating function is the Gauss series $\sum_{k \geq 0} \mu_k z^k = {}_2F_1(\frac{1}{2}, 1; s+1; -z)$, which by Lemma 23.5 has the Stieltjes continued fraction

$$\sum_{k \geq 0} \mu_k z^k = \frac{1}{1 - \frac{u_1 z}{1 - \frac{u_2 z}{1 - \frac{u_3 z}{1 - \ddots}}}}, \quad u_j = -\frac{j(j+2s-1)}{4(s+j-1)(s+j)},$$

with, explicitly, $u_1 = -\frac{1}{2(s+1)}$, $u_2 = -\frac{2s+1}{2(s+1)(s+2)}$, $u_3 = -\frac{3(s+1)}{2(s+2)(s+3)}$, $\dots$

Proposition 29.4. *For the Bessel-cosine moments $\mu_k = (-1)^k (\frac{1}{2})_k / (s+1)_k$ the classical Hankel determinant is, for all $n \geq 1$ and as an identity in $\mathbb{Q}(s)$,*

$$H_n(\mu) = \det(\mu_{i+j})_{0 \leq i, j \leq n-1} = \prod_{i=0}^{n-1} h_i^S = \prod_{i=0}^{n-1} \frac{(2i)! (s + \frac{1}{2})_i}{4^i (s+i)_i (s+1)_{2i}}.$$

This is the S -fraction form of Heilermann's formula (2.8) with coefficients u_j ; the product is that of the norms h_i^S of (23.6). The odd part $a_{2k+1} = (-1)^k (\frac{1}{2})_k / (t+1)_k$ of the same family gives the identical statement with t in place of s . The first values are, e.g.,

$$H_1 = 1, \quad H_2 = \frac{2s+1}{4(s+1)^2(s+2)}.$$

This — and, below, its two shifts — is the classical Hankel determinant of Jacobi (Beta) moments [26]: the entries $\mu_k = (-1)^k (\frac{1}{2})_k / (s+1)_k$ are the moments of a Beta weight, and multiplication by y keeps it Beta. The shifted sequence $\mu_{k+1} = a_{2k+2}$ gives, again by the odd companion of (2.8),

$$\det(\mu_{i+j+1})_{0 \leq i, j \leq n-1} = \prod_{i=0}^{n-1} h_i^S u_{2i+1}, \quad u_{2i+1} = -\frac{(2i+1)(s+i)}{2(s+2i)(s+2i+1)},$$

e.g. $H_1^{(1)} = -\frac{1}{2(s+1)}$ and $H_2^{(1)} = \frac{3(2s+1)}{16(s+1)^2(s+2)^2(s+3)}$. The double shift is clean as well: the doubly shifted sequence is again of Beta type,

$$\mu_{k+2} = a_{2k+4} = \frac{3}{4(s+1)(s+2)} (-1)^k \frac{(\frac{5}{2})_k}{(s+3)_k},$$

so that

$$\det(\mu_{i+j+2})_{0 \leq i, j \leq n-1} = \left(\frac{3}{4(s+1)(s+2)} \right)^n \prod_{k=0}^{n-1} \frac{k! (\frac{5}{2})_k (s+\frac{1}{2})_k}{(s+k+2)_k (s+3)_{2k}},$$

$$\text{e.g. } H_1^{(2)} = \frac{3}{4(s+1)(s+2)} \text{ and } H_2^{(2)} = \frac{45(2s+1)}{64(s+1)^2(s+2)^2(s+3)^2(s+4)}.$$

The Springer number family $1/(\cos x - t \sin x)^r$. Section 16 evaluates the *dilated* determinant of this family; its coefficients form, at $t = r = 1$, the Springer numbers

$$\frac{1}{\cos x - \sin x} = \sum_{n \geq 0} S_n \frac{x^n}{n!}, \quad S_n = 1, 1, 3, 11, 57, 361, 2763, \dots,$$

a moment sequence [39]. We record the *classical* Hankel determinant of the full moment sequence $a_n = n! [x^n](\cos x - t \sin x)^{-r}$ of the whole family, whose ordinary generating function has a particularly clean Jacobi continued fraction: the underlying orthogonal polynomials are, up to rescaling, the Meixner–Pollaczek polynomials.

Proposition 29.5. *Let $r > 0$. The ordinary generating function of $a_n = n! [x^n](\cos x - t \sin x)^{-r}$ has the Jacobi continued fraction*

$$\sum_{n \geq 0} a_n z^n = \frac{1}{1 - c_0 z - \frac{\lambda_1 z^2}{1 - c_1 z - \frac{\lambda_2 z^2}{1 - \ddots}}}, \quad c_n = t(2n+r), \quad \lambda_n = (1+t^2)n(n+r-1) \quad (29.2)$$

(so $\mu_0 = 1$ and, at $t = r = 1$, $c_n = 2n+1$, $\lambda_n = 2n^2$), and consequently, for all $n \geq 1$,

$$H_n = \det(a_{i+j})_{0 \leq i, j \leq n-1} = (1+t^2)^{\binom{n}{2}} \prod_{i=1}^{n-1} i! (r)_i.$$

Proof. Both c_n, λ_n and the moments a_n are rational in (t, r) — indeed $a_n \in \mathbb{Q}[t]$ of degree $\leq n$, by Faà di Bruno as in the proof of Theorem 16.1 — so it suffices to prove (29.2) for real t and real $r > 0$, a set with nonempty interior. Write $\phi = \arctan t$, so that $\cos x - t \sin x = \sqrt{1+t^2} \cos(x+\phi)$ and

$$f(x) := (\cos x - t \sin x)^{-r} = (1+t^2)^{-r/2} \sec^r(x+\phi).$$

By the Fourier evaluation of the Meixner–Pollaczek weight [25, §9.7],

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{2\psi\eta} \left| \Gamma\left(\frac{r}{2} + i\eta\right) \right|^2 d\eta = \Gamma(r) (2 \cos \psi)^{-r} \quad (|\psi| < \frac{\pi}{2}),$$

so the probability measure

$$d\mu(\xi) = \frac{1}{N} e^{\xi \arctan t} \left| \Gamma\left(\frac{r}{2} + \frac{i\xi}{2}\right) \right|^2 d\xi, \quad N = 4\pi \Gamma(r) 2^{-r} (1+t^2)^{r/2},$$

has moment generating function $\int_{\mathbb{R}} e^{x\xi} d\mu(\xi) = f(x)$ for $|x + \phi| < \frac{\pi}{2}$ (substitute $\xi = 2\eta$, $\psi = x + \phi$, and use $\cos(x + \phi)^{-r}$). Matching Taylor coefficients at $x = 0$ gives $a_n = \int_{\mathbb{R}} \xi^n d\mu(\xi)$: the a_n are the moments of μ .

Under the affine change $\xi = 2\eta$, μ is the Meixner–Pollaczek measure with parameters $\lambda = \frac{r}{2}$ and angle $\varphi = \phi + \frac{\pi}{2} \in (0, \pi)$, whose monic orthogonal polynomials satisfy $\hat{p}_{n+1} = (\eta - \beta_n)\hat{p}_n - \gamma_n\hat{p}_{n-1}$ with [25, §9.7]

$$\beta_n = -(n + \lambda) \cot \varphi, \quad \gamma_n = \frac{n(n + 2\lambda - 1)}{4 \sin^2 \varphi}.$$

Here $\cot \varphi = -\tan \phi = -t$ and $\sin^2 \varphi = \cos^2 \phi = 1/(1 + t^2)$, and the doubling $\xi = 2\eta$ multiplies β_n by 2 and γ_n by 4; hence the monic μ -recurrence has coefficients

$$c_n = 2\beta_n = t(2n + r), \quad \lambda_n = 4\gamma_n = (1 + t^2)n(n + r - 1),$$

which is (29.2) (the theorem of Stieltjes, (2.5)). Finally Heilermann’s formula (29.1) with $\mu_0 = a_0 = 1$ and $h_i = \prod_{k=1}^i \lambda_k = (1 + t^2)^i i! (r)_i$ gives $H_n = \prod_{i=0}^{n-1} h_i = (1 + t^2)^{\binom{n}{2}} \prod_{i=1}^{n-1} i! (r)_i$. $\square$

The product $\prod_{i=1}^{n-1} i! (r)_i$ is itself the classical Hankel determinant of $1/(1 - x)^r$ (moments $(r)_n$), so the evaluation mirrors Theorem 16.1 exactly, with the *dilated* factor $(t(t^2 + 1))^{\binom{n}{2}}$ there replaced by the *classical* $(1 + t^2)^{\binom{n}{2}}$ — the extra t dropping out because the classical determinant depends only on the λ_n , not on the c_n . At $t = r = 1$ it gives the Springer Hankel determinant

$$H_n = 2^{\binom{n}{2}} \prod_{i=1}^{n-1} (i!)^2 = 1, 2, 32, 9216, 84934656, \dots$$

This last is known — the Springer numbers are a Meixner–Pollaczek moment sequence [3, 39], and the values 1, 2, 32, 9216, ... are A091804 in [33]. The two-parameter family (29.2), by contrast, does not seem to have been recorded.

30. CONCLUDING REMARKS

We introduced the dilated Hankel determinant $\ddot{H}_n = \det(a_{2i+j})$ (Definition 1.1) in order to study the Chapoton–Han root conjecture (Section 26), and observed experimentally that for a surprisingly large collection of classical sequences it factors into small primes. Unlike the ordinary Hankel determinant, however, it admits no single universal evaluation: the even and odd parts of $\mathbf{a}$ interact through a connection determinant that collapses only under favourable circumstances. We therefore developed the six reductions **M1**–**M6** of Section 3 — four running through the paper, and two more specialised, the contiguous-relation method **M5** and the rank-one/matrix-determinant-lemma method **M6** — and with them evaluated, each as an explicit product, the Euler number family and its shifts, the Gaussian, secant, algebraic, Springer, elliptic and derivative-ladder families, and the Beta family; the reciprocal-sine function required, on top of **M2**, the new Catalan-determinant evaluation of Theorem 18.8.

What is still missing is a *theory*. We have no criterion that decides, in advance, whether a given generating function has a closed-form dilated Hankel determinant, and the boundary is delicate: an innocuous modification of a “nice” function usually destroys the product form, and we do not know why one perturbation succeeds where a neighbouring one fails. We record here the main questions this leaves open.

30.1. Further directions. Four problems seem especially natural.

Problem 30.1. Find further sequences $\mathbf{a}$ — or, more ambitiously, a criterion characterising them — whose dilated Hankel determinant $\ddot{H}_n$ has a closed form, that is, factors into a product of linear factors in n (equivalently, into small primes for every n).

Problem 30.2. The dilated Hankel determinant selects the rows $0, 2, 4, \dots$ of the infinite Hankel matrix. For an integer $r \geq 2$ one may equally form the r -step determinant $\det(a_{ri+j})$. For the Beta family this is again a closed product for every step r : the Vandermonde reduction **M1** of Section 3, which factors $a_{x+j} = w(x)Q_j(x)$ with Q_j triangular, applies verbatim to any set of row

indices (see also the determinant calculus of Krattenthaler [26, 27]). The question is whether any family *beyond* the Beta family has a closed-form r -step determinant for $r \geq 3$.

Problem 30.3. Give a combinatorial proof of our dilated Hankel determinant evaluations by means of the Lindström–Gessel–Viennot lemma (Section 28). The dilated Hankel determinants studied in this paper all have a simple closed form — a product of small integers. For the *classical* Hankel determinant, the Lindström–Gessel–Viennot lemma gives a nice explanation of this fact: it realises the determinant as a positive count of families of non-intersecting lattice paths, which visibly factors. Is there an analogous combinatorial model that explains the product-of-small-integers property of the *dilated* Hankel determinant?

Problem 30.4. Our reductions rest on orthogonal polynomials and the Jacobi continued fraction (J -fraction) of the even and odd moment sequences, which exists only when each is quasi-definite. The *Hankel continued fraction* (H -fraction) [17, 16] is more general: it exists for *every* sequence and reads off the Hankel determinants even when some vanish and the J -fraction breaks down. When the even or odd part of $\mathbf{a}$ has no J -fraction, can the dilated determinant still be evaluated by an H -fraction analogue of the biorthogonal reduction **M2**?

30.2. Data availability. The closed forms were discovered and all the identities of this paper were checked with the computer algebra system **SageMath**. The **SageMath** programs and the output data used to discover and check the results of this paper are available, for independent verification, in the GitHub repository

<https://github.com/GuoniuHan/dilated>.

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ON THE USE OF ARTIFICIAL INTELLIGENCE

An artificial-intelligence assistant based on a large language model was used to help find and write some of the proofs. All arguments were verified independently with **SageMath** and reviewed by the author, who takes full responsibility for the correctness and the content of the paper.

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